FRCEM Emergency Medicine High-Yield Review

Comprehensive Deduplicated Study Book

32 Chapters | 1,529 Source Questions
Exam Tricks, Algorithms, Pitfalls, and Brief Arabic Synthesis

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Educational disclaimer: This book is a revision resource, not a substitute for local protocols, specialist advice, or independent clinical judgment. Verify time-sensitive recommendations against current local guidance before clinical use.

CHAPTER 01: SURGICAL EMERGENCIES

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Chapter 01: Surgical Emergencies — Final (compact FRCEM study notes)

1. Clinical overview and pathophysiology

- Definition: Surgical emergencies = time-critical conditions where rapid diagnosis + source control (surgery/debridement/airway) ± resuscitation determine outcome.
- Common entities: body-packing/pack rupture (stimulant/opioid toxicity/ obstruction), necrotising soft-tissue infection (including Fournier's), intra-abdominal sepsis (perforation, ischaemia, appendicitis, cholecystitis), bowel obstruction/volvulus (sigmoid/caecal), Boerhaave's (transmural oesophageal rupture), abscesses needing formal I&D, major lower GI bleed, gallstone ileus, pilonidal disease.
- Pathophysiology keys:
- Toxin release from ruptured packages → rapid sympathetic storm (cocaine/ amphetamines) or opioid depression.
- Necrotising fasciitis → fulminant fascial spread with microvascular thrombosis → rapid tissue necrosis and sepsis.
- Volvulus/strangulation → vascular compromise → ischemia → perforation/sepsis.
- Boerhaave's → full-thickness oesophageal tear → mediastinitis, pleural contamination, sepsis.
- Management principle: ABCs first, then rapid surgical escalation for source control; imaging should not delay debridement or laparotomy when clinical suspicion is high.

> ملخص بالعربية:
> حالات الطوارئ الجراحية تحتاج تشخيصًا سريعًا وإزالة مصدر المرض (جراحة/تفريغ/تنفس). أمثلة شائعة: تهريب أو انفجار حزم مخدرة، التهاب لفافي نخرى، انفتال/انسداد أمعاء، تمزق المريء.
> الاستقرار الأولي ثم تحويل جراحى عاجل يغيّر النتيجة.

2. Recognition, differential diagnosis, and red flags

- Red flags requiring immediate escalation:
- Haemodynamic collapse, refractory shock, severe arrhythmia.
- Rapidly spreading soft-tissue pain ± crepitus, systemic toxicity → presume necrotising fasciitis.
- Severe chest/epigastric pain after vomiting, subcutaneous emphysema, pleural effusion/pneumomediastinum → Boerhaave's.
- Sudden peritonitis, rigid abdomen, free air on imaging → perforation → urgent laparotomy.
- Signs of packer rupture/toxicity: extreme agitation, diaphoresis, hypertension, tachycardia, hyperthermia (sympathomimetic) or respiratory depression, miosis (opioid).
- Bowel obstruction with peritonitis, persistent vomiting, rising lactate/leukocytosis → strangulation.
- Ongoing massive lower GI bleeding with instability → urgent localisation/IR/surgery.
- Key differentials (match sign → context):
- Coffee-bean sign (AXR) + elderly = sigmoid volvulus.
- Inverted-comma = caecal volvulus.
- Psoas sign (pain on hip flexion against resistance) = retrocaecal appendicitis; obturator sign (pain on passive internal rotation of flexed hip) = pelvic appendicitis.
- Rigler's sign = pneumoperitoneum (free intraperitoneal air).

> الإنعاش والعلامات الحمراء بالعربية:
> أعراض تستدعى إحالة فورية: صدمة غير مستقرة، ألم منفلت وسريع مع علامة نخرية، ألم صدري بعد قيء مع هواء في المنصف (Boerhaave)، علامات ثقب أو انسداد مع ارتفاع اللاكتات. اربط العلامة السريرية بسياق المريض (coffee-bean = انفتال القولون السيني، Rigler = هواء حر).

3. Diagnostic approach and interpretation

- Immediate priorities: ABCs, monitor, IV access ×2, NBM if surgical abdomen suspected, analgesia, antiemetic, group & save if bleeding.
- Rapid investigations:
- Erect chest X-ray (free subdiaphragmatic air, pneumomediastinum).
- Supine + erect abdominal X-rays (obstruction patterns, coffee-bean/

inverted-comma signs).
- Bloods: FBC, U&Es, LFTs, amylase, lactate (ischaemia marker), coagulation, blood cultures if sepsis.
- CT use:
- CT abdomen/pelvis with IV contrast = problem solver for undifferentiated acute abdomen (SBO transition point, volvulus whirl, pneumatosis, free air, collections).
- CT thorax with oral water-soluble contrast = best for suspected oesophageal leak/Boerhaave's.
- CT angiography = localise active lower GI bleeding; sensitivity detects rates ~0.3–0.5 ml/min (use if ongoing bleed and stable enough).
- Body-packer testing (RCEM guidance):
- Stable asymptomatic and consents → low-dose non-contrast CT abdomen/pelvis (high sensitivity) [RCEM 2020].
- Stable symptomatic → contrast CT to characterise complications.
- Unstable or clear toxicity/obstruction → urgent surgical removal/critical care [RCEM 2020].
- Adjuncts:
- Water-soluble contrast (Gastrografin) can be diagnostic and therapeutic in adhesive SBO.
- Do not delay theatre for imaging when necrotising fasciitis or peritonitis is clinically suspected.

> النهج التشخيصي بالعربية:
> الأولوية ABC ثم صيام المريض ووضع خط وريدي. صور صدر واقفة وبطن مستلقي/واقفة أولاً؛ CT مع تباين وريدي هو الفاصل لمعظم الحالات الحادة. CT صدري مع تباين عن طريق الفم لتحديد تسرب المريء. لا تؤخر الجراحة إذا كان الاشتباه السريري قوياً.

4. Emergency management algorithms

- Initial resuscitation (single-page algorithm):
1. ABCs → secure airway if compromised; C-line management: IV access ×2, monitor, ECG.
2. Oxygen to maintain SpO2 ≥94% (or individualised).
3. Give 500–1000 mL crystalloid bolus if hypotensive; reassess. Consider further boluses/vasopressors if persistent shock.
4. Analgesia/antiemetic; NBM; urine output monitoring; senior surgical call early.
- Suspected necrotising fasciitis / Fournier's:

- Immediate broad-spectrum IV antibiotics, aggressive fluid resuscitation, urgent extensive surgical debridement (do not delay for imaging), ICU level care, consider rapid CT if it will not delay theatre.
- Suspected Boerhaave's:
- NBM, high-flow IV fluids, broad-spectrum IV antibiotics covering oral flora and anaerobes (adjust per local protocol), analgesia, urgent CT thorax with oral water-soluble contrast, early surgical/thoracic referral; drain large pleural collections.
- Small-bowel obstruction (SBO):
- If peritonitis/strangulation → urgent laparotomy.
- If stable: NBM, IV fluids, NGT if vomiting, analgesia, CT abdomen/pelvis. Conservative trial up to 48–72 h if improving; use water-soluble contrast study to predict resolution. Operate if no improvement or deterioration.
- Volvulus:
- Sigmoid without peritonitis → endoscopic detorsion + decompression; elective resection later.
- Caecal volvulus → early surgery (endoscopic detorsion often unsuccessful).
- Body-packers/packers (summary algorithm per RCEM 2020):
- Unstable or signs of toxicity/obstruction → urgent surgical/critical care input and removal [RCEM 2020].
- Stable symptomatic → CT with IV contrast to assess complications [RCEM 2020].
- Stable asymptomatic + consents → low-dose non-contrast CT and observation [RCEM 2020].
- Stable asymptomatic + capacity but refuses imaging → document capacity/ refusal and follow local forensic/legal pathway (RCEM 2020).
- Consider whole-bowel irrigation for retained packages per local protocol (avoid if obstruction suspected).
- Major lower GI bleed:
- Resuscitate, transfuse as indicated, correct coagulopathy, consider urgent CT angiography if ongoing bleeding and stable; IR embolisation or surgery if localisation and patient suitable.

> الخوارزميات الطارئة بالعربية:
> أولاً: ABC ثم سوائل وريدية وملء مصدات وطلب جراحة مبكراً. التهاب اللفافة النخري يحتاج إزاحة جراحية فورية. تمزق المريء: صيام، مضادات حيوية واسعة، CT مع صبغة فموية، وإحالة

صدرية. انسداد الأمعاء: جراحة عند التهاب بريتوني أو احتشاء؛ تجربة محافظة 48–72 ساعة إذا كان المستقر.

5. Drugs, procedures, disposition, and follow-up

- Key drugs and considerations (verify local protocols/dosing):
- Broad-spectrum IV antibiotics for intra-abdominal sepsis/Boerhaave's/necrotising infection — choose hospital empiric regimens covering Gram-negatives and anaerobes (e.g., piperacillin–tazobactam per local guidance) and adjust for renal function.
- For suspected opioid toxicity from concealed packages: naloxone boluses and consider naloxone infusion if repeated doses needed (follow local toxicology guidance).
- Vasopressors per ICU protocols for refractory septic/cardiogenic/shock states.
- Key procedures to mobilise:
- Urgent surgical debridement for necrotising fasciitis/Fournier's (formal theatre, not delayed bedside I&D).
- Laparotomy/laparoscopy for perforation, ischaemia, failed conservative SBO, or pack removal when indicated.
- Endoscopic detorsion for sigmoid volvulus (if no signs of ischaemia).
- CT angiography ± IR embolisation for ongoing lower GI bleed.
- Pleural drainage (chest tube) for large pleural collections secondary to oesophageal rupture.
- Whole bowel irrigation (polyethylene glycol) for retained packages if indicated and no obstruction — follow local toxicology/forensic protocols.
- Manual rectal pack removal only with senior surgical/forensic advice and appropriate analgesia/anaesthesia.
- Disposition and follow-up:
- ICU/HDU for unstable, post-op, or toxic patients.
- Surgical in-patient admission for all confirmed necrotising infections, perforations, most SBOs unless fully resolved under observation.
- Asymptomatic body-packers: admit for observation per local/legal protocol; RCEM suggests observation and CT options [RCEM 2020]—document decisions and legal aspects if imaging refused.
- Outpatient/clinic follow-up after operative management as per surgical team; ensure wound care, antibiotics review, and social/forensic liaison where relevant.

الأدوية والإجراءات والمتابعة بالعربية:
استخدم مضادات حيوية وريدية واسعة الطيف وفق بروتوكول المستشفى. نفّذ إزالة جراحية عاجلة للالتهابات النخرية وتمزقات المريء والثغرات المعوية، وأنبوب صدري للانصباب الكبير. استخدم Naloxone في سمّيات الأفيونات وPEG للغسيل المعوي عند تعليمات التسمم. النزلاء المستقرون يتطلبون مراقبة جراحية؛ غير المستقرين إلى ICU.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية

- Quick exam mnemonics:
- Acute surgical abdomen workflow: ABC → NBM → IV fluids → analgesia/antiemetic → erect chest + AXR → CT if uncertain → surgical review.
- SBO checklist: NBM, fluids, NGT if vomiting, CT ± Gastrografin, operate for peritonitis/strangulation or failed conservative trial (48–72 h).
- High-yield signs to memorise:
- Coffee-bean sign = sigmoid volvulus (elderly, distension).
- Inverted-comma = caecal volvulus.
- Whirl sign (CT) = mesenteric torsion/volvulus.
- Rigler's sign = both sides of bowel wall seen = pneumoperitoneum.
- Psoas sign vs obturator sign: know which hip movement provokes pain and clinical context.
- Common pitfalls:
- Don't delay theatre for imaging in suspected necrotising fasciitis or peritonitis.
- Plain AXR/CXR can miss small oesophageal leaks or small pneumoperitoneum → low threshold for CT if clinical suspicion.
- Confuse packer vs stuffer: packers carry large, well-wrapped loads and higher rupture risk; management/legal approach differs — follow RCEM/local guidance [RCEM 2020].
- Do not perform blind rectal pack removal if ischaemia/perforation suspected — get surgical/forensic input.
- Beware discharging a patient who refuses imaging without documenting capacity and following local custody/legal policy.
- Exam technique tips:
- In MCQs, match age & classic sign (e.g., coffee-bean + elderly = sigmoid volvulus).

- Choose answers that prioritise resuscitation and surgical referral over investigations when instability or peritonitis present.

نصائح وعيوب شائعة بالعربية:
احفظ علامات الأشعة والمناورات السريرية (coffee-bean، Rigler، psoas/obturator). لا تؤخر الجراحة عند الاشتباه القوي في إفراز نخرى أو ثقب؛ لا تكتفِ بالأشعة البسيطة إذا الشك السريري مرتفع. تمييز packer عن stuffer مهم قانونياً وطبياً (RCEM 2020).

7. Arabic rapid-review summary

- الإجراءات العملية الفورية:
- استقر: ABC، O2، خطين وريديين، صيام، سوائل؛ استدعِ الجراحة مبكراً.
- علامات تضعك في المسار الجراحي: صدمة، التهاب بريتواني، ألم شديد متصاعد مع crepitus (نخر لفافي)، ألم صدر بعد قيء مع هواء في المنصف (Boerhaave).
- تصوير مفيد سريعًا: صورة صدر واقفة + بطن واقفة/مستلقي؛ CT مع تباين وريدياً هو الفاصل لمعظم الحالات.
- حزم المخدرات داخلية: مريض سمّي/غير مستقر → إزالة عاجلة؛ مريض سليم يوافق → CT منخفض الجرعة/مراقبة (RCEM 2020).
- انسداد الأمعاء: صيام، سوائل، أنبوب أنفي، CT ± Gastrografin؛ جراحة عند علامات احتشاء/فشل.
- نقاط تحذيرية: لا تؤجل تفريغ نخرى؛ لا تكتفٍ بالـAXR عند الشك في تمزق المريء؛ وثّق رفض الفحوص واحضر استشارات قانونية/شرطية عند الحاجة.

ملخّص سريع بالعربية:
ابتدِ بالإسعاف ثم قيّم للعمليات الجراحية العاجلة. استخدم CT بسرعة عندما يكون التشخيص غير واضح أو المريض مستقر. اتبع إرشادات RCEM بشأن تهريب المخدرات (2020) ودوّن كل قرار عند رفض الفحوص.

8. Question coverage map (concepts covered)

- Body-packing/internal drug concealment: recognition, toxicity patterns, imaging choices, RCEM algorithm for packers/stuffers, legal/consent issues [RCEM 2020].
- Necrotising soft-tissue infection / Fournier's: recognition, immediate debridement, antibiotics, ICU needs.
- Boerhaave's (oesophageal rupture): presentation, imaging (CT with oral contrast), antibiotics, drainage, surgical referral.

- Bowel obstruction / SBO: adhesive causes, initial conservative management, CT features, Gastrografin use, indications for surgery.
- Volvulus (sigmoid vs caecal): AXR signs (coffee-bean, inverted comma), endoscopic detorsion vs surgery.
- Perforation / pneumoperitoneum: Rigler's sign, urgency of laparotomy.
- Appendicitis/cholecystitis: focal signs (Rovsing, psoas, obturator, Murphy), imaging, early surgical referral.
- Abscess requiring I&D vs bedside drainage: indications for formal theatre vs simple drainage.
- Major lower GI bleed: resuscitation, CT angiography thresholds and IR/surgical options.
- Gallstone ileus, pilonidal disease: brief recognition and management pathways.
- Imaging signs and bedside clinical signs: coffee-bean, inverted-comma, whirl sign, Rigler, psoas/obturator, Rovsing.
- Practical resuscitation priorities and disposition (ED → surgical ward/OR/ICU).
(Note: question numbers not provided; map lists all concepts represented in source notes.)

Sources to verify
- RCEM: Management of Suspected Internal Drug Traffickers (December 2020) [RCEM 2020]. Also consult local hospital surgical and toxicology protocols, ERC guidance, NICE acute abdominal pain and sepsis guidance, and national critical care/IR guidance for local doses and pathways.

This material is educational revision only — not patient-specific advice. Verify local protocols and senior surgical/toxicology input for clinical decisions.

CHAPTER 02: PAIN AND SEDATION

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Chapter 02 — Pain and Sedation

1. Clinical overview and pathophysiology

Core: ED pain care uses a stepped approach — simple analgesics → systemic - opioids/inhaled analgesia → local/regional blocks → procedural sedation when ,immobility or deep analgesia required. Choose modality by pain severity, site comorbidity, airway risk and need for immobility. [RCEM]

Procedural sedation: aims to provide anxiolysis, analgesia and immobility - while preserving cardiorespiratory stability. Drugs act on CNS respiratory drive and airway reflexes — risk of hypoventilation/apnoea, aspiration and .haemodynamic effects

Regional/field blocks: deposit local anaesthetic near nerves or in fascial - planes to produce reversible nerve conduction block; reduce systemic opioid needs but carry risks (intravascular injection, nerve injury, LAST, masking .(compartment syndrome

Arabic: نظرة عامة — نهج طبقي لتسكين الألم؛ التخدير الإجرائي يؤثر على الجهاز التنفسي والانعكاسات الهوائية؛ الكتل العصبية تمنع التوصيل العصبي وتقلل الحاجة للأفيونات لكنها تحمل مخاطر خاصة.

2. Recognition, differential diagnosis, and red flags

Immediate sedation risks: hypoventilation/apnoea, hypoxia, airway - obstruction, aspiration, laryngospasm, haemodynamic instability, emergence .agitation (ketamine), vomiting. Monitor and act early

,Block-specific complications: LAST, local infection/haematoma, nerve injury - failed block, masking evolving compartment syndrome. Check anticoagulation status (not always absolute contraindication for fascial-plane blocks — follow .(local guidance

,High-risk patient factors: severe cardiorespiratory disease, pregnancy - ,hepatic/renal impairment, recent oral intake, extreme age, allergies .anticoagulation, poor cooperation

- Red flags requiring senior review/avoidance of ED sedation: ASA IV or unstable physiology for non-life-saving procedures; uncontrolled airway risk; suspected full stomach with urgent need for deep sedation without airway protection.

Arabic: التعرف وعلامات الخطر — راقب نقص التهوية، نقص الأكسجة، ولـLAST؛ اعتبر الأمراض المصاحبة والتخثر؛ ASA IV عادة يمنع التخدير الإجرائي الانتخابي بالقسم.

3. Diagnostic approach and interpretation

- Pre-procedure checks (mandatory): focused airway assessment, fasting history, current meds and allergies, baseline vitals, ASA classification, consent and plan for rescue/airway management. Document neurovascular baseline for limb procedures.
- Monitoring essentials: continuous pulse oximetry and capnography (if available), ECG and BP at least every 5 minutes during sedation. Post-block neurovascular checks and observation for systemic toxicity.
- Regional block planning: confirm anatomy/landmarks, consider ultrasound guidance to improve accuracy and reduce vascular puncture, check coagulation status and allergies.
- Interpretation pitfalls: do not assume return to SpO2 92% RA is safe for discharge — many ED protocols require return to baseline oxygen saturation and alertness. Verify local discharge criteria.

Arabic: النهج التشخيصي — فحوصات قبيل الإجراء: طريق هوائي، صوم، أدوية، ASA، موافقة؛ مراقبة مستمرة بالأكسيمتر والـcapnography؛ استخدم الموجات فوق الصوتية عند الإمكان.

4. Emergency management algorithms

A. Procedural sedation — concise algorithm

1) Prepare: consent, NPO check, IV access, monitoring (SpO2, capnography, ECG, BP), resuscitation kit, reversal agents (naloxone, flumazenil) available, defined team roles (operator, dedicated monitor, assistant). "Crisis plan audible." [AAGBI; RCEM]

2) Induction (commonly tested adult doses): ketamine 1.0 mg/kg IV bolus over 60–30 s; titrate small aliquots for effect; reduce dose in frail/elderly. (Use local protocol for alternatives/opiates/benzos).

3) Monitor continuously; reassess sedation depth, ventilation and circulation.

→ Manage airway/respiratory compromise: immediate airway manoeuvres (4 supplemental O2 → jaw thrust/adjuncts → bag-mask ventilation → escalate to advanced airway if refractory. Treat causes (opioid/benzo reversal vs supportive care for ketamine).

Emergence agitation (ketamine): calm environment, reassurance; consider (5 midazolam 0.03 mg/kg IV for severe agitation. [RCEM]

Recovery/discharge: patient alert and oriented at baseline, baseline vitals (6 and SpO2 on room air (many ED protocols require return to baseline; SpO2 RA often considered unsafe for discharge), tolerating oral fluids, minimal 92% nausea, and elapsed observation period (e.g., 2 hours after reversal agents per local protocol). Document discharge criteria met. [RCEM procedural sedation BPC 2022]

Arabic: خوارزمية التخدير الإجرائي — التحضير (أدوار، معدات، أدوية عكسية)، كيتامين 1 mg/kg اختبار شائع، مراقبة مستمرة، تدبير العجز التنفسي حسب الأولويات، ميدازولام 0.03 mg/kg للتهيج؛ لا تخرج المريض إلا بعد عودة القيم الأساسية.

B. Regional/field block — concise algorithm (example: fascia iliaca)

1) Indication/contraindication check: allergy, local infection (contra), coagulation status, patient cooperation. Anticoagulation may be relative — consult policy. [RCEM]

2) Final safety step: "Stop Before You Block" timeout — confirm patient, side, block, consent, equipment, resuscitation readiness.

3) Technique (landmark FIB): identify inguinal ligament; divide into thirds; point 1 cm below junction of lateral/middle third; needle passes fascia lata (first pop) /then fascia iliaca (second pop); deposit LA below fascia iliaca to bathe femoral ilioinguinal plexus. Use ultrasound when possible.

4) Dosing/monitoring: observe for LAST; monitor every 5 minutes for first 15 minutes and at 30 minutes; document neurovascular status. Typical FIB requires larger volume (often 30–40 ml) compared with femoral nerve block. Verify local maximum LA doses.

5) LAST immediate actions: stop injection, call for help, airway/O2, control seizures with benzodiazepines, treat arrhythmia per ACLS avoiding agents exacerbating toxicity, 20% lipid emulsion: bolus 1.5 ml/kg IV then infusion 0.25 ml/kg/min (follow ASRA/local protocol). [ASRA]

Arabic: خوارزمية الكتلة العصبية — تحقق من الموانع، نفّذ "Stop Before You Block"،

استخدم العلامات أو الموجات فوق الصوتية، راقب لاكتشاف LAST؛ مستحلب دهني 20%
حسب إرشادات ASRA في حالة سمية جهازية.

5. Drugs, procedures, disposition, and follow-up

Drugs — exam-important doses/limits (verify local formularies)

- Ketamine (procedural sedation adult): 1.0 mg/kg IV bolus commonly tested; titrate and reduce for frailty. [RCEM]
- Midazolam (for ketamine emergence agitation): 0.03 mg/kg IV commonly tested. [RCEM]
- Methoxyflurane (Penthrox): not licensed in many jurisdictions for <18 yrs; common limits cited in BNF: max 6 ml/24 h (2 × 3 ml vials) and 15 ml/week (5 vials). Patient exhales into carbon chamber to reduce staff exposure. [BNF]
- Local anaesthetics: lidocaine ≈ 4.5 mg/kg (without adrenaline) commonly quoted; with adrenaline ≈ 7 mg/kg. Bupivacaine/levobupivacaine commonly cited ≈ 2–2.5 mg/kg (sources vary) — verify local protocols. Avoid exceeding institutional maxima.
- LAST treatment: 20% lipid emulsion bolus 1.5 ml/kg then infusion 0.25 ml/kg/min (ASRA guidance). [ASRA]

Procedures — practical tips

- Wrist block: block median, ulnar, superficial radial nerves; landmarks — median between palmaris longus and FCR; ulnar between FCU and ulnar artery; superficial radial dorsolateral. Use ~3–5 ml per nerve (exam values approximate).
- Ankle block: five nerves — deep peroneal, superficial peroneal, posterior tibial, sural, saphenous. Remember common peroneal divides proximal to ankle.
- Fascia iliaca vs femoral block: FIB is further from neurovascular bundle (lower intravascular risk), needs larger volume and may give variable coverage; FNB uses lower volume but is closer to vessels.

Disposition and follow-up

- Observation: immediate post-procedure monitoring until return to baseline; specific timing per local policy (e.g., every 5 min initially for sedation; first 15 min then 30 min post-block checks).
- Discharge criteria: alertness at baseline, hemodynamic stability, SpO2 at/near baseline on room air (do not rely on an absolute 92% RA as safe), able to

tolerate oral fluids, controlled pain, minimal nausea/vomiting, escort or safe transport if required. Give written advice about block numbness and signs of infection/haematoma/LAST and when/how to seek help.

- Follow-up: analgesia plan, wound care review, review of anticoagulation management if modified, outpatient ortho/anaesthesia review for ongoing block or complications as needed.

Arabic: أدوية وإجراءات والمتابعة — جرعات شائعة للاختبار: كيتامين 1 mg/kg؛ ميدازولام 0.03 mg/kg؛ Penthr ox حدود BNF; ليدوكايين ≈4.5 mg/kg؛ بوبيفاكين ≈2–2.5 mg/kg (تحقق محلياً). راقب المريض حتى عودة القيم الأساسية وقدّم تعليمات واضحة عن المضاعفات والمتابعة.

6. High-yield FRCEM exam tricks and common pitfalls

- Ketamine dosing: 1.0 mg/kg IV is the usual exam answer for adult procedural sedation; midazolam 0.03 mg/kg for emergence agitation. [RCEM]
- Don’t accept “SpO2 92% RA is safe for discharge” as a blanket rule — many ED protocols require return to baseline; 92% RA is often flagged as unsafe in exam stems.
- Methoxyflurane (Penthrox): memorise 6 ml/24 h and 15 ml/week limits as exam fact (BNF) and remember age restriction (<18 yrs in many jurisdictions). [BNF]
- LAST management: lipid emulsion bolus 1.5 ml/kg → infusion 0.25 ml/kg/min is the tested sequence (cite ASRA). Know basic ACLS modifications (avoid vasopressin, high doses of local-anaesthetic-sensitising agents). [ASRA]
- Anticoagulation and nerve blocks: exam traps often present anticoagulation as absolute contraindication — for many fascial plane blocks this is not absolute; answer should reflect block-specific and policy-dependent considerations. [RCEM]
- Local anaesthetic dose variability: questions may offer bupivacaine limits of 2 mg/kg vs 2.5 mg/kg — flag that sources differ and select the option consistent with local/formulary guidance if in clinical context; if an exam requires a single best answer, choose the commonly cited 2 mg/kg only if that matches offered options.
- Entonox pitfalls: contraindicated in suspected pneumothorax, bowel — obstruction, recent middle ear surgery, or when gas expansion is hazardous — common exam distractors.

- Opioid conversion: codeine ≈ 10:1 oral morphine (exam shorthand) — useful for calculations but clinically unreliable due to CYP2D6 variability (and paediatric codeine restrictions). [MHRA]
Arabic: حيل الامتحان — كيتامين 1 mg/kg وميدازولام 0.03 mg/kg شائعان؛ لا تقبل 92% كحد آمن للخروج دون سياق؛ Penthr ox 6 ml/24h و15 ml/week؛ LAST: دهون 20% حسب ASRA؛ موانع Entonox شائعة في الامتحانات؛ تحويل الكوديين 10:1 قاعدة اختبارية وليست مطلقة.

7. Arabic rapid-review summary (concise)

- استخدم نهجاً طبقيّاً لتسكين الألم: مسكنات بسيطة → Entonox → كتلة عصبية → تخدير إجرائي.
- تحقّق قبل الإجراء: طريق هوائي، صوم، أدوية، حساسية، ASA، معدات إنعاش، أدوار الفريق.
- كيتامين 1 mg/kg IV (إجابة امتحان شائعة)؛ ميدازولام 0.03 mg/kg لعلاج اضطراب الخروج.
- Penthr ox (methoxyflurane): حد 6 ml/24h و15 ml/week؛ حذارٍ في <18 سنة حسب الترخيص المحلي. [BNF]
- حدود ليدوكايين ≈4.5 mg/kg؛ حدود البوبيفاكاين ≈2–2.5 mg/kg — تحقّق من بروتوكولك المحلي.
- LAST → أوقف الحقن، تنفس/أكسجين، بنزوديازيبين للتشنجات، مستحلب دهني 20%: دفعة 1.5 ml/kg ثم تسريب 0.25 ml/kg/min. [ASRA]
- لا تخرج المريض حتى يعود لمستواه الإدراكي والفيزيولوجي الأساسي وتحقق معايير المراقبة المحلية.
Arabic short take-home: اتبع فحوصات ما قبل التخدير، استخدم "Stop Before You Block"، راقب عن كثب، وتحقق من البروتوكول المحلي للجرعات والموانع.

8. Question coverage map (concepts covered in supplied stems)

EXAM TRICK / تريكة امتحانية

- RCEM procedural sedation guidance: pre-sedation checks, monitoring frequency, ketamine dosing, management of hypoventilation/airway, emergence agitation, discharge criteria (procedural sedation concepts). [RCEM procedural sedation BPC 2018–2022]
- AAGBI / Anaesthesia guidance: team roles, airway/rescue preparation, capnography use (sedation safety concepts). [AAGBI]
- BNF / MHRA stems: methoxyflurane (Penthrox) licensing, age limits, daily/weekly dose limits; codeine regulatory cautions. [BNF; MHRA]

- Regional block technique and safety: fascia iliaca block landmarks/technique, wrist block/anatomical landmarks, ankle block five-nerve list, comparative notes FIB vs FNB, anticoagulation considerations. (Field block concepts)
- Local anaesthetic pharmacology and dosing caveats: lidocaine, bupivacaine/levobupivacaine maximums and volume considerations (LA dose/last concepts).
- LAST recognition and management: immediate actions, benzodiazepine seizure control, lipid emulsion regimen (ASRA guidance).
- Entonox / inhaled analgesia: contraindications (pneumothorax, bowel obstruction, raised ICP), paediatric considerations (age/policy variations).
- Opioid conversion and prescribing pitfalls: codeine→morphine approximate conversion, CYP2D6 caveat, paediatric codeine restrictions.

Sources to verify: RCEM (procedural sedation and pain guidance), AAGBI, ASRA (LAST guidance), BNF (methoxyflurane), MHRA (codeine in children), local anaesthesia formularies and institutional sedation policies, ERC/NICE where relevant.

CHAPTER 03: PAEDIATRICS

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Chapter 03: Paediatrics — Final concise FRCEM revision chapter

1. Clinical overview and pathophysiology

- Principles
- Children = age-dependent physiology: small children decompensate fast (higher metabolic rate, larger head, smaller airway, higher fluid requirement). Use weight-based dosing and rapid estimation: WETFLAG / weight ≈ (2 × age) + 8 kg for quick calculations. [RCUK Mar 2022]
- ABC first: airway/oxygenation → circulation (perfusion, HR thresholds) → disability (GCS/seizures) → exposure/investigations.
- Prioritise time-critical diagnoses: neonatal/early-onset sepsis, airway foreign body, septic arthritis, intussusception, status epilepticus, severe asthma, DKA with cerebral oedema, cardiac arrest.
- Pathophysiology highlights
- Shock in infants may be masked (compensated shock): tachycardia, delayed capillary refill, normal BP until late.
- Respiratory distress: distinguish upper (stridor, drooling, epiglottitis/bacterial tracheitis) from lower (wheeze, bronchiolitis, asthma) airway disease.
- Cyanotic congenital lesions (TOF) → hypercyanotic (tet) spells from increased R→L shunt; knee-chest increases systemic vascular resistance and reduces R→L shunt.
- DKA cerebral oedema: osmolar/flow shifts during rehydration → rising intracranial pressure; early recognition critical. [BSPED 2020]
- IO samples are useful for urgent access but may be unreliable for some analytes (calcium, some electrolytes, WCC/platelets). [LitFL]

> Arabic (brief)
> - الأطفال لهم فيزيولوجيا خاصة: تدهور أسرع وجرعات على أساس الوزن (WETFLAG ≈ 2×العمر+8). الأولوية للجوْف والتنفس والدورة. تمييز انسداد فوقي/سفلي مهم. عينات العظم قد تخطئ لبعض التحاليل.

2. Recognition, differential diagnosis, and red flags

- Key presentations and discriminators
- Limping child: transient synovitis vs septic arthritis vs Perthes vs fracture. Red flags for septic arthritis: non-weightbearing, fever, high CRP/WBC/ESR → urgent ortho aspiration (Kocher criteria helpful; CRP most predictive). [MDCalc Kocher]
- Neonate with collapse/poor feeding: consider early-onset sepsis, congenital heart disease (duct-dependent lesions), metabolic disorder.
- High fever ≥5 days + conjunctivitis + mucosal changes + rash → Kawasaki disease (risk coronary aneurysm).
- Palpable purpura on buttocks/legs ± abdominal pain → HSP (monitor renal function).
- Sudden cyanosis/crying + hyperpnoea → tet spell (TOF).
- Sudden choking with unilateral decreased air entry → airway foreign body; immediate theatre if compromised.
- Paroxysmal very fast regular narrow-complex tachycardia → SVT.
- Barking cough, inspiratory stridor ± fever: croup (mild = no stridor at rest).
- Fever + oral vesicles + palm/sole rash → hand-foot-mouth (supportive).
- Acute severe asthma: severe tachypnoea, use of accessory muscles, poor air entry, hypoxia — escalate early.
- Status epilepticus with likely hyponatraemia → treat hyponatraemia to stop seizure.
- Red flags (must admit/escalate)
- Any infant <28 days with fever; any child with poor perfusion, persistent hypoxia, respiratory exhaustion, ongoing seizure after two benzodiazepine doses, suspected septic arthritis, lodged oesophageal button battery, signs of DKA cerebral oedema.

> Arabic (brief)
> - ملاحظات تحذيرية: الأطفال الصغار عرضة لعدوى خطيرة؛ الطفل الذي لا يضع وزنًا على الطرف أو الحمى المرتفعة أو CRP/WBC المرتفع يحتاج شكوكا في التهاب مفصلي خاثر. نباح الكحة = خناق؛ الاختناق المفاجئ قد يكون جسمًا غريبًا. أي رضيع <28 يوماً مع حمى يحتاج تقييم شامل.

3. Diagnostic approach and interpretation

- Principles: bedside glucose in any post-ictal/altered child; bloods (FBC, CRP) early in limp/fever; targeted imaging only when it affects immediate management.

- Specific pathways
- Limping hip: exam → weightbearing status. If non-weightbearing or fever → FBC + CRP ± ESR → pelvic AP + frog-leg XR; urgent hip US for effusion; if effusion/high probability → ortho referral + aspiration. (Kocher criteria / CRP best predictor). [MDCalc Kocher]
- Suspected foreign body aspiration: CXR for air-trapping or lobar collapse (AP + lateral) but do NOT delay theatre if unstable — prepare rigid bronchoscopy.
- Intussusception: abdominal ultrasound (target sign) is first-line; AXR insensitive (reserve XR for perforation signs or if US delayed).
- Febrile infant 1–3 months: low threshold for LP and full sepsis work-up if labs abnormal; send blood culture and CRP; if <28 days, treat empirically and consider LP before antibiotics if stable. [NICE NG143 / NG195]
- DKA: VBG/ABG, glucose, ketones, U&E, ECG; classify severity by pH/HCO3 to estimate fluid deficit (mild/moderate/severe guidance per BSPED). [BSPED 2020]
- Seizure workup: glucose first, then VBG/biochemistry, infection screen if indicated; urgent neuroimaging only if focal signs or trauma.
- IO samples: accept for urgent chemistry but confirm discrepant electrolytes/ calcium with peripheral sample when possible. [LitFL]

> Arabic (brief)
> - نهج تشخيصي عملي: قياس جلوكوز فورًا بعد النوبة؛ لعلاج الورك أعد FBC+CRP وصور وUS؛ الشُبهة لابتلاع جسم غريب تتطلب صورًا لكن لا تؤخر المناظير إذا الطفل غير مستقر. رضيع محمى <28 يوماً غالبًا يحتاج علاج ومراجعة شاملة.

4. Emergency management algorithms

(Select concise stepwise algorithms — apply local protocol where flagged)
- Limping hip (suspected septic arthritis)
1. ABCs, analgesia.
2. FBC + CRP ± ESR, pelvis XR, urgent ultrasound.
3. If effusion/high probability → urgent orthopaedic aspiration + start IV antibiotics. [MDCalc Kocher]
- Hypercyanotic (tet) spell
1. Knee-chest position; calm child; high-flow O2.
2. IV/IO fluids if shocked.
3. Give morphine (or ketamine per local congenital cardiac protocol) and urgent cardiology/surgical review.

- Airway foreign body (compromised)
- Immediate preparation for rigid bronchoscopy under GA; do not delay for imaging. If conscious with effective cough → encourage coughing. For infants: back blows then chest thrusts if ineffective; if unresponsive → CPR and theatre. [Resus guidance]
- Croup
- Single dose dexamethasone (commonly 0.15 mg/kg PO) ± nebulised adrenaline for moderate/severe disease; observe for rebound. [RCEM/NG91]
- Severe acute asthma (escalation)
1. Oxygen (target per local guidance), inhaled/nebulised salbutamol ± ipratropium, systemic steroid early.
2. If inadequate: IV salbutamol bolus 15 µg/kg over 5–10 min then infusion per local protocol (verify local practice) and/or aminophylline 5 mg/kg IV loading (follow local protocol). Consider IV magnesium sulphate 40 mg/kg (max 2 g) single dose for refractory severe attack. Escalate to PICU. [BNFC/NICE]
- Status epilepticus
1. ABC, oxygen, capillary glucose.
2. Give benzodiazepine (weight-based; e.g., lorazepam 0.1 mg/kg IV or buccal midazolam 0.3–0.5 mg/kg). Repeat once only.
3. If ongoing → second-line AED (levetiracetam ≈30–40 mg/kg IV over 5 min or phenytoin 20 mg/kg IV slow with ECG).
4. If refractory after second-line → secure airway, RSI and anaesthetic transfer to PICU. [Resus UK Mar 2022]
- Hyponatraemic seizure
- Give 3% sodium chloride bolus (commonly 2–3 mL/kg is used; some local protocols use 3 mL/kg) to terminate seizure while arranging definitive care — verify your local PALS/Resus guidance. [RCUK Mar 2022; flagged]
- SVT (unstable vs stable)
- If unstable → immediate synchronized cardioversion 1 J/kg (repeat 2 J/kg, escalate up to 4 J/kg).
- If stable → vagal manoeuvres (infants: ice to face; older: modified Valsalva) then adenosine per paediatric dosing (see Section 5). [RCUK Mar 2022]
- Cardiac arrest (shockable paediatric)
- Defibrillation 4 J/kg initial shock (Resus UK/RCUK guidance). [RCUK Mar 2022]
- DKA with suspected cerebral oedema
- Recognise early (falling GCS, headache, vomiting, Cushing signs). Immediate

hyperosmolar therapy per local/PICU guideline — BSPED commonly recommends 3% hypertonic saline bolus (e.g., 5 mL/kg) or mannitol 0.5–1 g/kg if HTS unavailable — verify local protocol. Urgent PICU. [BSPED 2020 — FLAG]

Arabic (brief)
- خوارزميات سريعة: التهاب مفصل خاثر → US+شفط+مضاد حيوي؛ نوبة "tet" → وضع ركبة-صدر، أكسجين، مهدئ وسوائل؛ جسم غريب مُعيق → منظار صلب فوري إذا يعاني الطفل؛ خناق → ديكساميثازون ± أدرينالين رذاذي؛ صرع مستمر → بنزوديازيبين ×2 → ليفيتيراسيتام/فينيتوين → RSI إذا مقاوم؛ جرعات HTS وادينوزين تحقق محليًا.

5. Drugs, procedures, disposition, and follow-up

- High-yield drug doses / procedural thresholds (well established or commonly used; verify local policies)
- Weight estimation: WETFLAG ≈ (2 × age) + 8 kg.
- Dexamethasone for croup: single dose commonly 0.15 mg/kg PO (range cited 0.15–0.6 mg/kg in different sources — verify local exam expectation). [RCEM]
- Nebulised adrenaline for croup: dose per local guideline (observe for rebound).
- Severe asthma: IV salbutamol loading 15 µg/kg over 5–10 min then infusion (local protocol); aminophylline loading 5 mg/kg IV if used. IV magnesium sulphate 40 mg/kg (max 2 g) single dose for refractory severe acute asthma. [BNFC/NICE]
- Status epilepticus: buccal midazolam 0.3–0.5 mg/kg or lorazepam 0.1 mg/kg IV; levetiracetam commonly 30–40 mg/kg IV (max 3 g); phenytoin 20 mg/kg IV slow with ECG. [Resus UK Mar 2022]
- Bradycardia: atropine 20 µg/kg (0.02 mg/kg) for vagal bradycardia; adrenaline for poor perfusion bradycardia 10 µg/kg (0.01 mg/kg) IV. [RCUK]
- SVT: adenosine dosing per Resus UK paediatric algorithm — 1–11 years initial 100 µg/kg rapid IV bolus (increase incrementally), 12–17 years may start 3 mg; infants <1 year have different dosing (e.g., 150 µg/kg) — confirm local guideline. [Resus UK Mar 2022]
- Defibrillation in paediatric arrest: 4 J/kg initial (RCUK). [RCUK Mar 2022]
- Hyponatraemic seizure: 3% NaCl bolus commonly 2–3 mL/kg (some use 3 mL/kg) — verify local PALS. [RCUK Mar 2022 — flagged]
- DKA cerebral oedema: BSPED commonly reports 3% HTS 5 mL/kg bolus (verify local policy). [BSPED 2020 — FLAG]
- Neonatal empiric antibiotics for suspected early-onset sepsis: benzylpenicillin + gentamicin (local neonatal guidance for age-specific cephalosporin cautions).

[NICE NG195]
- Analgesia: intranasal diamorphine (where available/authorized) ~0.1 mg/kg via atomiser for severe pain if IV not available — check local licencing. Avoid NSAIDs in significant renal impairment. (Regulatory variance flagged.)
- IVIG in ITP: regimes vary (e.g., 0.8–1 g/kg single dose or 0.4 g/kg ×2) — consult haematology/local protocol.
- Maintenance IV fluids: Holliday-Segar: 0–10 kg 4 mL/kg/hr; 10–20 kg 2 mL/kg/hr; >20 kg add 1 mL/kg/hr per kg above 20. [NICE NG29]
- Procedures / disposition
- Urgent orthopaedic theatre for septic hip aspiration/drainage if septic arthritis suspected. Admit and liaise with ortho.
- Rigid bronchoscopy in theatre for airway foreign body under GA.
- PICU for: refractory status epilepticus, intubated severe asthma, DKA severe or suspected cerebral oedema, neonatal sepsis unstable, post-resuscitation care.
- Admit and arrange follow-up: asthma discharge review by GP within 48 h; specialist review ~1 month for moderate-severe attacks. For AOM use watchful waiting unless indicated for antibiotics (otorrhoea, bilateral <2 yrs, very unwell) [NICE NG91].
- Safeguarding: unexplained injuries, inconsistent history, patterned bruises — follow local child protection procedures immediately.
- Follow-up investigations
- Septic arthritis: synovial fluid culture, blood cultures, CRP trend to guide therapy duration.
- HSP: monitor urine for haematuria/protein (baseline and follow-up).
- UTI (young child): renal ultrasound and specialist referral for atypical UTIs (per local pathways). [NICE CKS]

> Arabic (brief)
> - أدوية وإجراءات: WETFLAG للوزن؛ ديكساميثازون خناق ≈0.15 mg/kg؛ ربو حاد: سالبيوتامول وريدياً/أمينوفيلين/ماغنسيوم 40 mg/kg؛ صرع مستمر: بنزو ×2 ثم ليفيتيراسيتام 30–40 mg/kg أو فينيتوين 20 mg/kg؛ جرعات HTS وPGE1 وجرعة ديامورفين أنفي تحتاج تحقق محلي.
> إحالة عاجلة إلى ortho/ENT/PICU حسب الحالة. متابعة الربو خلال 48 ساعة مع المريض.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية

- Kocher criteria recall (original 4): non-weightbearing, fever >38.5°C, WBC >12 ×10^9/L, ESR >40 mm/hr — presence of all four → high probability septic arthritis; modern practice: CRP is more predictive than ESR. (Exam trap: CRP was not in original Kocher score.) [MDCalc Kocher]
- Imaging pitfalls
- Normal plain abdominal XR does NOT exclude intussusception — ultrasound is the diagnostic test of choice.
- AP chest/abdomen alone may misidentify stacked magnets as single — always get lateral view for suspected magnet ingestion (exam favourite). [RCEM magnet guidance 2021]
- Drug/management pitfalls
- Do NOT give insulin bolus in DKA; use 0.1 U/kg/hr infusion after initial resuscitation (BSPED). Exam trap: some stems try to lure you into giving bolus. [BSPED 2020]
- After two adequate benzodiazepine doses for an ongoing seizure, move promptly to second-line AEDs; avoid further large benzodiazepine boluses without airway protection. (Common error in exam stems.)
- Phenytoin infusion must be slow with ECG monitoring (cardiotoxic if rapid) — prefer fosphenytoin where available.
- In AOM do not reflexively give antibiotics — memorize exceptions (otalgia + otorrhoea, <2 yrs bilateral, systemically unwell). [NICE NG91]
- Foreign body airway: if child can cough effectively, do not attempt blind finger sweep or immediate intubation — encourage cough and call for theatre if ineffective.
- IO blood values: potassium/calcium/WCC/platelets may be unreliable — confirm with venous sample before critical decisions.
- Practical exam mnemonics
- Weight quick calc: WETFLAG = (2 × age) + 8 kg.
- ETT size (uncuffed) exam shortcut: (age/4) + 4 mm (subtract 0.5 mm for cuffed tube) — commonly tested.
- DKA severity for exam: mild pH 7.20–7.29, moderate 7.10–7.19, severe <7.10 (use to estimate fluid deficit per BSPED). [BSPED 2020]
- Frequent flagged items to double-check before answering in exam or acting

clinically
- Exact bolus dose of 3% NaCl for hyponatraemic seizure (2 vs 3 mL/kg) — local PALS/Resus guidance varies. [RCUK Mar 2022 — flagged]
- Ketamine absolute contraindication under 12 months is debated — check local procedural sedation guidance (RCEM Feb 2020 flagged).
- Neonatal cephalosporin caution (ceftriaxone in neonates with hyperbilirubinaemia) — check local neonatal guidance.

Arabic (brief)
- حيل الامتحان: Kocher الأصلي = 4 معايير (CRP ليس جزءًا منها لكن مفيد عمليًا). صورة بطنية طبيعية لا تنفي الانغلاف — استخدم الموجات. لا تعطَ جرعة إقلاع للإنسولين في DKA. بعد بنزو ×2 انتقل للأدوية الثانية. تحقق من جرعات HTS وكيتامين محليًا قبل اختيار الإجابة.

7. Arabic rapid-review summary (Modern Standard Arabic)

- مبادئ عامة: الأطفال = فيزيولوجيا خاصة؛ استعمل قواعد سريعة للوزن (WETFLAG = 2×العمر+8). ابدأ بـ ABCs.
- أهم الحالات الطارئة: رضيع محمى <28 يوماً، انسداد مجرى الهواء، التهاب مفصلي خاثر، انتفاخ أحمر (انغلاف معوي)، نوبة صرع مستمرة، ربو حاد، DKA مع وذمة دماغية.
- فحوص سريعة مفيدة: جلوكوز شعري فوراً بعد النوبة؛ FBC + CRP لتفريق الالتهاب النصفي عن التهاب المفصل؛ US للانغلاف؛ أشعة أمامية وجانبية للبطاريات/المغناطيس.
- علاجات مهمة: ديكساميثازون خناق ≈0.15 mg/kg؛ ماغنسيوم ربو حاد 40 mg/kg (حد أقصى 2 g)؛ ليفيتيراسيتام في الصرع ≈40 mg/kg؛ فينيتوين 20 mg/kg (ببطء مع ECG).
- ملاحظات سلامة: لا تعطِ جرعة إقلاع بالإنسولين في DKA؛ لا تفحص حلق المريض عند الشك في التهاب الحنجرة؛ أغلق على إزالة زر البطارية في المريء فوراً؛ تحقق محلياً من جرعات HTS وPGE1 وكيتامين وديامورفين الأنفي.
(مادة للمذاكرة فقط — راجع الأدلة والممارسات المحلية قبل الاستخدام السريري.)

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية
- Limping hip / septic arthritis vs transient synovitis (Kocher criteria; CRP)
- Neonatal sepsis / early-onset sepsis / PGE1 considerations
- Kawasaki disease recognition and coronary risk
- Henoch-Schönlein purpura (HSP) recognition and renal follow-up
- Tetralogy of Fallot — hypercyanotic (tet) spell management

- Foreign body airway obstruction (choking algorithms, rigid bronchoscopy) and button battery/magnet ingestion imaging rules
- SVT recognition and management (adenosine dosing, synchronized cardioversion)
- Croup (steroid + nebulised adrenaline)
- Acute severe asthma (nebulisers → IV salbutamol/aminophylline/magnesium → PICU escalation)
- Intussusception diagnosis (ultrasound) and reduction pathways
- Septic arthritis hip algorithm and orthopaedic aspiration
- Status epilepticus (benzodiazepines → second-line AEDs → RSI if refractory) and hyponatraemic seizure HTS bolus considerations
- DKA management and cerebral oedema recognition/management (BSPED fluid classification)
- Bradycardia algorithms (atropine, adrenaline thresholds)
- Paediatric resuscitation energy doses (defibrillation/cardioversion)
- Neonatal/paediatric analgesia and sedation issues (ketamine guidance, intranasal opioids)
- UTI in young child: urine testing, atypical features and imaging pathways
- ITP management (IVIG vs steroids) and HUS recognition
- IO sampling limitations and interpretation pitfalls
- Safeguarding: patterns prompting immediate referral

Sources to verify
- Resuscitation Council UK / Resus UK paediatric algorithms (Mar 2022) [RCUK]
- RCEM guidance (paediatric ketamine Feb 2020; magnet ingestion 2021) [RCEM]
- BSPED DKA guideline (2020) [BSPED]
- NICE guidelines: NG91 (Otitis media), NG29 (Fluids in children), NG143/NG195 (Fever/Neonatal infection), NG9 (Bronchiolitis) [NICE]
- BNFC / BNF for Children dosing references (salbutamol, aminophylline, magnesium) [BNFC]
- MDCalc Kocher criteria (paediatric septic arthritis) [MDCalc]
- LitFL intraosseous access review [LitFL]

Final note: This chapter collates high-yield exam and ED-practical points. Check local paediatric, neonatal and critical-care protocols for any variable/flagged doses, licensing differences (e.g., intranasal diamorphine), and exact HTS/ketamine/antibiotic recommendations before clinical use.

CHAPTER 04: TRAUMA

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Chapter 04 — Trauma

1. Clinical overview and pathophysiology

- Core principle: immediate life/limb/sight threats first. cABCDE (catastrophic haemorrhage, Airway with C-spine control, Breathing, Circulation, Disability, Exposure). Do not delay life-saving interventions for investigations. [ATLS 10th]
- Mechanism matters — energy transfer predicts injury patterns (e.g., anterior chest → RV contusion; high-energy pelvis → major pelvic haemorrhage).
- Shock types: hypovolaemic (tachycardia, cool/clammy), neurogenic (bradycardia, warm/dry after spinal cord injury above T1–L2), cardiogenic (signs of poor cardiac output).
- Trauma physiology: permissive hypotension/limit crystalloids in uncontrolled haemorrhage; early balanced transfusion and haemostasis prioritised (damage-control haemorrhage). [CRASH-2 / ATLS principles]
- Paediatric differences: ligamentous laxity → SCIWORA risk; different imaging and immobilisation thresholds; buckle (torus) distal-radius often managed with removable splint (evolving evidence — FORCE). [FORCE, NICE paediatric guidance]

> شرح موجز بالعربية:
> - الأولوية لإيقاف المخاطر المهددة للحياة فوراً باتّباع cABCDE. آلية الإصابة تحدد نمط الأذى. لدى الأطفال قواعد تصوير وعلاج مختلفة وكسور التورس غالباً تُعالج بجبيرة قابلة للنزع (دراسات جديدة).

2. Recognition, differential diagnosis, and red flags

- Airway compromise: blood pooling/gurgle, stridor, expanding neck haematoma, severe mid-face fracture → sit up, suction, prepare definitive airway. Avoid nasopharyngeal airway if basilar skull fracture suspected. [ATLS]
- Breathing threats:
- Tension pneumothorax: severe dyspnoea, hypotension, (late) tracheal deviation, absent breath sounds, hyperresonance → immediate decompression.
- Open (sucking) pneumothorax: visible chest wound with air entry/bubbling →

three-sided occlusive dressing then chest drain.
- Massive haemothorax: hypotension, dullness, poor air entry → large-bore chest tube and MHP.
- Circulation / vascular injury:
- Hard signs: pulsatile bleeding, expanding haematoma, audible bruit/thrill, absent distal pulses/acute limb ischaemia → urgent operative/angiographic management.
- Pelvic instability/haemodynamic compromise → pelvic binder at greater trochanters and activate MHP. [BOA]
- Neurology / head injury red flags (CT within 1 hour): GCS <14 at 2 hours, focal neuro deficit, post-traumatic seizure, suspected depressed/open skull fracture, basilar skull signs (CSF leak, Battle's sign), repeated vomiting. CT within 8 hours for anticoagulated/antiplatelet patients unless earlier indicated. [NICE NG232]
- Eye/orbit: proptosis, RAPD, decreased vision, resistance to retropulsion → retrobulbar haemorrhage requiring immediate lateral canthotomy/cantholysis. [EyeWiki]
- Limb threats: knee dislocation → popliteal artery injury (absent pulses → urgent reduction ± CTA/vascular input); compartment syndrome: pain out of proportion and pain on passive stretch (pulses late).
- Miscellaneous red flags: exposed bone (open fracture), evisceration, penetrating torso with peritonism, signs of spinal cord compression/herniation (one dilated pupil + dropping GCS).

> شرح موجز بالعربية:
> - علامات حمراء رئيسية: انسداد مجرى الهواء، ذبحة صدرية توتّريّة، رئة مفتوحة، نزف حوضى حاد، علامات عصبية بالجمجمة (تحتاج CT سريع)، نزيف خلف العين (يتطلب عمل جانبي عاجل)، خلع الركبة مع نبضات مفقودة وخطر متلازمة الحجرة (ألم زايد مع الشدّ السّلبى).

3. Diagnostic approach and interpretation

- General: primary survey with simultaneous resuscitation. Use eFAST + portable CXR for unstable patients. Do not send unstable patients to CT. [NICE NG39]
- Head CT timing (NICE NG232):
- Within 1 hour for red flags: GCS <13 initially, GCS <15 at 2 hours, focal neuro signs, suspected open/depressed skull fracture, seizure, repeated vomiting, basilar skull signs.
- Within 8 hours for patients on anticoagulants or with less urgent features where CT still indicated.

- C-spine: CT is first-line in adults with high-risk mechanism, midline tenderness or neurological signs; in children follow paediatric rules and consider MRI if neuro deficits with normal CT (SCIWORA).
- Vascular injury pathway:
- Hard signs → urgent OR/IR ± immediate exploration.
- Soft signs → CTA or serial exams; consider duplex/ABI as adjunct.
- Chest: unstable + suspected tension → immediate decompression (no imaging delay). Stable → CXR/eFAST then CT chest/CTA as indicated.
- Orbit/eye: do not delay lateral canthotomy if sight threatened; imaging after decompression if needed. [EyeWiki]
- Orthopaedic imaging: scaphoid suspicion → thumb axial loading positive = high specificity; if initial XR negative and high clinical suspicion immobilise and arrange repeat imaging or MRI/CT.
- Laboratory/POC: group & save, FBC/coag/ABG/lactate, ionised calcium during massive transfusion.

> شرح موجز بالعربية:
> - المسح الأولي متزامن مع الإنعاش؛ eFAST و CXR للمرضى غير المستقرّين. تصوير الرأس حسب قواعد NICE (1 ساعة أو 8 ساعات). علامات الأوعية الدموية الحادة تحتاج تدخلاً فورياً؛ الإنعاش بالدم فوري عند النزف الحاد.

4. Emergency management algorithms

(A) Airway in facial/neck trauma
- Immediate: sit upright/lean forward, suction, jaw-thrust with C-spine control.
- If cannot maintain airway or failing oxygenation → RSI with surgical airway back-up.
- If intubation fails and definitive surgical airway required:
- Adults/older children: emergency surgical cricothyroidotomy.
- Young children (<10–12 yrs) or anatomy/coagulopathy concerns: needle cricothyroidotomy as temporising measure until definitive airway (local protocols vary). [ATLS; LITFL flagged]

(B) Major haemorrhage / damage-control haemorrhage
1. Control external catastrophic bleeding (direct pressure, tourniquet for limb exsanguination).
2. Two large-bore IVs / IO if needed; limit crystalloids; activate MHP early for

ongoing hypotension.
3. Transfuse balanced blood products per local MHP; massive transfusion commonly defined as ≥10 units PRBC/24 h (definitions vary).
4. Give TXA if within 3 hours: 1 g IV over ~10 min then 1 g over 8 h (CRASH-2 regimen).
5. Apply pelvic binder at greater trochanter level for suspected unstable pelvic ring; then arrange pelvic X-ray/CTA/IR/OR as indicated. [CRASH-2; BOA]

(C) Raised ICP / impending herniation
- Immediate measures: head up 30°, ensure oxygenation/ventilation (avoid hypercapnia), brief hyperosmolar therapy while arranging neurosurgical care:
- Hypertonic saline 3%: bolus 3–5 mL/kg (institutional practice varies); OR
- Mannitol 0.5–1 g/kg if euvolaemic and renal function acceptable.
- Urgent neurosurgical notification and prepare for CT/surgery. Avoid prolonged hyperventilation unless imminent herniation.

(D) Tension pneumothorax / open pneumothorax
- Tension: immediate decompression — in trauma peri-arrest preferred open finger thoracostomy / tube thoracostomy at the bedside; needle decompression only temporising if nothing else available. Then insert chest drain.
- Open pneumothorax: three-sided occlusive dressing (or finger seal) immediately, high-flow oxygen, chest drain definitive.

(E) Limb vascular compromise (e.g., knee dislocation)
- Immediate gentle reduction of dislocation → reassess distal perfusion.
- If pulses absent or ischaemia persists → urgent vascular surgical input ± CTA/OR. Don’t delay reduction/transfer.

(F) Orbital compartment syndrome (retrobulbar haemorrhage)
- If proptosis, severe pain, falling vision, RAPD, resistance to retropulsion → immediate lateral canthotomy and cantholysis at bedside; do not await imaging. [EyeWiki]

(G) Compartment syndrome
- Remove constrictive dressings/cast; keep limb at heart level (avoid excessive elevation).
- If clinical concern persists → measure compartment pressures; delta pressure (diastolic BP − compartment pressure) <30 mmHg or absolute pressure >30

mmHg supports fasciotomy (local thresholds variably used). Urgent fasciotomy if confirmed.

(H) Traumatic cardiac arrest (TCA)
- Prioritise reversible causes: catastrophic haemorrhage (MHP), tension pneumothorax (bilateral finger thoracostomies if chest trauma + arrest), cardiac tamponade (consider ED thoracotomy in selected patients with witnessed arrest and short downtime), airway obstruction.
- Standard ALS drugs/compressions are less effective until reversible causes addressed. Resuscitative thoracotomy reserved for very select patients (penetrating chest with short prehospital CPR / local protocol criteria). [RCEM 2019]

(I) Burns & frostbite
- Burns first aid: cool running water (not ice) for 20 minutes (if within 3 hours), analgesia, cover with cling film, estimate BSA, arrange referral if indicated.
- Frostbite: rapid rewarming in circulating water 40–42°C until tissue pliable (≈15–30 min); avoid refreeze. [RCEM frostbite guidance]

> شرح موجز بالعربية:
> - خوارزميات رئيسية: تأمين المجرى الهوائي وتهيئة العملية الجراحية للمجرى في الكسر الوجهي/الرقبي، تفعيل بروتوكول النزف الكبير مع نقل متوازن ومنح TXA ضمن 3 ساعات، تفريغ الضغط الدماغي بالمحاليل عالية التوتر أو المانيتول، تفريغ الرئة التوتّري فورياً، تقليل الخلع لإعادة الدورة الدموية عند خلع الركبة، عمل جانبي للعين عند نزف خلف العين، وشقّ الحجرة وإحالة فورية إذا لزم.

5. Drugs, procedures, disposition, and follow-up

Drugs / key doses (well-established)
- Tranexamic acid (TXA): 1 g IV bolus over ~10 min then 1 g IV over 8 h — within 3 h of injury for major bleeding (CRASH-2). [CRASH-2]
- Hyperosmolar therapy for raised ICP:
- Mannitol 0.5–1.0 g/kg IV (use if euvolaemic, check renal function).
- Hypertonic saline 3% bolus ~3–5 mL/kg (institutional doses vary).
- Tetanus post-exposure: for unvaccinated with dirty/deep wound give human tetanus immunoglobulin (HTIG) + active vaccination per local schedules.
- Bite prophylaxis: co-amoxiclav first-line for human/dog bites with skin break (IV for deep/hand bites; oral for minor prophylaxis) — verify local dosing. [NICE CKS/NG184]

- Haemophilia with bleeding/head injury: urgent specific factor replacement (haematology guidance) and early specialist contact.
- Avoid nasal airways in basilar skull fracture.

Procedures (indications & cautions)
- Emergency surgical cricothyroidotomy: definitive airway if failed intubation and cannot oxygenate (adult).
- Needle cricothyroidotomy: temporising in small children or when surgical cricothyroidotomy contraindicated; provides oxygenation but limited ventilation — escalate early. (Age cutoffs vary; verify local policy.)
- Chest drain: insert in safe triangle (5th ICS mid-axillary line commonly) for pneumothorax/haemothorax; large bore for massive haemothorax per local practice. Beware diaphragmatic rupture — do not drain suspected intra-abdominal viscera in chest.
- Lateral canthotomy / cantholysis: bedside for orbital compartment syndrome; do not await imaging. [EyeWiki]
- Fasciotomy: for clinically suspected compartment syndrome (do not wait for pulseloss).
- Pelvic binder: apply at greater trochanters; remove only after stabilisation and definitive imaging/definitive plan. [BOA]

Disposition & follow-up
- Immediate OR/IR transfer: unstable torso with peritonism/evisceration, uncontrolled pelvic haemorrhage needing IR/OR, ongoing massive thoracic bleeding, ED thoracotomy indications.
- ICU/High-dependency: airway protection after major trauma, ongoing MHP, raised ICP needing monitoring, major burns, multi-system trauma.
- Observation/ward: minor injuries with safe discharge criteria and arranged follow-up (orthopaedics/plastics/ENT/eye/neurosurgery as indicated).
- Head injury: follow local observation protocols; repeat CT if neurological deterioration; discharged head-injury patients need clear return advice and arranged follow-up as required.
- Orthopaedics/plastics/vascular referrals: open fractures, tendon/nerve injuries (digital nerve repairs ideally early, <24 h for clean wounds), vascular injury, compartment syndrome.

> **EXAM TRICK / تريكة امتحانية**
> Pitfalls / safety notes

- Don’t remove pelvic binder pre-imaging/stabilisation unless necessary; binder can mask fracture displacement on CT—plan post-binder imaging. [BOA]
- Avoid excessive crystalloid in uncontrolled haemorrhage; early balanced transfusion is preferred.
- Verify antibiotic choices/durations and local MHP ratios — institutional variations exist.

شرح موجز بالعربية:
- أدوية رئيسية: TXA (1 g ثم 1 g/8h ضمن 3 ساعات)، مانيتول 0.5–1 g/kg أو 3% محلول ملح عالي التوتر 3–5 mL/kg لخفض الضغط داخل الجمجمة. إجراءات رئيسية: تريكوثيرويدوتومي للطوارئ، درن صدري للحالات المناسبة، شق جانبي للعين عند النزف خلف العين، وفاسيوتومي عند متلازمة الحجرة. فرّق التنويم بين نقل عاجل للعمليات، رعاية مكثفة، ومتابعة مسارية.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية

Exam tricks
- “Pelvic binder at greater trochanters” is a frequent MCQ answer — learn placement.
- Scaphoid: pain on axial thumb loading is highly specific (snuffbox tenderness is sensitive). Use “thumb axial loading” as exam cue.
- Anterior cord syndrome: motor + pain/temp loss with preserved proprioception/vibration — classic localisation.
- Retrobulbar haemorrhage: vision threat → immediate lateral canthotomy (don’t wait for imaging).
- For head injury CT timing memorise “1 hour vs 8 hours” split from NICE NG232 (1 h for red flags; 8 h for anticoagulated/less urgent features).

EXAM TRICK / تريكة امتحانية

Common pitfalls
- Don’t nasal intubate or insert a nasogastric tube with suspected basilar skull fracture.
- Pelvic binder can mask fractures on CT — plan imaging accordingly; don’t remove binder until stable.
- In traumatic cardiac arrest prioritise reversible mechanical causes (tension pneumothorax, tamponade, massive haemorrhage) over routine ALS emphasis

— RCEM emphasises cause-directed treatment. [RCEM 2019]
- Pulses may remain present in early compartment syndrome — the earliest sign is disproportionate pain and pain on passive stretch.
- Do not delay definitive airway in facial/neck trauma for imaging; prepare surgical airway early.
- Beware statements that “completely avoid immobilisation for paediatric buckle fractures” — FORCE shows evolving evidence but practice may vary; exam answers often expect removable splint/cast.
- Know that many numeric thresholds (ED thoracotomy time cutoffs, MHP pack composition, chest drain sizes) vary by local policy — flag such numbers in the stem and default to local protocol where required.

شرح موجز بالعربية:
- حِيَل امتحانية: وضع رباط الحوض على الشوك العظمي الأكبر، اختبار تحميل الإبهام للكسر الإسفيني، متلازمة الحبل الأمامية وصفاتها، عمل جانبي العين فوراً عند خطر البصر. مطبات شائعة: لا تدخل أنبوب أنفى عند كسر قاعدي، النبض لا ينفي متلازمة الحجرة، وركز على معالجة الأسباب القابلة للعكس في توقف القلب الرضحي.

7. Arabic rapid-review summary (ملف سريع)

- قم أولاً بتأمين المجرى الهوائي مع حماية العمود الفقري، ثم التنفّس والدورة الدموية (cABCDE).

- علّم تجنّب التصوير للمريض غير المستقر — عالج الأسباب القابلة للعكس فوراً (تفريغ صدرى توتّري، نزف حوضى، انصباب تامورى). [RCEM; NICE]
- تصوير الرأس: اتبع NICE NG232 — تصوير خلال ساعة للعلامات الحمراء؛ خلال 8 ساعات للمخاطر المتوسطة مثل تناول مضادات التخثر. [NICE NG232]
- نزف كبير: فعّل بروتوكول النزف الكبير، قلّل البلورات، أعطِ منتجات دموية متوازنة، وانقل TXA ضمن 3 ساعات (1 g + 1 g). [CRASH-2]
- إصابة خلف العين (retrobulbar) = عمل جانبي عاجل (lateral canthotomy) فوراً إذا انخفض البصر.
- خلع الركبة مع نبضات مفقودة → تقليل فوري وإحالة جراحية للأوعية الدموية إن استمرت الآفات.
- الحروق: تبريد جارى 20 دقيقة، تغطية بالحافظة، تقييم نسبة مساحة الجسم المصابة.
- لدغات الإنسان/الكلاب العميقة أو اليدية → co-amoxiclav مبدئياً وتقييم جرعة/طريق إعطاء محلى. [NICE CKS]

8. Question coverage map (concepts covered; original Q-numbers not supplied)

- Head injury imaging and timing (NICE NG232): CT within 1 hour vs 8 hours — concepts from head-injury stems.
- Traumatic cardiac arrest: priorities, indications for ED thoracotomy, withholding resuscitation context (RCEM 2019).
- Major haemorrhage / MHP / TXA dosing and timing (CRASH-2 / ATLS principles).
- Airway management in facial/neck trauma; surgical vs needle cricothyroidotomy and paediatric considerations (ATLS / LITFL notes).
- Thoracic trauma: tension pneumothorax, open pneumothorax, massive haemothorax, chest drain and thoracotomy indications (NICE NG39, RCEM).
- Orbital compartment syndrome / lateral canthotomy (EyeWiki).
- Pelvic fracture management and pelvic binder issues (BOA guidance).
- Paediatric fractures (buckle/torus distal radius) and FORCE trial implications.
- Frostbite rewarming (RCEM frostbite guidance).
- Compartment syndrome recognition and fasciotomy thresholds.
- Vascular injury pathway: hard vs soft signs, knee dislocation and popliteal artery risk.
- Bite management and tetanus prophylaxis (NICE CKS / NG184).
- Burns first aid and escharotomy indications.
- SCIWORA / paediatric spinal cord injury — MRI indications.
- Orthopaedic patterns (posterior shoulder/hip dislocations, scaphoid tests, Maisonneuve/ankle syndesmosis).
- Miscellaneous procedural cautions: avoid NPA with basilar skull fracture, avoid chest drain insertion into suspected diaphragmatic rupture without imaging.

(Note: specific original question numbers were not supplied in the chunks; mapping above summarises all distinct concepts present in the supplied material.)

Sources to verify
- RCEM guidance (Traumatic Cardiac Arrest, 2019; frostbite modules)
- NICE guidelines NG232 (Head injury), NG39 (Major trauma / thoracic), NG41 (Paediatric C-spine), CKS/NG184 (Bites)
- ATLS (10th edition) airway/initial resuscitation principles
- CRASH-2 and CRASH-3 trials (TXA in trauma/TBI)
- BOA guidance on pelvic binders/pelvic trauma
- FORCE trial (paediatric distal radius buckle fractures)
- EyeWiki (retrobulbar haemorrhage / lateral canthotomy)
- LITFL, WJES summaries and local Major Haemorrhage Protocols

Final note: this chapter is study material only — check local protocols for exact thresholds, device specifics, drug doses and institutional MHP composition.

CHAPTER 05: VASCULAR

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Chapter 05: Vascular

1. Clinical overview and pathophysiology

- Major vascular emergencies: acute aortic syndromes (dissection, rupture), ruptured abdominal aortic aneurysm (rAAA), acute arterial occlusion (embolus, in-situ thrombosis), extensive venous thrombosis/phlegmasia, penetrating/iatrogenic arterial injury and pseudoaneurysm, carotid/vertebral artery dissection, vascular injury from procedures/central lines.
- Pathophysiology essentials:
- Aortic dissection: intimal tear → false lumen propagation, organ malperfusion, aortic regurgitation, rupture. Type A needs surgery; Type B often medical unless complications.
- rAAA: contained → free rupture, massive haemorrhage and shock.
- Acute limb ischaemia (ALI): sudden arterial inflow loss → muscle/nerve ischemia; muscle viability commonly cited ~4–6 h.
- Phlegmasia cerulea dolens: massive venous outflow obstruction → oedema, venous gangrene risk.
- Iatrogenic arterial injury: hard signs (pulsatile/expanding haematoma, active arterial bleeding, bruit/thrill, ischaemia) predict need for urgent OR.

> Arabic: لمحة وشرح المرضي — حالات أوعوية رئيسية: تمزق الأبهر/تمدد منهار، انسداد شرياني حاد، تخثر وريدي واسع، إصابات شريانية رضّية أو ناتجة عن إجراءات. تلف العضلات في انسداد الطرف الحاد يحدث بسرعة (~4–6 س).

2. Recognition, differential diagnosis, and red flags

- Aortic dissection: sudden severe ‘tearing’ chest/back pain; inter-arm BP difference >20 mmHg; new diastolic murmur (acute AR); syncope/neurology; widened mediastinum on CXR suggestive but not sensitive. CTA = definitive imaging. [RCEM/RCR 2025]
- rAAA: sudden abdominal/back pain ± hypotension ± pulsatile abdominal mass; unstable → immediate operative management.

- ALI: the “6 Ps” — pain, pallor, pulselessness, paresthesia, paralysis, poikilothermia. Sensory change and severe pain occur before pulse loss. Look for AF/embolic source.
- Phlegmasia: massively swollen, painful, cyanotic limb; pulses may be present early; high risk of venous gangrene.
- Carotid/vertebral dissection: recent neck trauma/manipulation, unilateral neck pain/headache, Horner’s syndrome, focal cerebral ischaemia.
- Iatrogenic/penetrating arterial injury: hard signs → immediate OR. Soft signs (history of penetrating injury, expanding haematoma without clear bleeding) → urgent imaging/vascular review.

> Arabic: التعرف والتفريق — ألم صدري/ظهري تمزيقي لتمزق الأبهر، ألم بطني/ظهري شديد مع خفة ضغط للـ 6P، rAAA للانسداد الشرياني الحاد، ساق زرقاء ومؤلمة للـ phlegmasia، علامات «hard» للنزف الشرياني التداخلي تعني حاجة فورية للجراحة.

3. Diagnostic approach and interpretation

- Immediate bedside: ABC, ECG (search for AF/MI), focused vascular exam, handheld Doppler, ABPI (abnormal <0.9; borderline 0.9–1.0; non-compressible >1.4 suggests calcification).
- Imaging selection:
- CTA chest/whole-aorta: first-line for suspected thoracic aortic dissection and planning (fast, widely available). [RCEM/RCR 2025]
- TTE/TEE: TTE for pericardial effusion/AR; TEE more sensitive for proximal aortic dissection if available.
- Duplex US: first-line for femoral pseudoaneurysm and for lower-limb DVT; useful when CTA unavailable or for rapid bedside assessment.
- CT angiography (arterial phase): for mesenteric/ischaemic limbs and operative planning.
- Laboratory utility: D-dimer nonspecific — may be raised in dissection but cannot reliably exclude high-risk presentations; use only in low-risk algorithms. Baseline bloods for ALI/reperfusion: CBC, U&E, creatinine, CK, lactate, coagulation, blood type & screen.
- Practical caveats: do NOT delay life-saving operative transfer for CT in haemodynamically unstable rAAA; do not delay anticoagulation if ALI suspected.

Arabic: نهج تشخيصي — ECG، دوبلر يدوي، ABPI، CTA للأبهر، دوبلكس للأوردة/خراجات القسطرة؛ D-dimer محدود الفائدة في الحالات عالية الخطورة. لا تؤخر العمليات الطارئة من أجل التصوير.

4. Emergency management algorithms

A. Suspected acute aortic dissection
- Immediate:
1. ABC, high-dependency monitoring, IV access, analgesia (IV opiate).
2. Start IV beta-blocker (e.g., IV labetalol/esmolol) first to reduce shear; target HR <60, then add vasodilator (GTN/nicardipine) if SBP > target. Aim SBP ≈100–120 mmHg and HR <60 (guideline-dependent). [RCEM/RCR 2025]
3. Urgent vascular/cardiothoracic surgery for Type A; Type B medical unless malperfusion/rupture → urgent TEVAR/surgery.
4. If tamponade/rupture → immediate surgery.

B. Suspected ruptured AAA (unstable)
- Immediate:
1. ABCs, permissive hypotension (avoid large-volume crystalloids that may worsen bleeding) — target SBP ranges vary; aim to maintain perfusion while avoiding hypertension. [RCEM rAAA 2021]
2. Activate vascular/trauma/anaesthesia and transfer to OR/IR immediately; do not delay for CT if unstable.

C. Acute limb ischaemia (ALI)
- Immediate algorithm:
1. Start IV unfractionated heparin unless contraindicated (see section 5).
2. Urgent vascular/IR consult and preparation for revascularisation (embolectomy/thrombectomy, bypass, or catheter-directed thrombolysis depending on viability/time).
3. Analgesia; limb flat and warm; avoid elevation; frequent neurovascular checks.
4. Monitor for compartment syndrome → prepare for fasciotomy if clinical signs.

D. Phlegmasia cerulea dolens
- Immediate:
1. IV heparin.
2. Urgent vascular intervention for thrombectomy or catheter-directed

thrombolysis ± stenting.
3. Monitor perfusion and compartment pressures; fasciotomy if required.

E. Penetrating/iatrogenic arterial injury with hard signs
- Immediate surgical control: direct pressure, proximal control; urgent transfer to OR. Do not delay for imaging if limb or life threatened.

F. Pseudoaneurysm after arterial access
- Confirm with duplex; treat by ultrasound-guided thrombin injection or compression if appropriate; surgery if infected/large/expanding.

> Arabic: خوارزميات الطوارئ — تمزق الأبهر: خفض ضربات القلب والضغط، Type A جراحة فورية. rAAA: انخفاض ضغط مقبول ونقل فوري للعمليات. ALI/phlegmasia: هيبارين فوري ومراجعة وعائية لإعادة التروية. إصابة شريانية مع علامات «hard» → إلى غرفة العمليات فوراً.

5. Drugs, procedures, disposition, and follow-up

Drugs (well-established thresholds/doses where commonly used)
- Beta-blockers in dissection: IV labetalol/esmolol titrated to HR <60 (drug choice and doses per local guidance). Give beta-blocker before vasodilator. [RCEM/RCR 2025]
- Vasodilators: IV nicardipine/GTN added if BP remains above target after HR control.
- Unfractionated heparin (ALI/phlegmasia):
- Bolus commonly 70–100 units/kg IV; many EDs use 5,000 units if weight unknown.
- Infusion ~15–20 units/kg/hr, titrate to aPTT ~1.5–2.5× control per local protocol.
- Institutional protocols vary—verify local anticoagulation guidance.
- Reversal agents: PCC/Vitamin K for urgent warfarin reversal per haematology/ surgical advice.

Procedures
- Definitive: emergent thoracic aortic repair (Type A), TEVAR (complicated Type B), open repair or endovascular repair for rAAA, embolectomy/thrombectomy for ALI, catheter-directed thrombolysis or mechanical thrombectomy for phlegmasia, fasciotomy for compartment syndrome.
- Emergency OR vs imaging: hard signs of vascular injury or unstable rAAA/ tamponade → immediate OR without further imaging if required.

- Pseudoaneurysm: ultrasound-guided thrombin injection, ultrasound compression, or surgical repair.

Disposition and follow-up
- Admit to high-dependency or ICU for most aortic syndromes, rAAA postoperative care in critical care.
- ALI: urgent admission to vascular theatre/IR; postop/after revascularisation monitor for reperfusion injury (hyperkalaemia, acidosis, rhabdomyolysis, renal failure).
- Phlegmasia: urgent vascular intervention; inpatient anticoagulation and thrombosis workup after stabilization.
- Carotid/vertebral dissection: admit for stroke workup, antithrombotic therapy and vascular/neurology follow-up.
- Outpatient follow-up: vascular clinic, surveillance imaging for aneurysms/pseudoaneurysms, review anticoagulation strategy and secondary prevention (AF management, antiplatelet/statins as indicated).

> Arabic: أدوية وإجراءات ووجهة المريض — هيبارين: جرعات شائعة 70–100 وحدة/كجم (أو 5000 وحدة إذا مجهول الوزن)، ضخ ~15–20 وحدة/كجم/ساعة؛ بيتا-بلوكر أولاً في التمزق الأبهري؛ إجراءات نهائية: جراحة الأبهر، TEVAR، استئصال الصمة/تفتيت الخثرة، دعامات/فاسيوتومي حسب الحاجة. متابعة في وحدة عالية الاعتماد ومراجعة الأوعية/أمراض الدم/الأعصاب.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية
- Tricks:
- “6 Ps” for ALI; remember sensory change and severe pain precede absent pulses.
- Arm BP difference >20 mmHg is a strong red flag for dissection.
- Beta-blocker BEFORE pure vasodilator in dissection — exam favourite.
- If AF present + sudden limb pain → think embolus → start heparin.
- Common pitfalls:
- Normal CXR does NOT exclude aortic dissection—don’t be reassured by a normal film.
- Don’t give a vasodilator alone in suspected dissection (can increase shear).
- Do NOT delay UFH in ALI for imaging; imaging is for operative planning.
- ABPI >1.4 suggests non-compressible calcified arteries — false normal.

- D-dimer: may be raised in dissection but a negative result does not exclude high-risk presentations; use only in low-risk algorithms.
- In unstable rAAA/tamponade: do NOT delay for CT — prioritise transfer to theatre.
- Beware of reperfusion injury after limb revascularisation — monitor electrolytes, CK and urine output.

Arabic: خدع امتحانية ومزالق — 6P للانسداد الشرياني، فرق ضغط الذراع مفيد للتمزق، بيتا-بلوكر قبل موسع للأوعية، لا تؤجل الهيبارين من أجل التصوير، CXR طبيعي لا ينفي التمزق، D-dimer غير موثوق في الحالات عالية الخطورة.

7. Arabic rapid-review summary

- تمزق الأبهر: ألم صدري/ظهري مفاجئ تمزيقي، فرق ضغط بين الذراعين، ابدأ بيتا-بلوكر ثم اضبط الضغط؛ Type A تحتاج جراحة فورية. [RCEM/RCR 2025]
- تمدد أبهري متمزق: ألم بطني/ظهري وعلامات صدمة → ضغط مقبول ونقل فوري إلى العمليات؛ لا تؤجل للتصوير إن غير مستقر. [RCEM rAAA 2021]
- انسداد شرياني حاد: 6P؛ ابدأ هيبارين ورافق للمراجعة الوعائية فوريًا؛ نافذة إنقاذ العضلة ~4–6 س.
- تخثر وريدي واسع (phlegmasia): ساق زرقاء ومتوّرة → هيبارين وفعل وعائي عاجل (خطر غنغرينا).
- إصابة شريانية «hard» → إلى غرفة العمليات فورًا؛ كدمة/وريقة كاذبة بعد قسطرة → دوبلكس ثم حقن ثومبين أو جراحة حسب الحالة.
- تحذيرات: D-dimer وCXR محدودة؛ ABPI>1.4 يدل على أوعية متكلسة؛ راجع البروتوكولات المحلية للجرعات وحدود ضغط الدم.

8. Question coverage map (topics / source tags)

- Acute limb ischaemia (ALI) — clinical signs, immediate heparin, embolectomy/IR, reperfusion complications (Source: Q33; general ALI content)
- Aortic dissection — recognition (tearing pain, interarm BP difference), imaging (CTA/TEE), BP/HR targets, beta-blocker before vasodilator [RCEM/RCR 2025]
- Ruptured AAA — recognition, permissive hypotension, immediate operative management [RCEM rAAA 2021]
- Phlegmasia cerulea dolens — presentation, heparin, urgent thrombectomy/ thrombolysis

- Iatrogenic/penetrating arterial injury & pseudoaneurysm — hard signs → OR; duplex, thrombin injection/compression options
- Carotid/vertebral dissection — neck pain/headache after minor trauma, Horner's, stroke risk
- Diagnostic tests — CXR, CTA, duplex US, TTE/TEE, ABPI, D-dimer limitations
- Procedural/complication management — fasciotomy for compartment syndrome, reperfusion injury monitoring/treatment, central-line removal precautions
- Practical dosing/targets flagged for verification — UFH dosing, aPTT targets, BP ranges for dissection and rAAA (see notes above)

Final verification note
- Statements about exact numeric BP targets, universal fixed heparin boluses, and D-dimer utility are guideline-dependent and flagged for local verification.

> Sources to verify
> - RCEM rAAA guideline 2021; RCEM/RCR guidance 2025; national vascular surgery guidelines; European Resuscitation Council (ERC); NICE; institutional anticoagulation protocols; current vascular surgery and interventional radiology guidance.

This is study material, not patient-specific advice. Verify local protocols and latest guidelines before clinical use.

CHAPTER 06: UROLOGY

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Chapter 06 — Urology

1. Clinical overview and pathophysiology

- Time-critical urology emergencies: testicular torsion, ischaemic priapism, penile fracture, obstructing infected urinary stone — early action preserves function.
- Presentations cluster around severe localized pain, urinary obstruction, haematuria, swelling, or systemic infection. Pathology may be vascular (torsion, priapism), traumatic (fracture), obstructive (stone, catheter blockage) or infective/ inflammatory (epididymo-orchitis, prostatitis, balanitis).
- ED role: rapid recognition, bedside tests/imaging, prompt analgesia/antibiotics, urgent specialist referral or immediate procedural intervention when indicated.

> Arabic: لمحة وباثوفيزيولوجيا — الطوارئ البولية قد تكون وعائية أو رضّية أو عائقاً أو التهابياً؛ التدخل المبكر مهم للحفاظ على الوظيفة.

2. Recognition, differential diagnosis, and red flags

- Flank/groin pain: stone (colic) vs pyelonephritis vs AAA. Haematuria common with stones but not required. Red flags: sepsis signs, anuria, AKI, solitary kidney or bilateral obstruction.
- Acute scrotum: torsion = sudden severe pain, high-riding/ horizontal testis, absent cremasteric reflex, nausea/vomiting. Epididymo-orchitis = gradual onset, urinary symptoms, intact cremasteric reflex, may improve with scrotal elevation (Prehn’s sign unreliable). Blue-dot sign suggests torsion of testicular appendix. Use TWIST score to stratify risk [MDCalc]. Red flags: fever/sepsis, skin changes, testicular viability threatened.
- Priapism: ischaemic (painful, rigid erection >4 h) vs non-ischaemic (painless, partially rigid). Red flags: sickle cell disease, prolonged erection >4 h, severe pain — needs urgent decompression.
- Penile trauma: “pop”, immediate detumescence, swelling/ecchymosis = penile (tunica) fracture — surgical emergency.
- Balanitis/balanoposthitis: local erythema, white patches (candida), pruritus.

- Acute prostatitis: systemic/voiding symptoms + tender boggy prostate on DRE. Avoid prostatic massage.
- Suprapubic catheter (SPC) problems: blocked/new tract risk — do not change new SPC in ED without urology.

> Arabic: التعرّف والتمييز — ميز بين مغص كلوي، الصفن الحاد، بريابيْسِم ناقص التروية، وكسور القضيب؛ راقب علامات الخطر مثل الحمى، فشل الكلوي، أو قثطرة فوق العانة حديثة مسدودة.

3. Diagnostic approach and interpretation

- Renal/ureteric stone: low-dose non-contrast CT KUB = gold standard in adults; renal ultrasound for pregnancy/young females or to detect hydronephrosis if CT unsuitable [NICE CKS]. Urine dipstick for blood/leucocytes supportive but neither sensitive nor specific.
- Acute scrotum: high TWIST → immediate exploration; intermediate → urgent Doppler scrotal ultrasound; low → consider conservative management. Do not delay theatre if clinical suspicion high.
- Priapism: clinical diagnosis. Cavernosal blood gas differentiates low-flow (acidotic, hypoxic) from high-flow (oxygenated).
- Epididymo-orchitis: Doppler ultrasound to exclude torsion if uncertain; urine culture and STI testing guided by age/sexual history.
- Painless macroscopic haematuria in older adults → urgent urology referral for malignancy workup [NICE CKS].
- SPC blockage: try sterile saline flush first (do not change SPC <7 days old in ED). If flushing fails and tract matured, consult urology for exchange/cystoscopy.

> Arabic: الخطة التشخيصية — اعتمد CT منخفض الجرعة للحصى عند البالغين، الموجات فوق الصوتية للخصية عند الشك؛ الفحص السريري غالباً حاسم في تورشن وبريابيْسِم.

4. Emergency management algorithms

A. Renal/ureteric colic (suspected obstructing stone)
- Step 1: rapid assessment for sepsis, AKI, solitary kidney/bilateral obstruction. If present → urgent admission/urology.
- Step 2: analgesia immediately — NSAID first-line unless contraindicated (e.g., diclofenac PR 100 mg single dose or equivalent) [RCEM]. Antiemetic and IV fluids if vomiting/dehydration.

- Step 3: imaging (CT KUB) if diagnosis uncertain or to guide intervention. Admit if sepsis, uncontrolled pain/vomiting, obstructing solitary kidney, bilateral obstruction, or AKI risk [NICE CKS].

B. Acute scrotum (suspected torsion)
- Step 1: rapid clinical assessment (TWIST score). If high probability (TWIST ≥5) → immediate urology/surgical exploration; DO NOT wait for ultrasound.
- Step 2: intermediate probability → urgent Doppler US; negative/uncertain but persistent concern → explore.
- Step 3: if epididymo-orchitis suspected → analgesia, scrotal support, bed rest, antibiotics guided by age/sexual history (see section 5).

C. Ischaemic priapism (painful, >4 h)
- Step 1: analgesia + dorsal penile nerve block. Obtain cavernosal blood gas if available.
- Step 2: aspirate corpora cavernosa blood with saline irrigation. If detumescent → observe.
- Step 3: if persistent → intracavernosal phenylephrine titrated per protocol (see doses). Monitor BP/HR; continuous monitoring advised.
- Step 4: if refractory, urgent urology for shunt procedures; for sickle cell consider oxygen, IV fluids, urgent haematology/urology discussion for exchange transfusion.

D. Penile fracture
- Step 1: clinical diagnosis; immediate urology/surgical exploration and repair; do not delay for imaging if classic exam/history.

E. Paraphimosis
- Step 1: analgesia/local block, attempt manual reduction with compression/ osmotic measures (sugar, mannitol).
- Step 2: if fails → dorsal slit or other surgical release (urology).

F. SPC blockage
- Step 1: attempt sterile saline flush and repositioned patient; if SPC <7 days old do not change; contact urology if persists for exchange under guidance.

G. Post-obstructive diuresis
- Step 1: monitor urine output hourly, replace fluids/electrolytes to match losses, monitor electrolytes and renal function; consider observation/admission if large volumes.

> Arabic: خوارزميات الطوارئ — خطوات سريعة لكل حالة: مغص كلوي → NSAID أولاً + تصوير؛ الصفن الحاد → استكشاف فوري عند اشتباه قوي؛ بريابيْسِم ناقص التروية → تخفيف، شفط، فينيلإفرين، استدعاء المسالك البولية؛ كسر القضيب → جراحة عاجلة.

5. Drugs, procedures, disposition, and follow-up

Drugs (examples; verify local formularies)
- Analgesia for renal colic: NSAID first line (e.g., diclofenac PR 100 mg single dose) unless contraindicated [RCEM]. Opiates if NSAID contraindicated or inadequate.
- Epididymo-orchitis (likely STI, sexually active younger men): ceftriaxone 1 g IM single dose + doxycycline 100 mg PO twice daily for 10–14 days (age/sexual history dependent; check local guidance) [NICE CKS].
- Acute bacterial prostatitis: ciprofloxacin 500 mg PO twice daily or ofloxacin 200 mg PO twice daily for 14 days (verify local guidance and consider urine culture) [NICE CKS].
- Priapism: intracavernosal phenylephrine 100–500 µg every 3–5 min, titrate to effect with cardiovascular monitoring — follow local/urology protocol and contraindications (cardiovascular disease). Verify dosing/cumulative limits locally.
- Antiemetics/IV fluids as required.

Procedures (ED-performed vs specialist)
- Testicular exploration/orchidopexy: surgical — urgent urology/surgery. ED should arrange transfer; do not delay for ultrasound if high suspicion.
- Cavernosal aspiration and intracavernosal phenylephrine: may be initiated in ED with appropriate training and monitoring; always involve urology early.
- Penile fracture repair: surgical exploration/repair by urology.
- SPC flush: sterile saline flush in ED; exchange by urology when indicated.
- Doppler scrotal ultrasound: ED-requested urgent imaging when indicated.

Disposition and follow-up
- Admit: sepsis, obstructing infected stone, AKI/oliguria/anuria, solitary-kidney obstruction, uncontrolled pain/vomiting, priapism >4 h or refractory, penile fracture, failed SPC flush when urology input needed.
- Discharge with safety net: uncomplicated renal colic with analgesia, arranged outpatient urology/urology clinic, clear return precautions (fever, anuria, increasing pain, persistent erection). Provide STI testing/follow up where appropriate.

- Arrange urgent outpatient urology for painless macroscopic haematuria in older adults.

Arabic: أدوية وإجراءات ومتابعة — استخدم مضادات الالتهاب للألم؛ المضادات الحيوية وفق العمر وتاريخ جنسي؛ فينيلإفرين داخل الكهف وفق بروتوكول؛ إجراءات مثل استكشاف الصفن وكسر القضيب وجراحة الشنت هي لمتخصصي المسالك؛ ادخل حالات العدوى الحادة والانسداد والخطر الكلوي.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية

- TWIST score: useful triage tool — high score (≥5) → operate; do not let a “low” score delay action if exam strongly suggests torsion [MDCalc].
- Prehn’s sign: poor sensitivity/specificity — do not rely on it to exclude torsion.
- Haematuria: absence does NOT exclude ureteric stone; presence is non-specific. Avoid overreliance on dipstick alone.
- Imaging: CT KUB gold standard for adult stones but avoid in pregnancy — use ultrasound. Do not delay transfer for CT if immediate surgery required.
- Prostate massage: contraindicated in suspected acute bacterial prostatitis (risk of bacteraemia/sepsis).
- SPC care: do not replace a newly placed suprapubic catheter in ED; flush first and consult urology.
- Phenylephrine: effective but can cause hypertension/arrhythmia — monitor; verify dosing/cumulative limits locally.
- Antibiotic regimens: tailor to age and sexual risk — ceftriaxone + doxycycline is for likely STI-related epididymo-orchitis in younger men, not blanket therapy for all cases.
- Exam trick: “pop + immediate detumescence” strongly suggests penile fracture — surgical repair is required; don’t be reassured by absent external bleeding.

Arabic: حِيَل الامتحان والأخطاء — TWIST مفيد لكن لا تعوّل عليه فقط؛ لا تستبعد الحصى بغياب الدم؛ لا تمارس تدليك البروستاتا في التهاب حاد؛ لا تغيّر قثطرة فوق العانة حديثة في الطوارئ.

7. Arabic rapid-review summary

- توقعات سريعة: التواء الخصية، بريابيْسِم ناقص التروية، وكسر القضيب طوارئ جراحية زمنية؛ التصوير بالموجات فوق الصوتية للصفن أو CT منخفض الجرعة لحصى الكلية حسب الحالة.
- الإجراءات العاجلة: NSAID للمغص الكلوي، شفط الجوف الكهفي ثم فينيلإفرين لِـ priapism، استكشاف جراحي لكسر القضيب/التواء الخصية، غسيل قثطرة فوق العانة المسدودة أولاً.
- مضادات حيوية: اضبطها حسب العمر والتاريخ الجنسي (سيفترياكسون + دوكسيسيكلين للحالات المشتبهة بالـSTI؛ سيبروفلوكساسين للبروستاتيت؛ تحقق من البروتوكولات المحلية).
- تحذيرات: لا تجرّب تدليك البروستاتا عند الاشتباه بالتهاب حاد؛ لا تغيّر قثطرة فوق العانة حديثة في ED؛ راقب الضغط قبل/أثناء إعطاء فينيلإفرين.

Arabic: ملخص سريع — تعرف سريعًا، اعمل بسرعة، واستشر المسالك البولية عند الحاجة؛ اتّبع بروتوكولات المضادات والجرعات المحلية.

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية

- Renal/ureteric colic: presentation, analgesia, CT vs US, red flags (sepsis/AKI), disposition.
- Acute scrotum: torsion vs epididymo-orchitis vs appendage torsion, TWIST score, Doppler, exploration thresholds.
- Priapism: ischaemic vs non-ischaemic, cavernosal aspiration, intracavernosal phenylephrine, sickle-cell considerations.
- Penile fracture: recognition and urgent surgical repair.
- Paraphimosis and balanitis/balanoposthitis: recognition, conservative reduction, topical therapy.
- Acute bacterial prostatitis: recognition, DRE cautions, antibiotic choices.
- Epididymo-orchitis (STI vs non-STI): antibiotics and imaging.
- Suprapubic catheter issues: blocked SPC management, timing for exchange.
- Post-obstructive diuresis: monitoring and fluid/electrolyte replacement.
- Exam pitfalls/tricks: Prehn’s sign, haematuria interpretation, CT use in pregnancy, SPC management, phenylephrine cautions.

Sources to verify: RCEM guidance, NICE CKS (renal/ureteric colic; epididymo-orchitis; acute prostatitis; balanitis; suprapubic catheter management), MDCalc (TWIST score), ERC/local urology protocols, relevant urology guidance and NCBI/urology literature.

(Note: This chapter is study material, not patient-specific advice. Verify specific drug doses and procedural techniques with local/urology protocols before clinical use.)

CHAPTER 07: ULTRASOUND

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Chapter 07: Ultrasound

1. Clinical overview and pathophysiology

- Role: POCUS = focused bedside extension of clinical exam; ask one targeted question per scan (free fluid? tamponade? pneumothorax? AAA? DVT?). Use correct probe, window, and limited protocol (FAST/eFAST, FOCUS, lung, vascular). [POCUS101]
- Probes and windows:
- Curvilinear or phased-array (2–5 MHz): abdomen, FAST, cardiac (penetration).
- Linear high-frequency: pleura, superficial vessels, DVT compression.
- Standard cardiac windows: subxiphoid (subcostal), parasternal long-axis (PLAX), parasternal short-axis (PSAX), apical 4-chamber (A4C). [POCUS101]
- Orientation: pick one screen convention and learn it (e.g., in A4C left-sided chambers appear on the right of the screen). Marker orientation matters (PLAX marker → patient's right shoulder). [POCUS101]
- Physiology signposts:
- AAA: maximal transverse diameter measured outer wall → outer wall, perpendicular to vessel axis. Report maximal transverse diameter. [RCEM]
- RV dysfunction: TAPSE <17 mm suggests reduced RV longitudinal function (use clinically, not in isolation). [POCUS101]
- Lung sliding: present = pleural apposition; M-mode seashore sign = normal; barcode/stratosphere = absent sliding (pneumothorax or mimic). [POCUS101]

> Arabic (الفصحى):
> POCUS أداة مركّزة للتشخيص السريع؛ اختر المسبار والنوافذ المناسبة (قلب، بطن/FAST، رئة، أوعية). قِس AAA عبر القطر العرضي من جدار إلى جدار. تعرف نوافذ القلب وسمات انزلاق الرئة. [POCUS101] [RCEM]

2. Recognition, differential diagnosis, and red flags

- FAST/eFAST (trauma):
- Positive = anechoic free fluid in Morrison (RUQ), splenorenal (LUQ), pelvis

(rectovesical/rectouterine), or pericardial sac. RUQ most sensitive in supine trauma. False negatives: small volumes, retroperitoneal bleed, obesity, poor windows, immediate post-injury. [POCUS101]
- Pleural fluid vs empyema:
- Simple effusion: anechoic, dependent, meniscus; lung may float.
- Complex/empyema: echogenic fluid, septations, loculations, pleural thickening; aspirate/drain planning needs US. [POCUS101]
- Pneumothorax:
- Absent lung sliding + absent B-lines + barcode on M-mode = likely pneumothorax. Lung point (transition between sliding and no sliding) = diagnostic. Mimics: pleurodesis, extensive adhesions, mainstem intubation, apnea. [POCUS101]
- Cardiac tamponade:
- Pericardial effusion ± haemodynamic compromise. US signs: RA collapse (earliest), RV diastolic collapse (more specific), large effusion with swinging heart, plethoric IVC with reduced inspiratory collapse. Interpret with clinical context. [POCUS101]
- Pulmonary embolism (PE) sonographic clues:
- RV dilation, septal bowing (D-sign), reduced TAPSE (<17 mm), dilated non-collapsing IVC, possible visible right-heart thrombus — supportive but not diagnostic. [POCUS101]
- Vascular:
- DVT = non-compressible vein segment on compression US; Doppler waveform adjunct. Two-point scans miss calf DVTs. Gentle perpendicular pressure required. [POCUS101]
- Red flags:
- Hemodynamically unstable + positive FAST/pericardial effusion → urgent surgical/vascular intervention.
- Large AAA on bedside scan in unstable patient → do not delay definitive care for CT. [RCEM]
- Accidental arterial (e.g., carotid) cannulation with dilatation/large bore → do NOT remove catheter blindly; leave in situ and get vascular/IR advice. [POCUS101]

> Arabic (الفصحى):
> مظاهر إيجابية FAST: سائل أنيكويك في المرايا التشريحية المعروفة. استرواح الصدر يظهر بشريط الـM-mode (barcode) وغالبًا يتطلب إفراغًا عاجلاً إذا كان هناك ضغط توتري. الانخماص الأذيني

الأيمن والإفراط في توسع الوريد الأجوف السفلي توحيان بالانضغاط التاموري. [POCUS101] [RCEM]

3. Diagnostic approach and interpretation

- FAST algorithm (unstable trauma):
1. Scan RUQ (Morrison) → LUQ → pelvis (transverse + longitudinal fan) → subxiphoid pericardial.
2. If any view shows free intraperitoneal/pericardial fluid in unstable patient → activate surgical/trauma team and arrange immediate operative care/transfer. If FAST negative but instability persists → repeat FAST, extend views, or CT if stabilised. [POCUS101]
- Suspected ruptured AAA:
- Unstable + bedside US shows large/symptomatic AAA → minimise delays: urgent operative transfer/vascular surgery; do not delay for CT if diagnosis clear and immediate surgery available. Stable or equivocal → CTA for planning. [RCEM]
- Pneumothorax workflow:
- Absent sliding + barcode → search for lung point (specific). If tension physiology → immediate decompression (do not wait). If stable → chest X-ray/CT according to trauma pathway. [POCUS101]
- Cardiac tamponade assessment:
- Identify pericardial effusion → look for RA collapse (sensitive) and RV diastolic collapse (specific) → assess IVC (plethoric, low inspiratory collapse). Correlate with BP and clinical signs. Unstable → immediate pericardiocentesis or operative drainage per protocol. [POCUS101]
- DVT pathway:
- Whole-leg or at minimum common femoral and popliteal compression. Non-compressible → commence anticoagulation per local/NICE guidance and arrange follow-up imaging/vascular advice. Negative + high clinical suspicion → repeat in 5–7 days or perform whole-leg. [NICE]
- Image optimization basics:
- If dark → increase gain; if structure not seen → increase depth or change probe; for pleura use high-frequency linear probe; for deep abdominal/cardiac use curvilinear/phased array. Fan and rotate to sample recesses. [POCUS101]

Arabic (الفصحى):
خوارزمية FAST: RUQ → LUQ → حوض → تحت القص؛ إيجابيات في مريض غير مستقر تعني

> تحريك المريض للجراحة فورًا. AAA واضح في مريض غير مستقر → لا تؤخر العلاج الجراحي لإجراء CT. عند الاشتباه في استرواح صدر، ابحث عن lung-point لتأكيد التشخيص إن أمكن.
> [RCEM] [POCUS101]

4. Emergency management algorithms

- Unstable trauma with positive FAST:
- Immediate resuscitation (ABCDE), urgent surgical/trauma activation, prepare for theatre or transfer. Consider repeat FAST during resuscitation. [POCUS101]
- Ruptured/symptomatic AAA:
- Unstable + clear bedside diagnosis → expeditious vascular surgery/OR transfer; avoid CT delay. Use haemorrhage control strategies; permissive hypotension often used but exact BP targets vary—follow local protocols. [RCEM] (Flag: verify BP targets locally)
- Cardiac tamponade (compromised patient):
- Resuscitate (oxygen, IV access, IV fluids as temporising to improve preload), immediate definitive drainage:
- Medical/non-traumatic tamponade: ultrasound-guided pericardiocentesis (temporising) or surgical pericardial window as indicated.
- Traumatic tamponade with arrest/near-arrest (penetrating) → emergency department thoracotomy or operative sternotomy per trauma protocols. [POCUS101]
- Tension pneumothorax:
- Unstable: immediate decompression — current in-hospital guidance favors open thoracostomy/chest tube insertion (4th–5th ICS mid-axillary line) rather than needle decompression where possible; then formal chest drain. Stable → image and plan definitive management. [NICE NG39]
- Pleural empyema:
- Complex/loculated US appearance → chest drain placement ± surgical/IR referral; start appropriate antibiotics. [POCUS101]
- DVT:
- Non-compressible vein → start anticoagulation (DOACs first line for most adults per NICE/local guidance); pregnancy, severe renal impairment, or other contraindications → LMWH/warfarin per local protocols. [NICE]
- Accidental arterial catheterisation:
- If large bore/dilated catheter in carotid/large artery → leave catheter in place,

secure, and obtain immediate vascular/IR surgical advice; removal without expert support risks catastrophic bleeding/stroke. [POCUS101]

> Arabic (الفصحى):
> مريض مصاب + FAST إيجابي = تحضير عاجل للعمليات الجراحية. AAA ظاهر وغير مستقر = نقل فوري للجراحة الوعائية؛ لا تنتظر CT. الانصباب التاموري الحرج يتطلب تصريفًا فوريًا (بواسطة القثطرة الموجهة بالموجات أو جراحة الطوارئ حسب الآلية). استرواح صدر توتري = إفراغ فوري (قثطرة صدرية/فتحة صدرية) ثم درن. [POCUS101] [RCEM] [NICE NG39]

5. Drugs, procedures, disposition, and follow-up

- Drugs:
- Anticoagulation for DVT: DOACs are first-line for most uncomplicated DVTs per NICE/local guidance (select choice/dose from protocols). LMWH/warfarin for pregnancy, severe renal impairment, or other contraindications. [NICE]
- Fluids/pressors in tamponade/AAA: small IV boluses may temporise preload in tamponade; permissive hypotension used in ruptured AAA—specific BP targets vary and must follow local guidance. (Flag: verify BP targets locally.) [RCEM]
- Procedures (practical pearls):
- Ultrasound-guided pericardiocentesis: temporising in medical tamponade if immediate surgery not available — use subxiphoid or apical approach guided by effusion location. [POCUS101]
- Emergency department thoracotomy: consider for penetrating chest trauma with cardiac arrest in appropriate candidates per trauma protocols. [POCUS101]
- Chest decompression: open thoracostomy/chest tube at 4th–5th ICS mid-axillary line in hospital; needle decompression is temporising if necessary prehospital. [NICE NG39]
- DVT ultrasound: compress vein every few millimetres through CFV to popliteal; calf veins may require whole-leg scan. [POCUS101]
- AAA ultrasound measurement: transverse plane, outer-to-outer wall, perpendicular to vessel axis; report maximal outer diameter. [RCEM]
- Disposition:
- FAST positive + instability → theatre/specialist centre (surgical/vascular) or transfer per local trauma network.
- AAA unstable → immediate vascular operative care or urgent transfer to vascular centre. Stable AAA → CTA and planned definitive care. [RCEM]
- DVT stable → discharge with anticoagulation and outpatient follow-up unless

high-risk features. [NICE]
- Follow-up and documentation:
- Document scope of exam, views obtained, positive/negative findings, image saves, limitations (body habitus, windows), and clinical correlation. Escalate to senior/consultant when findings alter management. [POCUS101]

Arabic (الفصحى):
العلاج الدوائي الرئيسي للمُثلّ: مضاد التخثر لـ DVT حسب البروتوكول المحلي (غالبًا DOAC).
إجراءات الطوارئ: تصريف التامور بالموجات إرشاديًا كحل مؤقت، فتح صدر طارئ في الصدمات الانتقالية، درن صدري في الاسترواح التوتري. قِس AAA عبر عرضي من جدار إلى جدار. وثّق الصور والقياسات وحدود الفحص. [NICE] [RCEM] [POCUS101]

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية
- Quick exam tricks:
- FAST: RUQ (Morrison) = most sensitive in supine trauma; always fan and rotate pelvic probe (longitudinal then transverse). [POCUS101]
- AAA measurement: always measure maximal transverse diameter outer wall → outer wall perpendicular to axis. (Exam favourite) [RCEM]
- Cardiac views: in A4C, left heart appears on right of screen — memorise one convention. PLAX marker → patient right shoulder. [POCUS101]
- Lung M-mode: seashore = sliding; barcode = absent sliding. Look for lung point for high specificity. [POCUS101]
- DVT: compressibility is primary sign; gentle pressure and perpendicular probe to avoid false-positives. Two-point scan may miss calf DVTs (exam trap). [POCUS101]
- Common pitfalls:
- Absent lung sliding isn't specific for pneumothorax (adhesions, pleurodesis, mainstem intubation). Do not call pneumothorax without lung-point or contralateral comparison when clinical context permits. [POCUS101]
- False-negative FAST with small/retroperitoneal bleeds, early post-injury, obesity, operator inexperience. Repeat or move to CT when stable. [POCUS101]
- RA vs RV collapse timing: literature descriptions vary. RA collapse usually occurs during late diastole (sensitive); RV diastolic collapse is later and more specific — be cautious answering timing questions. (Exam trick: emphasise

clinical correlation.) [POCUS101]
- TAPSE and RV size: reduced TAPSE (<17 mm) suggests RV dysfunction but is not diagnostic for acute PE alone (chronic changes or poor windows can mislead). [POCUS101]
- Arterial catheterisation: removing a dilated arterial catheter (e.g., carotid) without vascular support is hazardous—leave in place and call vascular/IR. [POCUS101]
- Technical first steps: if image too dark → increase gain (first); if structure not visible due to depth → increase depth or change probe. Don't over-filter with Doppler when basic B-mode is inadequate.
- Exam practice tip: verbalise your clinical question, probe choice, views obtained, image optimisation steps, and limitations — examiners look for focused reasoning, not just images.

Arabic (الفصحى):
نصائح سريعة للممتحَن: قِس AAA من جدار إلى جدار عرضياً؛ RUQ هو النافذة الأكثر حساسية في FAST؛ استخدم M-mode لتأكيد انزلاق الرئة؛ لا تعتمد على عدم انزلاق الرئة وحدها لتأكيد استرواح صدر. احذر الإسقاط الخاطئ عند تفسير الانخماص الأذيني/البطيني. [POCUS101] [RCEM]

7. Arabic rapid-review summary

- POCUS = فحص موجات فوق صوتية موجّه وسريع بجانب السرير: أسئلة محددة لكل فحص (سائل؟ انصباب؟ استرواح؟ AAA؟ جلطة وريدية؟).
- المسابير: منحني/مصفوفة منخفضة للتعمق (قلب/بطن)، خطي عالي للتجاويف السطحية والرئة والأوعية.
- FAST: RUQ → LUQ → تحت القص → الحوض؛ إيجابيات في مريض غير مستقر = تحضير للعمليات.
- AAA: قِس القطر العرضي من جدار إلى جدار بزاوية عمودية على محوره.
- استرواح الصدر: barcode على M-mode = غياب انزلاق؛ lung-point تشخيصي.
- تامبوناد: انصباب حول القلب + انخماص الأذين الأيمن ± انخماص البطين الأيمن + IVC ممتلئ.
- DVT: عدم انضغاط الوريد = جلطة؛ ابدأ مضاد تخثر واتّبع بروتوكول المتابعة.
- مخاطر لا تتصرف بمفردك: لا تخرج قثطارًا شريانيًا كبيرًا دون استشارة جراحية؛ تحقّق دومًا من إرشادات المركز المحلية. [POCUS101] [RCEM]

8. Question coverage map (sources → concepts covered)

EXAM TRICK / تريكة امتحانية

- POCUS101 (cardiac, eFAST, lung, DVT modules): FAST sequence and windows; eFAST pneumothorax signs (sliding, M-mode seashore/barcode, lung-point); cardiac views and tamponade signs; RV dysfunction and TAPSE interpretation; DVT compression technique and pitfalls; image optimisation basics. [POCUS101]
- RCEM learning modules/guidance: AAA measurement technique (outer wall → outer wall, transverse), transfer/operative pathway decisions for ruptured AAA, training/competency notes. [RCEM]
- NICE NG39 / Major trauma guidance: chest decompression strategy in tension pneumothorax (chest drain/open thoracostomy site and approach), trauma assessment triage and imaging pathways. [NICE NG39]
- Exam/concept checkpoints included: FAST positive management; false-negative FAST scenarios; pneumothorax ultrasound signs and mimics; tamponade US signs and timing caveats; AAA recognition and immediate actions; DVT compression and anticoagulation pathway; accidental arterial cannulation management. (Use local Q-banks to map to specific question numbers.)

Arabic (الفصحى):
مخطط التغطية: وحدات POCUS101 تغطي FAST، رئة، قلب، DVT؛ RCEM يغطي قياس AAA ومسارات النقل؛ NICE NG39 يغطي استرواح صدر توتري وإدارة الصدمات الكبرى.

Sources to verify
- POCUS101 modules (eFAST, cardiac, pleural)
- RCEM guidance and learning modules (AAA, transfer, training)
- NICE guideline NG39: Major trauma (assessment and initial management)
- Local hospital/trauma/vascular/IR protocols and current resuscitation guidance (ERC/AHA) — verify procedural indications, permissive hypotension targets, and role of pericardiocentesis vs thoracotomy per institution.

CHAPTER 08: STATISTICS/ QUALITY IMPROVEMENT

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Chapter 08 — Statistics and Quality Improvement (compact FRCEM final chapter)

1. Clinical overview and pathophysiology

- Purpose: statistics/quality-improvement (QI) translate trial/test evidence into safe, reproducible ED decisions and system change.
- Core diagnostic measures (2×2): sensitivity = TP/(TP+FN); specificity = TN/(TN+FP); PPV = TP/(TP+FP); NPV = TN/(TN+FN). Likelihood ratios: LR+ = sens/(spec−1); LR− = (1−sens)/spec.
- Effect measures: absolute risk reduction (ARR) = AR_control − AR_treatment; NNT = 1/ARR (round up). Relative measures (RR, OR) need translation to absolute effects for clinical use.
- Precision & inference: 95% CI width = precision; p-values ≠ clinical importance; power = 1−β (target commonly ≈0.8).
- Diagnostic discrimination: ROC curve plots sens vs 1−spec; AUC = discrimination (0.5 = chance; 0.7–0.8 acceptable; 0.8–0.9 good; ≥0.9 excellent) [Radiopaedia].
- Meta-analysis visuals: forest plot = study boxes (point estimates) + horizontal CIs; diamond = pooled estimate; box size ∝ study weight (inverse variance); I^2 quantifies heterogeneity (guidance: 0–40% might not be important; 30–60% moderate; 50–90% substantial; 75–100% considerable) [Cochrane Handbook].
- Publication/small-study bias: funnel-plot asymmetry suggests bias or heterogeneity — interpret cautiously.
- Agreement/reliability: Cohen's kappa (two raters), Fleiss' kappa (>2); kappa affected by prevalence.
- Trial validity basics: randomisation, allocation concealment, blinding, intention-to-treat (ITT), handling of missing data.

Arabic (مختصر):

الغرض: تحويل الأدلة الإحصائية إلى قرارات سريرية وتحسينات نظامية. حساسية/نوعية من 2×2، ARR→NNT للفائدة السريرية، AUC لمنحنى ROC، مخطط الغابة والـI^2 للتغايرية، ومخطط القمع لتحيُّز النشر. التحقق من التعمية والـITT وجودة التصميم.

2. Recognition, differential diagnosis, and red flags

- Quickly identify plot type: forest (meta-analysis), ROC (diagnostic accuracy), funnel (publication bias), Cates pictogram (absolute effects).
- Forest-plot red flags: pooled diamond crosses null → non-significant; many wide/overlapping CIs → imprecision; small boxes with large effect → small study effect.
- Heterogeneity red flags: I^2 >50% → investigate differences in population/method/thresholds; inconsistent direction of effects → avoid simple pooling.
- ROC red flags: AUC near 0.5 or CIs spanning 0.5 → poor discrimination; chosen threshold not justified by Youden's index or clinical consequence.
- Funnel-plot red flags: asymmetry may mean publication bias, selective reporting, true heterogeneity, or chance — do not assume bias without extra checks.
- Study/population red flags: case-control diagnostic designs with extreme cases/healthy controls (spectrum bias), non-consecutive sampling, lack of blinded reference standard, large loss to follow-up, fixed small block randomisation without concealment (predictable allocation).

Arabic (مختصر):
تعرف نوع الرسم فوراً، وراقب إذا ما قطعت الماسة خط اللا تأثير، أو كان I^2 مرتفعاً أو AUC ضيقاً/غير كافٍ. عدم تماثل القمع ليس دليلاً قاطعاً على تحيُّز؛ تحرّ الدوافع الأخرى مثل التغايرية أو أخطاء منهجية.

3. Diagnostic approach and interpretation

- Stepwise appraisal checklist:
1. Question & population: is PICO clear and similar to your ED patients?
2. Design & internal validity: randomisation/allocation concealment/blinding/ITT (for trials); consecutive sampling and blinded independent reference standard (for diagnostic studies).
3. Metrics & precision: point estimates with 95% CI — does CI cross the null? compute/confirm sens/spec/PPV/NPV/LR+/LR−, AUC and CIs.
4. Heterogeneity & bias: I^2, forest-plot spread, funnel-plot symmetry, registered protocol vs report, selective outcome reporting.
5. Applicability: spectrum bias, setting (ED vs tertiary), resource/turnaround feasibility.

6. Clinical translation: convert RR → ARR → NNT/NNH; use LRs to convert pretest → posttest probability (Fagan nomogram or odds calculation).

- Practical 5-box quick algorithm:

Box 1: read abstract → identify population/outcome.

Box 2: meta-analysis? check forest + I^2; if I^2 >50% favour random-effects and explore subgroups.

Box 3: does pooled diamond cross no-effect? yes → non-significant.

Box 4: diagnostics → check AUC (use thresholds), chosen cut-off, LR+/LR−.

Box 5: translate relative to absolute (ARR → NNT/NNH), consider harms.

- Youden's index = sensitivity + specificity − 1 to select optimal ROC cut-off (but choose thresholds with clinical risk tolerance).

Arabic (مختصر):

قُم بفحص التصميم والصدق الداخلي ثم حجم الأثر وثقتَه (CI). تحقق من التغايرية/I^2 والقابلية للتطبيق. استخدم معاملات الإحتمال لتحويل الاحتمالات قبل/بعد الاختبار، وARR→NNT لتفسير الأثر السريري.

4. Emergency management algorithms

- Applying evidence/diagnostic tests in ED (decision algorithm):

1. Estimate pretest probability (clinical gestalt or decision rule).
2. If low and validated rule-out exists → discharge/avoid testing.
3. If intermediate → perform high-sensitivity test; use LR− to rule-out safely if ≤ threshold (LR− <0.1 strong rule-out; 0.1–0.2 moderate).
4. If high → proceed to definitive imaging/treatment; use high-specificity test or LR+ (>10 strong rule-in).

- New-test adoption algorithm:

1. Check published discrimination (AUC ≥0.8 preferred) and external validation in ED population.
2. Confirm acceptable LR+/LR− and decision thresholds.
3. Pilot locally (small prospective audit) if heterogeneity or different prevalence.
4. Implement with PDSA cycles, pre-specified safety/performance targets, and SPC/run charts for monitoring.

- QI (ED-friendly PDSA) quick steps:

Plan: define SMART aim and measures (process, outcome, balancing).

Do: small-scale change.

.Study: analyse using run-chart/SPC (distinguish common vs special cause)
.Act: adopt/adjust/abandon; repeat cycles
- Communicating risk: prefer absolute formats (ARR, NNT, Cates pictograms) .for shared decision-making

Arabic (مختصر):
في الطوارئ ابدأ بتقدير الاحتمال القبلي، استخدم اختبارات حساسة للسماح بالخروج واختبارات نوعية للتأكيد، واطبق بروتوكول اعتماد محلي (تحقق خارجي، تجربة أولية، PDSA، مراقبة بالرسوم البيانية الرقابية). استخدم الأثر المطلق لشرح الخطر للمرضى.

5. Drugs, procedures, disposition, and follow-up

- Drugs/procedures: no new numeric dosing thresholds to invent here; apply evidence by translating RCT results into ARR/NNT and balancing NNH and .harms. Present absolute benefits and harms with timeframes
- :Disposition rules driven by statistics
  - Rule-out safe discharge: high sensitivity test + low LR− + high NPV in local .prevalence (document pretest probability and threshold used)
  - Rule-in for invasive procedures: require high specificity and high LR+ or .confirmatory imaging
- :Procedural adoption/follow-up steps
  1. Pre-implementation: define clinical threshold and safety metrics (miss rate, .(adverse events
  2. Implementation: pilot with prospective data collection (consecutive patients), .blinded endpoints where feasible
  3. Post-implementation: monitor with run charts/SPC for outcome measures (time-to-diagnosis, missed cases, adverse events, resource use). Reassess ARR/ .NNT periodically
- Audit and reporting: record denominator (consecutive sampling), timings, inter-rater reliability (kappa), and unexpected harms; feed results back into .PDSA

Arabic (مختصر):
لا تُخترع عتبات جديدة؛ استخرج فوائد/أضرار مطلقة (ARR/NNT/NNH) من الأدلة. قرار الخروج يعتمد على اختبار عالي الحساسية/LR− منخفض وNPV مناسب للسياق المحلي. نفّذ التغييرات تجريبياً مع قياس متكرر ومراقبة للمخاطر والأداء.

6. High-yield FRCEM exam tricks and common pitfalls

- Memory mnemonics: SPIN (Specific → Positive rules IN); SNOUT (Sensitive → Negative rules OUT).
- Quick formulas to memorise:
- Sens = TP/(TP+FN); Spec = TN/(TN+FP).
- PPV = TP/(TP+FP); NPV = TN/(TN+FN).
- LR+ = sens/(1−spec); LR− = (1−sens)/spec.
- ARR = AR_control − AR_treatment; NNT = 1/ARR (round up).
- Forest plot exam cues: box = point estimate (size = weight), horizontal line = CI (width = precision), diamond = pooled estimate (tips = pooled CI). Diamond crossing null → pooled non-significant.
- Heterogeneity: quote Cochrane I^2 guidance; high I^2 → explore heterogeneity, do not pool blindly [Cochrane Handbook].
- ROC/AUC traps: AUC = discrimination across thresholds, not single-threshold accuracy. AUC 0.5 = useless; 1.0 = perfect [Radiopaedia]. Use LRs for bedside decisions.
- Kappa pitfalls: kappa can be negative; affected by prevalence — do not assert 1–0 range without caveat.
- NNT/NNH: compute from absolute risk difference (not from relative risk); always state time horizon and CI; round NNT up to whole patient.
- P-value cautions: $p<0.05$ ≠ clinical importance; non-significance ≠ no effect (possible type II error).
- Funnel-plot caution: asymmetry ≠ definitive proof of publication bias; consider heterogeneity and small-study effects.
- Jadad score: quick quality screen (randomisation, blinding, withdrawals) but limited — do not equate high Jadad with absence of all bias [Jadad 1996].

Arabic (مختصر):

احفظ SPIN/SNOUT والمعادلات الأساسية. في مخطط الغابة: حجم الصندوق = الوزن، الماسة تقطع خط اللا تأثير → غير ذا دلالة إحصائية. لا تعتمد على p-value وحدها؛ احسب NNT من الفرق المطلق ودوّن الإطار الزمني.

7. Arabic rapid-review summary (مراجعة سريعة باللغة العربية)

- المفاهيم الأساسية: الحساسية = TP/(TP+FN) للتساؤل عن الاستبعاد؛ النوعية = TN/(TN+FP) للتأكيد. PPV/NPV تتأثر بالانتشار. معاملات الإحتمال (LR) مفيدة لتحويل الاحتمال قبل/بعد

الاختبار.
- مخططات: مخطط الغابة لميتاتحليل وI^2 للتغايرية؛ منحنى ROC وAUC للدقة؛ مخطط القمع لتحيُّز النشر (تفسير حذر).
- تحويل الأثر إلى قرار: NNT/NNH ← ARR قبل تبنّي علاج/إجراء؛ استخدم عتبات معقولة (LR+>10 rule-in, LR−<0.1 rule-out).
- تحسين الجودة في الطوارئ: خطط SMART، اختبر تغييرات صغيرة (PDSA)، راقب بالرسوم البيانية الرقابية، وأكد فعالية الاختبارات في السياق المحلي قبل النصح بالاعتماد.
- نصيحة امتحان: لا تخلط بين الحساسية/النوعية وPPV/NPV؛ لا تفسر عدم تماثل مخطط القمع كدليل قاطع على تحيُّز؛ احسب NNT من الفرق المطلق ودوّن الفترات الزمنية.
(ملاحظة: مادة دراسية فقط — ليست نصيحة علاجية مخصصة للمرضى.)

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية

- Core diagnostic metrics and 2×2 interpretation (sensitivity, specificity, PPV, NPV, LR) — (concept cluster).
- ROC curve and AUC interpretation; Youden's index for cut-offs — (concept cluster).
- Meta-analysis visuals: forest plots, pooled diamonds, box size/weight, I^2 heterogeneity interpretation — (concept cluster) [Cochrane Handbook; Cochrane 2016].
- Publication bias and funnel plot caveats — (concept cluster).
- Effect measures and translation to clinical practice: ARR, RRR, NNT/NNH and time horizons — (concept cluster).
- Trial validity basics: randomisation, allocation concealment, blinding, ITT, Jadad score limits — (concept cluster) [Jadad 1996].
- Agreement/reliability measures: Cohen's/Fleiss' kappa and limitations — (concept cluster) [Landis & Koch 1977].
- Quality improvement methods in ED: PDSA, run charts/SPC, pilot implementation, local validation — (QI cluster).
- Common biases and pitfalls: spectrum bias, verification/incorporation bias, selection bias, measurement bias, small-study effects — (concept cluster).
- Exam tricks and calculation mnemonics: SPIN/SNOUT, formula recall, NNT rounding, interpretation of CIs & p-values — (exam skills cluster).

> (Notes: specific source Q-numbers were not supplied with the chunk notes; the map lists concepts represented by the supplied materials.)

Sources to verify
- Cochrane Handbook for Systematic Reviews (I^2 guidance) [Cochrane].
- Radiopaedia ROC/AUC guidance [Radiopaedia].
- Cochrane blog/tutorials (forest-plot interpretation) [Cochrane 2016].
- Jadad RCT reporting scale (1996) [Jadad 1996].
- Kappa interpretation conventions (Landis & Koch 1977) [Landis & Koch 1977].
- Local/national guidance and applicability: RCEM, ERC, NICE (and local health-board policies) — verify before applying to practice.

(Study material only — not patient-specific advice.)

CHAPTER 09: RESUSCITATION

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Chapter 09 — Resuscitation (compact FRCEM final chapter)

1. Clinical overview and pathophysiology (core points)

- Aim: secure airway, restore oxygenation/ventilation, maintain circulation/organ perfusion, reverse treatable causes (Hs & Ts), deliver time-critical interventions (defibrillation, thrombolysis, ECLS) when indicated. Prioritise physiology over tests. [Resuscitation Council UK; ERC/AHA]
- “Treat first what kills first”: hypoxia, massive haemorrhage, tension pneumothorax, tamponade, severe hyper/hypokalaemia, hypoglycaemia, massive PE, malignant hyperthermia.
- Special contexts modify thresholds and approach: neonatal, paediatric, pregnancy (>20 weeks → aortocaval compression concerns and perimortem hysterotomy), hypothermia (rewarming ± ECLS), tracheostomy patients, drowning, massive PE/arrest.
- Monitoring: continuous waveform capnography is gold standard for ETT confirmation and CPR quality (sudden ETCO2 rise → ROSC; sustained low ETCO2 → poor perfusion). [Resuscitation Council UK]

> خلاصة بالعربية: الهدف تأمين مجرى الهواء ثم دعم التهوية والدورة ومعالجة الأسباب القابلة للعكس. اضبط النهج طبقًا للفئات الخاصة (مواليد، حوامل، انخفاض حرارة الجسم، تراكيشومى). استخدم موجة ETCO2 لمراقبة الأنبوب والإنعاش.

2. Recognition, differential diagnosis, and red flags

- Airway obstruction (adult): effective cough = partial; ineffective cough/inability to speak or breathe = severe obstruction → intervene. Loss of consciousness raises arrest risk.
- Post-ROSC hypotension: consider myocardial dysfunction, ongoing ischaemia, hypovolaemia, tension pneumothorax, PE, tamponade, arrhythmia, sepsis. Initial MAP target ≥65 mmHg. [ERC/AHA]
- Massive PE: sudden dyspnoea ± pleuritic pain, tachycardia, hypoxia, hypotension (SBP <90 or large drop) — arrest often presents as PEA. Low ETCO2 supports but is

non-specific. Treat as unstable PE if clinical probability high.
- Neonates: poor tone/blue/gasping → prematurity, birth asphyxia, airway obstruction, surfactant deficiency, sepsis, anomalies. Initial rapid assessment: tone, breathing, heart rate (NOT SpO2).
- Hypothermia: bradycardia, J waves (Osborn), ventricular arrhythmias. Consider ECLS transfer if core <30°C with instability, arrest, or refractory arrhythmia.
- Post-intubation sudden desaturation: mainstem intubation, tube obstruction/ kink, cuff leak, equipment failure, pneumothorax, bronchospasm, pulmonary oedema.

> خلاصة بالعربية: علامات الخطر: سعال غير فعال في الاختناق، هبوط ضغط بعد ROSC يعنى اسباب ميكانيكية أو قلبية أو نقص حجم، صمة رئوية كبرى تعطي PEA غالبًا، وحديثي الولادة يقيمون باللون والتنفس والنبض لا SpO2. انخفاض حرارة الجسم الحاد يتطلب تفكير بتحويل ECLS.

3. Diagnostic approach and interpretation (rapid, goal-directed)

- Immediate bedside priorities: airway look/listen/feel, SpO2, ECG, capnography (if ventilated), glucose, IV/IO access, VBG/ABG, bedside echo/US for tamponade, RV strain, pneumothorax/FAST.
- Airway confirmation: direct visualisation + continuous waveform capnography. Don’t rely on misting or auscultation alone. [Resuscitation Council UK]
- Post-ROSC: ABG, 12-lead ECG (act on clear ischaemia; routine immediate cath not mandatory), lactate, glucose, temperature, haemodynamic support to MAP ≥65 mmHg. [ERC/AHA; Resus UK]
- Massive PE diagnostics: bedside echo for RV strain when unstable; CTPA if stable; treat immediately if unstable and diagnosis likely.
- Neonatal diagnostics: preductal SpO2 targets guide O2 titration; initial decisions use heart rate and breathing — inflation breaths first if apnoeic/gasping. Preterm <32 wks: plastic wrap + radiant heat.
- Ventilator alarm/ETCO2 patterns: sudden ETCO2 fall → disconnection, loss of circulation, massive PE; sudden rise → ROSC or CO2 retention; “shark-fin” waveform → bronchospasm/obstruction.

خلاصة بالعربية: فحوص فورية: موجة ETCO2 لتأكيد الأنابيب والإنعاش، جلوكوز سريع، ECG، صدى سريري للتامبوناد/صمة رئوية. حديثي الولادة تُقَيّم بقلب ونفس وليس SpO2 أولياً؛ القِمم قبلية للحجم الصغير.

4. Emergency management algorithms (practical concise algorithms)

- Adult choking (conscious)
1. Effective cough/able to speak → encourage cough.
2. Ineffective cough/airway compromise → up to 5 back blows; if ineffective → up to 5 abdominal thrusts (chest thrusts if pregnant/obese). If becomes unconscious → start CPR and inspect for visible obstruction before ventilations. [Resuscitation Council UK]
- Post-ROSC immediate care
1. Airway/oxygenation (capnography), ventilate as needed.
2. Circulation: IV/IO access, MAP target ≥65 mmHg (fluids ± vasopressors; noradrenaline first-line where needed). Treat reversible causes; obtain 12-lead ECG and act on clear ischaemia per local policy. [ERC/AHA]
- Massive PE with haemodynamic compromise
1. If SBP <90 or persistent hypotension → give systemic thrombolysis immediately unless contraindicated (alteplase per local protocol). If contraindicated or failed → consider surgical/percutaneous embolectomy or ECLS. After thrombolysis, continue CPR for prolonged periods (often ≥60–90 min) if indicated. [Resus UK]
- Tracheostomy emergency (green algorithm)
1. Call for help; look/listen/feel at mouth & stoma; attach capnography.
2. Remove speaking valve/cap, remove inner cannula, suction, deflate cuff if indicated.
3. If obstruction persists → remove tracheostomy tube and oxygenate via face/over stoma; consider oral intubation or surgical airway. [RCEM]
- Neonatal resuscitation essentials
1. Initial assessment: tone, breathing, heart rate.
2. If apnoeic/gasping: 5 inflation breaths (term ≈30 cmH2O, preterm lower per local protocol) with PEEP 5–6 cmH2O if available; start ventilation and reassess heart rate.
3. HR <60 after 30 s of effective ventilation → start chest compressions (3:1) and escalate per neonatal algorithm; adrenaline 20 µg/kg IV (1:10,000) via UVC preferred if available. [NLS 2021; Resus UK]

- Hypothermic arrest
1. Continue CPR while rewarming. Consider transfer to ECLS centre if core <30°C with instability, arrhythmia, or arrest. Limit futile defibrillations when severely hypothermic until rewarmed as per guidance. [Resus UK]
- Intubated patient sudden desaturation
1. Disconnect ventilator → bag-ventilate; assess compliance and bilateral breath sounds.
2. Check ETT depth (withdraw ~2 cm if right mainstem suspected), patency, cuff, circuit, oxygen source; look for pneumothorax and treat immediately (decompression) if suspected.
- Cardiac arrest (brief ALS)
1. Call for help → high-quality CPR → attach monitor/defib → shockable rhythms: defibrillate, continue CPR, give adrenaline/antiarrhythmics per ALS; use ETCO2 to guide quality/ROSC detection.
2. Consider reversible causes and advanced options (eCPR/PCI) in selected refractory cases.

> خلاصة بالعربية: خوارزميات رئيسية: الاختناق (تشجيع السعال إن فعال → 5 صفعات ظهر → 5 ضغطات بطنية → إغماء → CPR)، ما بعد ROSC (MAP≥65، ECG، علاج المسببات)، صمة رئوية كبرى → ثرومبوليز حتى تتوفر بدائل، ألم التراشي والتعامل الواضح بالخطوات المذكورة، رعاية حديثي الولادة بالتهوية أولًا و20 µg/kg أدرينالين IV عند الحاجة.

5. Drugs, procedures, disposition, and follow-up

Drugs / key, well-established doses
- Noradrenaline: first-line vasopressor for septic shock to target MAP ≥65 mmHg (titrate per local protocols).
- Adrenaline (neonatal): 20 µg/kg IV (1:10,000) via UVC if available. [NLS]
- Magnesium for Torsades: 2–4 g IV bolus (then consider infusion). (Well established.)
- Hyperkalaemia (principle): stabilise membrane first with IV calcium (gluconate or chloride per local stock), then shift K+ intracellularly (insulin 10 U IV + 25 g dextrose adjunct), nebulised β2-agonist as adjunct; arrange urgent dialysis for removal.
- Thrombolysis for massive PE: alteplase per local dosing (give immediately if unstable and no contraindication). Continue CPR prolonged if given.
- Dantrolene for malignant hyperthermia: 2.5 mg/kg initial IV (repeat as required)

per MH protocols.
- Hypoglycaemia: treat immediately with IV dextrose per local protocol (e.g., 10% 100–200 ml depending on local guidance) — follow local guidance and monitoring.
Procedures / priorities
- Airway: escalate mask → supraglottic device → ETT (RSI) → surgical front-of-neck access (CICO/cricothyroidotomy) when cannot oxygenate/ventilate. [DAS]
- Tracheostomy emergency steps: speaking valve off → inner cannula out → suction → cuff deflate → remove tube if obstruction persists → oxygenate and secure airway.
- Perimortem hysterotomy: consider when pregnancy >20 weeks and no ROSC by ~4 minutes; aim to deliver by 5 minutes when indicated. [Resus UK]
- ECLS transfer: consider for hypothermic arrest, refractory cardiac arrest, or massive PE when available.
Disposition & follow-up
- Post-ROSC: ICU/HDU admission for monitoring, hemodynamic support, targeted temperature management as per local protocols, investigate/treat cause (PCI if clear STEMI or local pathway). Routine immediate coronary angiography is NOT mandatory without ischaemic ECG/clinical indicators. [ERC/AHA]
- Neonates: escalate to NICU for ongoing support and monitoring; follow neonatal unit guidelines for oxygen targets and glucose monitoring.
- Tracheostomy patients: avoid discharge without specialist review; arrange ENT/ICU follow-up and tracheostomy care plan.
- Drowning: observe clinically stable asymptomatic patients ~6 hours with normal CXR/ABG prior to discharge per local guidance. [RCEM]

خلاصة بالعربية: أدوية أساسية: نورأدرينالين للصدمة الإنتانية (MAP≥65)، ماغنسيوم 2–4 غ لTorsades، أدرينالين حديثي الولادة 20 µg/kg IV، معايرة جرعات الكالسيوم والتحقق محليًا.
إجراءات رئيسية: إطار التسلسل للتهوية، خوارزمية التراشي، تفكير سَلَس لـ ECLS أو جراحة طارئة.
توجيه لمستويات أعلى ومتابعة في العناية المركزة بعد ROSC.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية
- Do NOT intervene if choking adult is coughing effectively or can speak — encourage cough (exam favourite distractor).
- Neonatal initial rapid assessment = tone, breathing, heart rate (SpO2 is NOT

for the initial decision). 5 inflation breaths then assess.
- Post-ROSC: routine immediate coronary angiography is not automatic — act on ECG/clinical signs.
- Suxamethonium pitfalls: causes transient bradycardia (first-dose) and can provoke life-threatening hyperkalaemia in burns/crush/denervation — exam trap to avoid sux in these contexts.
- Capnography: recognise normal four-phase waveform; “shark-fin” = bronchospasm/obstruction; sudden ETCO2 rise = ROSC; sudden fall = disconnection or loss of circulation.
- Tracheostomy exam sequence: speaking valve off → inner cannula out → try suction → deflate cuff → remove tube if obstruction persists. Failure to follow sequence is common pitfall.
- Right mainstem intubation: if unilateral breath sounds post-intubation, withdraw the ETT ~2 cm and reassess (don’t immediately extubate).
- Massive PE arrest: thrombolysis during CPR is indicated when PE is strongly suspected; expect prolonged CPR (≥60–90 min) — often tested.
- Verify calcium and other drug concentrations/volumes locally — exam banks vary; when asked state preparation (gluconate vs chloride) if specified.
- IO insertion absolute contraindication: fracture of target bone — common exam fact.
- APNOEA ventilator alarms during transfer are frequently connection/ETCO2 problems — “disconnect & bag” is standard immediate response.

خلاصة بالعربية: حيل الامتحان: اشجع السعال الفعال، لا تعتمد على SpO2 لحديثي الولادة في التقييم الأولي، احذر سوكس مع الحروق/التحلل، اتبع ترتيب خوارزمية التراشي، واستخدم ETCO2 لتقييم جودة الـCPR وROSC.

7. Arabic rapid-review summary (موجز سريع بالعربية)

- المبادئ: مجرى الهواء أولاً ثم التنفّس والدورة الدموية؛ عالج الأسباب القابلة للعكس فورًا (الصمة الرئوية، التوتر الرئوي، التامبوناد، نقص السكر، اضطراب الإلكتروليت).
- الاختناق: سعال فعال→ شجعه؛ سعال غير فعال→ 5 صفعات ظهر ثم 5 ضغطات بطنية؛ فقدان الوعي→ ابدأ CPR.
- بعد ROSC: هدف MAP ≥65 ممHg؛ ECG وفحص الأسباب؛ القسطرة التاجية ليست روتينية إلا بوجود دليل إقفاري.
- صمة رئوية كبرى: ثرومبوليز فورى إن كان المريض غير مستقر واللا مانع; استمر في الإنعاش

لفترة طويلة بعد العلاج.
- جهاز التنفّس/التنبيب: إذا هبوط الأكسجة فجأة افصل الجهاز وابدأ كبس كيس يدوي وقيّم الأنبوب والطنين/انفجارات الرئة (pneumothorax).
- ترشستومي: أزل الصمام/الأنبوب الداخلي وجرب الشفط وأزل الكفة؛ إذا انسداد استمر—أزل أنبوب التراشي وأمّن التهوية.
- حديثي الولادة: التنفّس أولًا (5 نفخات)، HR <60 بعد 30 ثانية تهوية فعالة → ضغطات صدرية 3:1 + أدرينالين 20 µg/kg IV عبر الحبل السري إن أمكن.
- جرعات مهمة: ماغنيسيوم 2–4 غ لTorsades؛ أدرينالين حديثي الولادة 20 µg/kg IV؛ نورأدرينالين لصدمة إنتانية (هادفو MAP≥65).
- تحقّق دائمًا من تحضيرات/تركيزات الأدوية محليًا (K ،+Ca2، ماغنيسيوم).

8. Question coverage map (sources / Q-numbers / concepts covered)

- Q503–Q505: malignant hyperthermia signs, suxamethonium effects and contraindications (bradycardia, hyperkalaemia risk).
- Q506, Q509, Q520: hyperkalaemia and calcium dosing issues; urgent membrane stabilisation and downstream therapies.
- Q507, Q526: refractory VF/pVT, defibrillation strategy, escalation and eCPR/PCI considerations.
- Q510: intraosseous access contraindications (fracture).
- Q511: Allen’s test for radial cannulation.
- Q512: hypothermia recognition, Osborn waves, ECLS transfer thresholds.
- Q513, Q515, Q525: ETCO2 interpretation, ventilator alarms (APNOEA/high pressure) and immediate responses (disconnect & bag).
- Q516, Q521, RCEM tracheostomy material: tracheostomy emergency (green algorithm) sequence and management.
- Q518: septic shock fluid resuscitation and vasopressor bridging (noradrenaline target MAP ≥65).
- Q520: hyperkalaemia algorithm (calcium, insulin+dextrose, nebulised salbutamol).
- Q522–Q524, Q504, Q523: neonatal resuscitation (inflation breaths, 3:1 compressions, preterm care) and NLS 2021 points.
- Q526, Q527: airway confirmation (capnography), failed intubation strategies (SAD, surgical airway) and DAS guidance.
- Q503–Q527 (general): drowning, post-intubation desaturation, chest decompression in tension pneumothorax, massive PE thrombolysis, perimortem

hysterectomy timing, MAP targets post-ROSC, magnesium for torsades, hypothermia algorithms, practical ventilatory rates during paediatric CPR.

Sources to verify
RCEM; Resuscitation Council (UK); European Resuscitation Council (ERC); American Heart Association (AHA); National Institute for Health and Care Excellence (NICE); Difficult Airway Society (DAS); Neonatal Life Support / PALS (2021).

CHAPTER 10: SLO7 - DEAL WITH COMPLEX OR CHALLENGING SITUATIONS IN THE WORKPLACE

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Chapter 10 — SLO7: Deal with complex or challenging situations in the workplace

1. Clinical overview and pathophysiology

- Systems safety > individual blame: recognise that iatrogenic and organisational failures (e.g. wrong gas connections) are system events requiring immediate patient stabilisation, remediation and formal reporting (PSIRF; never-event pathways). [PSIRF]
- Legal/ethical interfaces: confidentiality has lawful exceptions (e.g. name/address disclosure for alleged driving offences; terrorism, court order, public protection) — most other disclosures require legal basis or patient consent.
- Organ donation is a systems pathway triggered by catastrophic brain injury/ brainstem death; SNOD coordinates clinical assessment, registry check and family approach. [NHS Organ Donation]
- Safeguarding: domestic abuse and child protection are common complex presentations — pathophysiology is social/forensic rather than biological but causes repeated physical/psychological injury requiring multidisciplinary response.
- Capacity/consent: decision-specific capacity (Mental Capacity Act) — children assessed by Gillick competence; incapacity (reduced GCS, intoxication, severe mental disorder) obliges best-interests treatment.
- Forensic considerations: samples and evidence chain require FHCP involvement and lawful authority; do not bypass patient competence. [BMA/FFLM 2023]
- Major incident principles: scene safety, command/control and early declaration (METHANE) precede detailed clinical care.

> Arabic: التركيز على سلامة النظام أهم من اللوم الفردي؛ هناك استثناءات قانونية للسرية؛ عند إصابة دماغية قاتلة اتصل بـ SNOD؛ حماية الأطفال والعنف المنزلي مسؤولية متعددة التخصصات؛ قيّم الأهلية قراراً بقرار؛ لا تأخذ عينات جنائية دون صلاحية؛ أعلن الحادث الكبير وقدم تقرير METHANE.

2. Recognition, differential diagnosis, and red flags

- Immediate safety/red flags:
- Sudden hypoxaemia or unexplained deterioration after equipment change → suspect oxygen/equipment misconnection.
- Multiple casualties, hazards, access compromise, or overwhelmed resources → consider major incident and METHANE report.
- Reduced consciousness (GCS ↓), severe intoxication, incoherent answers → probable lack of capacity.
- Injuries of different ages, evasive answers, partner-presence blockade, threats/ weapon mention → suspect domestic abuse; always ask about children.
- Intubated/unstable patient with suspected RTC involvement → police may seek name/address only for driving offences; do not volunteer clinical info.
- Neurological exam compatible with brainstem death or catastrophic brain injury → trigger organ donation pathway; contact SNOD early. [FICM 2021; NHS Organ Donation]
- Pregnant suspected internal drug trafficker (body packing) → observe and image selectively; intervene only for clinical indications. [RCEM 2020]

> Arabic: راقب نقص التأكسج المفاجئ، الحوادث الكبرى، انخفاض الوعي، دلائل العنف المنزلي، طلبات الشرطة، الاشتباه في موت دماغي، والناقل الداخلي للمخدرات في الحامل.

3. Diagnostic approach and interpretation

- Immediate priorities: ABCs, stabilise life-threatening physiology, treat hypoxaemia before investigating equipment/connection errors.
- Equipment misconnection: correct supply, reassess, monitor SpO2 and observations; involve biomedical/senior staff and document sequence. Report via PSIRF. [PSIRF]
- Capacity assessment (adult): (1) Define specific decision; (2) Provide info and make reasonable adjustments; (3) Check understanding, retention, weighing, and ability to communicate choice. If unable → treat in best interests. For <16 apply Gillick competence.
- Police disclosure checklist: confirm officer identity/authority and reason for request; lawful minimal disclosure: name/address for alleged driver; otherwise consult Caldicott guardian/MDU/legal.
- Organ donation screening: on suspected brainstem death: contact SNOD; check

registry, residency (local rules), active absolute contraindications (verify current NHS list), infectious risk; SNOD leads family approach and donor suitability assessment. [NHS Organ Donation; FICM 2021]
- Forensic samples: if patient competent and refuses → do not take; advise police to arrange FHCP. If incapacitated follow legal/FFLM/BMA guidance for lawful sampling. [BMA/FFLM 2023]
- Domestic abuse assessment: private, non-judgemental history; document injuries with body maps/photographs if consented; ask about children; assess immediate risk and threshold for MARAC/IDVA referral.
- Toxicology/overdose: use standard tools—e.g. paracetamol single acute ingestion assessed by Rumack-Matthew nomogram and local NAC protocol. Do not invent thresholds beyond local guidance.

> Arabic: حافظ على ABC أولاً؛ أصلح الأخطاء بالأجهزة وسجّلها؛ قيّم الأهلية قراراً بقرار (أطفال ≡ غيليك)؛ تحقق من قانونية طلبات الشرطة؛ عند موت دماغي اتصل بـ SNOD وافحص السجل والموانع؛ عينات جنائية فقط عبر FHCP إن رفض المريض؛ للعنف المنزلي دوّن الإصابات واطلب عن الأطفال؛ استعمل منحنى Rumack لمعالجة الأسيتامينوفين.

4. Emergency management algorithms

- Oxygen/equipment misconnection (one-line algorithm): restore correct gas → immediate ABCs and monitoring → senior/biomed escalation → document events and remediate → report under PSIRF/never-event pathway. [PSIRF]
- Police request algorithm: verify officer and lawful basis → if alleged driving offence give name/address only (no clinical details) → if other request or doubt consult Caldicott/MDU/legal → document disclosure decision.
- Catastrophic brain injury / organ donation algorithm: stabilise & neuro-support → neurosurgery/ICU review → consider brainstem death testing per local protocol → contact SNOD early → SNOD checks registry/residency/contraindications → SNOD leads family discussion and donation logistics. [FICM 2021; NHS Organ Donation]
- Domestic abuse / safeguarding algorithm: see patient privately → assess immediate danger (weapons/threats? yes → call police) → document injuries/ timeframe → ask about children and immediate child safety → offer IDVA/MARAC referral if high risk → arrange follow-up and safeguarding referral.
- Capacity/consent algorithm (acute): is patient oriented & decision-specific capable? (Yes → respect choice) (No → treat in best interests; for minors consider parental consent if appropriate; for <16 assess Gillick). Emergently incapacitated

patients may be treated without consent to prevent harm.
- Forensic sampling algorithm: competent refusal → STOP; advise FHCP; incapacitated with lawful grounds → follow FFLM/BMA procedures for sampling and documentation; maintain chain-of-custody. [BMA/FFLM 2023]
- Jehovah's Witness / transfusion refusal: confirm competence/advance directive → discuss alternatives (cell salvage, topical haemostats, IV iron/EPO where appropriate) → recognise alternatives are slower — for life-threatening haemorrhage document discussions, escalate senior/legal, and follow local policies.

> Arabic: خوارزميات الطوارئ: أصلح إمداد الأكسجين وبلغ؛ للتحقيقات الشرطية أعطِ الاسم/العنوان فقط إن كان السائق المشتبه؛ عند إصابة دماغية قاتلة اتصل بـ SNOD؛ للعنف المنزلي اسأل خصوصاً عن الأطفال واطلب الشرطة إذا خطر فوري؛ عينات جنائية عبر FHCP أو طبق إجراءات FFLM؛ اشرح رفض نقل الدم وقدم بدائل ودوّن النقاش.

5. Drugs, procedures, disposition, and follow-up

- Drugs/antidotes:
- Paracetamol overdose → use Rumack-Matthew nomogram and give N-acetylcysteine per local protocol (follow toxicology guidance).
- Naloxone for opioid overdose as per standard resuscitation dosing (follow local/advanced life support protocols).
- Procedures:
- For suspected body packing: imaging (per local protocol) only if clinical deterioration or to confirm packing; avoid coercive invasive procedures without clinical need. [RCEM 2020]
- Brainstem death testing and donor management follow ICU/local protocols; SNOD coordinates organ preservation steps. [FICM 2021]
- Forensic sampling should be performed by FHCPs; if staff must collect, follow chain-of-custody and FFLM/BMA guidance. [BMA/FFLM 2023]
- Disposition:
- Immediate admission to HDU/ICU for catastrophic brain injury, airway compromise, major haemorrhage, or if required for forensic medical examination.
- Safe discharge considered only if no ongoing safety risk (domestic abuse planning in place; capacity intact; arrangements for follow-up and safeguarding).
- Follow-up:
- Safeguarding referrals, IDVA/MARAC, GP and community services as indicated.

- Governance reporting: PSIRF / local incident reporting for system errors; document contemporaneously and escalate to senior/governance.

> Arabic: الأدوية والإجراءات: استخدم منحنى Rumack وNAC للأسيتامينوفين؛ ناكسون عند جرعات الأفيون؛ تفحّص الناقلين بالمخدرات بالتصوير عند الحاجة؛ اختبارات موت الدماغ وإجراءات الحفظ بقيادة ICU/SNOD؛ العينات الجنائية حسب FFLM/BMA؛ ضع المريض في HDU/ICU إذا ضروري؛ أرسل تحويلات حماية ومتابعة للمجتمع ودوّن الحوادث للإدارة.

6. High-yield FRCEM exam tricks and common pitfalls

- Always report obvious system failures — never choose “do not report” for oxygen misconnection/never-events. [PSIRF]
- Police disclosure traps: exam vignettes often test the driver exception — name/address must be given for alleged driving offences; do not release clinical details without clear lawful basis.
- Organ donation traps: don’t over-simplify family “veto” — family views matter operationally but clinicians must follow SNOD/legal guidance; verify current NHS criteria for exclusions (policies change). [NHS Organ Donation]
- Domestic abuse Qs: always ask “are there children?”; immediate police contact is required if weapons/threats or immediate danger. MARAC is for high-risk referrals.

- Capacity in minors: apply Gillick competence — it’s decision-specific. Reduced GCS/intoxication → lacks capacity.
- Forensic sampling: clinical staff should avoid collecting forensic samples from competent refusers — request FHCP. Do not mix clinical care and forensic evidence collection without training. [BMA/FFLM 2023]
- Jehovah’s Witness: alternatives (IV iron/EPO) are slow — do not present them as immediate substitutes in major haemorrhage. Document discussions and escalate.
- Major incident questions: the tested sequence is declaration + METHANE → command → triage/treatment; don’t start detailed individual resource planning before declaring.

> Arabic: نصائح للامتحان: بلّغ الأخطاء النظامية؛ للشرطة أعطِ الاسم/العنوان فقط إن تعلق الأمر بالقيادة؛ تحقق من سياسات التبرع بالأعضاء؛ في العنف المنزلي اسأل عن الأطفال؛ غيليك قرارية؛ العينات الجنائية عبر FHCP؛ بدائل نقل الدم بطيئة؛ أعلن الحادث وقدم METHANE أولاً.

7. Arabic rapid-review summary

- أصلح المخاطر الحرجة فوراً (أكسجين/أجهزة) وبلغ عبر PSIRF.
- السرية لها استثناءات محددة — اسم/عنوان فقط للسائق المشتبه؛ استشر قانونياً في غير ذلك.
- عند موت دماغي اتصل بـ SNOD، افحص السجل وناقش العائلة بمرافقة SNOD.
- اسأل عن الأطفال في حالات العنف المنزلي؛ اتصل بالشرطة إذا خطر فوري؛ استخدم MARAC/IDVA.
- قيم الأهلية قراراً بقرار (غيليك للأطفال)؛ إن فقدت الأهلية عالج بمصلحة المريض.
- لا تاخذ عينات جنائية من مريض رافض إن كان قادراً؛ اطلب FHCP وإجراءات FFLM/BMA.
- أعلن الحادث الكبير وقدم تقرير METHANE قبل تفاصيل الإدارة السريرية.

Arabic rapid review: أصلح أخطاء النظام فوراً وأبلغ؛ للشرطة أعطِ الاسم/العنوان فقط؛ اتصل بـ SNOD عند موت دماغي؛ حماية الأطفال أولوية؛ عينات جنائية عبر FHCP؛ أعلن الحادث وقدم METHANE.

8. Question coverage map (source concepts covered)

- PSIRF / patient safety incidents: oxygen/equipment misconnection; never-events; incident reporting and remediation. [PSIRF]
- RCEM (2020): management and assessment of suspected internal drug trafficker (body packing) in pregnancy and adults. [RCEM 2020]
- FICM (2021) / organ donation: catastrophic brain injury, brainstem death pathway, SNOD contact, donor assessment and family discussion. [FICM 2021; NHS Organ Donation]
- BMA / FFLM (2023): forensic blood sampling, consent, incapacitated patients, chain-of-custody requirements. [BMA/FFLM 2023]
- Safeguarding/domestic abuse: identification, documentation, IDVA/MARAC referral, child protection triggers.
- Capacity/Gillick: adult capacity algorithm; minors' consent; best-interests decisions.
- Police disclosure/legal exceptions: disclosure of name/address for alleged driving offences; when to consult legal/MDU/Caldicott.
- Major incident: declaration, METHANE, command and triage sequencing.
- Clinical toxicology basics: paracetamol overdose assessment (Rumack-Matthew) and NAC use.

- Jehovah’s Witness / transfusion refusal: consent issues, alternatives, documentation.

Sources to verify
RCEM, Faculty of Intensive Care Medicine (FICM), Patient Safety Incident Response Framework (PSIRF) / NHS England, NHS Organ Donation, BMA, FFLM, Medical Defence Union guidance, ERC, NICE (where applicable).

CHAPTER 11: SEXUAL HEALTH

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Chapter 11 — Sexual Health

1. Clinical overview and pathophysiology

- Core syndromes in ED: urethritis/cervicitis (Neisseria gonorrhoeae, Chlamydia trachomatis), pelvic inflammatory disease (PID), trichomoniasis, genital/oral herpes simplex (HSV-1/2), acute HIV seroconversion, consequences of sexual assault (trauma, pregnancy risk, STI risk, forensic needs).
- Pathophysiology essentials: ascending infection → endometritis/adnexal inflammation/TOA in PID (C. trachomatis, N. gonorrhoeae ± anaerobes); urethral epithelial infection causes dysuria/discharge; trichomonas = protozoan vaginitis; HSV causes mucocutaneous vesicles→ulcers with neural latency; acute HIV — systemic viraemia before seroconversion.
- ED priorities: ABCs, pain control, pregnancy test, limit harm (emergency contraception + HIV PEP assessment), preserve evidence, start empiric therapy when indicated, arrange SARC/SANE and sexual-health follow-up. [NICE CKS; RCEM]

> ملخص عربي: المتلازمات الشائعة: التهابات الإحليل وعنق الرحم، PID، تريكوموناس، هربس، عدوى HIV الحادة، تبعات الاعتداء الجنسي. أولويات الطوارئ: استقرار الحالة، اختبار حمل، منع الحمل الطارئ/PEP للحماية، الحفاظ على الأدلة، وإحالة لمراكز متخصصة. [NICE CKS; RCEM]

2. Recognition, differential diagnosis, and red flags

- PID: clinical diagnosis — lower abdominal pain + cervical motion tenderness (CMT) ± uterine/adnexal tenderness, purulent discharge, fever. Differentials: ectopic pregnancy, appendicitis, ovarian torsion, UTI, Mittelschmerz. Red flags requiring admission/imaging: high fever, peritonitis, vomiting, suspected TOA, pregnancy, inability to tolerate oral therapy. [NICE CKS]
- Urethritis / epididymo-orchitis (men): dysuria, urethral discharge, scrotal/testicular pain — think gonorrhoea, chlamydia; consider urinary causes and torsion (acute severe pain).
- Genital ulcers: painful grouped vesicles/ulcers → HSV (primary more systemic);

painless single ulcer → syphilis differential.
- Trichomoniasis: frothy green-yellow discharge, vulvovaginal itch; wet mount shows motile flagellates.
- Acute HIV seroconversion: 2–4 weeks after exposure — fever, sore throat, lymphadenopathy, maculopapular rash; early Ab tests may be negative — consider RNA if high suspicion.
- Sexual assault: assess for acute life-threats, pregnancy risk, STI exposure, and need for forensic collection. Do not undertake forensic sampling unless trained; prioritise medical care and referral. [RCEM]

> ملخص عربي: PID = ألم أسفل البطن + حساسية حركة عنق الرحم؛ احذر الحمل خارج الرحم، التواء المبيض، وخراج المبيض. الهربس ألماني ومجموعة من الفقاعات؛ HIV الحاد قد يعطي أشباه أمراض فيروسية قبل إيجابية الاختبار. عند الاعتداء الجنسي: لا تجري فحصًا جنائيًا دون تدريب، قدّم الرعاية الطارئة وأحِل للـSARC. [NICE CKS; RCEM]

3. Diagnostic approach and interpretation

- Baseline ED tests: pregnancy test (urine/serum) for all reproductive-age females with pelvic/genital complaints; urine/first-void NAAT for chlamydia/gonorrhoea; genital swabs (endocervical/high-vaginal/prescription site) for NAAT/culture; wet mount microscopy for trichomonas/clue cells/yeast; HSV PCR from lesion swab; HIV 4th-generation Ag/Ab as baseline — consider HIV RNA PCR if very early suspected infection. Send gonococcal culture when possible for sensitivity. [NICE CKS]
- Imaging: pelvic/transvaginal ultrasound when ectopic pregnancy, tubo-ovarian abscess (TOA), or torsion suspected. TOA usually needs admission and IV therapy/intervention. [NICE CKS]
- Interpretation caveats: negative 4th-gen HIV does not exclude acute infection in the first days — use RNA if indicated; NAAT from first-void urine is reliable for urethral/C. trachomatis detection; wet mount sensitivity varies. Forensic evidence windows vary by exposure/site — local forensic guidance required. [RCEM]

> ملخص عربي: اختبارات أساسية: اختبار حمل، NAAT لعقد الكلاميديا/السيلان، مسحة HSV PCR، اختبار HIV رباعي الجيل (إذا اشتباه مبكر استخدم RNA). التصوير عبر السونار عند اشتباه خراج أو حمل خارج الرحم. احذر أن سلبية اختبار HIV المبكر لا تنفي العدوى الحادة. [NICE CKS; RCEM]

4. Emergency management algorithms

A. PID decision algorithm
- Suspected PID → pregnancy test + assess severity.
- If mild–moderate, tolerating oral, reliable follow-up → outpatient therapy: ceftriaxone 1 g IM once + doxycycline 100 mg PO bd ×14 days + metronidazole 400–500 mg PO bd ×14 days. Arrange sexual-health follow-up and advise abstinence until completed treatment. [NICE CKS]
- Admit if: TOA on imaging, pregnancy, severe systemic illness/peritonitis, vomiting/inability to take oral, high fever, or clinical deterioration. Inpatient: IV cephalosporin ± doxycycline ± metronidazole; surgical/IR input for abscess drainage. [NICE CKS]

B. Urethritis / chlamydia
- Empiric if clinical features: doxycycline 100 mg PO bd ×7 days (first-line) and NAAT. Partner notification and treatment. [BASHH]

C. Gonorrhoea
- Empiric parenteral therapy per current local guidance (commonly ceftriaxone IM single dose — check local dose). Send NAAT + culture for sensitivity; treat partners. [BASHH/local guidance]

D. Trichomoniasis
- Metronidazole 2 g PO single dose (or 400–500 mg PO bd ×7 days). Treat partner. [NICE CKS]

E. HSV (genital/oral)
- Symptomatic therapy: oral aciclovir (e.g., 200 mg five times daily for ~5 days or 400 mg TDS for 5 days). Severe or immunocompromised → IV aciclovir (specialist-led). [BNF/NICE]

F. Acute HIV exposure and PEP
- Assess exposure risk, time since exposure (PEP indicated ≤72 hours for significant risk) and baseline tests. Start local PEP protocol promptly if indicated and arrange urgent specialist follow-up. [local/RCEM]

G. Sexual assault immediate bundle
- ABCs → analgesia → pregnancy test → offer emergency contraception (copper IUD most effective ≤5 days; ulipristal 30 mg ≤120 h; levonorgestrel 1.5 mg ≤72 h) → assess HIV PEP eligibility (≤72 h) → offer STI prophylaxis per local protocol →

hepatitis B vaccination/IG if indicated → preserve evidence (avoid washing/urination if possible), document, contact SARC/SANE, arrange forensic exam by trained staff. [RCEM; NICE CKS]

> ملخص عربي: خوارزميات رئيسية: PID → قرر دخول المستشفى أم علاج عن طريق الفم (سيفترياكسون IM + دوكسيسيكلين + ميترونيدازول). الكلاميديا = دوكسيسيكلين 7 أيام؛ تريكوموناس = ميترونيدازول؛ الهربس = أسيكلوفير؛ PEP للـHIV في خلال 72 ساعة إذا مؤهل؛ اعتداء جنسي = حزمة طوارئ شاملة وإحالة متخصصة. [NICE CKS; RCEM]

5. Drugs, procedures, disposition, and follow-up

- Key drug regimens (well established in guidance):
- PID outpatient: ceftriaxone 1 g IM once + doxycycline 100 mg PO bd ×14 days + metronidazole 400–500 mg PO bd ×14 days. [NICE CKS]
- Chlamydia: doxycycline 100 mg PO bd ×7 days. [BASHH]
- Gonorrhoea: ceftriaxone IM single dose (confirm current local dose). [BASHH/local guidance]
- Trichomonas: metronidazole 2 g PO single dose or 400–500 mg PO bd ×7 days. [NICE CKS]
- HSV: aciclovir oral regimens (e.g., 200 mg five times daily ×≈5 days). [BNF]
- Emergency contraception: copper IUD (best ≤5 days), ulipristal 30 mg ≤120 h, levonorgestrel 1.5 mg ≤72 h. [NICE CKS]
- HIV PEP: follow local protocol; begin ≤72 h when indicated and arrange prompt specialist review. [local/RCEM]

- Procedures and practical actions: pregnancy testing; collect appropriate NAAT/swabs; send gonococcal culture for sensitivities; transvaginal ultrasound for TOA/ectopic/torsion; do not perform forensic sampling unless trained — preserve clothing and avoid washing when possible; provide analgesia and wound care for genital trauma; consider Hepatitis B vaccination/IG if indicated. [RCEM; NICE CKS]

- Disposition: admit for TOA, pregnancy, severe systemic illness, peritonitis, inability to tolerate oral, suspected ectopic, or failed outpatient therapy. Discharge with written advice, partner notification instructions, arranged sexual-health follow-up, and clear return advice for worsening features. [NICE CKS]

- Follow-up: arrange sexual health clinic review for test results, partner notification, counsel for safer sex, further HIV testing per window periods and local protocol,

specialist input for recurrent/complicated infections. Test-of-cure and timing vary by organism and local guidance—follow local recommendations. [BASHH; local guidance]

ملخص عربي: الأدوية الأساسية مذكورة أعلاه؛ إجراءات مهمة: فحوص الحمل، مسحات NAAT، زرع للسيلان، سونار للحوض، حفاظ على الأدلة عند الاعتداء، تطعيم التهاب الكبد B إذا لزم. الإحالة للمراكز المتخصصة والمتابعة ضرورية. [NICE CKS; RCEM; BASHH]

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية

- PID “triad” for exam stems: lower abdominal pain + cervical motion tenderness (CMT) ± adnexal tenderness — if present, treat empirically while excluding ectopic pregnancy. (Exam shorthand: CMT = cervical motion tenderness.) [NICE CKS]
- HIV testing trap: a negative 4th-gen test early after exposure does not exclude acute infection — consider RNA PCR if classic seroconversion syndrome.
- HSV trick: primary episodes are systemic and longer; recurrent episodes are milder. Don’t mistake aphthous ulcers (painful but non-viral) for HSV — HSV ulcers usually follow vesicles.
- Discharge vs admission pitfalls: do not discharge a woman with suspected TOA, high fever, peritonitis signs, pregnancy, or inability to tolerate oral meds.
- Drug resistance caveat: azithromycin is no longer preferred first-line for chlamydia in many updated guidelines — many exam stems still quote older regimens; check question date/context. Always check current gonorrhoea dosing in local guidance (may vary). [BASHH]
- Forensic pitfalls: do not collect forensic samples unless trained; do not delay urgent medical care for evidence; document consent and findings precisely.
- Microbiology interpretation: wet mount sensitivity is operator-dependent; negative microscopy does not exclude trichomonas — rely on NAAT when available.

ملخص عربي: نقاط للامتحان: CMT = علامة PID مهمة؛ الاختبارات المبكرة قد تعطي سلبية مزيفة للـHIV؛ تجنّب استخدام الأزيثرومايسين كخيار افتراضي دون مراجعة الإرشادات الحديثة؛ لا تفعل فحصًا جنائيًا دون تدريب. [NICE CKS; BASHH; RCEM]

7. Arabic rapid-review summary

- أولويات الطوارئ: ABC، اختبار حمل، تخفيف ألم، منع الحمل الطارئ، تقدير حاجة PEP للـHIV، حفظ الأدلة، وإحالة لمراكز الطوارئ الجنسي/الاعتداء.
- PID: تشخيص سريري (ألم أسفل البطن + CMT)؛ علاج خارجي: سيفترياكسون IM + دوكسيسيكلين 100 mg bd ×14 يوماً + ميترونيدازول ×14 يوماً؛ ادخل للمستشفى إذا خراج أو حالة شديدة.
- الكلاميديا: دوكسيسيكلين 100 mg bd ×7 أيام. السيلان: سيفترياكسون IM (راجع الجرعة المحلية). تريكوموناس: ميترونيدازول 2 g جرعة واحدة. الهربس: أسيكلوفير فموي.
- عند الاعتداء الجنسي: حافظ على الأدلة، عرض PEP خلال 72 ساعة إذا مؤهل، اعرض وسائل منع الحمل الطارئة، أحِل إلى SARC/SANE والمتابعة.
- تحقّق من التوجيهات المحلية والوطنية حول مقاومة الميكروبات ونوافذ الأدلة الجنائية.

الملخص العربي المختصر أعلاه يغطي الأدوية الأساسية والإجراءات الطارئة. [NICE CKS; RCEM]

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية

- Concepts covered (source Q-numbers not supplied):
- PID diagnosis, management, outpatient vs inpatient criteria, TOA.
- Urethritis/epididymo-orchitis (gonorrhoea, chlamydia) diagnosis and first-line therapy.
- Gonorrhoea — NAAT and culture, empiric parenteral therapy and resistance caveats.
- Chlamydia — doxycycline regimen.
- Trichomoniasis — clinical features and metronidazole regimens.
- Genital/oral HSV — recognition, aciclovir dosing, primary vs recurrent.
- Acute HIV seroconversion and testing windows; HIV PEP assessment and timing (≤72 h).
- Sexual assault — emergency bundle, evidence preservation, forensic considerations, emergency contraception options.
- Diagnostic tests: pregnancy test, NAAT, wet mount, HSV PCR, HIV 4th-gen and RNA, pelvic ultrasound indications.
- Forensic and medicolegal pitfalls and referral pathways (SARC/SANE).

- Note: specific Q-number ranges were not supplied with the chunk; map lists concepts likely tested across typical FRCEM question banks.

Sources to verify
RCEM; NICE CKS; BASHH; BNF; local public health / sexual-health guidance; SARC/SANE protocols; ERC for general emergency practice.

CHAPTER 12: RESPIRATORY

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Chapter 12 — Respiratory

1. Clinical overview and pathophysiology

- Core pattern recognition: respiratory emergencies cluster around
- Hypoxaemia (failure of oxygenation; low SpO2),
- Hypercapnia (ventilatory failure; rising PaCO2 ± acidosis),
- Airway compromise (obstruction, reduced effort, aspiration),
- Pleural complications (pneumothorax, effusion, tension physiology).
- Clinical approach principle: triage → stabilise physiology (ABC, targeted O2) → focused bedside tests (SpO2, PEFR, ABG) → imaging/definitive therapy.
- Physiology keys:
- V/Q mismatch → hypoxaemia (PE, consolidation, pleural effusion).
- Shunt → refractory hypoxaemia not corrected by supplemental O2.
- Hypoventilation → hypercapnia (neuromuscular disease, sedative overdose, severe COPD).
- Dynamic hyperinflation / auto-PEEP → obstructive disease complications (asthma/COPD) and risk during ventilation.
- Use validated pathways: BTS for pleural disease/NIV, NICE/BNF for asthma/COPD, CURB-65 for CAP severity.

> Arabic summary (مختصر بالعربية):
> - الطوارئ التنفسية تتمركز حول: نقص التأكسج، فشل التهوية، انسداد المجارى الهوائية ومضاعفات الجنب. مبدأ العمل: فرز → تثبيت فيزيولوجي → فحوص سريعة → علاج موجه.

2. Recognition, differential diagnosis, and red flags

EXAM TRICK / تريكة امتحانية

- Rapid red flags (require immediate action):
- Respiratory arrest, unresponsive, severe hypotension, silent chest (asthma), rising/normalising PaCO2 in severe asthma, SpO2 <90% refractory to O2 (condition-dependent), tension pneumothorax signs (tracheal deviation, severe hypotension), haemodynamic shock with suspected PE.

- Asthma severity (practical thresholds):
- Moderate: PEFR 50–75% predicted/best; speaks in sentences; RR <25; HR <110.
- Severe: PEFR <50% (eg ~40%); RR ≥25; HR ≥110; difficulty speaking.
- Life-threatening: SpO2 <92% (or per local target), silent chest, cyanosis, exhaustion, arrhythmia, hypotension, PaO2 <8 kPa, PaCO2 normalising or rising; PEFR <33% predicted. (NICE/BNF)
- COPD exacerbation: look for baseline hypercapnia, altered consciousness, marked dyspnoea, sputum purulence; hypercapnic acidosis suggests need for NIV consideration.
- Pneumothorax types and features:
- Primary spontaneous (young, tall, smokers) vs secondary (older, COPD, underlying lung disease).
- Symptoms: sudden pleuritic pain, unilateral reduced air entry, hyperresonance.
- High-risk features (favor intervention): underlying lung disease, age (guidelines commonly cite ≥50 with smoking history), physiological compromise.
- PE suspicion: acute dyspnoea ± pleuritic pain, tachycardia, risk factors (recent immobility, surgery, malignancy, OCP). Use pre-test probability → D-dimer/CTPA strategy.
- Pitfalls:
- Normal SpO2 does not exclude CO poisoning (pulse oximetry unreliable).
- Normal or rising PaCO2 in severe asthma = impending respiratory failure.
- “Small” pneumothorax definitions vary by measurement method—check the stem.

Arabic summary (مختصر بالعربية):
- علامات الخطر: الصدر الصامت، PaCO2 مرتفع أو طبيعي في الربو الشديد، نفخ صدري ضاغط، صدمة مع انصمام رئوي مشتبه. قيّم شدة الربو بالـPEFR، واضبط تدخلات النفاخ/الانسداد حسب وجود مرض رئوي أساسي.

3. Diagnostic approach and interpretation

- Immediate bedside tests: SpO2, PEFR (asthma), ABG if ventilatory failure suspected, ECG (PE/MI), CXR when it will change management (do not delay

life-saving therapy).
- PE pathway (stable patient):
- Assess stability → estimate pre-test probability (Wells/Geneva).
- Low/intermediate → D-dimer → if positive → CTPA.
- High pre-test probability → proceed to CTPA; in major delays consider anticoagulation per local policy. (NICE NG158)
- Pneumothorax:
- Unstable/tension → immediate decompression (needle/finger), then chest drain.
- Stable → erect CXR to estimate size; use guideline measurement method (apex or interpleural distance per local/BTS).
- Consider CT if complicated, recurrent or to guide chest drain.
- Asthma/COPD:
- Measure PEFR and SpO2 for triage.
- ABG if severe illness, poor response, decreased consciousness, or before/after NIV/intubation.
- NIV decision and monitoring:
- Indications: acute hypercapnic respiratory failure with acidosis (COPD) after initial medical therapy; monitor ABG ~1 hour after initiation.
- Contraindications: inability to protect airway, severe vomiting, facial trauma, respiratory arrest; relative: severe agitation, uncontrolled secretions.
- Cylinder duration quick formula: duration (min) = cylinder volume (L) ÷ flow (L/min). (Example UK CD ≈460 L → 460 ÷ 15 L/min ≈30 min) — verify local cylinder sizes.

> Arabic summary (مختصر بالعربية):
> - الفحوص السريعة: SpO2 وPEFR وABG. في الاشتباه بالانصمام استخدم تقدير الاحتمال المسبق قبل الـD-dimer. النفاخ: CXR إذا سيؤثر على الإدارة؛ النفاخ الضاغط لا ينتظر تصويراً.

4. Emergency management algorithms

- General respiratory failure (quick algorithm):
1. ABC: secure airway, high-flow O2 as indicated, treat shock.
2. Targeted SpO2: COPD 88–92% (unless other instruction); otherwise aim for normalisation as clinical context.
3. ABG if ventilatory failure suspected or severe disease.
4. Treat cause: bronchodilators/steroids (asthma), NIV/intubation (ventilatory failure), chest drain/aspiration (pneumothorax), antibiotics (CAP), thrombolysis

(massive PE).
- Acute severe asthma (triage → treatment ↑ escalation):
1. Immediate: high-flow inhaled β2 agonist (nebuliser or MDI+spacer), supplemental O2 to target (see above), systemic corticosteroid early.
2. Add ipratropium for severe attacks; give frequent β2 agonist dosing.
3. If poor response → IV magnesium sulphate 2 g over 20 min.
4. If life-threatening (silent chest, rising PaCO2, exhaustion) → prepare for intubation; ventilate with low Vt (~6 ml/kg), low RR, long expiratory time (I:E ≈1:4), consider neuromuscular blockade.
5. Consider IV salbutamol bolus (commonly 250 µg over 5–10 min) if inhaled therapy inadequate (verify local protocol).
- Pneumothorax decision tree (simplified):
- Stable small PSP, minimal symptoms → conservative outpatient observation and repeat imaging (48–72 h or local timeframe).
- Symptomatic or moderate PSP seeking rapid relief → attempt needle aspiration.
- Failed aspiration OR secondary pneumothorax OR large/physiologically compromised → insert small-bore Seldinger chest drain (commonly 10–14 Fr per BTS).
- Tension pneumothorax → immediate needle decompression then chest drain.
- Post-drain monitoring: immediate, 15 min, then every 30 min for 1 hour, then 4-hourly checks early on. (BTS pleural procedures)
- NIV initiation and escalation (COPD/hypercapnic failure):
1. Indication: hypercapnic acidosis after initial medical therapy (use local pH cutoffs).
2. Initial settings: IPAP ≈15 cmH2O / EPAP ≈3 cmH2O; increase IPAP to improve ventilation; adjust EPAP to improve oxygenation.
3. Reassess clinically and by ABG ~1 hour; if no improvement consider increasing IPAP (do not exceed IPAP ~30 cmH2O without specialist input) or escalate to invasive ventilation.
4. Contraindications/relative contraindications as above. (BTS NIV guidance)
- PE emergencies:
- Haemodynamically unstable with suspected massive PE → resuscitate and consider immediate systemic thrombolysis if no contra-indication.
- Stable patients follow pre-test stratification → imaging/anticoagulate per results.

Arabic summary (مختصر بالعربية):
- خوارزمية عامة: ABC ثم أهداف تشبع مؤقتة (COPD 88–92%) → ABG إن اشتبه بفشل تهوئي
→ علاج السبب (ربو: بروجنيستات+كورتيزون+ماغنسيوم، التهاب رئوي: مضادات حيوية، نفاخ: تفريغ/درين).
- NIV: IPAP بدءياً ~15 / EPAP ~3؛ زِدِ IPAP لتحسين إزالة CO2؛ راجع بعد ساعة؛ لا تتجاوز ~30 دون خبرة.

5. Drugs, procedures, disposition, and follow-up

- Key drugs and emergency doses (verify local policy):
- Oxygen: target sats by condition (COPD 88–92%; others higher as clinically indicated).
- Inhaled β2 agonists: frequent/nebulised or MDI+spacer.
- Ipratropium: add in severe asthma exacerbations.
- Systemic corticosteroids: give early in asthma and COPD exacerbations (route/dose per local guideline).
- IV magnesium sulphate: 2 g over 20 min for life-threatening asthma.
- IV salbutamol bolus: commonly 250 µg over 5–10 min when inhaled therapy inadequate (check local guidance).
- Antibiotics for CAP: guided by CURB-65; severe (CURB-65 ≥3) → hospitalisation and IV therapy (eg co-amoxiclav ± macrolide per NICE).
- Anticoagulation/thrombolysis: follow risk/benefit and NICE/locally agreed pathways for PE.
- Procedures:
- Needle aspiration for PSP (first-line for symptomatic primary in many guidelines).
- Small-bore Seldinger chest drain commonly 10–14 Fr for chest drainage (BTS guidance — confirm local practice).
- Urgent needle decompression for tension pneumothorax followed by chest drain.
- NIV (mask or helmet per local availability) with reassessment at ~1 hour.
- Intubation/controlled ventilation for impending respiratory arrest or failed NIV.
- Disposition:
- ICU/level 2 for: need for invasive ventilation, NIV requiring close monitoring, life-threatening asthma/near-fatal events, massive PE requiring thrombolysis.
- Admit for: secondary pneumothorax, CAP with CURB-65 ≥2 (local nuance), COPD exacerbation with failure of initial therapy.

- Consider outpatient follow-up for small PSP managed conservatively; plan imaging review (CXR/CT) and pneumothorax follow-up clinic.
- Follow-up specifics:
- Pneumothorax: repeat imaging at 48–72 h for conservative care; allow air travel only after radiological resolution plus guideline-recommended interval (commonly cited: 7 days after complete resolution; check BTS).
- COPD/asthma: arrange primary care/pulmonary follow-up, inhaler review, smoking cessation, vaccination as per guidance.
- After PE: anticoagulation follow-up, thrombophilia only if indicated; arrange follow-up imaging/echo for RV dysfunction if relevant.

> Arabic summary (مختصر بالعربية):
> - أدوية رئيسية: أكسجين وفق الهدف، موسعات براونكوسجيل، كورتيزون نظامي، ماغنسيوم 2 غ عبر 20 دقيقة، سالبوتامول وريدياً 250 ميكروغرام (تحقق من البروتوكول المحلي).
> - إجراءات: شفط إبرة للـPSP، درين صدر صغير 10–14Fr للثانوي أو الفاشل، تفريغ فوري للضغط.
> المتابعة بتصوير بعد 48-72 ساعة أو مكانياً حسب الدليل.

6. High-yield FRCEM exam tricks and common pitfalls

- Read stem for thresholds: age cutoffs and cylinder volumes vary — many questions use ≥65 for CURB-65 and commonly ≥50 as higher risk for pneumothorax with smoking history. Don’t assume 45 unless stated.
- Use the cylinder formula rather than memorising volumes: duration (min) = cylinder volume (L) ÷ flow (L/min). If a CD = 460 L is given in the stem, use it; otherwise verify.
- D-dimer only useful if pre-test probability is low/intermediate; high probability → image directly.
- Silent chest = life-threatening asthma irrespective of SpO2 readout.
- Rising or normalising PaCO2 in a worsening asthmatic patient is a poor prognostic sign — treat as impending failure.
- Do not delay NIV for CXR in a patient in severe distress; many exam stems test this principle.
- Small-bore chest drains (10–14 Fr) are commonly first line for PSP; large bore is often a distractor.
- In ventilated severe asthma allow permissive hypercapnia and long expiratory times — increasing RR to normalise PaCO2 causes dynamic hyperinflation.

- Beware of pulse oximetry in carbon monoxide poisoning and of over-reliance on SpO2 alone.

Arabic summary (مختصر بالعربية):
- خدع الامتحان الشائعة: راجع عمر المريض والحدود، لا تستخدم D-dimer عند احتمالية ما قبل الفحص عالية، لا تؤخر NIV بفعل تصوير صدر، الصدر الصامت في الربو حالة حرجة، استخدم حساب زمن الأسطوانة لا الحفظ.

7. Arabic rapid-review summary

- فرز سريع: ABC ثم تثبيت فيزيولوجي — تمارين: ABG، PEFR، SpO2.
- الربو الحاد: استنشاقي فوري + كورتيزون نظامي؛ صدري صامت أو PaCO2 متزايد = تهديد للحياة؛ ماغنسيوم 2 غ في الحالات الشديدة؛ تحضير للتهوية عند فشل العلاج.
- النفاخ (Pneumothorax): صغير ومستقر → مراقبة؛ معتدل → شفط إبرة؛ ثانوي/كبير/فشل → درين صدر صغير (10–14Fr عادةً)؛ نفاخ ضاغط → تفريغ فوري. تجنب السفر الجوي حتى الشفاء الشعاعي + فترة انتظار موصى بها.
- COPD وفشل CO2: جرّب NIV مبكراً (IPAP ~15 / EPAP ~3)، راجع بعد ساعة، لا تتردد بالتصعيد إن فشل.
- الانصمام الرئوي: استخدم تقدير الاحتمال المسبق ثم D-dimer إن مناسب أو CTPA مباشرة إذا احتمال عالٍ؛ الصدمة بسبب PE قد تتطلب ثرومبوليزا.
- تحقق محلياً من أحجام الأسطوانات، أحجام الدرين، وحدود IPAP القصوى.

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية
- Asthma: acute severity criteria, PEFR thresholds, life-threatening signs, bronchodilator/steroid/MgSO4 use, IV salbutamol bolus, intubation/ventilation strategies. (stems referencing NICE/BNF)
- COPD: hypercapnic respiratory failure, indication/initial settings/ contraindications for NIV, monitoring and escalation. (BTS NIV guidance)
- Pneumothorax: primary vs secondary distinctions, measurement-dependent size definitions, conservative vs aspiration vs chest drain (10–14 Fr), tension pneumothorax immediate decompression, post-procedure monitoring and air travel guidance. (BTS pleural procedures)
- Pulmonary embolism: pre-test probability, D-dimer use, CTPA pathway, massive PE and thrombolysis considerations. (NICE NG158)

- CAP: CURB-65 use for disposition and antibiotic thresholds. (NICE)
- Emergency diagnostics: use and timing of SpO2, PEFR, ABG, CXR/CT, co-oximetry for CO poisoning.
- Oxygen and equipment: oxygen cylinder duration formula (CD ≈460 L example), oxygen targets (COPD 88–92%).
- Ventilation physiology: auto-PEEP, permissive hypercapnia, ventilatory settings for severe asthma (low Vt, long expiratory time).
- Exam pitfalls: D-dimer misuse, cylinder volume traps, age cutoff traps, silent chest significance, “normal” PaCO2 in severe asthma.
Note: No original question numbers were supplied with the chunk notes; the map lists concepts/topics represented in the supplied material and typical FRCEM question areas.

Sources to verify
- BTS (British Thoracic Society) pleural procedures and NIV guidance; NICE guidance (asthma, CAP, PE — NG158); BNF; UKHSA antiviral / influenza guidance; RCEM and local hospital/ICU protocols for NIV and ventilator limits; Thorax/BMJ pleural and pneumothorax guidance.

CHAPTER 13: PHARMACOLOGY & POISONING

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Chapter 13 — Pharmacology & Poisoning (final)

1. Clinical overview and pathophysiology

- Core principle: ABC first (secure airway — intubate if GCS ≤8 or loss of airway reflexes), high-flow O2, circulation support; stop ongoing exposure (remove contaminated clothing, irrigate skin) before non-urgent interventions.
- Toxidrome-driven approach: patterns of autonomic/neuromuscular/mental state signs suggest groups of toxins and guide empiric antidote (opioid, cholinergic, anticholinergic, sympathomimetic, serotonergic, sedative-hypnotic, TCA, cyanide, CO, CCB/β-blocker).
- Mechanisms to recall:
- Membrane Na-channel blockade → QRS widening, arrhythmia risk (TCAs).
- ETC/mitochondrial inhibition → profound lactic acidosis with normal PaO2 (cyanide).
- Impaired O2 carriage (CO, methemoglobin) → normal PaO2 but tissue hypoxia.
- Mixed respiratory alkalosis + metabolic acidosis → salicylates.
- Antidotes/enhanced elimination: specific agents (NAC for paracetamol, digoxin-Fab, hydroxocobalamin for cyanide, methylene blue for methemoglobinaemia, fomepizole for toxic alcohols), MDAC for enterohepatic recirculators, haemodialysis for dialysable toxins.
- Monitoring essentials: ECG, ABG/VBG with lactate, glucose, electrolytes, renal/liver function, drug levels if available; continuous cardiac/respiratory monitoring for unstable patients.

> Arabic (موجز): ابدأ دوماً بـ ABC، تمييز التوكسدروم يوجّه العلاج الفعّال، واعلم ترياقات محددة (مثل NAC للباراسيتامول، Hydroxocobalamin للسيانيد، ميثيلين بلو للمِتهيموغلوبين).

2. Recognition, differential diagnosis, and red flags

- Major toxidromes (key discriminators):
- Opioid: pinpoint pupils, hypoventilation → naloxone.
- Cholinergic (organophosphate): SLUDGE, miosis, bronchorrhoea → atropine ±

oxime.
- Anticholinergic: dry, flushed, mydriasis, hyperthermia, delirium → physostigmine only if safe (no TCA cardiotoxicity).
- Sympathomimetic/cocaine/amphetamine: agitation, mydriasis, diaphoresis, hypertension → benzodiazepines.
- Serotonin syndrome: hyperreflexia, clonus (ocular clonus high-yield), hyperthermia.
- NMS: gradual onset, severe rigidity after neuroleptics — dantrolene.
- TCA: anticholinergic + seizures + hypotension + QRS widening (>100 ms risk; ≥160 ms severe).
- Cyanide (fire, burning plastics): very high lactate, shock, preserved PaO2/SpO2.
- Carbon monoxide (closed space): headache/confusion, normal SpO2/PaO2; measure COHb (co-oximetry).
- Methemoglobinaemia: cyanosis, chocolate blood, SpO2 ≈85% that does not correct with O2; normal PaO2 on ABG.
- Red flags (act urgently):
- GCS ≤8 or airway compromise → intubate.
- QRS >100 ms (TCA) — give sodium bicarbonate; ≥160 ms = high risk arrhythmia.
- Rapidly rising lactate in smoke inhalation → suspect cyanide.
- Severe hyperkalaemia with suspected digoxin toxicity (e.g., K+ >6.5 mmol/L) or life-threatening arrhythmia → consider digoxin-Fab.
- Refractory hypotension/bradycardia after CCB/β-blocker ingestion → escalate to HIET/ECMO.

> Arabic (موجز): تعرّف التوكسدرومات الأساسية؛ مؤشرات الخطر: QRS ≥100، GCS ≤8، المتّسع في TCA، حماض لبني كبير بعد حريق → سيانيد.

3. Diagnostic approach and interpretation

- Immediate tests: ECG (QRS/QTc, arrhythmias), capillary glucose, ABG/VBG with lactate, pulse oximetry + co-oximetry if CO/metHb suspected, electrolytes, renal/liver function, coagulation, CK.
- Targeted toxicology tests: paracetamol level (plot on nomogram if time of ingestion known; sample at ≥4 h post single acute ingestion), salicylate level, digoxin, lithium, COHb, methemoglobin, drug levels as available.
- Useful lab patterns:
- Mixed respiratory alkalosis + metabolic acidosis → salicylates.

- Very high lactate with normal PaO2 → cyanide.
- High anion gap metabolic acidosis + osmolar gap → consider methanol/ethylene glycol (toxic alcohols); look for visual symptoms (methanol) and renal/oxalate crystals (ethylene glycol).
- Chocolate-brown blood + SpO2 ~85% unchanged with O2 → methemoglobinaemia.
- ECG interpretation pointers: measure QRS width in overdose (TCA); identify hyperkalaemic changes and digitalis changes (arrhythmia types).
- Decontamination decision: activated charcoal generally useful within ~1 hour of ingestion for many oral drugs; MDAC for substances with significant enterohepatic/enteroenteric recirculation (e.g., carbamazepine, theophylline, phenobarbital, dapsone).

> Arabic (موجز): فحوص عاجلة: ECG، غازات، لاثات، مستويات دواء مخصصة (باراسيتامول، ساليسيلات، ديوكسين). نمط الـABG يساعد في التشخيص التفريقي.

4. Emergency management algorithms

A. General poisoned patient (algorithm)
1. Airway: protect; intubate if GCS ≤8 or airway reflex loss.
2. Breathing: high-flow O2 for all hypoxic/exposed patients; consider hyperbaric O2 for severe CO poisoning per local criteria (e.g., LOC, neuro deficit, cardiac ischaemia, pregnancy) [local protocols vary].
3. Circulation: IV fluids, titrated vasopressors (noradrenaline) for refractory hypotension.
4. Stop exposure: remove contaminated clothing; irrigate skin/eyes; protect staff.
5. Decontaminate: activated charcoal if appropriate (see section 3). MDAC for selected toxins.
6. Antidote/enhanced elimination give without delaying life-saving care when clearly indicated. Consult toxicology early.

B. Key toxin-specific algorithms (concise, include well-established doses only)
- Paracetamol (acetaminophen)
- If single acute ingestion and level above treatment line on nomogram → start NAC promptly. SNAP regimen commonly used: 100 mg/kg over 2 h then 200 mg/kg over 10 h (12 h total) [SNAP protocol]; 21-hour regimens still used in some centres — follow local guideline. [SNAP]

- Tricyclic antidepressant (TCA)
- QRS >100 ms or hypotension/arrhythmia/seizure → IV sodium bicarbonate boluses and infusion to narrow QRS and treat acidosis (standard teaching: 1–2 mEq/kg bolus or commonly cited 100 ml 8.4% in many references) — verify local dosing protocols. Benzodiazepines for seizures. Avoid phenytoin. [Toxbase]
- Opioid overdose
- Naloxone titrate IV: initial 0.4 mg IV, increase (0.8 mg etc.) to restore adequate ventilation not full wakefulness; monitor for recurrence. Local protocols vary.
- Organophosphate/cholinergic poisoning
- Decontaminate first; atropine IV titrated to drying of secretions/adequate ventilation (initial adult doses often 0.6–1 mg, larger/severe doses repeated rapidly until effect); give oxime (pralidoxime) when indicated. [Toxbase]
- Cyanide (suspected in smoke inhalation from burning plastics)
- Suspect with very high lactate/normal PaO2. Give hydroxocobalamin (first-line in many regions) and supportive care; sodium thiosulfate adjunct in some protocols. Early airway/circulatory support required. [source: hydroxocobalamin references]
- Carbon monoxide
- High-flow O2 immediately; consider hyperbaric oxygen per local criteria (availability/guidelines differ). Measure COHb via co-oximetry.
- Salicylates
- Early charcoal if indicated; IV sodium bicarbonate to alkalinise plasma/urine and enhance renal excretion. Indications for dialysis include very high levels or severe metabolic acidosis/coma/renal failure — thresholds vary by guideline (e.g., often cited severe level ≈900 mg/L but verify local thresholds).
- Calcium-channel blocker (severe)
- Initial: IV fluids, calcium (IV calcium gluconate/ chloride), vasopressors; for refractory cases start high-dose insulin-euglycaemia therapy (HIET) under ICU/ toxicology guidance (typical starting bolus 1 U/kg then 1 U/kg/hr titratable up to higher rates; monitor glucose/K+ closely) and consider lipid emulsion and ECMO if refractory. Dosing protocols vary — verify locally.
- Local anaesthetic systemic toxicity (LAST)
- Lipid emulsion 20%: bolus 1.5 ml/kg IV over 1 min, then infusion 0.25 ml/kg/min; repeat bolus 1–2× if needed; continue ACLS with modifications. [LitFL]
- Methemoglobinaemia
- Methylene blue 1–2 mg/kg IV over 5 min (repeat once if needed). Use caution in G6PD deficiency (risk hemolysis). Consider exchange transfusion or hyperbaric

oxygen if methylene blue contraindicated.
- Digoxin toxicity
- Digoxin-specific Fab for life-threatening arrhythmia, severe hyperkalaemia (e.g., >6.5 mmol/L) or haemodynamic instability — dose per product guidance and toxicology consult.
- Lipid rescue
- For severe lipophilic drug toxicity (e.g., local anaesthetic, some CCBs, β-blockers): Intralipid 20% bolus 1.5 ml/kg then infusion 0.25 ml/kg/min; repeat/extend per guidance. [LitFL]
- Toxic alcohols (methanol/ethylene glycol)
- Fomepizole first-line; urgent dialysis for severe acidosis, visual disturbance (methanol), renal failure; bicarbonate for severe acidosis.
- Isoniazid seizures
- Benzodiazepines; administer pyridoxine (vitamin B6) IV gram per gram of INH ingested when known, or empiric large dose (e.g., 5 g) in life-threatening seizures if ingestion unknown.
- Sulfonylurea hypoglycaemia
- Bolus glucose + continuous dextrose; give octreotide (e.g., 50 mcg) to prevent recurrent insulin secretion; monitor prolonged hypoglycaemia.
- Hemodialysis/extracorporeal
- Consider for dialysable toxins (e.g., lithium, severe salicylates, methanol/ethylene glycol, severe theophylline, phenobarbital, some valproate cases). Use EXTRIP and local toxicology to decide. [EXTRIP]

> Arabic (موجز الإجراءات): خوارزمية عامة: ABC → ازالة التعرض → الفحم المنشط إذا مناسب → تریاق محدد → غسيل دموي/MDAC حسب السم. جرعات محددة: NAC SNAP 100 mg/kg over 2 h then 200 mg/kg over 10 h؛ Intralipid 20% 1.5 ml/kg bolus + 0.25 ml/kg/min infusion؛ ميثيلين بلو 1–2 mg/kg.

5. Drugs, procedures, disposition, and follow-up

- Well-established emergency doses/starts (verify local protocol before use):
- NAC (SNAP) for paracetamol: 100 mg/kg IV over 2 h then 200 mg/kg over 10 h (12 h total) [SNAP].
- Naloxone: start 0.4 mg IV (titrate); escalate per local guidance.
- Atropine (organophosphate): initial adult 0.6–1 mg IV; repeat rapidly until atropinisation (severe poisoning often needs large/titrate doses) [Toxbase].

- Sodium bicarbonate (TCA): teach weight-based 1–2 mEq/kg bolus or commonly cited 100 ml 8.4% bolus in many references — confirm local dosing.
- Lipid emulsion (20% Intralipid): bolus 1.5 ml/kg then 0.25 ml/kg/min infusion [LitFL].
- Methylene blue: 1–2 mg/kg IV (repeat once if needed).
- Hydroxocobalamin for cyanide: regionally recommended first-line antidote (dose per product/local policy).
- Procedures and advanced support:
- Early intubation for airway compromise; RSI with appropriate drug choices (avoid suxamethonium if hyperkalaemia risk).
- Continuous cardiac monitoring; early ICU/HDU involvement for unstable patients or those requiring HIET, digoxin-Fab, or haemodialysis.
- Use extracorporeal life support (ECMO/CRRT) for refractory shock or severe poisonings not responsive to medical therapy where available.
- Disposition:
- Discharge only after toxicology assessment: asymptomatic low-risk single ingestions may be observed and discharged following local protocols.
- Admit to ICU for airway support, haemodynamic instability, arrhythmia, seizures, need for antidote infusion/HIET, or dialysis.
- Psychiatry/safeguarding referral for intentional self-poisoning as per local procedure; involve poisons/toxicology service early.
- Follow-up:
- Monitor for rebound toxicity (e.g., enterohepatic recirculation, delayed gastric emptying, extended-release preparations).
- Arrange hepatic/renal follow-up after paracetamol toxicity; repeat levels as indicated.
- Counsel regarding return precautions and link to mental-health services where relevant.

> Arabic (موجز): اعطاء الترياق والعناية المركزة حسب الحاجة؛ راجع سموم محلّي-وطني لجرعات تفصيلية؛ لا تخرج المرضى المختاة النفسية دون تقييم/إحالة.

6. High-yield FRCEM exam tricks and common pitfalls

> **EXAM TRICK / تريكة امتحانية**
> - Pulse oximetry pitfalls: normal SpO2 does NOT exclude CO poisoning;

methaemoglobinaemia often gives SpO2 ≈85% irrespective of PaO2. Use co-oximetry when suspected.
- Lactate clue: very high lactate with normal PaO2 in a fire victim → think cyanide, not primary hypoxia.
- TCA exam focus: QRS width is key. QRS >100 ms → increased risk of seizures/arrhythmia; treat with sodium bicarbonate. Avoid phenytoin for seizures. QRS ≥160 ms = severe.
- Salicylate exam stem: mixed respiratory alkalosis + metabolic acidosis (early) with tinnitus and hyperventilation. Aim urine pH >7.5 when alkalinising. Dialysis if very high levels or severe acidosis/renal failure — verify threshold locally.
- Cocaine chest pain: avoid non-selective β-blockers in sympathomimetic toxicity (unopposed α-effect). Use benzodiazepines and nitrates; labetalol is nuanced — check local guidance.
- MDMA: severe hyponatraemia from SIADH is a classic stem — use hypertonic saline for symptomatic severe hyponatraemia.
- Suxamethonium pitfall: prolonged paralysis/apnoea → pseudocholinesterase deficiency; continue ventilatory support. Avoid sux in hyperkalaemia risk states.
- Paracetamol timing: measure level at 4 h for single acute ingestion to use nomogram; if time unknown or staggered ingestion err on side of treatment.
- Activated charcoal: typical effective window ~1 h; some sustained-release formulations or large ingestions may extend this — agent specific. Do not give for corrosive or hydrocarbon aspiration risk.
- Methylene blue caution: can precipitate haemolysis in G6PD deficiency and may interact with serotonergic drugs (risk of serotonin syndrome) — weigh risks but do not delay life-saving treatment in severe metHb.

Arabic (موجز نصائح للامتحان): التمييز بين التوكسدرومات (SLUDGE، QRS في TCA، SpO2). مفاتيح للسرعة. QRS >100 ms → بيكربونات في TCA. حافظ على TIMING لباراسيتامول (4 ساعات).

7. Arabic rapid-review summary (مراجعة سريعة بالعربية)

- الأساس: ABC ثم إزالة المسبب → ترياق محدد → زيادة نزع السم (فحم منشط/MDAC/غسيل دموي) → مراقبة مستمرة.

- دلائل شائعة: QRS واسع = TCA → بيكربونات؛ دم بني شوكولا + SpO2 ~85% = ميت-هيموغلوبين → ميثيلين بلو؛ PaO2 طبيعي + حماض لبني عالي بعد حريق = سيانيد → Hydroxocobalamin.
- ترياقات ذات نفع فوري: NAC للباراسيتامول (SNAP أو 21h وفق المستشفى)، نالوكسون للأفيونات، أتروبين للكولينيرجي، فورمبيزول للكحوليات السامة، ميثيلين بلو للمِتهيموغلوبين، Intralipid للسموم الدهنية، ديفيروكسامين للحديد.
- تحقق من بروتوكولاتك المحلية قبل إعطاء الجرعات الدقيقة أو بدء غسيل دموي.

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية

- Paracetamol overdose and NAC regimens (SNAP vs 21-hour) — paracetamol nomogram timing.
- Tricyclic antidepressant cardiotoxicity — QRS thresholds and sodium bicarbonate.
- Cyanide poisoning in smoke inhalation — lactate clue, hydroxocobalamin.
- Carbon monoxide poisoning — co-oximetry and hyperbaric oxygen considerations.
- Organophosphate/cholinergic toxicity — atropine and oxime.
- Opioid overdose — naloxone dosing and monitoring.
- Calcium-channel blocker and beta-blocker poisoning — calcium, glucagon, HIET, vasopressors.
- Salicylate poisoning — mixed ABG, urine alkalinisation, dialysis indications.
- Methemoglobinaemia — methylene blue dosing and G6PD caution.
- Lipid emulsion therapy (LAST and lipophilic drug overdose).
- MDAC indications (theophylline, carbamazepine, dapsone, phenobarbital).
- Toxic alcohols (methanol/ethylene glycol) — fomepizole and dialysis.
- Digoxin toxicity and digoxin-Fab indications.
- Isoniazid seizures — pyridoxine.
- Malignant hyperthermia / NMS / Serotonin syndrome differentiation and antidotes (dantrolene, cyproheptadine).
- Metal toxicities: iron (deferoxamine), lead (EDTA/dimercaprol basics).
- Pseudocholinesterase deficiency after succinylcholine — prolonged apnea management.
- General items: ABC priorities, decontamination, key diagnostic patterns (anion/osmolar gaps), pulse oximetry pitfalls, activated charcoal timing,

> haemodialysis indications (EXTRIP).
> Note: specific exam question numbers were not supplied; mapping above lists all distinct concepts represented in the supplied material.

Sources to verify
- Toxbase (UK toxicology database) [Toxbase]
- Literature for lipid rescue (Life in the Fast Lane / LitFL) [LitFL]
- SNAP paracetamol regimen publications / local hospital protocols [SNAP]
- EXTRIP workgroup recommendations for extracorporeal treatments in poisoning [EXTRIP]
- RCEM guidance (nitrous oxide, stimulant/serotonergic toxicity) [RCEM 2023/2024]
- NICE Clinical Knowledge Summaries and relevant NICE guidance where applicable [NICE]
- Local toxicology/toxin centres and hospital protocols (antidote stocks, dialysis thresholds, hyperbaric oxygen criteria) — confirm before clinical use.

This chapter is study material, not patient-specific advice. Verify doses and thresholds with current local/national toxicology guidance before clinical application.

CHAPTER 14: PALLIATIVE AND END OF LIFE CARE

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Chapter 14: Palliative and End-of-Life Care — Final revision

1. Clinical overview and pathophysiology

- Core goals: prioritise patient comfort, symptom control and respect expressed wishes/advance decisions; avoid burdensome interventions unless they change outcomes or align with goals of care. Document treatment escalation plans. (BNF; NICE CKS; NHS England 2017)
- Symptom mechanisms — match to pharmacology:
- Breathlessness: hypoxaemia, airway obstruction (tumour/lymphangitis), pleural effusion, pulmonary oedema, bronchospasm, PE, anaemia, metabolic causes, anxiety-driven hyperventilation.
- Noisy breathing (terminal secretions): pooling of saliva/bronchial secretions in an obtunded patient rather than pulmonary oedema.
- Agitation/delirium: multifactorial — hypoxia, pain, medications, urinary retention, metabolic disturbance, CNS pathology.
- Nausea: vestibular/raised ICP, bowel obstruction, drug-induced, metabolic.
- Pain types: nociceptive vs neuropathic (burning, electric shocks, dermatomal).
- Principle: use lowest effective pharmacologic dose; prioritise non-pharmacological measures and reversible causes first.

> شرح موجز بالعربية:
> - الأهداف: راحة المريض والسيطرة على الأعراض واحترام رغبات المريض/العائلة. اربط الدواء بآلية العرض واستخدم أدنى جرعة فعّالة.

2. Recognition, differential diagnosis, and red flags

- Rapid recognition:
- Dyspnoea: assess SpO2, work of breathing, respiratory rate, speech, accessory muscle use. Red flags: severe hypoxaemia, stridor (upper airway obstruction), haemodynamic instability, signs of tension physiology.
- Noisy breathing: if oxygenation preserved and chest clear → terminal secretions; if crackles/oedema → pulmonary oedema.

- Agitation: look for reversible precipitants (hypoxia, pain, constipation/urinary retention, infection, drug withdrawal).
- Nausea/vomiting: red flags include signs of obstruction (colicky pain, distension), raised ICP (headache, focal neuro signs), uncontrolled vomiting causing dehydration.
- New focal neurological signs or rapidly progressive neurology = urgent imaging/ neurology review.
- Differential pearls:
- Anxiety vs organic breathlessness: preserved SpO2 and normal chest exam suggest anxiety; opioids treat refractory sensation but benzodiazepines helpful for anxiety component.
- Bowel obstruction: tinkling bowel sounds, distension; do not use prokinetics if high-grade mechanical obstruction.
- Legal/administrative red flags: verify presence/validity of advance decision to refuse treatment; confirm DNACPR/status of treatment escalation.

> شرح موجز بالعربية:
> - تمييز سريع للأعراض: تفحّص علامات الخطر (نقص أكسجة شديد، خَناق، علامات انسداد معوي أو ضغط دم غير مستقر). تحقق من وجود قرارات مسبقة موقعة.

3. Diagnostic approach and interpretation

- Focused bedside assessment only — investigations if they will change immediate care or align with goals:
- Immediate: ABCs, SpO2, vital signs, targeted exam (respiratory, cardiac, neuro, abdominal).
- Tests to consider selectively: CXR (pleural effusion/airway/aspiration), bedside ultrasound (effusions, pneumothorax, cardiac failure), ABG if respiratory failure suspected, point-of-care glucose, basic bloods if sepsis or reversible metabolic cause suspected.
- Avoid blanket tests in terminal phase that do not alter management; always weigh burden vs benefit.
- Interpretation tips:
- SpO2 threshold: consider supplemental O2 for symptomatic hypoxaemia (SpO2 ≤92% in most patients) [NICE CKS].
- Chest crackles + peripheral oedema → consider diuresis for pulmonary oedema

when consistent with goals.
- Normal O2 with noisy breathing → treat secretions, not oxygen.

> شرح موجز بالعربية:
> - اطلب فحوصات فقط إن ستؤثّر على الاختيارات العلاجية الفورية. عتبة مؤاتية للأكسجين عادة SpO2 ≤92% للمرضى المصابين بضيق التنفس الأعراضي.

4. Emergency management algorithms

- General initial steps for all palliative emergencies:
1. Assess goals of care, check advance decisions/DNACPR/treatment escalation plan.
2. ABC approach; stabilise reversible life-threats only if aligned with goals.
3. Treat symptoms for comfort while investigating reversible causes if appropriate.

Algorithm A — Dyspnoea in palliative patient
1. ABCs; treat severe hypoxaemia first (supplemental low-flow O2 for symptomatic hypoxaemia, usually SpO2 ≤92%). (NICE CKS)
2. If oxygenation adequate but distress persists → consider low-dose opioid for refractory breathlessness (first-line symptomatic therapy). Titrate to comfort; monitor sedation and secretions. (BNF/NICE)
3. If anxiety predominates (tachypnoea, fear, preserved SpO2) → add benzodiazepine (short acting).
4. Assess for reversible causes (treat infection, drain large pleural effusion if appropriate and desired, consider steroids for lymphangitic spread/airway inflammation).
5. If bronchospasm → bronchodilator therapy; if pulmonary oedema with goals permitting → diuretics.

Algorithm B — Noisy breathing (terminal secretions)
1. Confirm likely terminal secretion (unresponsive, normal oxygenation/clear chest).
2. Reposition patient; mouth care.
3. Antimuscarinic therapy (e.g., glycopyrronium) SC to reduce secretions; avoid routine suctioning in terminal phase.
4. Explain to family that sound ≠ necessarily distress.

Algorithm C — Agitation/delirium at end of life
1. Search and treat reversible causes (pain, hypoxia, urinary retention, constipation, medication effects).
2. For severe distress: use short-acting benzodiazepine (midazolam titrated SC/IV) for sedation; consider antipsychotic if delirium (haloperidol or other per local guidance) if consistent with goals.
3. Consider palliative sedation only with multidisciplinary and preferably prior consent/documentation.

Algorithm D — Nausea and vomiting
1. Identify mechanism: vestibular/ICP → cyclizine; bowel obstruction → anticholinergic (hyoscine butylbromide) and consider antisecretory measures; chemo/post-op → 5-HT3 antagonists or dopamine antagonists as indicated.
2. Avoid prokinetics in high-grade mechanical obstruction.

Algorithm E — Pain (including neuropathic)
1. Assess type: nociceptive → opioids titrated; neuropathic features → add gabapentinoid/antidepressant as adjuvant.
2. Consider route: oral if possible, else subcutaneous or syringe driver for continuous control.

> شرح موجز بالعربية:
> - نقاط عامة: تقيّم الأهداف أولًا. ضيق التنفّس: أكسجين إن نقص تشبّع، وإلا مرفين لخفض الشعور؛ إفرازات نهاية الحياة: غليكوبيررونيوم وموضعى/تموضع جيد؛ تيّبس/هيجان: استبعد الأسباب القابلة للعلاج ثم هدئ بالميدازولام؛ غثيان: ضع العلاج بحسب الآلية (سيكلزين للأسباب الدهليزيّة/ضغط الدماغ).

5. Drugs, procedures, disposition, and follow-up

- Key drugs and typical palliative starting ranges (verify local formularies):
- Oxygen: for symptomatic hypoxaemia (SpO2 threshold commonly ≤92%) [NICE CKS].
- Glycopyrronium (terminal secretions): commonly 200–400 mcg SC bolus (or syringe driver infusion per local protocol). (BNF)
- Hyoscine (butylbromide or hydrobromide): alternative antimuscarinic for secretions/antisecretory effect.
- Midazolam (acute severe agitation): typical palliative SC bolus 2.5–5 mg titrated to effect. (BNF)

- Lorazepam (anxiety-related dyspnoea): e.g., small oral doses (local variation).
- Opioids for refractory breathlessness: titrate low doses (local protocols/BNF).
- Cyclizine for vestibular/raised-ICP nausea; avoid prokinetic in mechanical obstruction.
- Gabapentin/pregabalin or amitriptyline/duloxetine for neuropathic pain adjuncts.
- Procedures/actions:
- Pleural drainage, paracentesis or bowel decompression only if aligned with goals and likely to relieve symptoms.
- Avoid invasive life-prolonging procedures if inconsistent with advance decisions.
- Consider SC syringe driver for continuous symptom control when repeated dosing needed and per local palliative team guidance.
- Disposition:
- If symptom control achieved and goals of care documented → consider discharge to hospice/inpatient palliative care or home with community support.
- If reversible cause requiring admission or intervention and aligned with goals → admit.
- If imminent dying and family support available → discuss community/hospice/ward management as appropriate.
- Follow-up:
- Ensure clear handover to palliative care team/community nursing; document medications given, response, and escalation preferences.

شرح موجز بالعربية:
- أدوية مفتاحية: غليكوبيررونيوم 200–400 ميكروغرام SC، ميدازولام 2.5–5 ملغ SC كبداية، استخدم الأفيونات لِتخفيف وطأة ضيق التنفُّس عند توفر الأوكسجين. الإجراءات الغازية فقط إن تتفق مع أهداف المريض. رتب متابعة مع فريق الرعاية الملطفة.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية
- Pitfalls:
- Don’t give oxygen reflexively for breathlessness if SpO2 adequate; oxygen often does not relieve subjective dyspnoea. (NICE CKS)
- Do not routinely suction terminal secretions — antimuscarinics are preferred.
- Avoid prokinetic antiemetics in high-grade mechanical bowel obstruction.
- Confirm validity of advance decisions/DNACPR before withholding

interventions; legal frameworks differ by jurisdiction (England specifics per NHS England 2017).
- Benzodiazepines may worsen respiratory drive — use cautiously; opioids for breathlessness are evidence-based when titrated carefully.
- Exam tricks:
- If SpO2 low → oxygen is first-line; if SpO2 normal but breathlessness persists → opioid is typical correct choice.
- “Death rattle” scenario + good oxygenation + clear chest → antimuscarinic (glycopyrronium/hyoscine) is frequently the tested answer.
- In nausea with raised ICP/vestibular signs → cyclizine preferred in many exam stems.
- For agitation due to reversible causes — always choose treat reversible cause first (pain, hypoxia, urinary retention) before sedation answers.
- When question mentions advance decision refusing life-sustaining treatment — choose comfort measures consistent with that decision.

شرح موجز بالعربية:
- أخطاء شائعة: لا تعطِ أكسجين لو تشبّع كافٍ؛ لا تمسح الإفرازات الروتينية؛ تحقق من صلاحية القرارات المسبقة. نصيحة امتحانيّة: «مرفين عند ضيق تنفّس متبقٍ ولوكسجين كافٍ» غالبًا صحيحة.

7. Arabic rapid-review summary (ملخص عربي سريع)

- أهداف: راحة، التحكم بالاعراض، احترام رغبات المريض. وثّق خطة العلاج.
- ضيق التنفّس: أكسجين إن SpO2 ≤92%؛ إن تشبع كافٍ → مرفين منخفض الجرعة؛ إذا قلق → بنزوديازيبين مضاد للقلق.
- أنفاس صاخبة: غالبًا إفرازات نهائية → غليكوبيررونيوم أو هيوسين؛ لا شفط روتيني.
- هيجان/هلع نهائي: اسْتبعد مسببات قابلة للعلاج ثم استخدم ميدازولام بحسب الحاجة.
- غثيان: سيكلزين إذا مشكلة دوُريّة/ضغط دماغي؛ تجنّب محفزات الحركة لو انسداد ميكانيكى.
- ألم عصبي: علاج تكميلي (جابابينتويد/مضاد اكتئاب) يضاف للأفيونات.
- فحوصات: اطلب فقط إن ستغيّر إدارة المريض.
- قانونيًا: تحقق من صحة القرارات المسبقة/DNACPR قبل وقف علاجات منقذة للحياة.

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية

- Dyspnoea: recognition, causes, SpO2 threshold, oxygen vs opioid management, anxiety component.
- Noisy breathing / “death rattle”: recognition, antimuscarinics vs suction.
- Agitation/delirium at end of life: reversible causes, midazolam, antipsychotics.
- Nausea/vomiting in palliative patients: ICP/vestibular vs bowel obstruction vs chemo/post-op; drug choices (cyclizine, hyoscine, ondansetron).
- Pain management: opioid titration, neuropathic adjuvants.
- Procedural decisions in palliative context: when to drain effusions, paracentesis, bowel decompression; aligning with goals.
- Legal/ethical: advance decisions to refuse treatment, DNACPR, documentation and verification.
- Diagnostics: focused bedside assessment, selective use of CXR/US/ABG/bloods.
- Common exam vignettes/tricks: mapping SpO2→oxygen vs opioid, death rattle→antimuscarinic, nausea mechanism→antiemetic class.

This chapter is study material, not patient-specific advice.

Sources to verify: RCEM, ERC, NICE CKS (palliative care: dyspnoea), NICE guidance, BNF (palliative prescribing), NHS England — Advance Decisions to Refuse Treatment (2017), local palliative care protocols and legal guidance.

CHAPTER 15: OPHTHALMOLOGY

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Chapter 15 — Ophthalmology (compact FRCEM final chapter)

1. Clinical overview and pathophysiology

- Triage principle: identify vision- or globe-threatening features early — new ↓vision, severe/boring pain, photophobia, corneal opacity/ulcer, hypopyon, fixed/mid-dilated pupil, very high IOP, proptosis/tight orbit, RAPD. Act immediately if present. [Eyewiki; NICE CKS]
- Major pathophysiologic clusters:
- Surface infections/ulceration → corneal epithelium breach → risk of stromal invasion, hypopyon, endophthalmitis.
- Angle closure → sudden blockage of trabecular outflow → rapid IOP rise → optic nerve/retinal ischemia.
- Orbital compartment syndrome (OCS)/retrobulbar haemorrhage → raised orbital pressure → globe ischemia and optic nerve compression.
- Penetrating injury/open globe → loss of globe integrity → infection risk, expulsive hemorrhage.
- Vascular occlusions (CRAO/CRVO) → retinal ischemia or venous congestion → acute painless vision loss.
- Inflammatory/autoimmune (scleritis, GCA) → deep pain, scleral necrosis or anterior ischemic optic neuropathy.
- Time-critical events: OCS (immediate canthotomy/cantholysis), AACG (rapid IOP lowering + ophth), endophthalmitis (intravitreal antibiotics), open globe (shield + urgent ophth). [Eyewiki; NCBI]

> موجز علمي: راجع العلامات التي تهدد الإبصار فورًا. المجموعات المرضية الرئيسية: قرنية (عدوى/قرحة)، زاوية العين (ارتفاع الضغط الحاد)، الحيز المحجري المضغوط، ثغرة غلوبية، انسدادات للأوعية، والتهاب مناعى (مثل GCA).

2. Recognition, differential diagnosis, and red flags

- Key discriminators:
- Dendritic fluorescein staining → HSV keratitis.

- Contact-lens user + severe pain / ring infiltrate or rapid worsening → Pseudomonas or Acanthamoeba keratitis.
- Corneal ulcer ± hypopyon → bacterial keratitis (urgent).
- Red painful eye + mid-dilated non-reactive pupil + cloudy cornea + N/V → acute angle-closure glaucoma (AACG).
- Sudden flashing lights/floaters then curtain → retinal detachment.
- Sudden painless profound loss with pale retina/cherry-red spot → CRAO (arterial occlusion).
- Proptosis, tense lid, restricted EOM, painful swelling, raised IOP → OCS / orbital cellulitis / retrobulbar haemorrhage.
- Periorbital swelling without EOM/vision involvement → preseptal cellulitis; with systemic features or EOM restriction → postseptal (orbital) cellulitis.
- Localised sectoral redness that blanches with phenylephrine → episcleritis; non-blanching, severe pain → scleritis.
- Red flags (act now): new ↓vision, severe or worsening pain, hypopyon, opaque cornea, fixed mid-dilated pupil, RAPD, proptosis/tension, penetrating injury signs, chemical burn.

> ملاحظات التعرف: افصل بين أسباب العين الحمراء بواسطة نقاط فحص أساسية: مظهر القرنية-الحجرة الأمامية-حدقة العين-حركة خارجية-بروتروز. أي علم يدل على تهديد الإبصار يحتاج إحالة فورية.

3. Diagnostic approach and interpretation

- Immediate bedside bundle: visual acuity (near & distance), pupils (APD/RAPD), extraocular movements, slit-lamp / torch exam, fluorescein staining, seidel test if trauma suspected, IOP if globe intact (tonometry), external exam for proptosis.
- Imaging/lab triggers:
- CT orbits non-contrast for trauma with suspected fracture, intraorbital foreign body, retrobulbar haemorrhage, orbital cellulitis surgical planning.
- B-scan ultrasound when fundus view obscured (dense hyphema, corneal opacity, vitreous haemorrhage) to assess retinal detachment.
- ESR/CRP and fast GCA pathway if temporal arteritis suspected with visual symptoms.
- Microbiology: corneal scrapings for microbial keratitis; vitreous/aqueous cultures in endophthalmitis — do not delay empiric therapy for sampling if vision threatened.

- Interpretation tips:
- Phenylephrine blanching differentiates episcleritis (blanches) from scleritis (does not).
- Dendritic pattern under fluorescein = HSV; pseudodendrites (VZV) often raised and less staining.
- Positive Seidel = globe leak/open globe → shield and urgent ophth; do not measure IOP.

> النهج التشخيصي: فحوص أساسية عند المنضدة، تصوير أو قياسات محددة عند اشتباه بالتهاب محجرى أو عدم وضوح القاع. لا تؤخر العلاج في حالات تهدد الإبصار لأجل زرع/عينات فقط.

4. Emergency management algorithms

A. Triage algorithm (ED short pathway)
- If any red-flag (see §2) → urgent ophthalmology contact + consider admission/ transfer.
- If no red-flag and simple conjunctivitis/allergic → outpatient tx and arranged follow-up.

B. Orbital compartment syndrome (OCS) algorithm
- Recognise clinically (rapid proptosis, tense lids/orbit, severe pain, RAPD, decreased vision, high IOP). Do NOT wait for imaging.
- Immediate: lateral canthotomy ± inferior cantholysis at bedside.
- Key steps: instil local anaesthetic (commonly 1 mL 1% lidocaine), crush lateral canthus ~1–2 min, cut lateral canthal tendon, divide inferior crus first; if no improvement divide superior crus; then urgent ophth. [Eyewiki; exam stepwise from Q900]
- After decompression: CT + definitive surgical/ophth care.

C. Acute angle-closure glaucoma (AACG) algorithm
- Immediate: analgesia, antiemetic, lower IOP emergently and contact ophthalmology for definitive therapy (laser iridotomy).
- ED IOP-lowering (common practice): IV acetazolamide 500 mg; topical beta-blocker (timolol); topical alpha-agonist; topical pilocarpine 2% may be given after some IOP reduction (timing varies in practice); consider IV mannitol if refractory. Admit/urgent ophth. [NICE CKS]
- Avoid treatments that increase inflammation or pressure inadvertently; verify sequence with local ophth.

D. Corneal ulcer/keratitis pathway
- Contact-lens wearers or central/vision-threatening ulcers: start intensive topical broad-spectrum antibiotic drops immediately (e.g., topical fluoroquinolone per local protocol hourly/near-hourly) and urgent ophthalmology. Do NOT patch. Avoid topical steroids until cleared by ophthalmology. Consider corneal scraping for culture if possible.

E. Endophthalmitis pathway
- Suspect after intraocular surgery/penetrating trauma with severe pain, ↓vision, hypopyon. Immediate ophthalmology; intravitreal antibiotics and vitrectomy decision per specialist. Obtain cultures if feasible.

F. Chemical eye injury
- Immediate copious irrigation (until pH neutral); measure surface pH after initial 5 min irrigation and repeat. Analgesia; topical anaesthetic short-term ok to facilitate irrigation. Ophth consult urgently.

G. Hyphaema (traumatic anterior chamber blood)
- Shield eye, head elevation, bed rest, cycloplegic (e.g., cyclopentolate), topical steroid per ophthalmology, avoid NSAIDs/anticoagulants. Monitor IOP; urgent ophth if IOP >30 mmHg, total hyphaema, corneal blood staining, or rebleed. [Eyewiki]

H. Preseptal vs orbital cellulitis
- Preseptal (mild, no EOM restriction): oral antibiotics outpatient if reliable follow-up.
- Orbital (EOM restriction/proptosis/vision loss/fever): IV broad-spectrum antibiotics, urgent imaging, admit, ophthalmology/ENT consult.

> خلاصة الخوارزميات: أي حالة مهددة رؤية → خوارزمية عاجلة مع اتصال بطبيب العيون، OCS يحتاج قِنطرة جانبية فورية، الزاوية الضيقة → خفض ضغط سريع ثم ليزر، قرحة قرنية لمرتدي العدسات → مضاد حيوي موضعي مكثف فوراً، إصابة كيميائية → ريّ فوري حتى التعادل.

5. Drugs, procedures, disposition, and follow-up

Drugs (ED-focused, commonly used)
- Acetazolamide IV 500 mg (single dose commonly used in AACG) — verify local protocol. [NICE CKS]
- Pilocarpine 2% (topical) for AACG after partial IOP reduction; some use 4% for

dark irides — practice varies.
- Topical broad-spectrum fluoroquinolone drops for suspected bacterial keratitis (hourly or per protocol); avoid topical steroids until ophth review.
- Topical ganciclovir ointment for HSV dendritic keratitis (e.g., 5×/day for 7–10 days in common regimens).
- Analgesia/antiemetics as required; topical cycloplegic (cyclopentolate) for hyphaema pain control.
- IV methylprednisolone 500–1000 mg daily is used in many centres for GCA with visual symptoms — verify local guidance. (Flagged for verification.)

Procedures (ED)
- Lateral canthotomy + inferior cantholysis for OCS (perform without imaging if clinical).
- Seidel test (fluorescein) for globe leak.
- Removal of superficial corneal foreign body by skilled operator; rust ring removal arranged within 24–48 h by ophth.
- B-scan ultrasound of globe when fundal view obscured.
- Do NOT patch suspected microbial keratitis, do NOT measure IOP if open globe suspected.

Disposition / follow-up
- Urgent ophthalmology or transfer: any vision-threatening sign, open globe, OCS, AACG, endophthalmitis, sight-threatening keratitis, orbital cellulitis, large/unclearing hyphaema, suspected retinal detachment/CRAO.
- Admit: orbital cellulitis, vision-threatening infections, uncontrolled IOP/hyphaema, post-op or penetrating trauma high-risk cases.
- Outpatient with timely follow-up: mild preseptal cellulitis, simple conjunctivitis (if no red flags), chalazion after conservative measures.
- Safety net: advise return for any worsening pain, ↓vision, increasing redness, or inability to open eye.

> توصيات العلاج والمتابعة بالعربية: جرعات ومبادئ عامة مدرجة أعلاه؛ إجراءات فورية تشمل قِنطرة جانبية للـ OCS، ريّ للبشرة الكيميائية، مضاد حيوي موضعى مكثف لعدوى القرنية لدى مرتدي العدسات. إحالة فورية لأمراض العيون عند أي علامة مهددة للإبصار.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية

- Phenylephrine blanching test: episcleritis blanches; scleritis does not — classic exam trick.
- Dendritic fluorescein staining = HSV keratitis; pseudodendrites (VZV) are raised and stain less.
- Always ask about contact-lens use — changes urgency and empiric therapy.
- Severe pain out of proportion to external findings → Acanthamoeba suspicion (contact-lens history) or deep corneal involvement.
- Don’t delay lateral canthotomy for CT if OCS suspected — many questions cue tense orbit + high IOP → perform canthotomy immediately. (See Q900 technique.)
- In suspected open globe (misshapen pupil, flat AC, Seidel +) — shield, nil by mouth, tetanus, urgent transfer; DO NOT measure IOP or apply pressure.
- Avoid topical steroids in suspected microbial keratitis until reviewed by ophthalmology — common pitfall that worsens infection.
- CRAO acute maneuvers (ocular massage, paracentesis) are of limited evidence — mark as low-evidence in exam answers; focus on urgent ophth and stroke pathway.
- Pilocarpine timing: may fail if given too late; many sources advise after partial IOP reduction — flag as exam caveat.

نصائح الامتحان بالعربية: اختبار فينيل يفرق الإبيسكليرا عن السكليرا؛ اسأل دائمًا عن العدسات؛ لا تؤخر قِنطرة جانبية عند المظاهر المميزة لـ OCS؛ تجنب الستيرويدات الموضعية في الشك بعدوى قرنية.

7. Arabic rapid-review summary

- استرشد بعلامات التهديد: فقدان بصر مفاجئ، ألم شديد، هيبوباين/هيفيمة، قرحة قرنية، حدقة ثابتة/متوسّطة، ضغط عين مرتفع، بروتروز، RAPD.
- OCS → قِنطرة جانبية + قطع وتر جفني فوري (لا تنتظر تصوير).
- AACG → خفض ضغط العين فوراً (أسيتازولاميد IV 500 mg شائع) ثم إحالة عاجلة للـ iridotomy.
- عدوى قرنية لدى مستخدمي العدسات → مضاد حيوي موضعي مكثف فوري؛ لا تضع رقعة، لا تعطي ستيرويد موضعي قبل مراجعة أخصائي.

- إصابة كيميائية → ري فوري حتى تعادل الأس الهيدروجيني.
- GCA بصرى → بدء ستيرويد نظامى فورى قبل الانتظار لتأكيد الاختبارات.
- مفاتيح عملية: فحص البصر والحدقة، فلورسئين، Seidel، قياس ضغط العين إن كان آمناً، تصوير أو م-الموجة عند الحاجة.

موجز بالعربية: ركّز على استبعاد الحالات المهددة للإبصار فورا، ابدأ التدخلات الأساسية المنقذة للحياة البصرية ثم إحالة متخصصة.

8. Question coverage map (sources / concepts covered)

- Q900 — lateral canthotomy / inferior cantholysis technique, indications (orbital compartment syndrome) [exam step reference].
- Corneal pathology (keratitis/ulcer/HSV/Acanthamoeba/Pseudomonas) — frequent FRCEM red-eye questions.
- Acute angle-closure glaucoma (recognition, ED IOP lowering, pilocarpine caveat).
- Orbital vs preseptal cellulitis; imaging and antibiotic escalation.
- Open globe / penetrating trauma (Seidel +, shielding, transfer).
- Chemical eye injury (irrigation, pH checks).
- Endophthalmitis (post-op/penetrating trauma, intravitreal antibiotics).
- Hyphaema and traumatic anterior chamber blood (monitoring, IOP thresholds).
- Retinal vascular events (CRAO/CRVO) and retinal detachment (photopsia/floaters/curtain) — acute recognition and referral.
- Scleritis vs episcleritis (phenylephrine test; systemic associations).
- Giant cell arteritis with ocular ischemia (urgent steroids and pathways).
(Use your local question bank index to map above topics to specific Q-ranges in your revision set.)

مصدر الربط للأسئلة: Q900 مذكور صراحة؛ بقية المواضيع مغطاة عبر بنوك أسئلة FRCEM القياسية حول العين الحمراء، الصدمة، والعدوى.

Sources to verify
- Eyewiki (topic pages cited) [Eyewiki]
- NICE CKS — Acute glaucoma guidance [NICE CKS]
- NCBI / Endophthalmitis reviews [NCBI]
- Local ophthalmology protocols and formulary for exact topical antibiotic choice/dosing and pilocarpine sequencing
- RCEM, ERC, NICE, AAO and local hospital ophthalmology guidelines for definitive recommendations and any dose/regimen variations

Final note: This chapter is high-yield ED guidance for exam preparation. Verify local formularies and national guidelines for exact drug preparations, concentrations, and institutional procedures before clinical use.

CHAPTER 16: NEUROLOGY

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Chapter 16 — Neurology (Compact FRCEM final chapter)

1. Clinical overview and pathophysiology

- Framework: categorise emergencies by mechanism — vascular (ischaemia, SAH, CVST, dissection), haemorrhagic (SDH, EDH, SAH), infective/inflammatory (meningitis, encephalitis, epidural abscess), compressive (cord compression, cauda equina), seizure disorders/status epilepticus, neuromuscular (GBS, myasthenia gravis), neuro-ophthalmic (optic neuritis, CRAO/AION, HZO), primary headache syndromes (cluster, migraine, IIH), functional (NEAD).
- Localisation rules (high-yield): cervical myotomes — C5 shoulder abduction; C6 elbow flexion & wrist extension; C7 elbow extension & wrist flexion; C8 finger flexors/thumb extension; T1 intrinsic hand. Dermatomes — C7 = middle finger; T10 = umbilicus. Lumbar LP landmark ≈ iliac crest ~ L4. Brown-Séquard: ipsilateral motor/proprioception loss, contralateral pain/temp loss.
- Pathophysiology short points:
- SAH — aneurysmal rupture → blood in subarachnoid space; thunderclap headache; risk of vasospasm days 3–14.
- Subdural haematoma — torn bridging veins, gradual evolution (hyperdense → isodense → hypodense). Elderly/atrophy/anticoagulated predispose.
- CVST — venous outflow obstruction → raised ICP, venous infarct/haemorrhage; risk factors postpartum, OCP, thrombophilia, COVID.
- GBS — postinfectious immune neuropathy; ascending weakness, areflexia; respiratory decline risk.
- Spinal epidural abscess — local infection → cord compression; back pain + fever + neuro deficit classic triad (may present incompletely).

> Arabic (مختصر): رتب الطوارئ العصبية حسب الآلية (وعائي، نزفي، التهابي، ضاغط، اعتلال عصبي). استخدم قواعد توطين العضلات والحسّ للتمييز الجذري/نخاعي. SAH = صداع رعدي؛ خراج فوق الجافية يسبب ألم ظهري وحمى وضعف عصبي.

2. Recognition, differential diagnosis, and red flags

- Red flags (act urgently):
- “Worst ever” thunderclap headache ± meningism → SAH (CT ± LP).
- Fever + neck stiffness ± photophobia ± reduced GCS → meningitis (start antibiotics immediately).
- New focal neuro deficit, progressive bilateral leg weakness, saddle anaesthesia, urinary retention → cauda equina (MRI lumbar urgently).
- Back pain + fever ± focal deficit → spinal epidural abscess (MRI spine).
- Seizure >5 min or repeated without recovery → status epilepticus (immediate BZD → second line).
- Rapidly progressive ascending weakness or FVC fall → Guillain-Barré (monitor FVC, ICU threshold).
- New visual disturbance ± jaw claudication ± scalp tenderness in elderly → Giant Cell Arteritis (start steroids urgently).
- New unilateral painful visual loss with painful eye movements and colour desaturation in a young patient → optic neuritis (MS association).
- Facial palsy with ear vesicles → Ramsay Hunt (antiviral + steroid). Forehead involvement: peripheral; forehead sparing: central.
- Posterior neck/head pain after trauma/manipulation with brainstem/cerebellar signs → vertebral artery dissection.

- Key differentials by presentation:
- Thunderclap headache: SAH vs reversible cerebral vasoconstriction vs migraine/other — use CT ± LP/CTA.
- Acute painless monocular vision loss → CRAO/AION; painful with movement → optic neuritis.
- Acute bilateral headache + papilloedema + normal neuroimaging → IIH (LP opening pressure high).
- New focal seizures or focal status → structural lesion/abscess/CVA; perform imaging.

> Arabic (مختصر): اشارات حمراء: صداع رعدي، حمى مع تيبس رقبة، ضعف تصاعدي مع انخفاض FVC، ألم ظهري مع حمى. شلل وجهي كامل = محيطي؛ حفظ الجبين = مركزي.

3. Diagnostic approach and interpretation

- General ED priorities: ABCs, oxygen, IV access, blood glucose, blood cultures if infection suspected, baseline bloods (FBC, U&Es, coagulation), ECG/oximetry. Consider drug/toxicology and anticonvulsant levels when relevant.
- Imaging & procedure algorithms (concise):
- Suspected bacterial meningitis:
1) Blood cultures → give IV ceftriaxone 2 g immediately. Add IV amoxicillin if Listeria risk (age >50, immunocompromised) [NICE NG240 2024].
2) If focal neurology, papilloedema, reduced GCS, or seizure → CT head before LP. If LP delayed, do not delay antibiotics/dexamethasone where indicated.
3) Interpret CSF: neutrophils ↓glucose ↑protein = bacterial; lymphocytes normal glucose = viral; xanthochromia = SAH if present after 12+ hours.
- Suspected SAH (thunderclap):
1) Non-contrast CT head immediately. If high-quality CT within 6 h and negative (interpreted appropriately) sensitivity is high [NICE NG228]; if CT negative but clinical suspicion remains → LP for xanthochromia (best ≥12 h after onset).
2) If CT positive → CTA/angiography to locate aneurysm; neurosurgical/ neuroradiology referral.
- Suspected stroke/dissection/posterior circulation:
- Non-contrast CT head ± CTA head/neck for dissection; MRI if available for posterior fossa.
- Suspected CVST: CT venogram (CTV) or MR venogram (MRV). Start anticoagulation promptly once diagnosis suspected/confirmed.
- Suspected spinal epidural abscess / cauda equina / cord compression: urgent MRI spine with contrast; do not wait for plain films.
- Status epilepticus: glucose ± point-of-care electrolytes immediately; urgent CT head if focal onset or prolonged/protracted events.
- Specific test interpretation tips:
- CT timing matters for SAH sensitivity — note time from headache onset.
- CSF opening pressure: >25 cm H2O supports IIH (confirm units locally).
- Distinguish central vs peripheral facial palsy by forehead involvement.

> Arabic (مختصر): فحص سريع: ثقافات دم، Ceftriaxone 2 غ للالتهاب السحائي، CT قبل LP
> إذا علامات ارتفاع ضغط أو بؤرية. صداع رعدي → CT فورًا؛ CT سلبي لكن اشتباه مستمر → بزل

> قطني بعد 12 ساعة للزانثوكروميا. MRI للعمود عند اشتباه خراج أو ذيل الفرس؛ CTV/MRV لـCVST.

4. Emergency management algorithms

- Meningitis (adult suspected bacterial):
- Algorithm: ABC → blood cultures → IV ceftriaxone 2 g immediately; add IV amoxicillin if Listeria risk; give dexamethasone where pneumococcal meningitis suspected and per local guidance (before/with first antibiotic) [NICE NG240 2024]. Do CT head before LP if any contraindication.
- SAH (acute thunderclap with CT positive):
- Immediate: admit/observation in high-dependency, nimodipine 60 mg orally q4h as soon as SAH diagnosed (start pre-transfer), control systolic BP to local target (common target <160 mmHg but follow local protocol), arrange urgent neurosurgical/neuroradiology definitive management (coiling/clipping), give seizure prophylaxis if indicated.
- Status epilepticus (adult):
- Step 1 (0–5 min): ABC, oxygen, IV access, check glucose; give benzodiazepine IV (lorazepam 4 mg IV bolus; repeat once if needed) — if no IV available use IN/IM/buccal midazolam (IN 0.2 mg/kg, max 10 mg) or rectal diazepam.
- Step 2 (5–20 min persistent): Give second-line IV antiseizure drug per local protocol — levetiracetam load commonly used (e.g., 60 mg/kg reported in sources; many centres use 1–2 g IV) or fosphenytoin/phenytoin/valproate depending on contraindications. (Verify local loading dose caps.)
- Step 3 (>30 min refractory): Prepare airway, request ICU, induce anaesthesia (propofol/thiopental/ketamine per protocol), continuous EEG if available.
- Spinal epidural abscess / cord compression / cauda equina:
- Urgent MRI spine → start empiric broad-spectrum IV antibiotics covering S. aureus (include MRSA per risk) → urgent neurosurgical review for decompression if cord compromise or progressive deficits; keep NPO and monitor neuro/urinary function.
- CVST:
- Start therapeutic anticoagulation (LMWH) promptly once suspected/confirmed even with haemorrhagic transformation (evidence supports treating thrombosis); arrange MRV/CTV and neurology/hematology input. Evolving recommendations on DOACs — verify local guidance.
- Guillain-Barré syndrome:

- Monitor serial FVC and bedside strength; ICU/ventilation if rapid decline or FVC thresholds approached (common trigger values 15–20 mL/kg); start IVIG or plasmapheresis early once diagnosis made.
- GCA with visual symptoms:
- Start high-dose corticosteroids urgently to reduce risk of permanent visual loss — route/dose regimens vary; confirm local protocol (many centres use high-dose IV/PO steroids immediately). Obtain ESR/CRP and urgent specialist review.
- Wernicke’s encephalopathy:
- Give high-dose parenteral thiamine immediately BEFORE glucose; follow local Pabrinex/BNF protocol.
- IIH:
- Neuroimaging (MRI/MRV) to exclude mass/venous sinus thrombosis; if normal and criteria met perform LP for opening pressure; treat with acetazolamide/ weight loss and urgent ophthalmology if vision threatened.

> Arabic (مختصر): خوارزميات سريعة — التهاب سحائي: ثقافات ثم Ceftriaxone 2 غ فورًا ± Amoxicillin عند خطر Listeria. SAH: نيموديبين 60 ملغ كل 4 ساعات و نقل جراحي عاجل. نوبة مستمرة: بنزو أولاً (لورازيبام 4 ملغ) → حمولة مضاد صرع IV → تخدير/ICU إذا مقاومة. خراج فوق الجافية: MRI عاجل، مضادات حيوية جدارية، جراحة.

5. Drugs, procedures, disposition, and follow-up

- Key drugs & well-established doses:
- Ceftriaxone adult meningitis empiric: 2 g IV immediately after cultures [NICE NG240 2024].
- Nimodipine for aneurysmal SAH: 60 mg orally q4h (start ASAP) [BNF/national practice].
- Benzodiazepines in status: lorazepam 4 mg IV (may repeat once); if no IV available use intranasal midazolam 0.2 mg/kg (max 10 mg).
- Second-line anti-seizure: levetiracetam commonly used as IV load (many protocols 1–2 g; some sources state 60 mg/kg — verify local caps). Phenytoin/ fosphenytoin or valproate are alternatives per local guidance. (Verify exact institutional doses.)
- IV acyclovir for encephalitis/HZO in immunocompromised or sight-threatening disease: 10 mg/kg IV q8h (verify local formulary).
- Prednisolone for Bell’s palsy: typical exam regimens ~50 mg PO daily for 7–10 days if within 72 h (local regimens vary).

- GBS: IVIG or plasmapheresis per neurology guidance (dose regimens per protocol).
- Procedures and key practical points:
- Lumbar puncture: check indications/contraindications (TAP AS IF mnemonic: Trauma, Age >60, Papilloedema, Altered mental status, Seizure, Immunocompromise, Focal deficit); use lateral decubitus for opening pressure measurement; use pencil-point needles to reduce PDPH when possible.
- Imaging: CT/CTA/CTV/MRI/MRV — choose modality targeted to pathology (CT first for bleed; MRI spine for cord/SEA; CTV/MRV for CVST).
- Airway management: anticipate early in refractory SE, GBS with falling FVC, myasthenic crisis. Document baseline FVC and bulbar function.
- Anticoagulation reversal: for intracranial haemorrhage on anticoagulants, follow local haematology/anticoagulation reversal protocols; reverse anticoagulation urgently for expanding SDH with neurosurgical input.
- Disposition & follow-up (general):
- Immediate neurosurgical/neuroradiology transfer: SAH with bleed or mass effect SDH/EDH, posterior fossa stroke with mass effect, spinal cord compression with progressive deficit.
- Admit HDU/ICU: status epilepticus not terminated, airway/ventilatory compromise, severe GBS, unstable SAH, septic/unstable meningitis.
- Urgent specialty review: ophthalmology for optic neuritis/HZO with ocular involvement, neurology for suspected GBS/CVST/encephalitis, ENT/ophthalmology for Ramsay Hunt/HZO, rheumatology/ophthalmology for GCA.
- Outpatient follow-up: IIH weight management and neurology/ophthalmology follow-up; Bell's palsy follow-up for incomplete recovery; post-SAH vascular clinic and neuropsychological support.

> Arabic (مختصر): أدوية ومداخلات مهمة: Ceftriaxone 2 غ للالتهاب السحائي، Nimodipine 60 ملغ/4س للـSAH، لورازيبام 4 ملغ IV للنوبة المستمرة. إجراءات: بزل قطني مع الانتباه لموانع (TAP AS IF)، MRI للحبل عند اشتباه خُراج أو ذيل الفرس. إحالة عاجلة لجراحة/ICU حسب الاستقرار والنتائج.

6. High-yield FRCEM exam tricks and common pitfalls

> **EXAM TRICK / تريكة امتحانية**
> - Localisation mnemonics and exam tricks:

- Key myotome checks: C6 = elbow flexion/wrist extension; C7 = elbow extension/wrist flexion; test middle finger sensation for C7. Dermatomes: T10 = umbilicus. Nipple = T4. L4 = iliac crest landmark for LP. [LITFL referenced]
- Facial palsy: forehead involvement → peripheral (Bell's/Ramsay Hunt); forehead sparing → central (UMN) — classic exam discriminator.
- Diplopia: if cover of either eye abolishes diplopia → binocular (neuro). If diplopia persists with cover of one eye → monocular (ocular).
- Seizure vs NEAD: resistance to eye opening during event, asynchronous limb movements, long duration without post-ictal state → suggests NEAD.
- Common exam pitfalls:
- Don't assume a negative CT rules out SAH if scan is delayed or image quality/documentation suboptimal; recall CT sensitivity falls with time — LP/CTA may still be required [NICE NG228 caveats].
- In suspected bacterial meningitis do not delay empiric antibiotics for LP/CT if obtaining imaging would delay treatment. Use TAP AS IF to decide CT before LP.
- In CVST do not withhold anticoagulation solely because of haemorrhagic parenchymal lesions — treat thrombosis after specialist input; check up-to-date guidance on DOACs.
- For status epilepticus, many exam stems test repeating benzodiazepine once before escalating — do not skip this. Prepare airway early.
- Be cautious with dose statements that vary by institution: levetiracetam 60 mg/kg loading and IV methylprednisolone 1 g/day for GCA are used in some centres — verify local guidelines before stating as universal.
- Cauda equina red flags: saddle anaesthesia, urinary retention, severe/progressive bilateral weakness; age 16–50 alone is not a red flag (NICE/CKS).
- Do not give aminoglycosides to myasthenic patients; watch drug lists that exacerbate neuromuscular weakness.
- Exam memory aids:
- TAP AS IF = reasons to do CT before LP (Trauma, Age>60, Papilloedema, Altered mental status, Seizure, Immunocompromise, Focal deficit).
- "Worst headache → CT; CT negative and high suspicion → LP for xanthochromia (≥12 h optimal)."

Arabic (مختصر): حيل الامتحان: فَرِّق بين شلل وجه محيطي/مركزي بواسطة الجبين؛ اختبار تغطية العين يميّز ازدواج أحادي/ثنائي؛ TAP AS IF لمعايير CT قبل LP. أخطاء شائعة: افتراض استبعاد SAH بواسطة CT متأخر؛ تأخير المضادّات في التهاب السحايا عند الحاجة.

7. Arabic rapid-review summary (مراجعة سريعة بالعربية الفصحى)

- قِسِّم الطوارئ العصبية حسب الآلية: وعائي، نزفي، التهابي، ضغطى، عصبي طرفي، أداء وظيفي.

- إشارات حمراء: صداع رعدي، حمى مع تيبس رقبى، ضعف تصاعدي مع تناقص FVC، ألم ظهري مع حمى أو ضعف عصبي، فقدان بؤري مفاجئ للرؤية، شلل وجهي جديد.
- إجراءات عاجلة: Ceftriaxone 2 غ IV فورًا عند اشتباه التهاب سحائي؛ CT فورًا عند صداع رعدي؛ لورازيبام 4 ملغ IV لنوبة مستمرة (كرر مرة) ثم حمولة مضاد صرع IV؛ MRI عند اشتباه خراج فوق الجافية أو ذيل الفرس؛ ابدأ علاج تجريبي إذا تأخرت الفحوص.
- نقاط عملية: راقب FVC في GBS، لا تؤخر المضادّات الحيوية من أجل صور/LP إذا التأخير قد يضر، ابدأ الستيرويد عالي الجرعة فورًا عند اشتباه GCA بأعراض بصرية، أجرِ LP لقياس ضغط الفتح إذا IIH مشتبه.
- تحقق محليًا من جرعات ليفتيراسيتام التحميلية ونماذج جرعات الستيرويد وجوانب استخدام DOACs في CVST.

8. Question coverage map (concepts covered)

- Meningitis/encephalitis — empiric antibiotics, CT-before-LP decision making, dexamethasone [NICE NG240 2024].
- Subarachnoid haemorrhage — thunderclap headache, CT/LP/CTA pathway, nimodipine and neurosurgical transfer [NICE NG228, BNF].
- Intracranial haemorrhage patterns — SDH/EDH evolutions and anticoagulation reversal.
- Status epilepticus — benzodiazepine first line, second-line antiseizure options, refractory management (ICU).
- Seizure vs NEAD differentiation (eye-opening resistance etc.).
- Spinal emergencies — cauda equina, spinal epidural abscess, cord compression (MRI urgent, antibiotics, decompression).
- Guillain-Barré syndrome — ascending weakness, FVC monitoring, IVIG/plasmapheresis.
- Cerebral venous sinus thrombosis — presentation, CTV/MRV, anticoagulation (LMWH), DOACs evolution.

- Cervical artery dissection/vertebral dissection — posterior neck pain, posterior circulation stroke, CTA.
- Neuro-ophthalmology — optic neuritis, CRAO/AION, HZO/Ramsay Hunt, Hutchinson sign.
- Headache emergencies and primary headaches — cluster headache oxygen/ sumatriptan, medication-overuse headache, IIH diagnosis (LP opening pressure).
- Peripheral facial palsy — Bell's palsy vs Ramsay Hunt; forehead involvement sign.
- Myasthenia gravis & myasthenic crisis — drug precipitants, thymoma screening.
- Wernicke's encephalopathy — give thiamine before glucose (Pabrinex references).
- ENT/ophthalmology intersections — HZO vision threats (IV acyclovir).
- Practical procedural items — LP technique/contraindications, needle choices, landmark L4/iliac crest.

Sources to verify
- NICE guidance NG240 (meningitis) 2024; NICE NG228 (SAH/CT timing) and CKS summaries; BNF; RCEM and ERC acute neurology/stroke/seizure guidance; LITFL (myotomes/localisation); CADISS trial (2015) for cervical artery dissection context; local hospital formularies and neurology/haematology policies for dosing, anticoagulation and steroid regimens.

(Study note: this chapter is high-yield summary material for exam preparation — verify local treatment protocols and up-to-date national guidance before clinical application.)

CHAPTER 17: OBSTETRICS & GYNAECOLOGY

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Chapter 17 — Obstetrics & Gynaecology (final)

1. Clinical overview and pathophysiology

- Priorities: maternal ABCs first; once mother stabilised, assess foetus (gestational age, CTG/USS). Many management choices depend on gestation, maternal haemodynamic status and placental/cord pathology.
- Pregnancy physiology alters baseline values and investigation interpretation (raised baseline D-dimer, increased blood volume, altered pharmacokinetics). Some drugs/imaging avoided or modified in pregnancy.
- Common immediate obstetric emergencies:
- Early pregnancy: ectopic pregnancy, miscarriage.
- Late pregnancy/third trimester: placenta praevia (painless PV bleeding), placental abruption (painful bleeding ± uterine tenderness), cord prolapse (acute foetal compromise), uterine rupture (prior scar + acute abdominal catastrophe), severe pre-eclampsia/HELLP, veno-thromboembolism, sepsis/TSS, postpartum haemorrhage (PPH).
- Key pathophysiology notes:
- Ectopic: implantation outside uterus → risk of tubal rupture and intra-abdominal haemorrhage.
- Abruption: premature placental separation → maternal haemorrhage, uterine tetany, foetal hypoxia.
- Placenta praevia: low-lying/covering os → painless major bleeding; digital vaginal exam contraindicated until excluded by ultrasound.
- HELLP: microangiopathic haemolysis + elevated liver enzymes + low platelets → rapid deterioration; delivery is definitive treatment.
- PPH: uterine atony most common; retained products, trauma, coagulopathy also causes.

Arabic summary:

الأولوية إنعاش الأم ثم تقييم الجنين. الحمل يغير القيم التشخيصية والعلاجية. انتبه للحمل خارج

الرحم، انفصال المشيمة، المشيمة النازحة، تسلل الحبل، تمزق الرحم، HELLP، ونزف ما بعد الولادة.

2. Recognition, differential diagnosis, and red flags

- Vaginal bleeding:
- First trimester: threatened miscarriage vs ectopic — pain, haemodynamic instability, adnexal tenderness/mass suggest ectopic.
- Second/third trimester: painless PV bleeding → placenta praevia until excluded; painful bleeding + uterine tenderness → abruption; sudden severe pain + loss of foetal heart sounds in scarred uterus → uterine rupture.
- Abdominal pain + positive pregnancy test: localised unilateral pain or adnexal mass → ectopic until proven otherwise.
- Reduced/absent foetal movements after 24 wks → urgent CTG (≥30 min) and consider bedside USS.
- Hypertension after 20 wks:
- New BP ≥140/90: pre-eclampsia.
- Severe features: BP ≥160/110 (two readings ≥4 h apart if time allows), severe headache, visual disturbance, epigastric/RUQ pain, oliguria, thrombocytopenia.
- HELLP red flags: RUQ/epigastric pain, nausea/vomiting, AST/ALT markedly raised, platelets <100×10^9/L, haemolysis (raised LDH).
- Shock + high fever + rash in menstruating tampon user → toxic shock syndrome (TSS).
- VTE: unilateral calf pain/swelling → DVT; dyspnoea/pleuritic pain/tachycardia → consider PE.
- Postpartum fever ± uterine tenderness → endometritis; mastitis presents with focal breast erythema/tenderness ± systemic illness.

Arabic summary:
نزف بلا ألم في الثلث الثالث يوجب استبعاد المشيمة النازحة. ألم مع نزف وحنان رحمى يوجّه للانفصال المشيمى. ضغط الدم ≥160/110 وعلامات عضوية يشيان بتسمم الحمل الشديد. توقف حركة الجنين بعد 24 أسبوعًا — راقب CTG فوراً.

3. Diagnostic approach and interpretation

- General:
- If unstable: treat first; do not delay definitive intervention for imaging.

- Labs: FBC, Group & Save/crossmatch, coagulation, LFTs, creatinine, urinalysis/ urine protein, quantitative β-hCG (early pregnancy), cultures if infection suspected.
- Early pregnancy bleeding:
- TVUS first-line to locate pregnancy; look for intrauterine gestational sac, yolk sac, free fluid.
- Serial quantitative β-hCG: exam rule — viable intrauterine pregnancy generally rises ≈≥66% in 48 h (approximate). Slow rise/plateau suggests ectopic/non-viable pregnancy.
- Suspected ectopic and unstable → immediate surgical management.
- DVT/PE in pregnancy:
- D-dimer is often raised in pregnancy — limited utility alone.
- DVT: compression duplex ultrasound of symptomatic leg(s) is first imaging.
- PE imaging algorithm: obtain CXR first.
- If CXR normal → V/Q scan often preferred (lower maternal breast dose).
- If CXR abnormal → CTPA preferred.
- Start therapeutic LMWH when clinical probability high and imaging delayed/ unavailable (unless contraindicated).
- Reduced foetal movements:
- ≥24 wks: CTG monitoring for ≥30 min.
- <24 wks: bedside USS (foetal heart activity).
- Infection exposures:
- Varicella exposure: immediate maternal VZV IgG serology; if non-immune and within window consider VZIG per local policy.
- Parvovirus B19: maternal IgM/IgG; if seroconversion → serial fetal USS for hydrops.
- Genital HSV: primary maternal lesions near term → caesarean consideration; consult obstetrics.
- Trauma ≥22 wks: CTG monitoring (minimum 1 h), abdominal USS for placenta/ abruption; check Rh status and consider Kleihauer to quantify FMH.

Arabic summary:
المريض غير المستقر يعالج فورًا. نزف الحمل المبكر يُقيّم بالموجات عبر المهبل وهرمون الحمل التكرارى. D-dimer محدود الفائدة فى الحمل؛ الاشتباه بالصمة الرئوية يوجّه إلى CXR ثم V/Q أو CTPA تبعًا للنتيجة.

4. Emergency management algorithms

- Algorithm A — Haemodynamically unstable pregnant patient with bleeding (any gestation)
1. ABCs, high-flow O2, two large-bore IVs, call senior obstetrics and anaesthetics/ theatre.
2. Send labs: FBC, coag, G&S ± crossmatch, LFTs, creatinine, β-hCG (if early).
3. Resuscitate: IV crystalloids judiciously, transfuse PRBC/FFP/platelets per massive haemorrhage protocol; give tranexamic acid 1 g IV early (within 3 h) for PPH if indicated.
4. Specific: first-trimester ongoing heavy bleeding → surgical evacuation; postpartum → uterotonics (oxytocin + adjuncts), bimanual massage, escalate to theatre if refractory.
- Algorithm B — Suspected ectopic (stable)
1. TVUS ± serial β-hCG.
2. If ectopic on USS or high clinical suspicion with abnormal hCG trend → urgent gynae consult: methotrexate (medical) if criteria met OR surgical management (laparoscopy/laparotomy).
3. If unstable → immediate transfer to theatre.
- Algorithm C — Cord prolapse with foetal compromise
1. Immediate: position mother head-down (Trendelenburg/left lateral or knee–chest), manual elevation of presenting part, avoid cord handling that compromises flow.
2. Call theatre for immediate delivery (usually emergent Caesarean).
3. Temporise: fill bladder or use tocolysis per local protocol if needed while preparing theatre.
- Algorithm D — Severe pre-eclampsia / HELLP
1. Stabilise: oxygen, IV access, labs, continuous fetal monitoring.
2. Seizure prophylaxis: start MgSO4 (see section 5).
3. BP control: give parenteral antihypertensive per local protocol; arrange urgent obstetric delivery.
4. HELLP: prepare for expedited delivery; correct coagulopathy, transfuse platelets if indicated.
- Algorithm E — Suspected PE in pregnancy
1. Assess stability. If unstable → resuscitate and consider immediate reperfusion per local guidance (consult specialists).
2. Obtain CXR; choose V/Q if CXR normal, CTPA if CXR abnormal or V/Q unavailable.

3. If high clinical suspicion and imaging delayed → start therapeutic LMWH.
- Algorithm F — Suspected maternal sepsis/TSS
1. ABC + sepsis resuscitation bundle: fluids, cultures, empiric antibiotics.
2. For suspected staphylococcal TSS: empiric vancomycin + clindamycin (clindamycin to inhibit toxin production); surgical source control if required.
- Algorithm G — VTE in pregnancy
1. Confirm DVT with compression ultrasound.
2. Treat confirmed DVT/PE with therapeutic LMWH throughout pregnancy (dose adjustments by weight and renal function).
3. DOACs = not recommended in pregnancy.

> Arabic summary:
> خوارزميات الطوارئ: نزف حاد → إنعاش، نداء الفريق، نقل دم، وإجراءات جراحية مبكرة حسب السبب؛ اشتباه بالمشيمة النازحة → لا تفحص مهبليًا قبل السونار؛ الشك بصدمة سمية → فانكوميسين + كليندامايسين.

5. Drugs, procedures, disposition, and follow-up

- Key well-established thresholds/doses (verify local protocols):
- Severe pre-eclampsia BP threshold: systolic ≥160 mmHg or diastolic ≥110 mmHg (two readings ≥4 h apart if time allows) → classed as severe disease.
- MgSO4 for seizure prophylaxis in severe pre-eclampsia: loading 4 g IV over 15–20 min, then maintenance infusion commonly 1 g/hr (monitor respirations, reflexes, urine output). (Follow local protocol.)
- Tranexamic acid for PPH: 1 g IV as early as possible within 3 hours (then per protocol).
- PPH definitions: minor 500–1000 mL; major ≥1000 mL in 24 h.
- Anti-D immunoglobulin: give after sensitising event within 72 hours; do not delay first dose while awaiting Kleihauer; quantify FMH (Kleihauer) to decide on additional dosing per local guideline.
- LMWH: first-line anticoagulation in pregnancy (therapeutic dosing for DVT/PE). DOACs not recommended.
- TSS empiric antibiotics: vancomycin + clindamycin initially (change to anti-staphylococcal β-lactam if MSSA proven).
- Procedures:
- TVUS for early pregnancy assessment and placenta localization.
- CTG for foetal monitoring ≥24–26 wks (or local viability threshold); continuous

monitoring if maternal/foetal concern.
- Surgical: laparoscopy/laparotomy for ruptured ectopic or uterine rupture; urgent Caesarean for cord prolapse with foetal compromise or other obstetric emergencies.
- Kleihauer–Betke or equivalent assay for FMH quantification in Rh-negative mothers after bleeding/trauma.
- Disposition and follow-up:
- Unstable patients → resuscitate + transfer to labour ward/theatre/ICU as appropriate; consultant obstetric/anaesthetic involvement early.
- Stable with ectopic → admit or organise urgent gynae follow-up for methotrexate or elective surgery.
- Confirmed DVT/PE → admit and continue therapeutic LMWH; arrange haematology/obstetric follow-up and postpartum thromboprophylaxis plan.
- PPH recovered → observe on labour ward, arrange haemoglobin check, and follow-up for anaemia and lactation support.
- Exposures (varicella, parvovirus, HIV/HSV) → document serology, liaise with obstetrics/ID, arrange neonatal planning and counselling.

Arabic summary:
نقاط علاجية: سلفات المغنيسيوم 4 جم تحميل ثم صيانة 1 غ/ساعة (تحقق من البروتوكول المحلى). ترانكساميك أسيد 1 غ IV مبكرًا فى PPH. مضاد-D خلال 72 ساعة بعد حدث حسّاس—لا تؤخر الجرعة الأولية. LMWH للعلاج الوقائى والعلاجى للجلطات أثناء الحمل؛ DOACs غير مستحبة.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية
- hCG dynamics: rise <66% in 48 h suggests failing/ectopic pregnancy; fall suggests miscarriage — commonly tested.
- Never perform digital/vaginal examination for bleeding after 24 weeks until placenta praevia is excluded by ultrasound — classic pitfall question.
- D-dimer is frequently elevated during pregnancy—don’t use a single positive D-dimer to rule in VTE; rely on imaging (compression US/CTPA/V-Q).
- Imaging for suspected PE: start with CXR. If normal → V/Q scan often preferred; if abnormal → CTPA. This CXR-guided algorithm is a frequent exam stem.

- In cord prolapse with persistent severe variable decelerations, immediate manual elevation/positioning and transfer for emergency delivery are exam focal points — do not wait for imaging.
- TSS: addition of clindamycin is to inhibit toxin production — an often-tested justification for combination therapy (vancomycin + clindamycin).
- Anti-D: do not delay giving initial anti-D for a potentially sensitising event pending Kleihauer results — exam-style trap.
- VTE questions in pregnancy: LMWH is usually the correct answer; DOACs are common distractors.
- Methotrexate for ectopic: eligibility criteria (hCG cutoffs, sac size) vary — expect “contraindications include haemodynamic instability, suspected rupture, high hCG or significant hepatic/renal disease”.
- Hyperemesis: doxylamine + pyridoxine first-line oral therapy in many guidelines; ondansetron is effective but may appear as a distractor due to teratogenicity debates.

Arabic summary:
خدع الامتحان: ارتفاع هرمون الحمل أقل من 66% خلال 48 ساعة يثير الحمل خارج الرحم؛ لا تفحص المهبل بنزف بعد 24 أسبوعًا قبل السونار؛ D-dimer قد يكون مرتفعًا فلا يعتمد عليه وحده؛ CXR يحدد اختيار V/Q أو CTPA.

7. Arabic rapid-review summary (concise)

- أولوية إنقاذ الأم ثم تقييم الجنين.
- النزف فى الثلث الثالث بلا ألم → استبعد المشيمة النازحة قبل الفحص المهبلي.
- حمل خارج الرحم: ألم بطني أحادي الجانب + كتلة مبيضية محتملة + هرمون حمل يرتفع ببطء.
- صدمة أو نزف حاد → إنعاش ABC، عزل وريدي واسع، فصيلة وG&S، مناداة فريق التوليد.
- PPH كبير ≥1000 مل: مساج، أوكسي توسين/أدوية منظفة الرحم، ترانكساميك أسيد 1 غ IV مبكرا، ومنظومة نقل دم.
- تسمم الحمل الشديد/MgSO4 → HELLP للوقاية من التشنجات (4 غ تحميل) + ضبط الضغط والتحضير للولادة.
- جلطة وريدية/صمة: استخدم LMWH؛ تجنب DOACs أثناء الحمل.
- التعرض لجدري الماء: فحص IgG وتعامل بمضاد-فيروس أو VZIG حسب الحساسية والبروتوكول.

- لا تؤخر جرعات مضاد-D الأولية بعد حدث حساس; استخدم اختبار Kleihauer لتحديد جرعات إضافية.

8. Question coverage map (source Q-number ranges / concepts covered)

EXAM TRICK / تريكة امتحانية

- Q1–Q4: Early pregnancy bleeding, ectopic pregnancy diagnosis and management (hCG dynamics, TVUS, methotrexate vs surgery).
- Q5–Q8: Placenta praevia vs placental abruption; vaginal bleeding in 2nd/3rd trimester; uterine rupture recognition.
- Q9–Q11: Cord prolapse and immediate manoeuvres; foetal distress management and urgent delivery decision.
- Q12–Q15: Pre-eclampsia and HELLP (diagnostic thresholds, MgSO4, BP control, delivery indications).
- Q16–Q18: Postpartum haemorrhage (definitions, uterotonics, tranexamic acid, MHP activation).
- Q19–Q22: VTE in pregnancy (DVT/PE diagnostics, D-dimer caveats, V/Q vs CTPA algorithm, LMWH therapy).
- Q23–Q25: Maternal infections and exposures (varicella/VZIG, parvovirus B19, herpes/HIV neonatal considerations).
- Q26–Q28: Maternal sepsis and TSS (empiric antibiotics including clindamycin rationale, source control).
- Q29–Q31: Trauma in pregnancy (CTG ≥22 wks, FMH/Kleihauer, anti-D timing), and obstetric imaging decisions.
- Q32–Q34: Medications/procedures (MgSO4 dosing, tranexamic acid, anti-D timing, LMWH contraindications).
- Q35–Q37: Breast infections, mastitis vs abscess management, breastfeeding advice in HIV-positive mothers.
- Q38–Q40: Exam pitfalls and common MCQ traps (D-dimer, vaginal exam in 3rd trimester bleeding, imaging choices).

Arabic summary of coverage map:

خريطة الأسئلة تغطى: نزف الحمل المبكر والحمل خارج الرحم؛ النزف فى الثلث الثالث؛ تدبير اختناق الحبل والولادة الطارئة؛ تسمم الحمل/HELLP؛ نزف ما بعد الولادة؛ الجلطات الوريدية

والصمة الرئوية؛ العدوى المتخصصة (جدري الماء، بارفو، هربس)؛ التسمم السمي/TSS؛ إصابات الحوامل؛ الأدوية والإجراءات الحيوية.

Sources to verify
- Royal College of Obstetricians & Gynaecologists (RCOG) guidance; Royal College of Emergency Medicine (RCEM) clinical resources; National Institute for Health and Care Excellence (NICE) guidance; British National Formulary (BNF); local obstetric/hospital guidelines and national obstetric infectious disease guidance.

CHAPTER 18: ONCOLOGICAL EMERGENCIES

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Chapter 18 — Oncological emergencies (compact FRCEM final chapter)

1. Clinical overview and pathophysiology

- Definition: time-critical consequences of cancer or its treatment causing rapid organ compromise — common syndromes: metastatic spinal cord compression (MSCC), superior vena cava obstruction (SVCO), neutropenic sepsis, tumour lysis syndrome (TLS), chemotherapy-induced nausea (febrile) and vomiting (CINV) and treatment-related complications.
- Pathophysiology (core ideas):
- MSCC: extradural tumour (metastasis or local extension) or vertebral collapse → cord/cauda compression → vascular congestion and ischaemia → progressive neurological loss.
- SVCO: malignant mediastinal mass compresses SVC → impaired venous return from head/neck/upper chest → raised venous pressures, oedema, airway compromise risk.
- Neutropenic sepsis: chemotherapy-induced marrow suppression → low neutrophils → impaired infection control; small infections can cause fulminant sepsis.
- +TLS: massive tumour cell lysis (spontaneous or after therapy) → release of K, phosphate, nucleic acids → hyperkalaemia, hyperphosphataemia, hyperuricaemia → secondary hypocalcaemia and AKI.
- CINV: drug-specific CNS/enteric receptor activation (5-HT3, NK1, dopamine) → acute/delayed/anticipatory patterns.
- Arabic summary: الطوارئ الأورامية هي مضاعفات سريعة ومهددة للحياة لورم أو علاجه؛ آلياتها الرئيسية: ضغط الحبل الشوكي، انسداد الوريد الأجوف العلوي، تعفن مع قلة العدلات، تحلل سريع للخلايا الورمية واضطرابات القيء بعد العلاج.

2. Recognition, differential diagnosis, and red flags

- MSCC — red flags:
- Progressive local back pain (worse supine/night), new focal motor weakness,

.sensory level, gait disturbance, urinary retention/incontinence or constipation
- Differentials: degenerative spine disease, vertebral fracture, epidural abscess (fever, ESR/CRP), primary spine tumour.
- SVCO — recognition:
  - Facial/neck/upper chest swelling, venous distension, dyspnoea/orthopnoea, cough, hoarseness, stridor/airway compromise.
  - Differentials: SVC thrombosis, heart failure, mediastinitis.
- Neutropenic (febrile) sepsis — red flags:
  - Any fever in recent chemotherapy patient → assume neutropenic sepsis until proven otherwise. Watch for hypotension, tachycardia, hypoxia, altered mental status.
  - Differentials: drug fever, tumour fever, other infections.
- TLS — red flags:
  - Recent cytotoxic therapy or high-grade haematologic malignancy + electrolyte abnormalities (K↑, PO4↑, uric acid↑, Ca↓) or oliguria/AKI.
  - Differentials: rhabdomyolysis, kidney injury from sepsis, other metabolic causes.
- CINV — recognition:
  - Temporal relation to chemo: acute (within 24 h), delayed (>24 h), anticipatory.
  - Assess regimen emetogenicity and prior response.

Arabic summary: راقب ألم الظهر المتقدم، ضعف عصبي أو خلل بولي في MSCC؛ وورم/وذمة وجهية وصعوبة تنفس في SVCO؛ وحمى لدى متلقّي الكيماوي تعني تعفن محبّ للعدلات؛ واضطراب الإلكتروليتات بعد العلاج يلمّح إلى TLS.

3. Diagnostic approach and interpretation

- Immediate general tests (for most oncological emergencies): FBC, U&E, creatinine, Ca, PO4, Mg, uric acid (if TLS suspected), LFTs, lactate, blood cultures x2, ECG (if electrolytes abnormal), CXR if respiratory signs.
- MSCC:
  - Urgent MRI whole spine (not just symptomatic level). Do not delay steroid start if clinically indicated.
  - Plain XR/CT only if MRI unavailable; urgent neurosurgical/radiation oncology review.
- SVCO:
  - Contrast CT chest for diagnosis, level and cause; consider venography if

.planning stent
:Neutropenic sepsis -
Blood cultures before antibiotics if does not delay therapy; urine culture, CXR -
.if chest signs. Do not wait for neutrophil count to start antibiotics
:TLS -
Serial electrolytes, renal function, uric acid, urine output, ECG monitoring for -
.hyperkalaemia/hypocalcaemia changes
:CINV -
Reconcile chemotherapy agents, previous antiemetic prophylaxis and timing -
.(acute/delayed)
Arabic summary: فحوصات دم شاملة وثقافتان للدم قبل المضادات الحيوية إن أمكن؛ MRI كامل للعمود في MSCC؛ CT صدر في SVCO؛ مراقبة متسلسلة للكهارل في TLS؛ وسجل بروتوكولات مضادات الغثيان للجلسات السابقة.

4. Emergency management algorithms

:MSCC (time-critical algorithm) -
Suspect MSCC → immediate high-dose IV corticosteroid (start without .1
.(waiting for MRI if high suspicion
Common ED regimen used in exams: dexamethasone 16 mg IV stat, then 16 -
.mg/day (verify local guideline) [NICE NG234]
Urgent MRI whole spine while maintaining analgesia and hemodynamic .2
.stability
/Urgent multidisciplinary discussion: neurosurgery vs. spinal oncology .3
radiation oncology → definitive therapy: surgical decompression (for
.radioresistant disease, spinal instability) or urgent radiotherapy
Support: bladder catheterisation if retention, physiotherapy, venous .4
.thromboprophylaxis as appropriate
:SVCO (algorithm) -
.Assess airway/oxygenation; sit patient upright; give oxygen and analgesia .1
Give corticosteroid for symptomatic relief (e.g., dexamethasone) if .2
.malignancy suspected
.Obtain urgent contrast chest CT; oncology/radiology consult .3
Definitive: tumour-directed therapy (radiotherapy for radiosensitive .4
tumours; chemotherapy for chemosensitive tumours). Endovascular stenting
for severe airway compromise or rapid progression, or when diagnosis

.pending
.Do not anticoagulate empirically unless thrombosis proven .5
:Neutropenic sepsis (1-hour rule) -
.Any febrile chemotherapy patient → treat as neutropenic sepsis .1
.Take blood cultures x2 and other relevant samples if does not delay care .2
Start IV broad-spectrum anti-pseudomonal antibiotic within 1 hour (common .3
first-line: piperacillin–tazobactam; meropenem for allergy/resistance — follow
.local policy) [NICE CKS]
Give IV fluids, oxygen, monitor MAP/urine output; escalate per sepsis .4
.criteria; admit and involve oncology/haematology
:TLS (recognition + management) -
.Risk stratify pre-treatment; high-risk patients need prophylaxis .1
,Established/suspected TLS: aggressive IV fluids (to maintain urine output) .2
.continuous cardiac monitoring
Manage hyperkalaemia per emergency hyperkalaemia protocols .3
treat ,(insulin+glucose, beta-agonist, consider calcium if ECG changes)
phosphate with binders, avoid routine calcium replacement unless
.symptomatic/ECG-indicated
Consider rasburicase for high-risk/established TLS; check G6PD before use .4
.when possible (contraindicated in G6PD deficiency)
Renal replacement therapy if refractory hyperkalaemia, severe AKI or fluid .5
.overload
:CINV (practical algorithm) -
.Identify emetogenicity of regimen and previous control .1
Prophylaxis: acute — 5-HT3 antagonist (ondansetron); add dexamethasone .2
.and NK1 antagonist (aprepitant) for highly emetogenic regimens
,Treat breakthrough: use antiemetic of a different class (metoclopramide .3
.(prochlorperazine, olanzapine per local protocols

Arabic summary: خطوات قابلة للتطبيق:

- MSCC: ديكساميثازون فورًا → MRI كامل → جراحة أو إشعاع عاجل.
- SVCO: وضع رأسي، أكسجين، ستيرويد، CT صدر ثم علاج موجه/دعامة إذا لزم.
- تعفن محبّ للعدلات: ثقافتان + مضاد حيوي واسع الطيف داخل ساعة (مثلاً بيبيراسيلين/تازوباكتام).
- TLS: سوائل متسلسلة، مراقبة كهارل، rasburicase للحالات عالية الخطر مع فحص G6PD.
- CINV: بروتوكول مضادات القئ حسب درجة التسبب بالغثيان.

5. Drugs, procedures, disposition, and follow-up

- Key drugs and pragmatic notes:
  - Dexamethasone: commonly used in MSCC/SVCO (exam regimen often quoted: 16 mg IV stat then 16 mg/day — verify local guidance) [NICE NG234].
  - Empiric antibiotics for neutropenic sepsis: start within 1 hour; common choice piperacillin–tazobactam (or meropenem for allergy/high resistance) — follow local antimicrobial policy [NICE CKS].
  - TLS: rasburicase for high-risk or established TLS; allopurinol for prophylaxis in lower-risk patients. Check G6PD before rasburicase where possible.
  - Hyperkalaemia: emergency measures per standard algorithms (insulin+glucose, nebulised salbutamol, consider IV calcium if ECG changes).
  - Antiemetics: ondansetron (5-HT3), dexamethasone, aprepitant (NK1); olanzapine for refractory/anticipatory nausea in some protocols.
- Procedures:
  - MRI whole spine (urgent) for MSCC.
  - CT chest with contrast for SVCO; endovascular stenting if indicated.
  - Central venous access may be needed for IV therapy; avoid device insertion in active bacteraemia unless necessary.
  - ICU-level support: for haemodynamic instability, severe TLS requiring dialysis, respiratory failure.
- Disposition and follow-up:
  - Admit all patients with neutropenic sepsis, TLS, symptomatic SVCO and MSCC.
  - MSCC/SVCO require urgent oncology/haematology and specialist referral and clear (neurosurgery/radiation/interventional radiology) documentation of neurologic baseline.
  - Arrange inpatient oncology/haematology follow-up and outpatient rehabilitation/physiotherapy for MSCC survivors.
- Arabic summary: أدوية رئيسية: ديكساميثازون (MSCC/SVCO)، مضاد حيوي واسع الطيف داخل ساعة لتعفن قلة العدلات، rasburicase في TLS مع فحص G6PD. الإجراءات: MRI كامل للعمود، CT صدر، دعامة وريدية عند الحاجة. قبول بالمستشفى دائمًا في الحالات المذكورة.

6. High-yield FRCEM exam tricks and common pitfalls

- MSCC:
  - Trick: MRI whole spine — don’t scan a single level; start steroids immediately

.if clinical suspicion high — timing matters for neurological outcome
- Pitfall: dismissing progressive neurological signs as chronic pain without imaging.
- Neutropenic sepsis:
- Trick: any post-chemotherapy fever = neutropenic sepsis until proven otherwise. Start antibiotics within 1 hour — do not delay waiting for neutrophil count or seniors.
- Pitfall: choosing antibiotics without checking local resistance/allergy policy.
- TLS:
- Trick: classic biochemical pattern — K↑, PO4↑, Ca↓ (secondary), uric acid ↑. Think TLS with oliguria + rising creatinine after chemo.
- Pitfall: giving calcium for asymptomatic hypocalcaemia (risk calcium-phosphate precipitation). Rasburicase needs G6PD consideration.
- SVCO:
- Trick: steroids give rapid symptomatic improvement in malignant oedema — buy time for definitive therapy, especially with lymphomas.
- Pitfall: assuming anticoagulation is always indicated — don't empirically anticoagulate unless thrombotic cause proven.
- CINV:
- Trick: acute CINV responds best to 5-HT3 antagonists; refractory requires change to a different pharmacologic class (olanzapine useful).
- Pitfall: repeating same-class antiemetic for breakthrough.

Arabic summary: نقاط الاختبار:

- MSCC: ديكساميثازون فورًا وMRI كامل.
- حمى بعد العلاج الكيميائي = مضاد حيوي فوري.
- TLS: تجنّب إعطاء الكالسيوم بدون مبرر؛ فكر في rasburicase بعد فحص G6PD.
- SVCO: الستيرويد مفيد مؤقتًا؛ تجنب ممانعة التثبيت دون تشخيص.

7. Arabic rapid-review summary (المراجعة السريعة بالعربية)

- MSCC: ألم ظهر متقدم + علامات عصبية → ديكساميثازون فورًا → MRI كامل → جراحة أو إشعاع عاجل.
- SVCO: وذمة وجهية/عنق، ضيق تنفس → اجلس مريضًا، أكسجين، ديكساميثازون، CT صدر، علاج موجه أو دعامة وريدية.
- Neutropenic sepsis: أي حمى بعد الكيماوي → ثقافتان + مضاد حيوي واسع داخل ساعة → قبول وإشراك الأورام/الدم.

- TLS: فرط PO4، K+، حمض اليوريك → سوائل، مراقبة كهارل، rasburicase إذا عالي الخطر، تجنّب كالسيوم إلا عند الحاجة.
- CINV: استخدم HT3-5 للحاد، أضف ديكساميثازون/NK1 حسب خطورة النظام، لعلاج كسر القئ استخدم فصلية دوائية مختلفة.
خلاصة: لا تؤخر العلاجات الطارئة انتظار نتائج المخبر؛ اعرف بروتوكول المستشفى.

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية

- (No explicit Q-numbers provided in supplied material.) Covered FRCEM-relevant concepts:
- MSCC: recognition, MRI-first principle, steroid timing/dosing, surgical vs radiotherapy choices.
- SVCO: clinical features, CT diagnosis, steroid use, stenting indications.
- Neutropenic (febrile) sepsis: definition, 1-hour antibiotic rule, sample-taking, first-line antibiotic classes.
- Tumour lysis syndrome: electrolyte pattern, acute management, rasburicase and G6PD considerations, dialysis indications.
- CINV: acute/delayed/anticipatory types, prophylaxis regimens and breakthrough strategies.
- Diagnostics: essential blood tests, cultures, ECG, imaging priorities.
- Practical pitfalls and exam-focused mnemonics/tricks.
- Use this map to cross-reference FRCEM question bank topics (MSCC, SVCO, neutropenic sepsis, TLS, CINV, emergency imaging, sepsis antibiotics).

Sources to verify
- NICE Guideline NG234 (MSCC corticosteroids) [NICE NG234]
- NICE Clinical Knowledge Summary — Neutropenic sepsis management [NICE CKS]
- BMJ Best Practice — Tumour lysis syndrome [BMJ BestPractice]
- Royal College of Emergency Medicine (RCEM) guidance, European Resuscitation Council (ERC) and local oncology/haematology protocols for institution-specific antibiotic and steroid dosing.

This chapter is study material only — adapt to local protocols and the latest national guidance.

CHAPTER 19: NEPHROLOGY

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Chapter 19 — Nephrology (final)

1. Clinical overview and pathophysiology

- Acute kidney injury (AKI): abrupt fall in GFR → rise in serum creatinine and/or fall in urine output. Use KDIGO criteria for diagnosis. [KDIGO 2012]
- KDIGO (concise): ↑ creatinine ≥26.5 µmol/L (0.3 mg/dL) within 48 h OR ≥1.5× baseline within 7 days OR urine <0.5 mL/kg/h for 6 h.
- AKI categories: pre-renal (reduced perfusion), intrinsic (ATN, glomerulonephritis, AIN), post-renal (obstruction). Pathophysiology drives initial management (restore perfusion vs treat inflammation vs relieve obstruction).
- Nephritic syndrome: haematuria (cola/tea urine), RBC casts, hypertension, oedema, reduced urine output — immune/inflammatory glomerular injury (e.g., post-streptococcal GN).
- Nephrotic syndrome: heavy proteinuria → hypoalbuminaemia, severe peripheral/anasarca, hyperlipidaemia; urine often “foamy.”
- Rhabdomyolysis: muscle breakdown → myoglobinuria → direct tubular injury + renal vasoconstriction → AKI risk. CK very elevated; dipstick + blood but no RBCs.
- Renal replacement therapy (RRT) indicated for life-threatening complications summated by AEIOU: Acidosis, Electrolytes (refractory hyperkalaemia), Intoxications (dialysable), Overload (refractory), Uraemia (pericarditis/encephalopathy).

> ملخص بالعرَبيّة: الفشل الكلوي الحاد = نقصان مفاجئ في وظيفة الكُلى؛ التمييز بين أسباب قبل كلوية/كلوية داخلية/بعد كلوية يوجّه العلاج. متلازمة نفريتية = بول دموي/ضغط مرتفع؛ نفروتية = بروتينية شديدة. اعتلال الربد العضلي يسبب بول داكن وCK مرتفع ويهدد بالفشل الكلوي. AEIOU لتحديد الحاجة إلى غسيل.

2. Recognition, differential diagnosis, and red flags

Recognition
- Presentations: oliguria/anuria, non-oliguric AKI, fluid overload (pulmonary oedema), confusion, nausea, arrhythmia (hyperkalaemia).

- Red flags prompting urgent action: ECG changes of hyperkalaemia, refractory pulmonary oedema, uremic pericarditis/encephalopathy, rapidly rising creatinine, bilateral obstruction, anuria with suspected obstruction.

Key differentials
- Pre-renal: dehydration, haemorrhage, sepsis, reduced cardiac output, drugs (ACEi/ARB/NSAID).
- Intrinsic: acute tubular necrosis (ischaemic/toxic), glomerulonephritis (nephritic picture), acute interstitial nephritis (fever, rash, eosinophilia, drugs).
- Post-renal: bladder outlet obstruction, bilateral ureteric obstruction — consider prostate disease, stones, external compression.

Red-flag exam clues
- Cola/tea urine after sore throat → post-strep GN (nephritic).
- Massive oedema + foamy urine → nephrotic syndrome.
- Dipstick “blood” + few/no RBCs on microscopy → myoglobinuria from rhabdomyolysis.
- Rapidly progressive oliguria with hyperkalaemia and ECG changes → urgent RRT consideration.

> الشرح بالعربية: علامات التحذير: تغيرات ECG بسبب فرط بوتاسيوم، وذمة رئوية مقاومة، التهاب التامور اليوريمى، انسداد ثنائي المجرى البولي. فرط الكرياتينين سريع أو انعدام التبول يستلزمان تقييم فوري.

3. Diagnostic approach and interpretation

Immediate bedside
- ABCs; ECG early if hyperkalaemia suspected (peaked T, wide QRS).
- Urine dipstick (blood, protein), bedside glucose. Insert urinary catheter if accurate hourly urine output needed.

Initial laboratory panel
- Serum: creatinine, urea, electrolytes (K+, Na+, Ca2+, PO4), venous/arterial blood gas if acidotic, CK, LFTs, CBC (look for eosinophilia), coagulation as indicated.
- Urinalysis: dipstick + microscopy (RBC casts, WBCs, eosinophils), urine protein quantification (ACR/UPCR) where possible.
- Urine indices: urine Na, FENa or FEUrea can help pre-renal vs intrinsic but are unreliable after diuretics or in CKD — interpret cautiously.

Imaging and specialty tests
- Renal ultrasound urgently if obstruction suspected (hydronephrosis) or to look for chronicity (small echogenic kidneys).
- Specific serology if GN suspected: ANCA, ANA, complement levels (C3/C4), anti-GBM, anti-streptococcal titres — arrange early but manage emergently if life-threatening.
- Rhabdomyolysis: CK (often >>1,000 U/L; markedly high values >5,000 U/L increase risk but thresholds vary), urinalysis showing dipstick blood without RBCs.

> **EXAM TRICK / تريكة امتحانية**
> Interpretation pitfalls
> - Don’t rely on a single creatinine without baseline.
> - FENa misleading after diuretics; CKD alters indices.
> - Dipstick cannot distinguish haemoglobin vs myoglobin vs RBCs; microscopy is essential.

الشرح بالعربية: فحص البول الشريطي سريع ومفيد لكن تأكّد من الميكروسكوب. تصوير بالموجات فوق الصوتية للكِلى ضروري عند الاشتباه بالانسداد. حسابات الـ FENa لا تُعتمد بعد استخدام المدرّات أو في CKD.

4. Emergency management algorithms

Algorithm A — AKI in ED (compact)
1. ABC → monitor ECG if K+ suspected high.
2. Stop nephrotoxins (NSAIDs, ACEi/ARB, diuretics, aminoglycosides).
3. Assess volume status:
- Hypovolaemia → give isotonic crystalloid (bolus) and reassess.
- Euvolemic → search intrinsic causes (labs, urine studies, GN screen).
- Hypervolemia → oxygen, consider loop diuretic bolus/infusion if responsive; if refractory or life-threatening pulmonary oedema → urgent RRT.
4. Insert catheter for accurate urine output.
5. Urgent nephrology if severe/uncertain cause or likely RRT.

Algorithm B — Life-threatening hyperkalaemia
- If ECG changes or K+ markedly high: do steps in parallel
1. Stabilise membrane: IV calcium gluconate 10 mL of 10% (1 g) IV over 2–5 min.
2. Shift K+ intracellularly: insulin 10 U IV + 25 g glucose IV (e.g., 50 mL 50%);

consider nebulised salbutamol (10–20 mg) and IV bicarbonate if severe acidosis.
3. Remove K+: consider sodium polystyrene (limited role in emergencies) and arrange urgent RRT for refractory hyperkalaemia or very high K+ (esp. K+ >6.5 mmol/L or with ECG changes).
4. Continuous cardiac monitoring; check K+ frequently.

Algorithm C — Rhabdomyolysis
1. Suspect: muscle pain/weakness + dark urine or dipstick blood without RBCs.
2. Immediate actions: aggressive IV isotonic crystalloids — initial bolus (commonly 1–2 L unless contraindicated) then high-rate infusion to maintain brisk urine output; insert catheter and monitor.
3. Monitor CK, creatinine, electrolytes, urine colour/output; correct hyperkalaemia and acidosis per protocols.
4. Consider nephrology if CK very high, rising creatinine, oliguria, hyperkalaemia, or severe acidosis. Bicarbonate or mannitol are controversial — consult nephrology.

Algorithm D — Obstruction
- If bladder outlet obstruction → catheterisation (urgent).
- If suspected bilateral ureteric obstruction or solitary kidney obstruction → urgent urology ± decompression (stent/nephrostomy) and renal ultrasound.

Notes on RRT indications
- AEIOU mnemonic for urgent dialysis: Acidosis (severe/refractory), Electrolytes (refractory hyperkalaemia), Intoxications (dialysable), Overload (refractory pulmonary oedema), Uraemia (pericarditis/encephalopathy). Clinical context and nephrology input determine timing.

> الشرح بالعربية: خوارزميات سريعة — وقف السموم الكلوية، تقييم السوائل وتعديلها، مراقبة البول وECG، علاج فرط بوتاسيوم طارئ (كالسيوم، أنسولين+غلوكوز)، وبدء سوائل قوية في الربد العضلي. انسداد الحالب يحتاج تفريغ عاجل.

5. Drugs, procedures, disposition, and follow-up

Drugs and doses (well established)
- Calcium gluconate 10 mL of 10% (1 g) IV over 2–5 min — for ECG protection in hyperkalaemia.
- Insulin for hyperkalaemia: 10 U regular IV + 25 g glucose IV.

- Nebulised salbutamol 10–20 mg can shift K+ intracellularly (adjunct).
- Stop nephrotoxins immediately (NSAIDs, ACEi/ARB, diuretics, aminoglycosides).

Procedures
- Urinary catheter for output measurement and bladder decompression.
- Renal ultrasound for obstruction or chronicity.
- Urgent urology for decompression (stent/nephrostomy) if obstruction not relieved by catheter.
- Vascular access for dialysis/CRRT if RRT anticipated (early nephrology/renal unit involvement).

Disposition and follow-up
- Admit all patients with AKI except mild, clearly pre-renal reversible cases with outpatient follow-up.
- Urgent nephrology review for: rapidly rising creatinine, oliguria/anuria, refractory hyperkalaemia, severe acidosis, suspected GN, need for RRT, or uncertainty.
- Arrange outpatient renal follow-up if AKI resolves but baseline abnormality or persistent proteinuria/haematuria.
- Document and communicate medication changes (hold ACEi/ARBs/NSAIDs) and reason for hold to primary care.

Special notes
- Bicarbonate or mannitol for rhabdomyolysis: evidence inconsistent — use after nephrology advice.
- Avoid automatic numeric cutoffs for dialysis (use AEIOU + clinical picture). pH thresholds for dialysis vary by guideline; clarify local practice.

> الشرح بالعربية: جرعات رئيسية: كالسيوم جلوكونات 1 غرام IV لعلاج فرط بوتاسيوم الحاد؛ أنسولين 10U + 25 غرام سكر لنقل البوتاسيوم داخل الخلايا. قم بوضع قسطرة بولية وموجات فوق صوت للكِلى عند الاشتباه في انسداد. راجع الكُلى عاجلاً عند حالات شديدة أو حاجة لغسيل الكِلى.

6. High-yield FRCEM exam tricks and common pitfalls

> **EXAM TRICK / تريكة امتحانية**
> High-yield tricks
> - KDIGO quick recall: 0.3 mg/dL (26.5 µmol/L) rise in 48 h or 1.5× baseline in 7 days or urine <0.5 mL/kg/h for 6 h. Exam favourite. [KDIGO 2012]
> - AEIOU mnemonic = dialysis indications — classic exam staple.

- Dipstick “blood” + no RBCs on microscopy = myoglobinuria (rhabdomyolysis).
- Cola/tea-coloured urine after sore throat = post-streptococcal GN (nephritic).
- Foamy urine + heavy proteinuria = nephrotic syndrome.
- Hyperkalaemia immediate sequence: Calcium → Insulin+glucose → Removal (nebulised salbutamol, bicarbonate if acidotic, RRT). In exams, remember calcium first for ECG changes.

EXAM TRICK / تريكة امتحانية

Common pitfalls

- Overreliance on single creatinine value without baseline—may miss AKI or misclassify CKD.
- Using FENa after diuretics or in CKD — can be misleading. FEUrea may be more useful with diuretics but still imperfect.
- Assuming fixed numeric dialysis cutoffs (e.g., creatinine > X) — context matters; AEIOU + nephrology.
- Forgetting to stop nephrotoxins early (NSAIDs, ACEi/ARBs) — common cause of preventable AKI.
- Treating hyperkalaemia without continuous ECG monitoring or without definitive potassium removal when indicated.

نصائح للامتحان بالعربية: تذكّر رقم 0.3 mg/dL في تعريف AEIOU، KDIGO للغسيل، وُجهة العلاج لفرط بوتاسيوم: كالسيوم ثم أنسولين+جلوكوز ثم إزالة/غسيل. لا تعتمد على FENa بعد المدرّات.

7. Arabic rapid-review summary (Modern Standard Arabic)

- تعريف: الفشل الكلوي الحاد = ارتفاع مفاجئ في الكرياتينين أو نقصان في إخراج البول؛ استخدم معايير KDIGO.
- الأسباب: قبل كلوية (نقص تروية/أدوية)، داخلية (ATN، التهاب كبيبات)، بعد كلوية (انسداد).
- علامات حمراء: تغيرات ECG بفرط بوتاسيوم، وذمة رئوية مقاومة، التهاب غشاء التامور اليوريمى، انسداد ثنائي.
- فحوصات عاجلة: ECG، تحليل بول شريطي + ميكروسكوب، كرياتينين/إلكتروليتات، CK، تصوير كِلى بالموجات فوق الصوتية عند اشتباه انسداد.
- علاج طارئ: أوقف السموم الكلوية، عدّل حالة السوائل، غلوبولات/أنسولين+غلوكوز لفرط

بوتاسيوم؛ سوائل قوية للربد العضلي؛ تصريف انسداد فوراً.
- إحالة عاجلة للكُلى عند شدة الحالة أو احتمال الحاجة لغسيل.

خلاصة قصيرة: قيّم ABC، أوقف المسببات، عالج فرط بوتاسيوم وفق الأولوية (كالسيوم → أنسولين+غلوكوز → إزالة)، واطلب مساعدة الكُلى مبكراً.

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية

- AKI definition and KDIGO criteria; urine output thresholds.
- AKI aetiology: pre-renal vs intrinsic vs post-renal; drug causes (NSAIDs, ACEi/ARB).
- Initial ED workup: urine dipstick/microscopy, electrolytes, creatinine, CK, ECG, renal ultrasound.
- Use and limitations of urine indices (FENa/FEUrea).
- Hyperkalaemia recognition and emergency treatment sequence (calcium, insulin+glucose, salbutamol, RRT indications).
- AEIOU indications for dialysis and caveats about numeric cutoffs.
- Rhabdomyolysis: diagnosis (dipstick blood/no RBCs, CK), immediate aggressive IV fluids, monitoring, nephrology referral.
- Obstructive uropathy management: catheterisation, urology decompression.
- Nephritic vs nephrotic clinical clues.
- Practical ED algorithms for AKI, hyperkalaemia, rhabdomyolysis, obstruction.
- Exam tricks: KDIGO 0.3 mg/dL rule, dipstick vs microscopy, AEIOU mnemonic, FENa pitfalls.

Sources to verify
- KDIGO Clinical Practice Guideline for Acute Kidney Injury (2012).
- McMahon score literature for rhabdomyolysis risk stratification.
- Local and national renal/dialysis protocols; RCEM, ERC, NICE renal and hyperkalaemia guidance as applicable.

(Study material only — not individual patient advice.)

CHAPTER 20: MUSCULOSKELETAL (NON-TRAUMATIC)

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Chapter 20: Musculoskeletal (non-traumatic)

1. Clinical overview and pathophysiology

- Common ED categories: inflammatory (septic arthritis, gout/pseudogout, bursitis, septic bursitis, tenosynovitis), overuse/tendinopathy (lateral/medial epicondylitis, De Quervain, rotator cuff), systemic inflammatory myopathies/PMR, degenerative/fragility (osteoarthritis, osteoporosis, adhesive capsulitis), neuropathic/entrapment (radial tunnel, PIN palsy, carpal tunnel).
- Pathophysiology focus:
- Septic arthritis: bacterial invasion → synovial inflammation, purulence, cartilage destruction; rapid diagnosis essential to prevent irreversible joint damage and sepsis.
- Crystal arthropathies: monosodium urate (gout) or calcium pyrophosphate (pseudogout) provoke neutrophil-mediated synovitis; crystals may coexist with infection.
- Tenosynovitis/hand infections: confined sheath/space infection → compartment-like pressure, tendon ischemia, potential flexor tendon necrosis; Kanavel signs indicate flexor sheath involvement.
- Tendinopathy/overuse: microtears, degenerative tendinosis, peritendinous inflammation; mechanical load and repetitive use predominate.
- Adhesive capsulitis: capsular fibrosis with global loss of capsular volume → progressive active and passive ROM restriction.

> المفهوم والفيزيولوجيا المرضية (عربي مختصر)
> - حالات الطوارئ: العدوى داخل المفصل وغمد الأوتار واحتشاء اليد ونقائل العمود الفقري.
> الآليات تشمل التهاب قيحي للمفصل، التهاب بلّوري بلّوري بالنقش الكريستالي، وضمور ليفي للمفصل (الكتف المتجمد).

2. Recognition, differential diagnosis, and red flags

- Monoarticular hot swollen joint: always exclude septic arthritis first. Red flags → fever, rigors, severe pain, inability to weight-bear (large joint), systemic sepsis signs. DDx: septic arthritis vs gout/pseudogout vs haemarthrosis vs acute osteomyelitis adjacent to joint.
- Hand/phalangeal infections: paronychia, felon, pyogenic flexor tenosynovitis (Kanavel's 4 signs: fusiform swelling, tenderness along flexor sheath, pain on passive extension, flexed posture). Any Kanavel sign → urgent surgical/hand consult.
- Shoulder pain:
- Painful arc 60–120° → subacromial pathology (supraspinatus tendinopathy/ impingement).
- Global active + passive restriction → adhesive capsulitis.
- Bilateral proximal stiffness with raised inflammatory markers → polymyalgia rheumatica (PMR); screen for GCA symptoms.
- Elbow/wrist distinctions:
- Lateral epicondylitis (tennis elbow): point tenderness at lateral epicondyle, pain on resisted wrist extension.
- Radial tunnel: more diffuse pain 3–5 cm distal; pain on resisted middle-finger extension; PIN palsy causes motor weakness without sensory loss.
- De Quervain: radial-styloid pain, positive Finkelstein (thumb inside fist + ulnar deviation). Intersection syndrome = painful crepitus 4–6 cm proximal, different location.
- Red flags across MSK: rapidly progressive weakness, neurovascular compromise, overlying skin infection with spreading cellulitis, systemic sepsis, back pain with neurological deficits or cancer history (spinal metastasis/cord compression).

التعرّف والتمييز والعلامات الحمراء (عربي مختصر)
- مفصل أحادي ساخن → استبعد التهاب مفصلي صديدي.
- علامات كانافيل → استدعاء جراحة يد عاجل.
- تمييز دقيق لموقع الألم في الرسغ/المرفق يساعد التفرقة بين دقرفان، intersection syndrome، ومتلازمة النفق الشعاعي.
- فكر بـ PMR وGCA في كبار السن مع ألم كتف/حوض صباحي.

3. Diagnostic approach and interpretation

- General principles: focused history (onset, trauma, systemic features, immunosuppression, joint prosthesis, animal/human bite), targeted exam tests (Kanavel, Finkelstein, painful arc, Neer/Hawkins, Empty-can, lift-off), directed baseline labs and imaging.
- Acute hot joint algorithm:
1) Rapid assessment (vitals, sepsis screen).
2) Blood cultures if febrile/septic.
3) Urgent arthrocentesis BEFORE antibiotics when feasible — send Gram stain, cell count with differential, culture, crystal analysis, glucose, protein.
4) Labs: CBC, CRP/ESR (CRP correlates with infection/inflammation), renal function (for drug dosing), consider uric acid only as baseline (not diagnostic in acute attack).
5) Imaging: plain X-ray to exclude fracture/advanced OA; US to guide aspiration or detect effusion; MRI if deep infection/adjacent osteomyelitis suspected.
- Synovial fluid interpretation:
- Non-inflammatory: <2,000 WBC/mm^3.
- Inflammatory: ~2,000–50,000 WBC/mm^3.
- Suspicious for septic arthritis: commonly >50,000 WBC/mm^3 with predominance of neutrophils, but overlap exists — clinical context and Gram stain/culture are decisive.
- Hand infections: X-ray for foreign body/osteomyelitis; ultrasound to detect abscess/foreign body; consider CT/MRI for deep space or fascial plane involvement.
- Tendon/nerve diagnostics: ultrasound for tendon tears/tenosynovitis; EMG/nerve conduction for persistent nerve palsies; MRI for complex rotator cuff pathology or preoperative planning.
- PMR vs polymyositis: PMR → raised ESR/CRP, normal/mildly raised CK; polymyositis → markedly raised CK, proximal muscle weakness, EMG/ autoantibodies.

> النهج التشخيصي (عربي مختصر)
> - ابدأ بالتاريخ والفحص الموجّه. بزل المفصل ضروري عند مفصل ساخن. تحليل سائل المفصل (خلايا، صبغة جرثومية، بلورات، مزرعة) يحدد العدوى مقابل النقرس. استخدم الأشعة والسونار لتحديد الحاجة للجراحة أو لتوجيه البزل.

4. Emergency management algorithms

- Suspected septic arthritis (monoarthritis) — concise ED algorithm:
1) ABCs, analgesia, immobilize joint; septic work-up (blood cultures, lactate if septic).
2) Urgent arthrocentesis (cell count, Gram stain, culture, crystals) — do before antibiotics if safe.
3) Start empiric IV antibiotics after aspirations and blood cultures; choose agents per local guidance to cover Staphylococcus aureus (consider MRSA where prevalent) and Streptococci; add Gram-negative coverage based on risk factors.
4) Early orthopedic/rheumatology/surgical discussion for washout (arthroscopic/open) especially for large joints, prosthetic joints, or failure to respond.
5) Monitor for sepsis; tailor antibiotics to culture results.
- Hot swollen joint where crystals identified:
- If crystals present and low clinical suspicion of sepsis, treat for gout/pseudogout: NSAIDs (if not contraindicated), low-dose colchicine regimens per current local guidance, or systemic/intra-articular corticosteroid if appropriate. Do not withhold aspiration if concern for infection.
- Flexor tenosynovitis / deep hand infection:
- Immediate IV antibiotics (cover Staph/Strep; add Pasteurella coverage for cat bites; add anaerobic/human-bite coverage as indicated) and urgent surgical/hand service consult for washout/drainage. Do not delay operative management if Kanavel signs or fluctuance.
- Overuse/tendinopathy (ED management):
- Rest/activity modification, relative immobilization (appropriate splint), ice, NSAIDs, provider-directed physiotherapy referral (eccentric strengthening). Consider corticosteroid injection only when infection excluded and usually after conservative measures fail; surgical referral for refractory cases.
- De Quervain:
- Thumb-spica splint, rest, NSAIDs; consider corticosteroid injection at 4–6 weeks if persistent; surgical release if refractory.
- Adhesive capsulitis:
- Early physiotherapy and analgesia; consider intra-articular steroid injection or hydrodilatation for severe functional loss; refer for formal physio/orthopedics.
- PMR:
- Start prednisolone 15 mg daily if clinical picture consistent and arrange urgent rheumatology follow-up; assess and treat for giant cell arteritis if GCA features

present (higher dose steroids needed).
- Bites:
- Clean, consider prophylactic antibiotics for human/cat/dog bites and puncture wounds; amoxicillin-clavulanate commonly recommended for Pasteurella and mixed flora (verify local guidance). Tetanus status updated per local policy.

> الادارة الطارئة (عربي مختصر)
> - مفصل مشتبه بالتهاب إنتاني: بزل عاجل، مزراعة دم، بدء المضاد الحيوي الوريدي بعد البزل، استدعاء جراحة/مفصل.
> - غمد أوتار فلكسوري/خراج اليد: مضاد حيوي وريثمي وعمليّة عاجلة.
> - دقرفان: جبيرة إبهام وراحة، حقن ستيرويد إذا استمرت الحالة.
> - PMR: بريدنيزون 15 ملغ يومي ومتابعة روماتولوجية طارئة.

5. Drugs, procedures, disposition, and follow-up

- Drugs (ED-relevant highlights):
- Analgesia: paracetamol ± opioid for severe pain; avoid NSAIDs if contraindicated (renal failure, anticoagulation, peptic ulcer disease) — choose per patient risk.
- NSAIDs: first-line for many inflammatory/overuse conditions if not contraindicated; verify local dosing.
- Colchicine: use current low-dose regimens per local/national guidance — older high-dose regimens are toxic (VERIFY before prescribing).
- Steroids: intra-articular or systemic for gout/PMR/adhesive capsulitis as indicated; do not inject into infected fields.
- Bites: amoxicillin-clavulanate typical empiric choice for cat/dog/human bites (covers Pasteurella, anaerobes) — check allergy status and local guidance.
- Procedures:
- Arthrocentesis: sterile technique, record appearance, send appropriate tests; ultrasound guidance if difficult or small joint. Do before antibiotics if possible.
- Incision and drainage/operative washout: for abscesses, deep hand space infection, septic joints not controlled by aspiration alone — arrange urgent surgical care.
- Splinting: thumb spica for De Quervain; wrist splint for ECU/intersection syndromes; sling for acute significant shoulder pain.
- Disposition:
- Admit: confirmed/probable septic arthritis, deep hand infection requiring surgery, systemic sepsis, uncontrolled pain, significant neurovascular

compromise, PMR needing assessment when GCA suspected, or significant comorbidity affecting outpatient care.
- Discharge with arranged follow-up: uncomplicated tendinopathies, stable gout/pseudogout after ED treatment and clear safety net, non-infected olecranon bursitis with advice. Ensure analgesia, immobilization plan, and timely outpatient physiotherapy/orthopedic/rheumatology follow-up.
- Follow-up:
- Orthopedics/Hand surgery urgent for septic joints, deep hand infections, prosthetic joint involvement.
- Rheumatology for PMR, recurrent crystal arthritis, or unclear inflammatory arthropathy.
- Primary care/physio for tendinopathy rehabilitation and osteoporosis assessment (DEXA referral where indicated).
- Practical ED notes:
- Don’t delay arthrocentesis for imaging unless image guidance needed to avoid iatrogenic injury.
- If starting empiric antibiotics before aspiration is possible only when aspiration delayed by logistic/safety issues — still perform aspiration ASAP.

الأدوية، الإجراءات، والمراقبة (عربي مختصر)
- الأدوية: مسكّنات، NSAIDs عند الاقتضاء، كولشيسين بجرعات منخفضة حسب الإرشادات الحالية (تحقق)، أموكس-كلاف للمضغات.
- الإجراءات: بزل المفصل، تصريف/تنظيف جراحي لالتهابات اليد، جبائر مناسبة.
- الخروج أو القبول: ادخال للمستشفى لحالات التهاب مفصلي إنتاني، عدوى يد عميقة أو ضعف نظامي؛ إحالة متابَعَة للروماتولوجي/جراحة يد/أورثوبيدكس حسب الحالة.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية
- Never rely on serum uric acid alone to diagnose gout — synovial fluid crystal analysis is diagnostic. (Exam trick frequent in SBAs.)
- Synovial fluid WBC >50,000 and neutrophil predominance strongly suggests septic arthritis, but overlap exists; presence of crystals does NOT exclude concurrent infection. Always correlate clinically.
- Kanavel’s 4 signs = flexor tenosynovitis → assume need for urgent surgical review even if single sign present. Negative signs do not fully exclude early

disease.
- Finkelstein technique — ensure the patient makes a fist with thumb inside the fist; examiner ulnar-deviates wrist. Confusion between Eichhoff and true Finkelstein is a common exam trap.
- Painful arc interpretation: 60–120° → subacromial (supraspinatus/impingement); pain at extremes only → glenohumeral/AC pathology. Use exact arc numbers in answers.
- Differentiate lateral epicondylitis vs radial tunnel by location of maximal tenderness (epicondyle vs 3–5 cm distal) and resisted middle-finger extension test. PIN palsy = motor deficits without sensory loss.
- Do not delay arthrocentesis for routine X-ray when infection suspected — culture yield falls after antibiotics. Exam scenarios often test timing.
- Bites: first-line prophylaxis for cat/human/dog puncture wounds often amoxicillin-clavulanate — exam scenarios expect this. Check allergy options (e.g., doxycycline for penicillin allergy in adults per local guidance).
- Watch for outdated/unsafe statements in stems (older colchicine regimens, specific NSAID dosing like unusual naproxen schedules, absolute statements about aspirin and gout) — flag and verify with current guidelines.

نصائح للامتحان وأخطاء شائعة (عربي مختصر)
- اليوريك أسيد المرتفع وحده لا يثبت النقرس.
- >50,000 خلية/مم^3 في سائل المفصل يشير بشدة للعدوى لكن يوجد تداخل.
- تذكر تفاصيل فحص فينكلشتاين وصيغة علامات كانافيل.
- لا تؤجل البزل بسبب صورة روتينية إذا اشتبهت بعدوى.

7. Arabic rapid-review summary

- أي مفصل أحادي ساخن → بزل فوري وإرسال لعزل الجراثيم والبلورات، مزرعة دم إذا حمى، وبدء مضاد حيوي وريدي بعد البزل إذا الشك مرتفع.
- علامات كانافيل → افترض التهاب غمد الأوتار الفلكسوري واطلب رأي جراحة يد عاجل؛ الخراج → تصريف.
- دقرفان → فinkelstein إيجابي: جبيرة إبهام وراحة وNSAID؛ فشل التحسن → حقن ستيرويد أو جراحة.
- الكتف: القوس المؤلم 60–120° → التهاب أوتار supraspinatus؛ فقدان الحركة النشط والسلبي → كتف متجمد → علاج طبيعي وحقن ستيرويد عند الحاجة.
- PMR → بريدنيزون 15 ملغ يومياً مع متابعة روماتولوجية وتقييم لعلامات GCA.

- لدغات/عضات: أموكس-كلاف عادة لتغطية Pasteurella والميكروبات المختلطة — تحقق من التوصيات المحلية.

خلاصة سريعة بالعربية (موجزة)
- بزل المفصل عند وجود مفصل ساخن، علاج جراحي للمضاعفات اليدوية، جبائر وأدوية تحفظية للأوتار، وستيرويد منخفض الجرعة للـ PMR مع متابعة.

8. Question coverage map (concepts → typical FRCEM question foci)

- Q1–Q6: Acute monoarticular joint — septic arthritis vs gout/pseudogout; timing of arthrocentesis, synovial fluid interpretation.
- Q7–Q9: Flexor tenosynovitis / Kanavel signs, hand infections, bite management and antibiotic choices.
- Q10–Q12: De Quervain vs intersection syndrome; Finkelstein test technique and splinting/indications for steroid injection.
- Q13–Q16: Rotator cuff impingement tests (painful arc, Neer/Hawkins, Empty-can), adhesive capsulitis features and management.
- Q17–Q19: Lateral elbow pain — tennis elbow vs radial tunnel vs PIN palsy; localization and special tests.
- Q20–Q22: PMR recognition, initial steroid dosing and GCA screen; polymyositis vs PMR differentiation (CK, EMG, autoantibodies).
- Q23–Q25: Synovial fluid thresholds, limitations of serum uric acid, co-existence of crystals and infection.
- Q26–Q28: Imaging choices (X-ray, US, MRI) for joint effusion, tendon pathology, and when to use guided aspiration.
- Q29–Q31: Bite wounds and empiric antibiotic selection (Pasteurella/human bite coverage), tetanus considerations.
- Q32–Q34: Emergency disposition decisions — who to admit vs discharge, orthopedics/hand/rheumatology referral thresholds.

Sources to verify
- RCEM, NICE, ERC, local antimicrobial guidelines, national rheumatology/orthopaedics guidance, and current formularies (for colchicine dosing and specific antibiotic regimens).

CHAPTER 21: MENTAL HEALTH

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Chapter 21: Mental Health

1. Clinical overview and pathophysiology

- Scope: ED mental-health emergencies include acute behavioural disturbance (ABD)/agitation, acute psychosis/mania, self-harm/suicidality, intoxication/withdrawal (alcohol, stimulants, GHB), and delirium from medical causes. Prioritise safety and exclude medical mimics.
- Pathophysiology (brief): agitation/ABD arises from toxicologic (sympathomimetic, anticholinergic, sedative withdrawal), neurologic (delirium, head injury, seizures), endocrine/metabolic (thyrotoxicosis, hypoglycaemia, electrolyte disturbance), and primary psychiatric processes (psychosis, mania). Severe agitation → hyperadrenergic state, hyperthermia, rhabdomyolysis, metabolic collapse.
- Legal/ethical: capacity is decision- and time-specific; never assume incapacity from diagnosis. MHA pathways (Section 136, Section 2) determine detention and assessment logistics in the UK. [MHA 1983; RCEM Best Practice 2024]

> Arabic summary:
> - الطيف: ABD، ذهان، انسحاب/تسمم، اضطراب تأخّر عقلاني. استبعد الأسباب الطبية أولاً.
> - القدرة العقلية تتغير ويجب تقييمها لكل قرار؛ قوانين الاحتجاز (MHA) تحدد المسارات.

2. Recognition, differential diagnosis, and red flags

- Presentation spectrum: mild agitation → violent ABD (biting/spitting, disorientation, inability to record observations). Look for clues: pupils, temperature, sweating, tremor, focal neurology, history of substance use, medication changes.
- Differential (key distinctions):
- Intoxication/withdrawal: variable consciousness, autonomic signs, rapid onset.
- Delirium/medical: acute onset, fluctuating cognition, abnormal vitals, often older patients.
- Primary psychosis/mania: gradual change, chronic social decline, persistent hallucinations/delusions.

- Neurologic lesion: focal signs, post-traumatic onset.
- Red flags requiring immediate medical stabilisation:
- Unable to obtain vital signs
- Respiratory compromise or hypoxia
- Convulsions
- Profuse agitation with exhaustion or hyperthermia
- Suspected acute poisoning/overdose
- Active suicidal intent/plan or command hallucinations to self/other

> Arabic summary:
> - تميّز التسمم/الانسحاب، الهذيان، والذُهان بالسرعة، العلامات الحيوية، والعلامات العصبية.
> - أعلام حمراء: فقدان القدرة على قياس العلامات، صعوبات تنفسية، نوبات، فرط حرارة، نيّة انتحارية فورية.

3. Diagnostic approach and interpretation

- Immediate priorities: scene safety → brief focused assessment: ABCs, GCS, capillary glucose, pulse oximetry, temperature, ECG (if antipsychotic/droperidol anticipated), blood pressure.
- Initial tests: finger-prick glucose; ECG (QTc baseline if possible); bloods as indicated — U&E, CK, LFTs, VBG/ABG if respiratory or metabolic compromise, paracetamol level if overdose suspected; pregnancy test for women of childbearing potential.
- Focused history: onset, substance use, meds, prior psychiatric history, previous suicide attempts, current intent/plan.
- Imaging/other: CT head if trauma, focal signs, acute confusion of unknown cause. Toxicology screens helpful but do not delay treatment.
- Psychiatric liaison: obtain psychiatric input concurrently with medical workup where indicated—do not delay psych assessment until “medically clear.” [RCEM Mental Health Toolkit 2021]
- Legal checks: if patient absconds and returns, repeat triage, vitals and risk assessment. Section 136 clock runs from entry to place of safety (24 h, with possible 12-h extension). Section 2 requires two doctor recommendations + AMHP. [MHA 1983; RCEM Best Practice 2024]

> Arabic summary:
> - فحص سريع: ABC وقياس سكر، ECG قبل مضادات الذهان إن أمكن، وتحاليل موجهة (CK،

وظائف كبد/كلية).
- تقييم نفسي متزامن مع الفحص الطبي؛ أعِد تقييم المريض إذا عاد بعد هروب. القوانين تحدد أطر الاحتجاز.

4. Emergency management algorithms

Algorithm A — ABD (compact)
1) Safety and environment: secure low-stimulus area, remove hazards, one-to-one observation, call security/police if necessary.
2) Verbal de-escalation: calm tone, simple requests, time-limited offers. Document attempts.
3) If imminent harm or restraint required despite de-escalation:
- Use minimum necessary, proportionate physical restraint; avoid prolonged prone restraint; reposition to supine/side-lying.
- If restraint >10 minutes → proceed to rapid tranquilisation. [NICE NG10]
4) Rapid tranquilisation (see Section 5 for doses). Monitor airway, breathing, circulation continuously.
5) Post-sedation: continuous obs, ECG if QT-prolonging drugs used, repeat assessment, escalate to critical care for airway compromise, unstable haemodynamics, refractory agitation, or failed sedation.
6) Concurrently: search for reversible causes and treat (hypoglycaemia, infection, intoxication, electrolyte disturbances).

Algorithm B — Suicidal/high-risk self-harm
1) Immediate safety: remove means, constant observation, one-to-one if high risk.
2) Rapid risk assessment (prior attempts, plan/intent, access to means, substance use).
3) Medical workup as indicated (tox screen, paracetamol level).
4) Urgent psychiatric liaison; admit if persistent high risk or inability to secure safety.

Arabic summary:
- مسار ABD: أمان الموقف → تهدئة لفظية → تقييد متناسب → إذا تجاوزت القيود 10 دقائق أعطِ تهدئة سريعة → مراقبة مستمرة.
- مسار الانتحار: أزل الوسائل → رصد دائم → تقييم المخاطر → تواصل مع الطب النفسي.

5. Drugs, procedures, disposition, and follow-up

Drugs (high-yield doses per RCEM/NICE):
- Ketamine IM 4 mg/kg — first-line rapid tranquilisation for severe ABD where indicated. [RCEM Oct2023 / May2025]
- Droperidol IM 5–10 mg — alternative first-line; monitor QTc. [RCEM Oct2023]
- Midazolam IM 5–10 mg — alternative (caution: prolonged sedation, elderly). [RCEM]
- Lorazepam IM 4 mg — alternative benzodiazepine option.
- Haloperidol 5 mg IM ± lorazepam 2 mg IM — effective combo; watch QTc and extrapyramidal effects.
- Diazepam IV maintenance 0.3 mg/kg (titrate) or lorazepam IV 0.03 mg/kg for escalation in monitored setting.
Practical notes:
- Adjust doses for frailty, comorbidity, and prior IM agents (reduce subsequent IV doses).
- Large IM volumes: plan multiple injection sites or higher concentration preparations.
- Avoid combining multiple sedatives without airway plan — increased respiratory depression risk.
Airway/RSI:
- In anticipated controlled airway need, avoid suxamethonium where risk of hyperkalaemia exists (severe exertional rhabdomyolysis, renal impairment) — prefer rocuronium for paralysis. Avoid routine opioid pre-treatment in ABD. [RCEM Oct2023]
Specific scenarios:
- Serotonin toxicity: consider specialist toxicology input; some guidance references chlorpromazine IV/IM with hypotension risk — verify locally.
- Alcohol/benzodiazepine withdrawal: benzodiazepines are first line.
- GHB/GBL withdrawal: consider baclofen per toxicology guidance (TOXBASE).
Procedures and monitoring:
- Continuous cardiorespiratory monitoring after sedation, regular obs, temperature, CK and renal function if hyperthermia or prolonged agitation.
Disposition and follow-up:
- Admit to medical/critical care for airway compromise, unstable haemodynamics, ongoing medical cause, severe overdose, or failed ED sedation.
- Psychiatric admission/liaison if high suicide risk, severe psychosis, inability to

ensure safety, or need for MHA assessment.
- If discharged: clear safety plan, follow-up arranged, documented capacity assessment, and restraints/medications recorded.

Arabic summary:
- أدوية التهدئة السريعة: كيتامين IM 4 mg/kg أو دروبروليدول 5–10 mg IM؛ بدائل بنزوديازيبينية وهالوبيريدول ± لورازيبام.
- في التنبيب السريع تجنّب سوكساميثونيوم عند خطر فرط بوتاسيوم؛ استخدم روكورونيوم.
- ادخل المريض إلى العناية المكثفة إذا وجود مشاكل في المسالك الهوائية، عدم استقرار، أو فشل التهدئة.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية
- Exam tricks:
- Strongest predictor of future suicide = prior suicide attempt.
- Restraint >10 minutes → indicates need for rapid tranquilisation (NICE NG10). Cite drug/dose choices and monitoring.
- Section 136 detention: clock starts on entry to place of safety (24 h, possible 12-h extension); Section 2 needs two medical recommendations + AMHP. [MHA 1983]
- Common pitfalls:
- Equating orientation with capacity — perform structured capacity test (understand, retain, weigh, communicate) and document. Diagnosis ≠ incapacity.
- Delaying psychiatric assessment until “medically clear” — psychiatric input should be concurrent for self-harm/overdose.
- Forgetting medical causes — always exclude hypoxia, hypoglycaemia, sepsis, head injury, metabolic disturbance before attributing to primary psychiatric illness.
- Prone restraint risk — avoid prone positioning to prevent positional asphyxia.
- QTc and drug interactions — obtain ECG where possible prior to droperidol/haloperidol and correct electrolytes.
- RSI drug caveats — don’t reflexively use suxamethonium in vigorous/exertional rhabdomyolysis or suspected hyperkalaemia risk.
- Dosing and accumulation — account for prior IM doses when giving IV

sedation.
- Documentation: record de-escalation attempts, capacity assessment, rationale for detention, exact drugs/doses/times, and plan.

Arabic summary:
- حيل للامتحان: المحاولة الانتحارية السابقة أهم مؤشر للخطر؛ إذا تجاوز التقييد 10 دقائق استخدم تهدئة سريعة.
- أخطاء شائعة: عدم تقييم القدرات بشكل منهجي، تأخير تقييم الطب النفسي، تهميش الأسباب الطبية للاضطراب، وتجاهل مخاطر وضعية التقييد.

7. Arabic rapid-review summary (Modern Standard Arabic)

- ABD حالة طارئة: ابدأ بالبيئة الآمنة والتهدئة اللفظية، وقيّم ABCs ووعي المريض وسكر الدم.
- تهدئة سريعة: كيتامين 4 mg/kg IM أو دروبروليدول 5–10 mg IM؛ بدائل: ميدازولام 5–10 mg IM، لورازيبام 4 mg IM، هالوبيريدول 5 mg ± لورازيبام 2 mg IM. راقب التنفس وQTc.
- RSI: تجنّب سوكساميثونيوم إذا خطر فرط بوتاسيوم؛ روكورونيوم مفضل. لا تخلط مهدئات متعددة بلا خطة مجرى هوائي.
- قانونياً: Section 136 يبدأ عند دخول مكان الأمان (24 ساعة + 12 ساعة تمديد محتمل)؛ Section 2 يتطلب توصيتين طبيتين وAMHP.
- راجع دائمًا الأسباب الطبية وقم بتوثيق تقييم القدرة والقرارات العلاجية.

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية
- ABD recognition and priorities: agitation spectrum, red flags, safety sequence (de-escalation → restraint → rapid tranquilisation). [RCEM Oct2023; NICE NG10]
- Rapid tranquilisation: ketamine 4 mg/kg IM, droperidol 5–10 mg IM, benzodiazepine/antipsychotic alternatives, monitoring and complications. [RCEM Oct2023/May2025]
- Airway and RSI in ABD: rocuronium vs suxamethonium, opioid pre-treatment cautions. [RCEM Oct2023]
- Medical causes and diagnostics: ABCs, glucose, ECG, U&E/CK/LFTs, paracetamol level, CT head indications. [RCEM; NICE]
- Legal/ethical: capacity principles, Section 136 timing, Section 2 requirements, management after absconding. [MHA 1983; RCEM Best Practice 2024]
- Suicidality assessment and disposition: risk factors, removal of means,

observation, psychiatric liaison. [RCEM Mental Health Toolkit 2021]
- Special scenarios: serotonin toxicity, alcohol/benzodiazepine withdrawal, GHB/GBL withdrawal, specific antidotes/agents (refer to toxicology). [TOXBASE; RCEM notes]
- Practical pitfalls/exam pearls: documentation, ECG before QT-prolonging drugs, restraint positioning, dosing adjustments after IM drugs.

Sources to verify
- RCEM Acute Behavioural Disturbance in Emergency Departments (Oct 2023; May 2025 updates)
- RCEM Mental Health Toolkit (2021)
- RCEM Best Practice: The Patient Who Absconds (2024)
- NICE NG10: Violence and aggression (short-term management)
- Mental Health Act 1983 (UK)
- TOXBASE / local toxicology and anaesthesia/airway guidelines; ERC, local hospital protocols

(Study material — not individual patient advice. Verify drug availability, concentrations, and contentious/flagged items against local formularies, toxicology, and up-to-date guidelines.)

CHAPTER 22: MAXILLOFACIAL/DENTAL

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Chapter 22 — Maxillofacial / Dental (compact FRCEM final chapter)

1. Clinical overview and pathophysiology

- Core risks: airway compromise (oral blood, swelling, midface instability), major haemorrhage, dental/occlusal disruption, cranial nerve sensory deficits (V1/V2/V3 distribution). Always start with ABC, C-spine control as indicated.
- Typical injuries: anterior mandibular (TMJ) dislocation, mandibular fractures (inferior alveolar nerve risk → lower lip/chin numbness), orbital floor (blowout) fracture with infraorbital nerve injury/rectus entrapment, zygomatic/maxillary/Le Fort fractures (I/II/III), tooth avulsion (permanent vs primary management differs).
- Anatomy clinically useful: supratrochlear (medial forehead/medial eyelid), supraorbital (~2–3 cm above rim, mid-pupil line), infraorbital (~1 cm below midpoint inferior orbital rim). Use these for sensory assessment and nerve blocks.

> Arabic: مخاطر أساسية: انسداد مجرى الهواء، نزف، اضطراب الأسنان وفقدان الإحساس العصبي. الإصابات الشائعة: خلع الفك الأمامي، كسور الفك، كسر أرضية الحجاج، كسور Le Fort، اقتلاع الأسنان.

2. Recognition, differential diagnosis, and red flags

- Anterior mandibular dislocation: sudden mouth opening/yawning then jaw locked open, protruded mandible, severe preauricular pain, inability to close. Differentiate from mandibular fracture: fractures more often give malocclusion, trismus, step-off, mobility.
- Mandibular fracture: pain, malocclusion, mobile tooth/segment, chin deviation, numbness lower lip/chin (inferior alveolar nerve).
- Orbital floor (blowout): peri-orbital swelling/ecchymosis, infraorbital numbness, vertical diplopia (esp. on upgaze), enophthalmos, restricted ocular movement — entrapment suggested by diplopia + positive forced duction.
- Le Fort patterns:
- I: transverse maxillary mobility, malocclusion.
- II: pyramidal midface mobility, infraorbital sensory loss.

- III: craniofacial dissociation, flattened midface, CSF rhinorrhoea, high airway/bleeding risk.
- Avulsed tooth: identify as permanent vs primary (do NOT replant primary). Permanent teeth: best prognosis with immediate replantation (ideally <60 min).
- Red flags requiring urgent specialist input: airway compromise/stridor, expanding facial/neck haematoma, uncontrolled bleeding, CSF rhinorrhoea, diplopia with entrapment, visual loss, open fractures with contamination.

> Arabic: مؤشرات الخطر: صعوبة في التنفس، نزف مستمر، تسرب سائل شفاف من الأنف (CSF)، ازدواجية الرؤية مع احتباس العضلة، فقدان البصر. لا تعيد زرع أسنان لبنية.

3. Diagnostic approach and interpretation

- Initial priorities: ABCs, C-spine precautions if high mechanism. Control obvious haemorrhage; secure IV access.
- Focused facial/dental exam: inspect occlusion, open/close jaw, palpate mandible/zygoma for step/crepitus, assess CN V sensory branches, ocular exam (visual acuity, pupils, EOMs, look for enophthalmos, globe injury), look for CSF rhinorrhoea (β2-transferrin if diagnostic doubt).
- Imaging:
- CT maxillofacial (thin cuts) = gold standard for facial fractures (Le Fort, orbital floor, zygoma, mandible).
- Panoramic OPG useful for mandible if CT not available.
- Orbital CT with coronal slices for blowout and inferior rectus entrapment.
- Dental avulsion assessment: identify tooth type (permanent vs primary), check for associated alveolar/mandibular fracture, evaluate socket integrity; time since avulsion and storage medium critical.
- Ancillary: tetanus status, baseline labs in major trauma, consider pregnancy/vision documentation.

> Arabic: فحص مركّز: فحص الإطباق، تحرّك الفك، حساسية الأعصاب الوجهيّة، فحص العينين. التصوير المثالي: CT وجهي بقطع رفيعة؛ OPG مفيد للفك السفلي إن لم يتوفر CT.

4. Emergency management algorithms

A. Initial (first minutes)
1. Airway: prepare for difficult airway. Maintain C-spine if indicated. Anticipate RSI

or surgical airway. Avoid nasal intubation if suspected Le Fort/basal skull fracture.
2. Breathing/ventilation: oxygen, treat tension pneumothorax if present.
3. Circulation: direct pressure, packing for facial haemorrhage, large-bore IV/ consider blood products per major haemorrhage protocol.
4. Disability/exposure: neuro, ocular, dental check. Rapid CT once resuscitated.

> Arabic: إجراءات سريعة: حماية مجرى الهواء مع توقع صعوبات، تجنّب التنبيب الأنفي عند الشبه بكسر قاعدة الجمجمة، إيقاف النزف بالضغط والتعبئة.

B. Anterior mandibular dislocation — ED reduction algorithm
1. Confirm anterior dislocation (jaw locked open, protrusion) and exclude mandibular fracture.
2. Analgesia + sedation (procedural sedation or local + opioid ± benzodiazepine). Pad thumbs.
3. Technique: patient seated/supine. Place wrapped thumbs on occlusal surface/ ramus; apply steady downward pressure to disengage condyle inferior to articular eminence, then posterior/upward (reduce into glenoid fossa). Expect muscle resistance—use steady force; warn patient.
4. Post-reduction: assess occlusion, neurovascular status; soft diet, avoid wide opening; consider Barton bandage/support 48–72 h. If unreduced or recurrent: consider GA with muscle relaxation or operative management (OMFS).
Reference: DynaMed.

> Arabic: خفض خلع الفك الأمامي: تسكين/تخدير، إبهامات محفوظة على الأضراس السفلية، ضغط لأسفل ثم للخلف، دعم بعد الخفض؛ إذا فشل، استخدام GA/استشارة جراحة الفكين.

C. Mandibular fracture management (ED)
- Do not attempt definitive fixation. Secure airway. Analgesia, cold packs, soft diet, avoid blind reductions. Open fractures → IV antibiotics and urgent OMFS. Immobilize (soft support/bandage); intermaxillary fixation only by specialists.

> Arabic: كسور الفك: لا تحاول تثبيت نهائي في الطوارئ؛ استدع OMFS، أعطِ مسكنات ومضادّات حيوية إذا مفتوح.

D. Orbital floor (blowout) management
- Protect globe (shield), avoid nose-blowing, give antiemetic/analgesic. Urgent CT; urgent ophthalmology/OMFS/ENT for entrapment or large defect.
- Indications for urgent surgery: muscle entrapment with positive forced duction,

persistent diplopia in primary position, large defect causing enophthalmos, oculocardiac reflex signs (bradycardia), orbital compartment syndrome (urgent lateral canthotomy).
- Consider prophylactic antibiotics if sinus communication (commonly amoxicillin-clavulanate per local guidance) and check tetanus.

> Arabic: كسر أرضية الحجاج: حماية العين، منع النفخ بالأنف، CT عاجل، جراحة عاجلة إذا احتبست العضلة أو فقدان رؤية أو قصور حركي واضح.

E. Tooth avulsion (permanent) algorithm
1. Confirm permanent tooth. If primary → do NOT replant; discard.
2. If patient cooperative: replant immediately into socket (hold in place). If not possible, store tooth in HBSS > milk > saline > patient's saliva. Avoid dry storage; best if <60 min.
3. After replantation: flexible splint and urgent dental/OMFS follow-up within hours. Tetanus status and consider systemic antibiotics per dental trauma guidance.
4. Document tetanus, advise soft diet; confirm radiographs for root/alveolar fracture and follow-up for pulp necrosis/endo as advised.

> Arabic: سنّ دائم مقتلع: أعد زراعته فوراً إن أمكن (<60 دقيقة أفضل)، أو خزّنه في محلول مناسب/حليب. لا تعيد زرع الأسنان اللبنية.

5. Drugs, procedures, disposition, and follow-up

Drugs / prophylaxis
- Analgesia and sedation: follow local procedural sedation protocols; calculate LA and sedative doses per weight and guidelines.
- Antibiotics: use for open fractures, sinus communication, or contaminated wounds. Common choices in many protocols: amoxicillin-clavulanate for sinus-exposed orbital fractures; for tooth replantation antibiotics are often recommended (amoxicillin in children; doxycycline in adults in some guidance) — confirm current guidance (IADT/BAOMS/local).
- Tetanus prophylaxis as per immunisation status.
- Anti-emetics to prevent Valsalva in orbital fractures.

Procedures (ED)
- Jaw reduction (anterior dislocation) under sedation; pad thumbs; consider GA if

unsuccessful.
- Local nerve blocks (infraorbital, supraorbital, supratrochlear) for analgesia—use 1–3 mL typical per site; aspirate and inject slowly; calculate max LA dose per agent/weight.
- Lateral canthotomy/cantholysis for orbital compartment syndrome (vision-threatening).
- Packing and surgical control for severe facial haemorrhage; early OMFS/ENT/vascular surgery input for uncontrollable bleeding.

Disposition & follow-up
- Immediate OMFS/ENT/ophthalmology referral for: airway compromise, Le Fort III, open/depressed fractures, diplopia with entrapment, globe injury, large facial wounds, complex dental avulsion needing operative management.
- Replanted permanent tooth: urgent dental/OMFS follow-up within hours, splint removal per dental advice (commonly ~7–14 days but variable).
- Stable, simple soft tissue facial injuries: ED wound care, antibiotics/tetanus as indicated, arrange outpatient OMFS/plastics follow-up.
- Admit if airway concern, uncontrolled bleeding, multi-system trauma, or need for operative management.

> Arabic: الأدوية: مسكنات، مضادات حيوية عند وجود اتصال جيب أنفي/جرح مفتوح، وتطعيم ضد الكزاز حسب الوضع. إجراءات: خفض الفك تحت تسكين، بلوكات عصبية محلية، شق جوانب الحدقة للحدقة المدارية إذا لزم. إحالة فورية لجراحة الفم/العين عند مؤشرات الخطر.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية
- Never replant a primary (deciduous) tooth — classic trap.
- Permanent tooth: “milk is good, HBSS best; <60 minutes ideal for replantation” — mnemonic for storage/time.
- Anterior jaw dislocation = jaw locked open + protrusion; fracture = malocclusion/trismus/step-off.
- Infraorbital numbness → think orbital floor/zygoma fracture (V2); supraorbital vs supratrochlear localization is lateral vs medial forehead respectively.
- Avoid nasal intubation if suspected basal skull or Le Fort fractures — frequent exam/clinical pitfall.
- When doing nerve blocks, remember to aspirate before injection; do not

inject deep to periosteum; calculate LA dose by agent/weight — exam questions may test landmarks (supraorbital 2–3 cm above rim; infraorbital ≈1 cm below rim).
- In blowout fractures: diplopia + positive forced duction → entrapment → urgent surgery. Enophthalmos may be delayed; absence of enophthalmos does not exclude significant fracture.
- Jaw reduction: pad thumbs; warn about biting—examiners like the “wrap thumbs in gauze” detail.

Arabic: نقاط للامتحان: لا تعيد زرع الأسنان اللبنية؛ الحفظ في الحليب أو المحلول مناسب؛ تجنّب التنبيب الأنفي في كسور قاعدة الجمجمة؛ تنفيس قوة العضلة في كسور أرضية الحجاج يعني احتباسًا ويستدعي جراحة عاجلة.

7. Arabic rapid-review summary (ملخص سريع بالعربية)

- ابدأ بالـ ABCs وحماية فقرات العنق عند الآلية العالية.
- فكّ الفك الأمامي: فم مفتوح مقفل + بروز الفك → تقليل يدوياً (ضغط لأسفل ثم للخلف) بعد تسكين؛ احذر الإصبع.
- كسر أرضية الحجاج: خدّ عين المريض، منعه من النفخ بالأنف، CT عاجل، جراحة عاجلة إذا احتبست العضلة أو فقدت الرؤية.
- سنّ دائم مقتلع: أعد زرعه فوراً إن أمكن أو خزّنه في حليب/محلول مناسب؛ لا تُزرع الأسنان اللبنية.
- تجنّب التنبيب الأنفي عند الشبهة لكسر Le Fort أو كسر قاعدة الجمجمة.
- استدع OMFS/ENT/ophthalmology بسرعة في حالة النزف الشديد، CSF rhinorrhoea، مشاكل بصرية أو احتباس عضلي.

8. Question coverage map (concepts covered; source Q-number ranges/ concepts)

EXAM TRICK / تريكة امتحانية
- Covered concepts (mapped to typical FRCEM question topics):
1. Anterior mandibular (TMJ) dislocation — recognition, ED reduction technique, pitfalls.
2. Mandibular fracture — features (malocclusion, inferior alveolar nerve), initial ED management.
3. Orbital floor (blowout) fracture — signs (diplopia, infraorbital numbness),

forced duction/entrapment, indications for surgery.
4. Le Fort fractures (I/II/III) — clinical features, airway risk, CSF rhinorrhoea considerations.
5. Tooth avulsion — permanent vs primary management, storage media, timing, splinting/follow-up.
6. Facial nerve sensory landmarks and practical nerve blocks (supraorbital, supratrochlear, infraorbital).
7. Airway and haemorrhage priorities in facial trauma and contraindications (nasal intubation).
8. Imaging strategy — CT facial bones, OPG for mandible.
9. ED procedures: lateral canthotomy for orbital compartment syndrome; indications for GA/muscle relaxant for failed reductions.
10. Ancillary: tetanus, antibiotics (sinus communication), documentation and disposition criteria.
- Source Q-number ranges: not supplied in the chunk notes. Use the above concept map to cross-reference your question bank.

Arabic: خريطة التغطية: خلع الفك، كسور الفك، كسر أرضية الحجاج، كسور Le Fort، اقتلاع الأسنان، حواجز المسار الهوائي والنزف، بلوكات الأعصاب والـ CT/OPG. أرقام الأسئلة لم تُزوَّد.

Sources to verify
- RCEM guidelines, ATLS (latest), IADT (International Association of Dental Traumatology), BAOMS/OMFS guidance, DynaMed (mandibular dislocation), NICE/UK trauma guidance, ERC trauma airway guidance, local OMFS/ENT/ophthalmology protocols.

CHAPTER 23: INFECTIOUS DISEASES

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Chapter 23 — Infectious Diseases (final)

1. Clinical overview and pathophysiology

- Core principle: rapid recognition and empiric treatment of life-threatening infections saves lives; do not delay antibiotics/antivirals for severe disease while waiting for confirmatory tests. Include infection control/public-health actions (notification, contact prophylaxis) as part of ED management where indicated.
- Common ED syndromes and key pathogens:
- Acute bacterial meningitis: N. meningitidis, S. pneumoniae; Listeria risk in ≥50 y or immunosuppressed.
- Severe malaria: Plasmodium falciparum most critical—high parasitaemia causes multi-organ failure.
- Dengue: plasma leakage in critical phase → haemodynamic compromise; bleeding risk from thrombocytopenia.
- Varicella zoster (shingles): viral reactivation in dorsal root ganglia → dermatomal vesicles; risk of dissemination in immunosuppression.
- Skin/soft tissue and zoonoses: impetigo, scabies, orf (parapox), animal bites; many are self-limiting but need systemic therapy if spread or host immune compromise.
- Healthcare exposures: needlestick risks—HIV/HBV/HCV; PEP and HBV immunoprophylaxis are time-sensitive.
- C. difficile: toxin-mediated diarrhoea after antibiotics → risk of severe colitis, leukocytosis, AKI.

> Arabic summary (موجز فسيولوجي/سريري):
> القاعدة: العلاج المبكر لعدوى مهددة للحياة ضروري؛ تشمل الإدارة الطوارئ الإبلاغ والسيطرة الوبائية عند الحاجة.

2. Recognition, differential diagnosis, and red flags

- Meningitis: fever + headache + neck stiffness ± photophobia/altered mental status/seizure → meningitis pathway. Red flags: reduced GCS, focal neurology,

papilloedema (consider CT before LP).
- Severe malaria: recent travel to endemic area + fever ± jaundice, confusion, shock, severe anaemia, AKI, high parasitaemia, thrombocytopenia.
- Dengue: high fever followed by defervescence; watch for warning signs — abdominal pain, persistent vomiting, mucosal bleeding, rising transaminases, falling platelets, narrowing pulse pressure.
- Varicella zoster: dermatomal unilateral vesicular rash with prodromal neuropathic pain; red flags = immunosuppression, dissemination, visceral involvement.
- Skin infections:
- Impetigo non-bullous: localized honey-coloured crusts, no systemic upset → topical therapy.
- Scabies: intense nocturnal pruritus, interdigital burrows; treat all contacts.
- Orf: single nodular lesion after sheep/goat exposure — usually self-limiting.
- Needlestick exposures: index case risk, type of exposure (percutaneous > mucous membrane), host vaccination status.
- C. difficile: profuse watery diarrhoea post-antibiotics, leucocytosis, rising creatinine — suspect toxin-mediated disease.

> Arabic summary (التعرف والعلامات الحمراء):
> التمييز يعتمد على العرض العام، السفر، والمظاهر الحادة (مثلاً تغير الوعى، صدمة، تدهور وظائف الأعضاء). تعامل مع أي حالة شديدة كعدوى مهددة للحياة حتى تثبت العكس.

3. Diagnostic approach and interpretation

- Guiding principle: stabilise (ABCs) → take microbiology (blood cultures, swabs, baseline serology) → start empiric therapy in severe cases; confirm with targeted tests.
- Meningitis: blood cultures then immediate empiric IV antibiotics; CT brain before LP only if focal neurology, papilloedema, or low GCS. CSF: cell count, Gram stain, culture, PCR.
- Malaria: thick and thin blood films = gold standard (species ID, % parasitaemia); RDTs useful for screening but do not quantify parasitaemia and do not replace microscopy in severe disease.
- Dengue: NS1 antigen early; IgM/IgG later; management guided by clinical severity and serial labs (haematocrit, platelets, LFTs).
- C. difficile: stool toxin assay/PCR; interpret with clinical severity (WBC, creatinine,

albumin).
- Needlestick: baseline recipient serology (HIV, HBsAg/anti-HBs, HCV); test source if possible.
- Skin lesions and outbreaks: generally clinical diagnosis; send swabs/cultures only if systemic signs, cellulitis, atypical features, or outbreak investigation.

> Arabic summary (النهج التشخيصي):
> استقر أولاً، خذ مزروعات/فحوصات أساسية، وابدأ العلاج التجريبي في الحالات الشديدة؛ اختبارات محددة لتأكيد وتوجيه العلاج لاحقاً.

4. Emergency management algorithms

- Meningitis (ED algorithm):
1. ABCs; resuscitate as needed.
2. Blood cultures; give empiric IV ceftriaxone 2 g IV immediately. Add amoxicillin if ≥50 y or immunosuppressed to cover Listeria (do not delay antibiotics for LP/CT).
3. Assess need for CT prior to LP (focal signs, papilloedema, low GCS). If CT required, still give antibiotics beforehand.
4. Perform LP when safe; send CSF (cell count, Gram stain, culture, PCR).
5. Notify public health if meningococcal suspected; arrange chemoprophylaxis for contacts per public-health guidance.
- Suspected severe malaria:
1. ABCs, correct hypoglycaemia, oxygen, IV fluids as needed.
2. Take blood films + RDT, but give parenteral artesunate immediately for severe features (do not wait for microscopy).
3. Admit to HDU/ICU; monitor parasitaemia, glucose, renal function, Hb/platelets; treat complications (transfusion, dialysis, ventilation).
- Dengue with warning signs/critical phase:
1. Monitor haemodynamics and fluid balance closely; use judicious IV crystalloids for shock and guided by clinical markers (pulse pressure, urine output, haematocrit).
2. Avoid NSAIDs/antiplatelets; transfuse platelets for bleeding per local policy.
- Needlestick exposure:
1. Immediate wound care (wash), report, and assess exposure risk.
2. Draw baseline recipient bloods; test source where possible.
3. Start HIV PEP promptly for high-risk exposures; consider HBIG + vaccine for HBV non-immune recipients if source HBsAg positive.

4. Arrange occupational health/infectious disease follow-up and serial serology.
- C. difficile:
1. Stop inciting antibiotics where possible.
2. Start oral vancomycin 125 mg QID for 10 days as first-line for initial episode.
3. Escalate (fidaxomicin, surgery) for fulminant disease per local guidance.
- Skin infections:
- Impetigo localized non-bullous without systemic features → topical hydrogen peroxide 1% cream TDS for 5 days (NICE); oral antibiotics for extensive/systemic disease.
- Scabies → permethrin 5% cream to whole household; consider oral ivermectin for outbreaks/widespread disease.
- Varicella zoster → immunocompetent with early shingles (≤72 h) and ≥50 y: oral acyclovir 800 mg five times daily for 7 days (or equivalent valaciclovir/famciclovir). Immunocompromised/disseminated: IV acyclovir 10 mg/kg q8h (adjust renal).
- Orf → supportive care; rarely requires antivirals.

> Arabic summary (خوارزميات الطوارئ):
> اعطِ العلاج التجريبي فور الاشتباه في حالات مهددة للحياة (السحايا، الملاريا الشديدة، الحزام الناري المنتشر) واتبع خطوات التشخيص والإحالة والمراقبة المذكورة أعلاه.

5. Drugs, procedures, disposition, and follow-up

- Key ED drugs/doses (well-established):
- Ceftriaxone 2 g IV immediately for suspected bacterial meningitis (add amoxicillin if ≥50 y or immunosuppressed) [NICE NG240].
- Contact prophylaxis (meningococcal): ciprofloxacin 500 mg PO single dose commonly used — confirm local public-health guidance [UKHSA].
- Severe malaria: start parenteral artesunate immediately per WHO/local formulary (admit to HDU/ICU).
- Varicella zoster: oral acyclovir 800 mg 5×/day for early cases in older adults; IV acyclovir 10 mg/kg q8h for immunocompromised/disseminated disease.
- C. difficile: oral vancomycin 125 mg QID for 10 days first-line [NICE NG199].
- Scabies: permethrin 5% cream applied overnight to all contacts; ivermectin oral as alternative in outbreaks.
- Needlestick PEP: initiate HIV PEP promptly for high-risk exposures — typical strategy = two NRTIs + integrase inhibitor (follow local occupational health protocols). For HBV exposure in unvaccinated recipients: give HBIG and start

vaccine series immediately.
- Procedures and disposition:
- Admit all suspected bacterial meningitis, severe malaria, dengue with warning signs/critical phase, and disseminated zoster to appropriate monitored care (HDU/ICU as indicated).
- Isolate patients with likely airborne or droplet transmissible infections as per local IPC; notify public health for notifiable diseases (e.g., meningococcal disease).
- For needlestick exposures, ensure urgent occupational health/ID follow-up and serial serology at recommended intervals.
- Outpatient follow-up: arrange primary/ID review for results, vaccination status (HBV), and antibiotic stewardship concerns (C. difficile recurrence).
- Practical notes:
- Stop inciting antibiotics when C. difficile suspected.
- For dengue, fluid management is guided by clinical markers—avoid overresuscitation.
- For impetigo, use topical agents to limit systemic antibiotic use unless extensive.

> Arabic summary (الأدوية والإجراءات والمتابعة):
> تذكر الجرعات المؤسسة (سيفتر/أرتيسونات/acyclovir/vancomycin) ووجه المرضى ذوي المخاطر العالية إلى الملاحظة/الاستشفاء؛ راجع صحة التطعيم واتبع تعليمات الصحة العامة للمخالطين.

6. High-yield FRCEM exam tricks and common pitfalls

- Do not delay empiric therapy in life-threatening infections to obtain confirmatory tests — exams often test immediate IV ceftriaxone for suspected meningitis vs waiting for LP.
- Malaria: RDT positive is not a reason to avoid parasite quantification; for severe disease treat parenterally regardless of pending film.
- Meningitis contacts: remember to notify public health and offer chemoprophylaxis (ciprofloxacin 500 mg single dose commonly tested) — this is often examined as a systems/management step.
- Shingles: age ≥50 with early disease → oral antivirals; immunocompromised → IV acyclovir. These distinctions are frequent exam points.
- Impetigo: identify “localized non-bullous without systemic features” → topical hydrogen peroxide 1% first-line (NICE change) — avoid reflex oral antibiotics in exam answers unless systemic features.

- C. difficile: oral vancomycin is now first-line for initial episode; metronidazole is downgraded. Exam makers test this change.
- Needlestick PEP: obtain baseline serology but do not delay starting PEP in high-risk exposures — expected answer pattern.
- Bed bugs vs vectors: bed bugs cause bites and distress but are not proven disease vectors — commonly used as an exam reassurance/triage point.
- Watch for “traps”: asking for exact PEP drug doses — state local protocol and that PEP is time-sensitive; avoid inventing non-standard doses unless guideline cited.

> Arabic summary (نصائح للامتحان والمحفوفات):
> اجب بتفضيل العلاج المُباشر للحالات المهددة للحياة، تذكر تغييرات الإرشاد الحديثة (vancomycin للـC. difficile، hydrogen peroxide للـimpetigo) ولا تتأخر في بدء PEP عند الضرورة.

7. Arabic rapid-review summary (مراجعة سريعة بالعربية)

> - علاج عاجل للحالات الشديدة: لا تؤخر المضاد الحيوي/الدواء المضاد للفيروسات إذا كانت العدوى مهددة للحياة.
> - التهاب السحايا: سيفترياكسون 2 غ IV فور الاشتباه؛ أضف أموكسيسيلين إذا ≥50 أو مضعف مناعة؛ أبلغ الصحة العامة وعالج المخالطين.
> - الملاريا الشديدة: اعطِ أرتيسونات وريدي فوراً؛ لا تنتظر الشريحة.
> - الحمى الضنك: راقب السوائل والصفائح والإنزيمات، لا تعطِ NSAIDs.
> - الجدري/الحماق: مضعف المناعة → acyclovir وريدي؛ كبار السن المبكرون → acyclovir فموي.
> - قرح الجلد/القشور: القليل يتطلب علاجًا جهازياً؛ impetigo محدودة → هيدروجين بيروكسيد 1% موضعي.
> - لسعات الإبرة: اغسل الجرح، افحص المصدَر والمتلقي، ابدأ PEP/HBIG/لقاح للـHBV حسب الحالة، واتبع الفحوصات المتسلسلة.

8. Question coverage map (concepts covered)

- Bacterial meningitis: recognition, immediate empiric antibiotics, CT vs LP, public-health notification, contact prophylaxis.
- Severe malaria: recognition, role of RDT vs microscopy, parenteral artesunate, HDU/ICU care.
- Dengue: critical phase recognition, fluid management, avoidance of NSAIDs/antiplatelets.

- Varicella zoster (shingles): indications for oral vs IV antivirals, immunocompromised management.
- Impetigo: localized non-bullous management (topical hydrogen peroxide vs systemic antibiotics).
- Scabies and household management; institutional control and ivermectin use.
- Needlestick exposures: wound care, baseline serology, HIV PEP strategy, HBV prophylaxis (HBIG + vaccine).
- C. difficile: suspicion after antibiotics, stool testing, oral vancomycin first-line, stop inciting antibiotics.
- Zoonotic skin infections (orf) and bedside reassurance (bed bugs).
- Infection control and public-health actions (notification, contact prophylaxis) across relevant conditions.

Sources to verify
- NICE guidelines (NG240 meningitis; NG199 C. difficile; NG153 and CKS where applicable), UKHSA meningococcal guidance (2024), WHO severe malaria and dengue guidance, UK Green Book (HBV), local occupational health/PEP protocols, RCEM/ERC for resuscitation/ED care.

CHAPTER 24: HAEMATOLOGY

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Chapter 24: Haematology — Final concise FRCEM review

1. Clinical overview and pathophysiology

- Major categories
- Bleeding from anticoagulation (VKA/warfarin; DOACs — dabigatran [direct thrombin], factor Xa inhibitors [rivaroxaban/apixaban/edoxaban]); heparin (UFH/LMWH). Reversal is agent-specific. [NICE CKS]
- Consumptive coagulopathy (DIC): systemic activation of coagulation → simultaneous bleeding + thrombosis; associated with sepsis, obstetric catastrophe, trauma.
- Thrombotic microangiopathies (TTP/HUS): platelet microthrombi → MAHA, thrombocytopenia; TTP often neuropredominant, ADAMTS13 deficiency; HUS renal dominant (typical = diarrhoeal).
- Immune cytopenias: ITP (isolated severe thrombocytopenia), HIT (anti-PF4/heparin antibodies → thrombosis).
- Congenital/structural bleeding disorders: haemophilia A/B (deep bleeds, prolonged aPTT), VWD (mucocutaneous bleeding, variable lab pattern), HAE (bradykinin-mediated angioedema).
- Transfusion complications: acute haemolytic, anaphylaxis, febrile non-haemolytic, TRALI (noncardiogenic pulmonary oedema), TACO (volume overload).
- Practical guidance: prioritise ABCs, stop exposure (anticoagulant/transfusion), give specific antidote if available, use haemostatic resuscitation guided by physiology and POC tests (TEG/ROTEM) in massive haemorrhage [LiTFL].
- Pathophysiology highlights
- DIC: ↑PT/aPTT, ↓platelets, ↓fibrinogen, ↑D-dimer.
- TTP: ADAMTS13 deficiency → uncleaved vWF multimers → platelet aggregation; coagulation screen usually normal but there is MAHA with schistocytes.
- HIT: immune-mediated platelet activation ~5–10 days after heparin exposure; thrombosis risk despite thrombocytopenia.

ملخص سريري/فيزيولوجى (مبسّط عربي)
- فئات رئيسية: نزف بسبب مضادات التخثر، ITP/HIT، TTP/HUS، DIC، اضطرابات خلقية (الهيموفيليا/VWD/HAE)، ومضاعفات نقل الدم.
- أولويات: ABC → إيقاف المعرض المسبب → علاج معاكس محدد → إعادة التوازن التخثرى بدليل TEG/ROTEM عند النزف الشديد. [LiTFL]

2. Recognition, differential diagnosis, and red flags

- Anticoagulant-related major bleed
- Red flags: hypotension/tachycardia, expanding haematoma, intracranial signs, rapid Hb fall, recent/ongoing anticoagulant dose.
- Identify agent and time of last dose.
- DIC
- Bleeding + thrombosis in sepsis/OB/trauma; labs as above. Red flag: ongoing oozing, multiorgan dysfunction.
- TTP
- Pentad (not always complete): MAHA (schistocytes), thrombocytopenia, renal impairment, fever, neurological signs. Clotting times usually normal — do not delay PEX.
- HUS
- Predominant renal failure, often diarrhoeal prodrome (typical).
- HIT
- Platelet fall ≈5–10 days after heparin, new thrombosis; use 4T pretest probability.
- ITP
- Isolated severe thrombocytopenia with mucocutaneous bleeding; otherwise well.
- Haemophilia A/B
- Spontaneous deep soft-tissue or joint bleeding; markedly prolonged aPTT with normal PT/platelets.
- VWD
- Mucosal bleeding, menorrhagia; platelet count usually normal.
- TRALI vs TACO
- TRALI: acute hypoxaemia, hypotension, fever, noncardiogenic oedema within 6 h of transfusion.
- TACO: hypertension, pulmonary oedema, volume overload features, responds to diuretics.
- Transfusion reactions red flags

- Immediate fever/back pain/haemoglobinuria → acute haemolytic transfusion reaction (stop transfusion and resuscitate).
- Urticaria/bronchospasm/hypotension → anaphylaxis (treat as anaphylaxis).

التعرّف والتمييز والتنبيهات (مبسّط عربي)
- تحقق من تاريخ الدواء، وقت الجرعة الأخيرة، وعلامات الصدمة أو نزف جوفى/مخى.
- DIC = نزف + تخثر؛ TTP = شستوسايتس + صفيحات منخفضة + زمن تخثر طبيعى.
- TRALI يظهر كوذمة رئوية غير قلبية خلال 6 ساعات؛ TACO يصاحب علامات حمولة حجمية.

3. Diagnostic approach and interpretation

- Immediate ED investigations (while resuscitating)
- CBC (Hb, platelets), coagulation screen (PT/INR, aPTT), fibrinogen, D-dimer, peripheral blood film (schistocytes), G&S/type & cross, renal function, lactate, bilirubin/LDH/haptoglobin (haemolysis), urine for haemoglobin.
- POC: capillary/Hb, blood gas ± lactate, TEG/ROTEM in major haemorrhage.
- TEG/ROTEM interpretation (key exam mapping)
- R time (clotting time) ↑ → clotting-factor deficiency → give FFP (or PCC in VKA/DOAC context).
- Alpha angle ↓ → low fibrinogen or slow fibrin build-up → give cryoprecipitate or fibrinogen concentrate.
- MA (maximum amplitude) ↓ → platelet dysfunction/low platelet count → give platelet transfusion.
- Hyperfibrinolysis (LY30 high; threshold institutionally variable, commonly cited >7.5%) → give tranexamic acid. [LiTFL]
- Specific tests (do not delay treatment)
- ADAMTS13 activity for TTP (send urgently but start PEX without waiting).
- Anti-PF4/heparin ELISA (and functional assay) for HIT; calculate 4T score first.
- DOAC-specific assays: dabigatran-sensitive thrombin time/ diluted thrombin time; anti-Xa for Xa inhibitors (availability limited; time since last dose useful).
- Factor assays for haemophilia; vWF antigen/activity for VWD.
- Imaging
- CT head for suspected intracranial bleed; CXR/CTPA if pulmonary pathology suspected (ACS/PE/TRALI).

النهج التشخيصى والتفسير (مبسّط عربي)
- اختبارات فورية: Hb/صفائح/INR/aPTT/فيبرينوجين/مسحة دم للكشف عن شستوسايتس.

- FFP → TEG: R↑ ؛α↓ → فيبرينوجين/كرايو؛ MA↓ → صفائح.
- ارسل ADAMTS13 وanti-PF4 لكن عالج دون انتظار النتائج الحرجة.

4. Emergency management algorithms

A. Life-threatening bleeding in anticoagulated patient — rapid algorithm
1. ABCs, secure airway, high-flow O2, IV large-bore access, blood bank notified, activate MHP if massive.
2. Local haemostasis (direct pressure/surgical/IR) and haemodynamic resuscitation (blood products guided by physiology/TEG).
3. Identify anticoagulant and timing.
4. Specific reversal:
- Dabigatran → idarucizumab 5 g IV (2 × 2.5 g) emergently. [well established]
- Factor Xa inhibitors (rivaroxaban/apixaban) → andexanet alfa per local/ manufacturer dosing (bolus + 2-hr infusion) if available; if unavailable or for other Xa agents, give 4-factor PCC (follow local dosing; commonly 25–50 IU/kg range). Flag licensing/availability.
- Warfarin with major bleeding → 4-factor PCC (rapid) + IV vitamin K (5–10 mg) (give PCC first if life-threatening).
- UFH severe bleed → protamine sulphate (approx. 1 mg neutralises ~100 U heparin; adjust for time since last dose; follow local protocol).
5. Supportive adjuncts: tranexamic acid (1 g IV) per local protocol, platelets/FFP/ cryoprecipitate guided by labs/TEG.
6. Monitor and admit to appropriate level (ICU/HD).

B. Elevated INR without major bleeding
- INR 5–8, no bleeding: withhold warfarin ± oral vitamin K (typical IV/oral dose decisions per local guidance; common oral 1–5 mg). [NICE CKS]
- INR >8 without bleeding: give oral or IV vitamin K (often 5 mg oral; IV 1–5 mg slow depending on situation) and withhold warfarin; repeat INR monitoring.
- Major bleeding on warfarin: PCC + IV vitamin K.

C. Suspected TTP
1. Recognise MAHA + thrombocytopenia ± neuro/fever/renal signs.
2. Do not wait for ADAMTS13. Urgent plasma exchange (PEX) + high-dose steroids (IV methylprednisolone/ oral prednisone ~1 mg/kg as per local practice); involve haematology/ICU. Avoid platelet transfusion unless life-threatening bleeding.

D. Suspected HIT
1. Calculate 4T score. If intermediate/high:
- Stop all heparin products (including flushes), send anti-PF4 ELISA and functional assay.
- Start alternative anticoagulation (argatroban, bivalirudin, fondaparinux) per local protocol; do NOT give platelets unless bleeding.
- Once HIT excluded, resume appropriate anticoagulation.

E. ITP (acute severe bleeding or platelet <30×10^9/L with bleeding)
- Severe bleeding or platelet critical:
- IVIG 1 g/kg ×1–2 and steroids (prednisone ≈1 mg/kg or equivalent). Platelet transfusion only for life-threatening haemorrhage.
- No major bleeding and moderate thrombocytopenia: oral prednisone first-line; urgent haematology follow-up.

F. Hereditary angioedema (HAE) acute attack
- Airway risk → early C1 esterase inhibitor concentrate IV (first-line) or icatibant 30 mg SC (bradykinin B2 antagonist) if licensed/available. If unavailable, consider FFP as rescue with haematology guidance.
- Adrenaline/antihistamines largely ineffective (bradykinin mediator) — classic exam distractor.

G. Transfusion reactions — immediate management
- Always: STOP transfusion, keep IV line patent with saline, call for help, send implicated unit and post-reaction samples to transfusion lab.
- Anaphylaxis: IM epinephrine 0.5 mg (500 µg) lateral thigh immediately; airway/oxygen/IV fluids; H1/H2 antihistamines and steroids as adjuncts.
- Acute haemolytic: aggressive IV fluids, maintain urine output, monitor for DIC/renal injury; ICU.
- TRALI: supportive oxygen/ventilation; do not rely on diuretics; consider critical care.
- TACO: stop transfusion, O2, upright position, IV loop diuretics (furosemide 40 mg IV adult initial dose), haemodynamic support.

H. TEG-guided massive haemorrhage protocol (quick)
- R ↑ → FFP/PCC.
- Alpha angle ↓ → cryoprecipitate or fibrinogen concentrate.
- MA ↓ → platelets.
- Hyperfibrinolysis (LY30 ↑) → tranexamic acid 1 g IV.

الخوارزميات الطارئة (مبسّطة عربي)
- نزف مهدد للحياة على مضاد تخثر: ABC → إعلام بنك الدم → idarucizumab 5 g (dabigatran) أو andexanet/PCC (Xa inhibitors) أو PCC+فيتامين K (warfarin) → دعم مختبري/TEG-موجّه.
- INR مرتفع بلا نزف: INR>8 → فيتامين K (فمياً/IV حسب السياق)؛ نزف كبير على warfarin → PCC + فيتامين K IV.
- TTP → ابدأ تبادل البلازما فوراً + ستيرويدات (لا تنتظر ADAMTS13).
- HIT → أوقف الهيبارين فوراً، ابدأ ضد-هيبارين بديل وأرسل anti-PF4.
- تفاعلات نقل: أوقف النقل، حافظ على وريد بالمحلول الملحي، إعطاء أدرينالين للصدمة التأقية، سوائل للصدمة الهيمولايتية، ومدير للتعامل مع TRALI/TACO.

5. Drugs, procedures, disposition, and follow-up

- Key acute drugs/doses (well established)
- Idarucizumab 5 g IV (2 × 2.5 g) — dabigatran reversal. [established]
- Andexanet alfa — reversal for apixaban/rivaroxaban per manufacturer/local protocol (bolus + infusion; dosing complex and variable) — verify local availability/licence.
- 4-factor PCC for life-threatening warfarin/DOAC bleeding (typical range 25–50 IU/kg; local protocols vary) — follow local dosing.
- Vitamin K IV 5–10 mg for major bleeding on warfarin; oral 1–5 mg commonly used for high INR without bleeding. [NICE CKS]
- Protamine sulphate for UFH (≈1 mg protamine per 100 U heparin; adjust for timing; check local limits).
- Tranexamic acid 1 g IV bolus (trauma/hyperfibrinolysis) — time sensitive.
- Prednisone ≈1 mg/kg for ITP; IVIG 1 g/kg ×1–2 for urgent platelet rise.
- IM epinephrine 0.5 mg for anaphylaxis.
- Furosemide 40 mg IV initial dose for TACO (adult).
- Procedures
- Emergency plasma exchange (PEX) for TTP (urgent haematology/ICU).
- Surgical/IR haemostasis for expanding haematomas.
- Exchange transfusion/simple transfusion in severe sickle chest (discuss with haematology/ICU).
- Factor concentrate replacement for haemophilia/VWD per local haematology protocols.
- Airway management and ENT involvement for HAE with laryngeal oedema.

- Disposition and follow-up
- Admit to HDU/ICU for life-threatening bleeding, ongoing transfusion requirement, respiratory failure, or ongoing haemodynamic instability.
- Haematology and transfusion services involvement early for PEX, factor concentrates, and complicated reversal.
- Arrange specialist outpatient haematology follow-up for confirmed ITP, TTP survivors, VWD/haemophilia management plans, anticoagulation restart planning.
- Document and report transfusion reactions to transfusion service and pharmacovigilance.
- Safety/verification notes
- PCC dosing and andexanet licencing vary by jurisdiction and product — always verify local protocol before administration.
- IO samples have limitations: reliable for Hb and glucose but unreliable for WBC, platelets, and electrolytes [LitFL].

الأدوية والإجراءات والتنويم والمتابعة (مبسّط عربي)
- أدوية رئيسية: idarucizumab 5 g (dabigatran)، andexanet (rivaroxaban/apixaban حسب البروتوكول)، PCC (25–50 IU/kg شائع)، فيتامين K IV 5–10 mg للنزف على warfarin.

- إجراءات: تبادل البلازما لـTTP، علاج جراحى/قسطرة للنزف المستمر، بدائل للدم/ترميم عامل فى الهيموفيليا.
- التنويم: HDU/ICU حسب الاستقرار؛ إحالة هيماتولوجى مبكرة؛ توثيق/تبليغ تفاعلات النقل.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية
- Reversal mapping memorise:
- Idarucizumab = dabigatran only. Do NOT use for Xa inhibitors.
- Andexanet alfa = licensed for rivaroxaban/apixaban (complex dosing; check local policy). PCC is common alternative if andexanet unavailable. (Flag licensing). [Exam trap]
- Distinguish DIC vs TTP:
- DIC → prolonged PT/aPTT, low fibrinogen, high D-dimer.
- TTP → normal clotting times, MAHA with schistocytes and severe thrombocytopenia.
- HIT: use 4T score before testing; if intermediate/high, stop all heparin and

start non-heparin anticoagulant.
- Platelet transfusion pitfalls:
- Do not transfuse platelets routinely in ITP or TTP — generally ineffective/harmful unless life-threatening bleeding.
- TRALI vs TACO:
- TACO = volume overload, hypertension, responsive to diuretics.
- TRALI = hypotension, fever, noncardiogenic pulmonary oedema, poor diuretic response.
- TEG exam mapping:
- R ↑ → FFP; α ↓ → fibrinogen/cryoprecipitate; MA ↓ → platelets. Hyperfibrinolysis → TXA.
- IO laboratory pitfalls: Hb and glucose reasonable; do not rely on IO for platelets/WBC/electrolytes in exams.
- HAE distractors: antihistamines/adrenaline often ineffective; C1 esterase inhibitor/icatibant are targeted therapies.
- Sickle chest syndrome:
- Exam often expects antibiotics + oxygen + consider transfusion; avoid reflex steroid use as default—check scenario and local guidance.
- Licensing and local guidance caveats:
- Andexanet alfa, PCC dosing, exact O2 targets in ACS vary—state “follow local protocol” in answers if asked.

نصائح للامتحان ومطبات شائعة (مبسّطة عربي)
- تذكر: idarucizumab فقط لدا بيغاتران؛ andexanet لـapixaban/rivaroxaban (تحقق محلياً).
- DIC = اختبارات تخثر مطوّلة؛ TTP = زمن تخثر طبيعى + شستوسيتس.
- لا تعطى صفائح فى ITP/TTP إلا عند نزف يهدد الحياة.
- خريطة TEG مهمة: R→FFP؛ α→فيبرينوجين؛ MA→صفائح.

7. Arabic rapid-review summary (مختصر سريع بالعربية)

- استعمل نهج ABC أولاً، ثم حدد السبب: مضاد تخثر؟ تفاعل نقل؟ TTP؟ DIC؟ ITP؟
- نزف مهدد: idarucizumab 5 g لـdabigatran؛ andexanet أو PCC لـXa inhibitors (تحقق محلياً)؛ PCC + فيتامين K لل warfarin؛ بروتامين للهيبارين.
- TTP = تبديل بلازما فوري (لا تنتظر ADAMTS13).
- HIT = أوقف الهيبارين وابدأ مضاد-هيبارين بديل واطلب فحص anti-PF4.

- ITP شديدة = IVIG 1 g/kg ± ستيرويد (1 mg/kg).
- توقف كل نقل دم مشبوه فوراً، حافظ على وريد بالمحلول الملحي، أرسل العينات وبنك الدم.
- استخدم TEG/ROTEM لتوجيه منتجات الدم: α ،FFP→ ↑R ↓→فيبرينوجين/كرايو، MA ↓→صفائح.
- تحقّق محلياً من تراخيص andexanet/PCC وجرعاتها قبل الاستخدام.

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية

- Anticoagulant bleeding and reversal: dabigatran (idarucizumab), factor Xa inhibitors (andexanet alfa vs PCC), warfarin (PCC + vitamin K), protamine for heparin. (DOAC reversal)
- Massive haemorrhage and haemostatic resuscitation: TEG/ROTEM interpretation and product mapping; TXA use. (MHP/TEG)
- DIC vs TTP vs HUS differential and immediate management (PEX for TTP). (Microangiopathies)
- HIT recognition (4T score), testing (anti-PF4) and management (non-heparin anticoagulants). (Immune thrombocytopenias)
- ITP acute management: steroids, IVIG, platelet transfusion threshold. (ITP)
- Transfusion reactions: acute haemolytic, anaphylaxis, TRALI, TACO — recognition and immediate steps. (Transfusion)
- Haemophilia/VWD acute bleed management: factor replacement, DDAVP, VWF concentrate. (Inherited bleeding disorders)
- Hereditary angioedema acute therapy: C1 inhibitor, icatibant, role/limits of FFP. (HAE)
- Sickle cell acute chest syndrome: recognition and ED management including transfusion considerations. (Sickle cell)
- Practical lab approach: immediate tests (CBC, PT/aPTT, fibrinogen, D-dimer, blood film), ADAMTS13/anti-PF4/DOAC assays, IO sample limitations. (Diagnostics)
- Emergency drugs and doses frequently tested: idarucizumab, IV-vitamin K, PCC ranges, protamine rule of thumb, TXA, IVIG/steroids, IM adrenaline, furosemide for TACO. (Drugs/doses)
- Exam-style pitfalls: idarucizumab wrongly chosen for Xa inhibitors, platelet

> transfusion in ITP/TTP, TRALI vs TACO differentiation, TEG mapping, IO lab limits. (Exam traps)

Final note

- This is study material only, not patient-specific advice.

Sources to verify

- RCEM guidelines/teaching resources, ERC/ERC resuscitation guidance, NICE (CKS and NG guidance) — e.g., NICE CKS (Warfarin), NICE NG158; local transfusion service and haemostasis protocols; LiTFL/LitFL TEG/IO primers; institutional DOAC/andexanet/PCC policies; haematology society guidelines (local/national).

CHAPTER 25: ENVIRONMENTAL EMERGENCIES

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Chapter 25: Environmental Emergencies

1. Clinical overview and pathophysiology

- Core idea: environmental emergencies = physiologic injury from heat/cold, pressure changes (diving/altitude), electricity/lightning, toxins (CO, cyanide, chemicals), radiation, blast. Treat physiology first (ABCs, oxygen, circulation) and reverse offending exposure (remove from heat/contaminant, descent or recompression for diving).
- Mechanisms:
- Heat: direct thermal cellular injury → systemic inflammatory response, coagulopathy, rhabdomyolysis, multi-organ failure.
- Cold: (not expanded in chunk notes) hypothermia causes coagulopathy, arrhythmias, altered drug kinetics — manage per hypothermia algorithms.
- Diving: pressure changes → inert gas supersaturation → bubble formation (DCS) and barotrauma; arterial gas embolism from lung over-expansion.
- Altitude: hypobaric hypoxia → AMS, HACE, HAPE via hypoxic pulmonary vasoconstriction and cerebral/ pulmonary oedema.
- Electricity/lightning: direct tissue/nerve/heart injury, thermal burns, muscle necrosis → rhabdomyolysis, arrhythmias (AC more VF risk; DC more asystole/ tetany — simplification).
- CO: binds haemoglobin (COHb) and impairs O2 delivery; pulse oximetry unreliable.
- Cyanide (smoke inhalation): inhibits cellular respiration → very high lactate, rapid collapse.
- HAZMAT/radiation: local and systemic toxicity; radiation causes prodrome (nausea/vomiting) and lymphocyte kinetics predict dose.
- Initial ED priorities: scene safety (PPE, hot/warm/cold zone), remove contaminated clothing, ABCs, high-flow O2 for hypoxia/CO/DCS/AGE, IV access, monitor, early specialist alerts (hyperbaric, toxicology, HAZMAT, radiation).

Arabic summary (مختصر):
- الطوارئ البيئية تنتج عن حرارة/ضغط/تيار/سموم/إشعاع؛ البدء بالحماية ثم ABCs وإزالة التعرض؛ الأكسجين 100% مهم مبكراً لبعض الحالات (CO، غوص، انصمام فقاعي).

2. Recognition, differential diagnosis, and red flags

- Diving
- Nitrogen narcosis: euphoria, impaired judgment, incoordination at depth (classically >≈30 m); improves on ascent.
- Decompression sickness (DCS): after ascent — musculoskeletal “bends”, skin changes, fatigue, delayed neurologic signs (minutes–hours).
- Arterial gas embolism/pulmonary barotrauma: immediate chest pain, dyspnoea, sudden neurologic deficits (stroke-like) after ascent.
- Inner-ear barotrauma: vertigo, tinnitus, vomiting during descent.
- Heat illness
- Heat exhaustion: temp <40°C, preserved mental status, profuse sweating, cramps.
- Heat stroke: core temp ≥40°C with CNS dysfunction (confusion, seizures, coma) — emergency.
- Red flags: rapid deterioration, coagulopathy, very high CK, oliguria/AKI, DIC, severe acidosis.
- Electrical/lightning
- Lightning: Lichtenberg figures (fern-like), keraunoparalysis (transient limb paralysis/pallor/coolness), possible arrhythmia.
- Electrical injury: burns at entry/exit, myoglobinuria, fractures, delayed arrhythmias.
- CO and smoke inhalation
- Nonspecific symptoms (headache, nausea, confusion); SpO2 often misleading (normal). Cyanide: rapid collapse, severe lactic acidosis.
- Red flags: LOC, neuro deficits, cardiac ischemia, very high lactate.
- HAZMAT/chemical
- Clues: multiple victims, unusual odors, visible contamination, ocular irritation, rapid skin damage. Caustic exposures require wet decontamination.
- Radiation and blast
- Radiation: early nausea/vomiting may indicate significant exposure.

- Blast: primary blast injures air-filled organs (lungs, ears); initial normal exam possible — consider 24-hour observation.

Arabic summary (مختصر):
- ملاحظة الوقت (أثناء الغطس مقابل بعد الصعود) مهمة للتفريق بين narcosis وDCS؛ قياس COHb مهم عند الاشتباه في تسمم أول أكسيد الكربون؛ ضربة الشمس = حرارة ≥40° مع خلل عصبي.

3. Diagnostic approach and interpretation

EXAM TRICK / تريكة امتحانية
- Immediate: ABCs, continuous cardiac monitoring, SpO2 (note limitations), core temperature (rectal/oesophageal preferred), blood glucose, neuro exam.
- Targeted tests:
- CO suspicion: arterial/venous blood gas with CO-oximetry for COHb (pulse oximetry unreliable).
- Smoke inhalation: ABG, CO-oximetry, lactate (for cyanide), ECG (ischemia/arrhythmia), CXR.
- Heat stroke: electrolytes, CK, creatinine, LFTs, coagulation, urinalysis for myoglobin.
- Diving/neuro deficits: CT brain ± CT angiography if stroke suspected; CXR for barotrauma/pneumothorax; urgent hyperbaric evaluation.
- Radiation: baseline FBC and serial lymphocyte counts q6–12 h to estimate dose.
- HAZMAT: decontaminate first; consult poison control; collect exposure history and substances.
- Imaging: CXR (pneumothorax, pulmonary contusion, blast lung), CT brain/spine for suspected AGE or focal deficits, FAST/CT for trauma as indicated.
- Interpretation pitfalls:
- Normal pulse oximeter does NOT exclude CO poisoning.
- Early normal CXR/CT does not exclude blast lung or delayed DCS manifestations.

Arabic summary (مختصر):
- فحوصات رئيسية: CO-oximetry لقياس COHb، CK ومؤشرات الكلوى لضربات الشمس/الصدمة العضلية، تكرار عد الدم للحالات الإشعاعية، تصوير للصدر/المخ حسب الأعراض.

4. Emergency management algorithms

- General immediate steps (all environmental exposures)
1. Scene safety → PPE, establish hot/warm/cold zones.
2. Remove patient from exposure → remove clothing (removes ~80–90% contamination).
3. Airway, 100% O2 if hypoxia/inhalation/CO/DCS/AGE, cervical spine as indicated.
4. IV access, cardiac monitor, temperature, neuro monitoring; early specialist alerts (hyperbaric/toxicology/HAZMAT).
- Heat illness
- Heat exhaustion: cool, hydrate (oral/IV), monitor.
- Heat stroke (core ≥40°C + CNS dysfunction):
1. ABCs; intubate if altered airway.
2. Rapid cooling: ice-water immersion is gold standard for exertional heat stroke. If immersion unavailable use evaporative cooling (spray + fans) and cold packs to groin/axillae.
3. IV fluids, monitor for rhabdomyolysis, DIC, electrolytes; avoid antipyretics as primary therapy.
- Diving (narcosis/DCS/AGE)
- Nitrogen narcosis: ascent to shallower depth, O2, supportive.
- DCS: 100% O2, immobilize, IV fluids, analgesia, urgent hyperbaric recompression (definitive).
- AGE/pulmonary barotrauma: 100% O2, consider supine or left lateral decubitus for suspected venous air embolism (traditional); urgent hyperbaric consult; treat tension physiology as per trauma.
- Electrical/lightning
- ABCs, monitor ECG (risk of arrhythmia); check CK, renal function, urine output target >1 mL/kg/hr initially for rhabdomyolysis; manage burns; consider fasciotomy for compartment syndrome.
- Lightning-specific: expect possible transient paralysis (keraunoparalysis) — monitor and supportive care.
- CO and smoke inhalation
- 100% high-flow O2 immediately.
- CO: measure COHb by co-oximetry. Consider hyperbaric O2 for severe cases (LOC, neuro deficits, cardiac ischemia, severe acidosis, pregnancy) per local protocol [thresholds vary — verify locally].
- Cyanide: if suspected (smoke inhalation + high lactate, rapid deterioration), give

hydroxocobalamin 5 g IV (Cyanokit) empirically while arranging further care [Cyanokit].
- HAZMAT/chemical/radiation
- Establish zones; remove clothing; dry decontamination for non-caustic powders; wet decontamination for caustics/unknown/CBRN in warm zone; staff PPE.
- Radiation: decontaminate, baseline FBC, serial lymphocytes; KI only for confirmed/likely radioactive iodine exposure per public-health guidance.
- Altitude illnesses
- AMS: descent, rest, acetazolamide (commonly 125 mg PO twice daily for prophylaxis/tx) and oxygen as needed [CDC Yellow Book 2024].
- HACE: immediate descent, dexamethasone 8 mg then 4 mg q6h, oxygen, urgent evacuation.
- HAPE: descent, oxygen, nifedipine SR (e.g., 30 mg q12h or 20 mg q8h) and consider PDE5 inhibitors/salmeterol in selected cases per guidance [CDC Yellow Book 2024]. Verify specific dosing locally.

> Arabic summary (مختصر للخوارزميات):
> - خطوات عامة: سلامة المكان → إزالة الملابس → ABCs → أكسجين 100% → إعلام اختصاصي.
> - ضربة الحرارة المجهدة: غمر بارد (غمر بالماء المثلّج) أسرع علاج؛ سيانيد مشتبه = هايدروكسيوكوبالامين 5 جم وريدياً؛ غوص مع DCS/AGE = أكسجين 100% ثم إعادة ضغط عاجلة.

5. Drugs, procedures, disposition, and follow-up

- Key drugs/doses (established/commonly cited):
- Oxygen 100% for CO, DCS/AGE, smoke inhalation, heat stroke as needed.
- Hydroxocobalamin 5 g IV for suspected cyanide poisoning (Cyanokit).
- Acetazolamide for AMS prophylaxis/treatment: commonly 125–250 mg PO twice daily (prophylaxis usually started 24–48 h prior) [CDC Yellow Book 2024]. (Pediatric dosing noted in some sources — verify locally.)
- Dexamethasone for HACE: 8 mg IV/IM/PO once, then 4 mg q6h [CDC Yellow Book 2024].
- Nifedipine SR for HAPE prevention/treatment: 30 mg q12h or 20 mg q8h (examples in CDC Yellow Book 2024) — verify local practice.
- Sildenafil/tadalafil reported as adjuncts for HAPE prevention (sildenafil 50 mg q8h / tadalafil 10 mg bid reported in literature) — evidence variable; verify locally.
- Procedures:

- Rapid cold-water immersion for exertional heat stroke.
- Decontamination: remove clothing; dry decon first for powders; wet decon for caustics/unknown in warm zone.
- Hyperbaric recompression for DCS/AGE and selected CO cases — urgent transfer to specialist centre.
- Airway control and intubation for altered consciousness/compromised airway.
- Chest decompression/chest tube for tension pneumothorax/persistent barotrauma as per trauma protocols.
- Fasciotomy for compartment syndrome from electrical injury if indicated.
- Disposition:
- Heat stroke: admit to critical care; monitor for organ failure.
- DCS/AGE: urgent transfer to hyperbaric centre; consider ICU post-recompression.
- Electrical injuries: observe for 24 hours for arrhythmia if significant exposure/syncope; admit if CK rise/renal risk/burns.
- CO: consider admission for symptomatic or high COHb, pregnancy, cardiac ischemia, neuro signs; local hyperbaric criteria apply.
- HAZMAT: admit as per toxicity and exposure; decontaminated patients may still need monitoring.
- Blast: consider 24-hour observation even if initially asymptomatic.
- Follow-up:
- Neuro/cognitive follow-up after CO or AGE/DCS due to delayed neurologic sequelae.
- Renal follow-up if rhabdomyolysis or AKI.
- Occupational/public-health notification for significant HAZMAT/radiation incidents.

> Arabic summary (مختصر الأدوية/الإجراءات/التنويل):
> - أدوية رئيسية: هايدروكسيوكوبالامين 5 جم للسيانيد؛ أسيتازولاميد 125–250 مج مرتين يومياً للوقاية/علاج AMS؛ ديكساميثازون للـ HACE.
> - إجراءات رئيسية: غمر بارد لضربة الحرارة المجهدة، ديكونتامينيشن منظم، وإحالة عاجلة للغرف الضاغطة عند DCS/AGE.

6. High-yield FRCEM exam tricks and common pitfalls

- Timing clues: narcosis/inner-ear barotrauma occur at depth/descent; DCS/AGE present after ascent (AGE immediate; DCS may be delayed) — use timing to choose diagnosis.

- Pulse oximetry is unreliable in CO poisoning — exam answers that rely on normal SpO2 to exclude CO are incorrect.
- “Hot/dry” vs “sweaty” skin: classic teaching for classic vs exertional heat stroke is not reliable — use core temp + CNS signs.
- Decontamination: dry decon first for powders/non-caustic solids; wet decon for caustics/unknown/CBRN — do NOT wet powder before removing it.
- Lightning specific: Lichtenberg figures pathognomonic; do not mistake for thermal burns.
- Rhabdomyolysis: urine output target >1 mL/kg/hr initially, monitor CK, treat with aggressive IV fluids to prevent AKI.
- Hyperbaric oxygen: many exam stems ask about indications — remember LOC, neuro deficits, cardiac ischemia, pregnancy are commonly cited triggers for urgent hyperbaric consultation but local thresholds vary.
- Altitude exam tips: acetazolamide is the “prophylaxis” answer; dexamethasone is the immediate drug for HACE; nifedipine is for HAPE.
- Radiation: early vomiting/nausea within hours is the best early indicator of significant whole-body exposure — exam may test lymphocyte kinetics (fall in lymphocytes).
- Electrical exposures: AC (50–60 Hz) more likely to cause VF; DC more likely to produce asystole/tetany — exam may test rhythm associations.
- Don’t assume normal initial imaging excludes blast or DCS — observation and re-evaluation are often required.

> Arabic summary (نصائح للامتحان):
> - استخدم توقيت الأعراض للتفريق بين narcosis وDCS؛ لا تعتمد على oximeter لرفض تسمم CO؛ أسيتازولاميد للوقاية من AMS، ديكساميثازون للـ HACE.

7. Arabic rapid-review summary (مراجعة سريعة بالعربية)

> - مفاهيم أساسية: الطوارئ البيئية تشمل الحرارة، الغوص، الكهرباء/البرق، السموم (CO/سيانيد)، الإشعاع، والانفجارات. العلاج الأولي = سلامة الموقع → ABCs → أكسجين 100% عند الشك → إزالة الملابس/الديكونتامينيشن → تنبيه الأخصائيين.
> - حرارة: ضربة الشمس = حرارة ≥40°C + خلل عصبي → غمر بالماء البارد (مجهدة) أو تبريد تبخيري (كلاسيكية)، مراقبة للـ rhabdo وDIC.
> - غوص: narcosis يحدث في الأعماق ويزول بالصعود؛ DCS بعد الصعود يعالج بالأكسجين 100% وإعادة ضغط عاجلة.

- كهرباء/برق: فحص لـ ECG، CK، رصد البول لصالح تجنّب الفشل الكلوى؛ كيرونوباراليسيس قد يكون عابراً.
- CO/دخان: أعطِ أكسجين 100% فوراً؛ استخدم co-oximetry لقياس COHb؛ هايدروكسيوكوبالامين 5 جم وريدياً عند اشتباه بسيانيد.
- HAZMAT: جفف أولاً للمواد الغير كاوية؛ اغسل بالماء للمواد الكاوية/غير المعروفة؛ افصل المناطق (hot/warm/cold).
- ارتفاعات: أسيتازولاميد للوقاية؛ ديكساميثازون للـ HACE؛ نيفيديبين للـ HAPE — راجع إرشادات محلية للجرعات.

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية

- Diving: nitrogen narcosis, decompression sickness (DCS), arterial gas embolism (AGE), pulmonary/inner-ear barotrauma.
- Heat illness: heat exhaustion, exertional heat stroke, classic heat stroke, rhabdomyolysis complications.
- Electrical and lightning injuries: arrhythmias, keraunoparalysis, rhabdomyolysis, burns.
- Inhalational toxins: carbon monoxide poisoning (COHb interpretation), smoke inhalation, cyanide (hydroxocobalamin).
- HAZMAT/chemical: decontamination principles (hot/warm/cold zones, dry vs wet decon), PPE.
- Radiation: acute radiation syndrome basics, lymphocyte kinetics, decontamination, KI for radioactive iodine per public-health guidance.
- Blast injuries: primary blast effects (lungs/ears/GI), observation principles.
- Altitude medicine: AMS, HACE, HAPE — acetazolamide, dexamethasone, nifedipine and adjuncts.
- Diagnostics and pitfalls: CO-oximetry vs pulse oximetry, CK/renal monitoring for rhabdo, imaging for barotrauma/neurology.
- Practical emergency algorithms: immediate O2, scene safety, urgent hyperbaric referral, rapid cooling for heat stroke, hydroxocobalamin for suspected cyanide.

Final checks / flagged items (verify locally)

- Pediatric acetazolamide dosing cited in some sources — confirm with paediatric guidelines.

- Sildenafil/tadalafil roles and exact dosing for HAPE prevention/treatment — evidence variable; verify specialty guidance.
- Exact numeric COHb thresholds for hyperbaric oxygen vary between centres — follow local hyperbaric protocols.
- Positioning for gas embolism (left lateral/Trendelenburg) is traditional; modern emphasis is immediate 100% O2 and urgent recompression — verify local resuscitation guidance.

Sources to verify
- RCEM Learning, CDC Yellow Book 2024 (altitude guidance), Cyanokit (hydroxocobalamin), RCEM/UK, European Resuscitation Council (ERC), NICE, local hyperbaric centre protocols, national HAZMAT and public-health guidance.

CHAPTER 26: GASTROENTEROLOGY & HEPATOLOGY

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Chapter 26: Gastroenterology & Hepatology

1. Clinical overview and pathophysiology

- Upper GI bleeding (UGIB)
- Common causes: peptic ulcer disease (most common), variceal haemorrhage (portal hypertension, cirrhosis), Mallory-Weiss tear, oesophagitis. Pathophysiology: mucosal erosion, portal-hypertension variceal rupture.
- Risk stratify with Glasgow-Blatchford Score (GBS).
- Variceal haemorrhage
- Portal hypertension → variceal formation and rupture; large-volume haematemesis, high rebleeding/mortality risk.
- Management includes vasoactive therapy, antibiotics, urgent endoscopic therapy.
- Spontaneous bacterial peritonitis (SBP)
- Gut bacterial translocation in cirrhosis with ascites (commonly E. coli). Diagnostic paracentesis essential.
- Ascending cholangitis
- Biliary obstruction → infection (Charcot triad: RUQ pain, fever, jaundice; Reynolds pentad adds hypotension, altered mentation).
- Acute liver failure (ALF)
- Rapid loss of hepatic function (coagulopathy, encephalopathy); paracetamol common cause.
- Metabolic/other hepatology emergencies
- Alcoholic ketoacidosis (AKA): binge ethanol + starvation → high AG metabolic acidosis with ketones.
- Refeeding syndrome: phosphate, magnesium, potassium shifts on re-feeding malnourished patients.
- C. difficile colitis: recent antibiotics → pseudomembranous colitis.
- Oesophageal disease: eosinophilic oesophagitis (progressive dysphagia), candida (odynophagia in immunosuppressed).
- Food bolus impaction: commonly meat bolus in oesophagus; risk of complete obstruction/aspiration.

موجز سريري/فيزياء المرض
- نزف علوي: القرحة الشائعة؛ دوالي المريء نتيجة ارتفاع الضغط البابي.
- SBP: عبور بكتيري من الأمعاء إلى البريتوني.
- التهاب القناة الصفراوية: انسداد مُعدٍ للقناة الصفراوية.
- الفشل الكبدي الحاد: فقد سريع للوظيفة مع اضطراب تخثر واعتلال دماغي.
- متلازمات أيضية: AKA وإعادة التغذية وفيروس/بكتيريا معوية مميزة.

2. Recognition, differential diagnosis, and red flags

- UGIB recognition
- Presentation: haematemesis ± melaena, syncope, hypotension, tachycardia. Red flags: haemodynamic instability, ongoing large-volume haematemesis, fresh PR bleeding suggesting massive bleeding.
- Differentiate variceal vs non-variceal: cirrhosis/alcohol/portal HTN → variceal likely.
- SBP
- Suspect in any cirrhotic with ascites + fever, abdominal pain, encephalopathy, unexplained deterioration. Red flags: peritonism, severe focal pain → consider secondary peritonitis/surgical abdomen.
- Cholangitis
- Suspect with RUQ pain + fever ± jaundice. Red flags: hypotension and confusion (Reynolds pentad) → septic shock, urgent drainage.
- ALF
- New encephalopathy + coagulopathy after acute insult. Red flags: grade 3–4 encephalopathy, rapid INR rise, acidosis.
- Other red flags
- Food bolus with drooling/complete obstruction → urgent airway/endoscopy.
- Massive haematemesis → early intubation for airway protection.
- Refeeding: early hypophosphataemia, arrhythmias/seizures.

التعرّف والتفرقة والعلامات الحمراء
- علامات حمراء: نزيف كبير، اختلال ضغط الدم، اعتلال دماغي كبدي حاد، صدمة إنتانية في التهاب القناة الصفراوية، ألم بطن بموضع محدد (استبعد التهاب بطني ثانوي).

3. Diagnostic approach and interpretation

- General priorities: ABC → monitor; targeted history (cirrhosis, NSAIDs, alcohol, varices, recent antibiotics, foreign-body ingestion).
- UGIB
- Immediate: two large-bore IVs, type & cross if significant bleed, labs (Hb, platelets, coagulation, urea, LFTs, group & save), GBS for risk stratification (GBS = 0 → consider outpatient management).
- Imaging/endoscopy: urgent endoscopy for unstable or suspected variceal bleed (within 12 h). Stable non-variceal: endoscopy within 24 h [NICE CG141].
- Note: Urea rises with upper GI blood.
- SBP
- Diagnostic paracentesis in ED for any cirrhotic with signs of infection or decompensation. Send cell count, culture, albumin. PMN ≥250 cells/mm3 diagnostic.
- Differentiate from secondary peritonitis by focal peritonism, high neutrophil count, imaging.
- Cholangitis
- Blood cultures, LFTs, WCC, bilirubin. Abdominal ultrasound to detect dilation; CT if diagnostic uncertainty. ERCP for therapy once stabilised.
- ALF
- Labs: INR, bilirubin, creatinine, arterial blood gas, lactate, ammonia, paracetamol level. Apply King's College Criteria for transplant need.
- Other diagnoses
- C. difficile: stool testing and clinical severity guide therapy; recent antibiotic exposure is key.
- Oesophageal candida: visual/clinical diagnosis; endoscopy if atypical.
- Food bolus: bedside assessment (can patient swallow saliva?), lateral neck/chest X-ray rarely helpful — endoscopy if complete obstruction or failed conservative measures.

التشخيص والتفسيرات
- خط أساس: ABC ثم فِحوصات موجهة. SBP يُشخّص عبر بزل البطن وPMN≥250. جراحية إذ وُجد ألم موضعي شديد. قيّم حاجة زرع للكبد باستخدام معايير King's College.

4. Emergency management algorithms

- UGIB (concise algorithm)
1. ABC, cardiac monitoring, two large bore IVs, resuscitate with crystalloids.
2. Activate transfusion/major haemorrhage protocol if unstable or ongoing loss.
3. Restrictive transfusion strategy: transfuse if Hb <7 g/dL (higher threshold if ischaemic heart disease) — follow local policy.
4. If cirrhosis/variceal suspicion:
- Start IV antibiotics (e.g., ceftriaxone per local guidance) AND vasoactive therapy (terlipressin or local equivalent) immediately.
- Arrange urgent endoscopy (within 12 h).
5. If massive haematemesis and airway at risk: intubate for airway protection before endoscopy or Sengstaken-Blakemore insertion.
6. Balloon tamponade (Sengstaken-Blakemore) only as a temporising bridge to definitive therapy:
- Insert orally with patient intubated; inflate gastric balloon first; oesophageal balloon inflation ~30–40 mmHg; remove ≤24–48 h; arrange transfer to definitive care.
7. For coagulopathy in major bleed: FFP 15 mL/kg for INR correction during massive haemorrhage.
8. Routine TXA not recommended for UGIB (HALT-IT trial) [HALT-IT 2020].
- SBP
1. Perform paracentesis promptly (before antibiotics if possible).
2. If clinically suspected, start empiric IV third-generation cephalosporin (e.g., ceftriaxone) pending results.
3. Consider albumin infusion per local guideline in high-risk patients (verify hospital protocol).
- Ascending cholangitis
1. Resuscitate, fluids, broad-spectrum IV antibiotics covering gram negatives/anaerobes.
2. Early biliary decompression (ERCP) once patient stabilised; urgent if severe/septic.
- Acute liver failure (paracetamol)
1. Admit to critical care; start hepatoprotective measures (e.g., N-acetylcysteine per local protocol).
2. Apply King's College Criteria; discuss urgent transplant referral when criteria met.

- Alcoholic ketoacidosis
1. IV crystalloid resuscitation, high-dose thiamine before glucose, correct electrolytes, monitor acid–base.
- Refeeding syndrome prevention
1. Identify high-risk patients, start low calories, give thiamine, and monitor/replace phosphate, magnesium, potassium closely.
- Food bolus impaction
1. If able to manage secretions → conservative escalation: carbonated drink trial → antispasmodic (hyoscine/Buscopan) → consider glucagon/GTN/nifedipine trial (variable efficacy).
2. If complete obstruction, drooling, or failed conservative measures → urgent endoscopy.

خوارزميات علاجية طارئة
- نزيف علوي: ABC → استقرار دوراني → مضادات حيوية + عامل وعائي لدوالي → منظار عاجل.
لا تُستخدم أنبوب Sengstaken إلا كجسر بعد تنبيب المريض.
- SBP: بزل ثم مضاد حيوي ثالث الجيل عاجل.
- التهاب القناة: سوائل ومضادّات حيوية ثم تفريغ القناة (ERCP) سريعًا.
- فشل كبدي حاد: NAC واعتبارات زرع حسب معايير King's College.
- إعادة التغذية: بدء متدرّج ومراقبة الفسفور والماغنسيوم والبوتاسيوم.

5. Drugs, procedures, disposition, and follow-up

- Key drugs/principles
- Vasoactive agents for variceal bleed: terlipressin (local dosing protocols vary) — start early.
- Antibiotics for cirrhotic UGIB: ceftriaxone commonly used; adjust for local resistance.
- IV third-generation cephalosporin for suspected SBP (e.g., ceftriaxone).
- N-acetylcysteine for paracetamol ALF per local protocol.
- Oral vancomycin or fidaxomicin for C. difficile per local guideline.
- Thiamine before glucose in malnourished/alcoholic patients.
- Key procedures
- Diagnostic paracentesis (ED) for ascites + infection suspicion.
- Urgent endoscopy (OGD) for major UGIB or failed conservative management.
- ERCP for biliary decompression in cholangitis.
- Sengstaken-Blakemore tube as temporary measure for refractory variceal

bleeding (intubate first).
- Consider interventional radiology (TIPS or embolisation) if endoscopy fails for variceal or non-variceal bleeding.
- Disposition and follow-up
- Admit unstable UGIB, variceal bleeds, SBP, cholangitis, ALF to appropriate monitored bed/critical care.
- GBS = 0 may be considered for outpatient management with prompt endoscopy arrangement.
- SBP: hospital admission, complete antibiotic course, follow ascitic fluid cultures; consider secondary prophylaxis.
- Cholangitis: admission, definitive ERCP, follow-up hepatobiliary surgery/ gastroenterology.
- ALF: critical care + early transplant centre referral if criteria met.
- Food bolus managed successfully: arrange gastroenterology follow-up and investigation for underlying oesophageal disease (e.g., EoE, strictures).
- Practical notes
- Confirm local terlipressin/antibiotic dosing and PPI policies; hospital protocols vary.
- For coagulopathy in massive bleed, use FFP 15 mL/kg; consider PCC for warfarin reversal as per guideline.

الأدوية والإجراءات والمآل والمتابعة
- أدوية أساسية: تيرليبريسين للدوالي، سيفترياكسون لـSBP/وقاية نزفيّة لدى تليفى الكبد، NAC لفشل كبدي دوائي.
- إجراءات: بزل البطن تشخيصي، منظار عاجل، ERCP للتفريغ الصفراوي، Sengstaken كحل مؤقت بعد تنبيب.
- التنويم: حالات خطرة أو فشل علاجي—وضع مراقبة/عناية مركزة، إحالة مبكرة لمراكز الزرع عند الحاجة.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية
- Paracentesis rule: any cirrhotic with ascites + fever/abdominal pain/ encephalopathy → do paracentesis (exam classic).
- GBS trick: GBS = 0 is the classic safe outpatient discharge answer in exam vignettes [NICE CG141].

- Sengstaken tube pitfalls: never insert without securing airway — always intubate first in massive haematemesis scenarios.
- Variceal bleed bundle: antibiotic + vasoactive agent + urgent endoscopy — omission is a common negative exam trap.
- TXA in UGIB: HALT-IT trial showed no routine benefit — answer exam questions asserting “not recommended routinely.” [HALT-IT 2020]
- FFP vs PCC: FFP 15 mL/kg used in massive bleed coagulopathy; PCC used for rapid warfarin reversal per local policy — read the vignette for anticoagulant context.
- PPI pre-endoscopy: guideline variability—NICE does not recommend routine IV PPI for undifferentiated UGIB; exam distractors may exploit this — check stem clues for suspected non-variceal bleed.
- Refeeding syndrome: phosphate falls first — do not wait for severe symptoms; monitor and replace early.
- Food bolus: attempt conservative measures only if no airway compromise; exam will often have drooling/stridor as red flags for immediate endoscopy.
- Alcoholic ketoacidosis vs DKA: exam distinction relies on history (alcohol, starvation) and relatively low glucose in AKA; management prioritises IV fluids and thiamine.
- ALF transplant threshold: remember King's College Criteria — pH <7.3 single criterion for paracetamol ALF; else combination of INR/creatinine/encephalopathy.

نصائح للامتحان ومزالق شائعة
- بزل البطن دائمًا عند الاشتباه بـSBP في مريض تليفي.
- GBS=0 إجابة شائعة للمناظير الخارجيّة.
- تنبيب المجرى التنفسي قبل أي محاولة لإدخال أنبوب Tamponade.
- HALT-IT: لا تعتمد على TXA روتينيًا في النزف الهضمي العلوي.

7. Arabic rapid-review summary (مراجعة سريعة بالعربية)

- نزف علوي: قيّم ABC؛ استخدم GBS؛ دوالي → مضادّ حيوي + تيرليبريسين + منظار عاجل. (NICE CG141)
- SBP: بزل البطن تشخيصي؛ PMN ≥250 خلية/مم3؛ ابدأ سيفترياكسون مبدئيًا.
- التهاب القنوات الصفراوية: حمى + ألم RUQ + يرقان → سوائل، مضادّات حيوية ثم ERCP للتفريغ.

- فشل كبدي حاد بعد باراسيتامول: تطبيق معايير King's College → إحالة زرع مبكرة.
- متلازمة إعادة التغذية: راقب واستبدل الفسفور والماغنسيوم والبوتاسيوم؛ أعطِ الثيامين قبل الجلوكوز.
- انسداد طعام بالمريء: جرّب مشروبات غازية، مرخِّيات؛ منظار عاجل إذا فشل أو اكتمال انسداد.
- تحذيرات: لا تُدخل Sengstaken دون تنبيب؛ TXA ليس روتينياً في UGIB (HALT-IT).

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية

- UGIB & risk stratification: GBS, transfusion strategy, major haemorrhage, endoscopy timing (concepts often in Qs on UGIB/upper GI haemorrhage).
- Variceal haemorrhage: antibiotics, vasoactive therapy, urgent endoscopy, Sengstaken tamponade, airway protection.
- SBP: diagnostic paracentesis, PMN ≥250, empiric third-generation cephalosporin.
- Ascending cholangitis: Charcot triad, Reynolds pentad, resuscitation + ERCP.
- Acute liver failure (paracetamol): King's College Criteria, transplant referral, NAC.
- Alcoholic ketoacidosis vs DKA: recognition and initial management (fluids, thiamine).
- Refeeding syndrome: phosphate-centric electrolyte management.
- C. difficile colitis: recognition and first-line agents (vancomycin/fidaxomicin per local guideline).
- Oesophageal disorders & food bolus: eosinophilic oesophagitis, candida, conservative escalation, endoscopy indications.
- Coagulopathy in bleeding: FFP dosing during massive haemorrhage, PCC vs FFP considerations.
- Exam/pitfall topics: TXA (HALT-IT), PPI pre-endoscopy variability, Sengstaken tube airway rule.

خريطة تغطية الأسئلة (المفاهيم)
- تشمل كل مفاهيم النزف العلوي، دوالي المريء، SBP، التهاب القناة الصفراوية، الفشل الكبدي الحاد، متلازمات أيضية مهمة، ومداخلات الطوارئ الأساسية.

Sources to verify
- NICE CG141 (upper GI bleeding), King's College criteria for paracetamol ALF, HALT-IT trial (Lancet 2020) re: TXA,

local hospital protocols for terlipressin/antibiotics/ERCP, RCEM/ED/critical-care transfer pathways, BSG/EASL guidance where applicable.

(Study note: This chapter is for exam preparation and general emergency management principles. Always follow local protocols and consult specialist services for individual patient care.)

CHAPTER 27: ENDOCRINOLOGY

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Chapter 27 — Endocrinology (compact FRCEM final chapter)

1. Clinical overview and pathophysiology

- Core idea: endocrine emergencies present as metabolic instability ± volume/electrolyte disturbance, acute neuro or cardiovascular compromise. Rapid recognition and hormone replacement/supportive care are often life-saving.
- Key syndromes: adrenal crisis (primary/secondary), diabetic ketoacidosis (DKA) including euglycaemic DKA (SGLT2i), hyperosmolar hyperglycaemic state (HHS), severe hyponatraemia / SIADH, central diabetes insipidus (post-op), pituitary apoplexy, myxoedema coma, thyroid storm, pheochromocytoma crisis, severe hypercalcaemia, toxic high anion gap acidoses (eg. pyroglutamic), toxic alcohols.
- Physiologic rules to memorise:
- Primary adrenal insufficiency → hypotension + hyponatraemia + hyperkalaemia.
- DKA → high anion gap metabolic acidosis, ketonaemia, total body K+ depleted despite serum K+ often normal/high initially.
- SIADH → hypotonic euvolemic hyponatraemia with inappropriately concentrated urine (Uosm >100) and urine Na >30–40 mmol/L.
- Central DI → large polyuria (>3 L/day), hyperosmolar plasma, dilute urine.
- Pituitary apoplexy → sudden severe headache, visual field loss, ocular palsies, hypotension from secondary adrenal failure.
- Exam relevance: fluid/electrolyte patterns + specific triggers (surgery, infection, drugs like SGLT2i or DDAVP) point to diagnosis.

> Arabic (مختصر): الطوارئ الهرمونية تسبب اضطراب استقلابي وحجم/إلكتروليتات أو اختلال عصبي/قلبي؛ التعرف السريع وتعويض الهرمونات ضروريان للنجاة.

2. Recognition, differential diagnosis, and red flags

- Adrenal crisis:
- Presentation: acute hypotension, vomiting, abdominal pain, weakness, confusion.
- Key labs: hyponatraemia, hyperkalaemia (primary), very low random cortisol.

- Differentials: sepsis, GI bleed, cardiac causes.
- Red flag: shock unresponsive to fluids → give hydrocortisone immediately.
- DKA / euglycaemic DKA:
- Presentation: polyuria, polydipsia, dehydration, Kussmaul breathing, abdominal pain, confusion.
- DKA labs: ketonaemia/ketonuria, metabolic acidosis (pH <7.3 or HCO3 <15), hyperglycaemia (usually >11 mmol/L). Euglycaemic DKA: same acidosis + ketones with normal/modest glucose — suspect SGLT2 inhibitors.
- Red flag: falling serum K+ after insulin → arrhythmia risk.
- HHS:
- Very high glucose, hyperosmolarity, little/no ketones, severe dehydration, altered mental state.
- Hyponatraemia / SIADH:
- Classify by tonicity and volume status. SIADH: euvolemic, hypotonic serum, Uosm >100, UNa >30–40, normal adrenal/thyroid.
- Red flags: seizures, severe obtundation → urgent hypertonic therapy.
- Central DI:
- Large-volume dilute urine, hypernatraemia/high plasma osmolality, recent pituitary surgery/trauma.
- Red flag: rapid sodium change after desmopressin → monitor closely.
- Pituitary apoplexy:
- Sudden severe headache, bitemporal field loss, ocular palsies, hypotension. Red flag: visual deterioration, shock → urgent steroids + neurosurgical review.
- Thyroid storm / myxoedema coma:
- Thyroid storm: high fever, severe tachycardia/AF, delirium, GI dysfunction.
- Myxoedema coma: hypothermia, bradycardia, hypoventilation, hyponatraemia, hypoglycaemia; treat steroids first if adrenal insufficiency not excluded.
- Pheochromocytoma:
- Paroxysmal headache, sweating, palpitations, labile/severe hypertension.
- Toxic/metabolic differentials:
- Pyroglutamic acidosis: high AG metabolic acidosis in elderly/female on chronic paracetamol/sepsis/malnutrition.
- Toxic alcohols: osmolar gap present.

Arabic (مختصر): اذكر علامات الخطر: صدمة لا تستجيب للسوائل → هيدروكورتيزون؛ صرع/اضطراب وعي بفرط ناقص الصوديوم → سالين مفرط التركيز؛ DKA بدون سكر مرتفع → فكر SGLT2i.

3. Diagnostic approach and interpretation

- Immediate bedside/urgent tests:
- Capillary and serum glucose; VBG/ABG (pH, HCO3); serum electrolytes (Na, K, Cl, HCO3), urea, creatinine, calcium; serum osmolality (measured if possible); urine osmolality and urine Na; blood ketones; random cortisol; ECG.
- Calculations and key rules:
- Anion gap = Na − (Cl + HCO3). Normal ≈ 8–12 mmol/L.
- Estimated serum osmolality = 2×Na + urea + glucose (all mmol/L). Osmolar gap = measured − calculated; gap >10–20 suggests toxic alcohols.
- Winter's formula for expected pCO2 in metabolic acidosis: pCO2 ≈ 1.5×HCO3 + 8 ±2 (use to detect mixed disorders).
- Syndrome diagnostic cues:
- DKA (JBDS DKA guidance): ketones positive, pH <7.3 and/or HCO3 <15 mmol/L, plus hyperglycaemia (commonly >11 mmol/L). Anion gap supportive but not required.
- SIADH: hypotonic serum, urine osmolality inappropriately >100 mOsm/kg, urine Na >30–40 mmol/L, exclude adrenal/thyroid/renal causes.
- DI: dilute urine despite hyperosmolar plasma; formal water-deprivation and desmopressin tests in non-acute settings; in post-op acute DI treat empirically.
- Pituitary apoplexy: urgent MRI with contrast (CT may show haemorrhage but MRI more sensitive) [Society for Endocrinology 2010].
- Practical interpretation tips:
- Serum K in DKA may be misleadingly normal/high; always correct K before insulin if <3.3 mmol/L.
- In suspected pheochromocytoma crisis, plasma/urine metanephrines confirm diagnosis but do not delay acute stabilisation.

Arabic (مختصر): فحوص فورية: غلوكوز، غازات دموية، إلكتروليتات، كيتونات، أوزمولالية/بولية. حسابات: Anion gap و osmolar gap و 'Winters تُفيد التفريق بين الاضطرابات.

4. Emergency management algorithms

(Keep ABCs and monitor ECG throughout; call endocrine/ICU/neurosurgery as indicated.)

Adrenal crisis (suspected/confirmed)
1. ABC, IV access, fluids (0.9% NaCl).
2. Give hydrocortisone 100 mg IV stat. Then either 200 mg/24h as continuous infusion OR 50 mg IV q6h. Do not delay steroids for testing. [Society for Endocrinology 2010]
3. Replace fluids/electrolytes, monitor glucose and K+. Add fludrocortisone later if primary adrenal insufficiency once stable.

Pituitary apoplexy
1. ABC, IV fluids.
2. Give IV hydrocortisone 100 mg immediately, then 50 mg q6h or infusion.
3. Urgent endocrine + neurosurgical review; urgent MRI. Surgical decompression for visual/neurological compromise.

DKA (adult) — initial algorithm (JBDS 2023)
1. ABCs, cardiac monitoring, IV access, bloods (glucose, VBG, electrolytes, ketones).
2. Fluid resuscitation: isotonic crystalloid bolus if hypovolaemic; then guided replacement.
3. Potassium: if K <3.3 mmol/L → replace and HOLD insulin; if 3.3–5.5 → replace per protocol while giving insulin; if >5.5 → withhold K+ and monitor.
4. Insulin: fixed-rate IV infusion 0.1 U/kg/hr. If glucose fall <3 mmol/L/hr, increase by 1 U/hr. When glucose ≈14 mmol/L, add dextrose and continue insulin until acidosis resolves.
5. Transition: give subcutaneous insulin and overlap before stopping infusion once clinically improving and eating.

Euglycaemic DKA (SGLT2 inhibitor)
- Treat as DKA: fluids + fixed-rate insulin infusion + dextrose early if glucose <14 mmol/L so insulin can continue. Stop SGLT2 inhibitor. [RCEMLearning]

HHS
- Aggressive IV fluids (restore volume); insulin later at lower doses after initial rehydration; monitor Na and osmolality; high VTE risk — thromboprophylaxis.

Severe hyponatraemia (severe neuro signs)
1. Give small boluses of hypertonic saline (eg 100–150 mL of 3% saline) per local protocol to raise Na by ~4–5 mmol/L to control life-threatening features.
2. Then slow correction targeting ≈8 mmol/L in 24 h (local protocol dependent). Monitor Na hourly initially.
3. If SIADH: stop offending drugs, fluid restrict, consider demeclocycline/vaptans only with specialist input.

Central DI (acute, post-op)
- If symptomatic: give desmopressin (IV/SC in acute setting) and replace fluids; monitor Na and urine output hourly; beware delayed hyponatraemia.

Thyroid storm
1. Supportive care, cooling, oxygen, fluids.
2. Beta-blocker for rate control (propranolol common; esmolol if short-acting required) unless contraindicated.
3. Give antithyroid drug (PTU preferred in storm for T4→T3 conversion block) then iodine after antithyroid drug.
4. Give glucocorticoids (eg hydrocortisone) and identify precipitant.

Myxoedema coma
- Give IV hydrocortisone 100 mg stat (cover possible adrenal insufficiency), then IV levothyroxine (lower dose if elderly/cardiac disease). Supportive care and treat precipitant.

Pheochromocytoma crisis
- Start alpha blockade (eg phentolamine bolus/infusion or oral/IV selective alpha-1 blockade); only add beta blockade after adequate alpha blockade. Urgent ICU/ specialist input.

Hypercalcaemia (severe/symptomatic)
- IV isotonic saline to restore intravascular volume and promote calciuresis; consider loop diuretics after euvolaemia; IV bisphosphonates for sustained reduction; calcitonin for short-term effect. Specialist input.

Toxic high anion gap acidosis (pyroglutamic / toxic alcohols)
- Treat underlying cause, supportive care, consider fomepizole/ethanol and dialysis for toxic alcohols. Pyroglutamic: consider stopping paracetamol and treating sepsis/malnutrition contributors; specialist toxicology advice.

Arabic (مختصر): خوارزميات العملية: أزمة أدريال → هيدروكورتيزون 100 mg IV فوري؛ DKA → سوائل، إنسولين ثابت 0.1 U/kg/h، مراقبة بوتاسيوم؛ نقص صوديوم شديد → دفعات صغيرة سلفاتى مركز 3% ثم تصحيح بطيء؛ تجنب إعطاء حاصرات بيتا وحدها في فيئوكروموسايتوما.

5. Drugs, procedures, disposition, and follow-up

EXAM TRICK / تريكة امتحانية

- Key emergency doses (widely used/established):
- Hydrocortisone in adrenal crisis/pituitary apoplexy/myxoedema cover: 100 mg IV stat → then 200 mg/24 h either infusion or 50 mg IV q6h [Society for Endocrinology 2010].
- DKA insulin: fixed-rate infusion 0.1 U/kg/h (JBDS DKA 2023).
- Potassium thresholds: if K <3.3 mmol/L → replace before insulin; 3.3–5.5 mmol/L → replace while giving insulin; >5.5 mmol/L → withhold replacement and monitor.
- Severe hyponatraemia: hypertonic saline small boluses (eg 100–150 mL 3% saline) per local protocol to raise Na by 4–5 mmol/L initially.
- Desmopressin: IV/SC/IN depending on acuity; use IV in acute post-op DI.
- Phentolamine bolus often cited (acute pheochromocytoma): specialist/ICU dosing (eg 2–5 mg IV bolus) — verify local guidance.
- Procedures:
- Urgent MRI pituitary for suspected apoplexy; CT if MRI delayed.
- ICU admission indications: shock, need for vasoactive support, ventilatory support, severe altered mental state, refractory electrolyte disturbance (e.g., severe DKA/HHS), severe thyroid storm, pheochromocytoma crisis.
- Endocrine specialty review for all adrenal crisis, pituitary apoplexy, severe hypercalcaemia, persistent DI/SIADH, or complex replacement decisions.
- Disposition & follow-up:
- Admit DKA/HHS patients to monitored ward or ICU per severity; arrange diabetes specialist input and clear sick-day advice.
- Adrenal crisis survivors: endocrinology follow-up, education on steroid sick-day rules, steroid card.
- Pituitary apoplexy: early endocrine + neurosurgical follow-up; visual field testing and pituitary function tests when stable.
- SIADH/hyponatraemia: identify cause, arrange endocrinology/medical follow-up and consider slow outpatient correction if stable.

- Pheochromocytoma suspicion: specialist follow-up for biochemical testing and surgical planning after acute control.
- Practical procedural pitfalls:
- Do not delay steroids for cortisol testing in unstable patients.
- Obtain baseline labs (including cortisol, thyroid function) prior to definitive therapy when feasible, but stabilise first.

Arabic (مختصر): جرعات رئيسية: هيدروكورتيزون 100 mg IV فوري ثم 200 mg/24h؛ إنسولين DKA 0.1 U/kg/h؛ دفعات صغيرة من 3% صوديوم لمصابين بنقص صوديوم حاد؛ مراجعة إندكرينولوجى و/أو ICU حسب الحاجة.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية

- Do not delay empiric IV hydrocortisone in suspected adrenal crisis or pituitary apoplexy — exam frequently tests this.
- In DKA:
- Remember fixed-rate insulin 0.1 U/kg/h (JBDS 2023) and K+ management: if K <3.3 → replace before insulin; do not let insulin precipitate hypokalaemia.
- Euglycaemic DKA is a trap — normal glucose does not exclude DKA (think SGLT2i).
- In pheochromocytoma: always alpha-block before beta-block. Exam scenario trick: starting beta-block alone precipitates hypertensive crisis.
- Thyroid storm/myxoedema: steroid coverage before giving thyroid hormone if adrenal insufficiency possible.
- Hyponatraemia: correction speed is commonly examined — avoid overcorrection (target ≈8 mmol/L/24 h unless local guideline states otherwise).
- Calculations: practise anion gap, osmolar gap, and Winters' quickly — often required in written/short answer questions.
- Diagnostic order: remember which tests to do immediately (glucose, VBG, ketones, electrolytes, urine osm, cortisol) vs. specialist tests (plasma metanephrines, pituitary MRI).
- Watch for delayed effects: desmopressin can cause delayed hyponatraemia; correcting Na too fast → osmotic demyelination days later.
- When asked to prioritise: stabilise ABC and correct immediately

life-threatening biochemistry (eg hypoglycaemia, adrenal crisis, severe hyponatraemia with seizures) before definitive investigations.

Arabic (مختصر): حيل للامتحان: اذكر إعطاء الستيرويد فوراً في اشتباه أزمة قشرية؛ إنسولين DKA 0.1 U/kg/h ومراقبة بوتاسيوم؛ لا تتوقف على سكر طبيعي لاستبعاد DKA (فكر SGLT2i).

7. Arabic rapid-review summary

- النقاط الأساسية السريعة:
- الطوارئ الهرمونية = اضطراب استقلابي + اختلال حجم/إلكتروليتات أو خلل عصبي/قلبي؛ افعل ABC ثم الفحوص الأساسية.
- أزمة قشرية: هيدروكورتيزون 100 mg IV فوري + سوائل؛ لا تؤجل الستيرويد للفحوص.
- DKA: سوائل، إنسولين ثابت 0.1 U/kg/h، مراقبة بوتاسيوم؛ في SGLT2i قد يكون السكر طبيعياً — افحص الكيتونات والحموضة.
- نقص صوديوم حاد مع أعراض عصبية: دفعات صغيرة من 3%؛ ثم تصحيح بطيء (≈8 mmol/L/24h حسب بروتوكول).
- DI مركزي ما بعد عملية: ديسموبريسين IV/SC + مراقبة صوديوم وكمية البول.
- Pituitary apoplexy: هيدروكورتيزون فورى + MRI عاجل + جراحة عصبية عند تدهور بصري/عصبي.
- في فيبوكروموسايتوما: احكم ألفا قبل بيتا؛ تفادى إعطاء بيتا بمفرده.
- احفظ حسابات Anion gap وOsmolar gap وWinter’s formula لتطبيق سريع في الامتحان.

8. Question coverage map (concepts covered)

- Note: original chunk provided concepts rather than numbered question lists. Map of concepts likely to appear in FRCEM coverage:
- DKA (including euglycaemic DKA, JBDS DKA 2023) — diagnosis & management
- HHS — differentiation from DKA and initial resuscitation
- Adrenal crisis and steroid sick-day rules — immediate management
- Pituitary apoplexy — recognition, steroids, imaging, neurosurgical indications
- Thyroid storm and myxoedema coma — initial sequence and treatments
- SIADH and hyponatraemia — classification, emergency hypertonic therapy, correction limits
- Central diabetes insipidus (post-op) — diagnosis & desmopressin use
- Pheochromocytoma crisis — alpha then beta blockade, acute control

- Hypercalcaemia — fluids, bisphosphonates, calcitonin
- Toxic metabolic acidoses: pyroglutamic acidosis, toxic alcohols — AG/osmolar gap & dialysis considerations
- Interpretation skills: anion gap, osmolar gap, Winter’s formula, ECG changes with electrolytes
- If mapping to exam question banks: these topics commonly span multiple short-answer and SBAs across endocrine and metabolic sections — ensure practice across DKA/acid-base, electrolyte emergencies, pituitary emergencies, and thyroid crises.

> Arabic (مختصر): خريطة التغطية: DKA/ euDKA، HHS، أزمة قشرية، نزف النخامية، أزمة درقية/كوما الدرقية، نقص صوديوم/DI، SIADH مركزي، فيوكروموسايتوما، فرط كالسيوم، الحموضات السامة — حسابات Anion/osmolar/Winters.

Sources to verify
- JBDS DKA guideline (2023), Society for Endocrinology guidance (pituitary apoplexy 2010), RCEM / RCEMLearning resources, NICE and ERC acute care guidelines, local hospital endocrine protocols, and toxicology resources (eg. Life in the Fast Lane / LitFL for pyroglutamic). Always check your local trust guideline for exact drug doses and infusion regimens.

(Study material — not patient-specific advice.)

CHAPTER 28: ELDERLY CARE

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Chapter 28 — Elderly Care

1. Clinical overview and pathophysiology

- Age-related change summary: immunosenescence, reduced fever response, altered physiology (less physiologic reserve), higher prevalence of multimorbidity and polypharmacy → atypical presentations (delirium, falls, reduced oral intake), higher baseline prevalence of asymptomatic bacteriuria.
- Catheter-related: chronic indwelling catheters increase bacteriuria risk, biofilm formation, and specific phenomena (purple urine bag syndrome = bacterial tryptophan metabolism producing indigo/indirubin pigments). [BMJ Case Rep 2018]
- Pharmacology/haemodynamics: common offending drugs for orthostatic hypotension include alpha-1 blockers (e.g., tamsulosin), other antihypertensives, diuretics; autonomic dysfunction (eg, diabetic neuropathy) and volume depletion magnify risk.
- Capacity: decision-specific; dementia ≠ automatic lack of capacity. Always assess capacity for the specific treatment/consent decision at hand.

> Arabic:
> موجز فسيولوجي: التقدم في العمر يقلّل الدفاعات المناعية ويغيّر عتبات الأعراض؛ الأمراض المتعددة والأدوية المتعدية تؤدي لعرض غير نمطي مثل الارتباك والسقوط. القثاطر المزمنة تزيد البكتيريّة وتسبب ظاهرة كيس البول البنفسجي (استقلاب التريبتوفان). تقييم الأهلية قرار-محدّد الخرف لا يعني فقدان الأهلية تلقائياً.—

2. Recognition, differential diagnosis, and red flags

- Atypical infection presentation: delirium, new/acute decline in mobility/function, anorexia, falls, in absence of fever or urinary symptoms. Differential: asymptomatic bacteriuria, medication effect, constipation, dehydration, sepsis from non-urinary source, heart failure, pneumonia, metabolic disturbance.
- PUBS: purple discoloration of catheter/bag in long-term catheter users suggests heavy bacteriuria; treat based on clinical infection, not colour alone. [BMJ Case Rep

2018]
- Orthostatic symptoms: consider medications, volume depletion, cardiac arrhythmia, anaemia, autonomic failure. Measure orthostatic BPs (see diagnostic section). Diagnostic thresholds: fall ≥20 mmHg systolic or ≥10 mmHg diastolic on standing.
- Red flags requiring urgent action: signs of sepsis (hypotension, tachycardia, high lactate), hypoxia, altered level of consciousness, inability to maintain oral meds/fluids, perineal/urosepsis signs, obstructive uropathy.

:Arabic
التعرّف والتمييز: لدى المسنين قد تكون العدوى بدون حمّى؛ توقع الارتباك والسقوط كعلامات. كيس البول البنفسجي مؤشر لبكتيريّة عالية ولكن القرار العلاجي يعتمد على الأعراض النظامية. هبوط الضغط الانتصابي يُشتبه به بتراجع ≥20/≤10 ممHg. علامات الخطر: صدمة إنتانية، نقص أكسجين، تغيّر الوعي، عدم القدرة على تناول السوائل/الأدوية.

3. Diagnostic approach and interpretation

- Initial steps: ABC, sepsis screen if unwell, focused history (catheter, recent antibiotics, functional baseline, medications), medication review.
- Urinary testing:
- Do bedside urine dipstick (LE/nitrite) but interpret cautiously — dipstick alone is unreliable in older adults; false positives/positives from asymptomatic bacteriuria common (verify local guidance, age threshold often ≥65).
- Send urine culture (catheter specimen if catheter present) when clinical infection suspected; do not treat culture-positive but asymptomatic patients (exceptions: pregnancy, pre-urological procedure).
- Blood tests: CBC, CRP, renal function, electrolytes; blood cultures if febrile/septic; serum lactate if sepsis suspected.
- Imaging: renal tract ultrasound/CT only for complicated/recurrent infections or suspected obstruction/abscess.
- Orthostatic BP measurement technique: measure supine after 5 minutes, then standing measurements at 1 and 3 minutes. Record symptoms with drops.
- PUBS: obtain catheter urine culture, change catheter/bag, examine for systemic infection before prescribing antibiotics. [BMJ Case Rep 2018]

:Arabic
النهج التشخيصي: ابدأ بتقييم ABC وشاشة الإنتان، راجع الأدوية وحالة القثطار. شرائط البول

مفيدة مبدئياً ولكن غير كافية عند كبار السن؛ أرسل مزرعة بول عند الشك السريري. تحاليل دمّية ومزرعة دم إذا اشتبهت بإنتان نظامي. التصوير مخصص للمضاعفات أو الانسداد. قياس ضغط الدم الانتصابي بعد الاستلقاء 5 دقائق ثم عند 1 و3 دقائق. عند الكيس البنفسجي خذ مزرعة وغيّر القثطار وقيّم وجود أعراض نظامية.

4. Emergency management algorithms

- Algorithm A — Suspected UTI in elderly (compact):
1. ABC + sepsis screen (vitals, lactate if available).
2. If sepsis criteria or haemodynamic instability: give immediate resuscitation (oxygen as needed, IV access, fluids if hypovolaemic), blood cultures, lactate, empiric IV antibiotics promptly per local protocol; change/remove infected catheter if source. Admit to appropriate monitored setting.
3. If not septic but symptomatic (dysuria, fever, new urinary frequency, suprapubic pain, or atypical features with clinical suspicion): send urine culture, consider oral antibiotics based on severity/local guidance; arrange follow-up and review culture results.
4. If asymptomatic bacteriuria: do not treat unless clear indication (pregnancy/urology).
- Algorithm B — Capacity assessment for treatment decisions:
1. Presume capacity. Explain decision, options, risks, benefits.
2. Assess understanding, retention, weighing, and ability to communicate decision.
3. If capacity present, follow patient's informed decision. If lacking, use best-interests process (advance directives, nominated proxy, multidisciplinary input). Document thoroughly.
- Algorithm C — Orthostatic dizziness/falls:
1. Assess ABCs, check injuries. Measure supine (after 5 min) then standing BP at 1 and 3 min.
2. If diagnostic drop: review and stop/hold offending meds (alpha-blocker, diuretic, antihypertensive), assess for hypovolaemia and give fluids if appropriate, institute fall precautions, consider senior/geriatric/neurology referral for recurrent cases. Specialist agents (midodrine/fludrocortisone) only after specialist review.

Arabic:
خوارزميات طارئة:
A — عدوى المسالك: افحص الإنتان أولاً؛ إذا صدق الإنتان أعطِ resuscitation، مزرعات، لا

تتأخر في المضاد الحيوي وريدي وغير القثطار إذا مصدر العدوى. في الحالات الخفيفة أرسل مزرعة واعتبر مضاداً فموياً حسب الدليل المحلي. لا تعالج بكتيريا البول بدون أعراض.
B — تقييم الأهلية: افترض الأهلية ثم قيّم الفهم والقدرة واتبع رغبة المريض أو مبدأ المصلحة الفضلى إذا فقد الأهلية.
C — دوار النهوض: قِس الضغط بعد الاستلقاء و1 و3 دقائق؛ أوقف الأدوية المسببة، عوّض الحجم إذا لازِم، اتخذ إجراءات منع السقوط وأحال عند التكرار.

5. Drugs, procedures, disposition, and follow-up

- Drugs:
- Empiric antimicrobials: follow local microbiology guidance; obtain cultures before antibiotics when possible in non-critical patients. Avoid treating asymptomatic bacteriuria.
- Medications to withhold/stop if orthostatic hypotension suspected: alpha-1 blockers (tamsulosin), diuretics, excessive antihypertensives — individualise.
- Specialist pharmacotherapy for chronic orthostatic hypotension (midodrine, fludrocortisone) only after specialist review.
- Procedures:
- Obtain urine culture (catheter sample if present), blood cultures if sepsis.
- Change or remove indwelling catheter if suspected catheter-related infection or for PUBS management (replace catheter and bag). [BMJ Case Rep 2018]
- Imaging only when indicated (obstruction, recurrent/complicated infection, suspected abscess).
- Disposition:
- Admit: any sepsis/haemodynamic instability, hypoxia, altered consciousness, inability to maintain oral intake/meds, complicated UTI, obstructive uropathy, recurrent falls with risk.
- Consider ED discharge with safety net and oral antibiotics when stable, tolerating PO, able to self-care or with reliable carers, and follow-up arranged.
- Follow-up and prevention:
- Review urine culture results and de-escalate/stop antibiotics accordingly.
- Review and rationalise medications (polypharmacy review). Consider fall-prevention measures, post-falls assessment, community/GP/geriatric follow-up, catheter care education, and treatment plans for recurrent UTIs.
- Document capacity assessment, best-interests decisions, and escalation plan.

Arabic:

الأدوية والإجراءات والتصرّف والمتابعة:
الأدوية: ابدأ مضاداً حيوياً وفقاً لدليل الأحياء الدقيقة المحلي بعد أخذ المزروعات؛ لا تعالج بكتيريا البول بدون أعراض. إذا شُكّ في هبوط الانتصاب أوقف مثبطات ألفا-1 ومدرات البول ومخفضات الضغط المناسبة مؤقتاً. الأدوية المتخصصة تُستخدم بتوجيه أخصائي.
الإجراءات: مزرعة بول/دم، تغيير القثطار عند الاشتباه بعدوى أو الكيس البنفسجي، تصوير للمضاعفات فقط.
التصرّف: دخول المستشفى عند الإنتان، عدم الاستقرار أو عدم القدرة على الرعاية بالمنازل؛ تصريف خارجي عند الاستقرار مع تعليم ومتابعة. المتابعة: تعديل الأدوية، مراجعة ثقافات البول، خطط منع السقوط، إحالة لطب الشيوخ والمجتمع.

6. High-yield FRCEM exam tricks and common pitfalls

- Do not equate dementia with incapacity — exam vignettes often try to trap with that assumption. Always answer using the decision-specific capacity framework.
- Do not treat positive dipstick or urine culture alone in elderly patients — look for clinical signs of infection. Treating asymptomatic bacteriuria is a common error. (Verify local age thresholds/guidance.)
- PUBS: the purple bag is a clue to bacteriuria but not an automatic indication for antibiotics — change catheter/bag and assess systemic features first. [BMJ Case Rep 2018]
- Orthostatic BP measurement: measure at 1 and 3 minutes — immediate only readings can miss delayed orthostatic drops.
- Always ask “is there a catheter?” early in the assessment; review medications as a frequent reversible cause of symptoms.
- Documentation: clear record of capacity assessment, rationale for treating or not treating bacteriuria, and decision to change or leave catheter are scored in viva/OSCEs.

Arabic:

حيل الامتحان والمصائد:
لا تفترض فقدان الأهلية بسبب الخرف. لا تعالج نتائج البول الموجبة دون دليل سريري. الكيس البنفسجي دليل فقط—غيّر القثطار وقَيّم النظامي قبل العلاج بالمضاد الحيوي. قِس الضغط عند 1 و3 دقائق لتفادي الأخطاء. اسأل مبكراً عن وجود قثطار وراجع الأدوية. وثّق قرارات الأهلية والعلاج جيداً.

7. Arabic rapid-review summary

- المسنّون يعرضون عرضاً غير نمطي للعدوى (ارتباك، سقوط، قلة تناول).
- شرائط البول وحدها غير كافية عند كبار السن؛ مزرعة بول ضرورية إذا تُعالج العدوى.
- لا تعالج بكتيريا البول بدون أعراض ما لم توجد موانع/استثناءات واضحة (حمل، تدخلات بولية).
- كيس البول البنفسجي يدل غالباً على بكتيريويا في القثطار — غيّر القثطار وقيّم وجود أعراض نظامية قبل العلاج بالمضادات الحيوية. [BMJ Case Rep 2018]
- هبوط الضغط الانتصابي يُقاس بعد الاستلقاء 5 دقائق ثم عند 1 و3 دقائق؛ راجع الأدوية وغيّرها إذا لزم.
- قيّم وتوثّق القدرة لقرار العلاج بشكل محدّد لكل قرار.

8. Question coverage map (concepts covered)

- Concepts covered (source Q-numbers not provided):
- Atypical infection presentations in the elderly (delirium, falls).
- Asymptomatic bacteriuria vs UTI; limitations of urine dipstick in ≥65.
- Catheter-associated issues including purple urine bag syndrome (PUBS) — diagnosis and management. [BMJ Case Rep 2018]
- Orthostatic hypotension: measurement, causes (drugs, volume, autonomic), immediate ED management.
- Capacity assessment principles—decision-specific model.
- Sepsis recognition and initial resuscitation/antibiotic principles in elderly patients.
- Practical disposition decisions (admit vs discharge, follow-up).
- Note: no original question numbering was supplied with the chunk notes; map therefore lists covered concepts rather than Q-ranges.

Sources to verify
- RCEM guidelines, ERC/European Resuscitation Council, NICE guidance (urinary tract infection and sepsis), Surviving Sepsis Campaign, IDSA guidance on asymptomatic bacteriuria, BMJ Case Reports 2018 (purple urine bag syndrome), local microbiology and trust antibiotic policies.

CHAPTER 29: EAR, NOSE AND THROAT

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Chapter 29 — Ear, Nose and Throat (ENT)

1. Clinical overview and pathophysiology

- Core priorities: secure airway, control life-threatening haemorrhage, treat invasive infection, prevent cartilage necrosis (auricle, septum), preserve hearing. ENT emergencies often threaten airway, cause major bleeding, or progress to sepsis/osteomyelitis.
- Common pathophysiologic clusters:
- External ear: otitis externa → local cellulitis; malignant (necrotizing) otitis externa → Pseudomonas invasion of soft tissue → skull-base osteomyelitis in diabetics/immunocompromised.
- Middle/inner ear and hearing: TM perforation allows middle-ear infection; cochlear injury causes sensorineural loss. SSNHL = rapid cochlear/nerve dysfunction.
- Vestibular: canalithiasis (BPPV) = brief position-triggered vertigo; vestibular neuritis = prolonged peripheral vertigo without cochlear signs; central causes (stroke, tumour) produce atypical nystagmus/focal signs.
- Nasal and oropharyngeal bleeding: anterior (Kiesselbach) vs posterior (sphenopalatine/arterial) sources; post-operative bleeds risk major haemorrhage.
- Soft-tissue/space infections: peritonsillar, parapharyngeal, retropharyngeal, Ludwig's — risk airway compromise and septic thromboembolism (Lemierre's).
- Traumatic/cartilage injuries: auricular or septal haematomas disrupt perichondrial blood supply → cartilage necrosis if not drained.

> شرح عربي موجز:
> - الأولوية: تأمين مجرى الهواء، إيقاف النزف الخطير، معالجة العدوى الغازية، ومنع نخر الغضروف والحفاظ على السمع.
> - الفشل في تصريف الهيماتوما أو تأخير علاج التهابات الأذن الخارجية في مرضى السكر قد يؤديان إلى مضاعفات عظمية خطيرة.

2. Recognition, differential diagnosis, and red flags

- Ear:
- Otitis externa: ear pain, pruritus, otorrhea after swimming; granulation tissue + severe pain in diabetic → malignant otitis externa (Pseudomonas) → red flag: cranial nerve palsy, severe refractory pain.
- Sudden sensorineural hearing loss (SSNHL): rapid unilateral loss (≥30 dB across ≥3 contiguous frequencies within 72 h) — ENT emergency.
- Mastoiditis: post-otitis pain, swelling behind auricle, displaced pinna, systemic signs.
- Vertigo syndromes:
- BPPV: brief (<1 min) position-triggered vertigo, latency then crescendo/decrescendo nystagmus.
- Vestibular neuritis: single prolonged attack (days), severe vertigo, horizontal unidirectional nystagmus, no hearing loss.
- Ménière's: episodic vertigo + fluctuating hearing loss, tinnitus, aural fullness.
- Central (posterior circulation stroke): acute vertigo with focal neuro signs, direction-changing or vertical nystagmus, normal head impulse; red flag: vascular risk factors.
- Throat/neck:
- Peritonsillar abscess: trismus, muffled "hot-potato" voice, uvula deviation away from lesion.
- Lemierre's syndrome: preceding oropharyngeal infection, sepsis, IJV tenderness/thrombosis, pulmonary septic emboli.
- Deep neck space infection or post-op neck haematoma: progressive airway compromise → immediate escalation.
- Nose:
- Anterior epistaxis: bright brisk anterior bleed from Little's area; posterior bleed: sudden profuse hemorrhage, older/comorbid patients.
- Tracheostomy/tracheo-innominate fistula: sentinel bleeding from stoma → massive hemorrhage risk; red flag: brisk arterial haemorrhage, haemodynamic collapse.

> شرح عربي موجز:
> - العلامات الحمراء: شَلَل عصبي محيطي أو مركزي مع ألم أذني شديد، نقص سمع مفاجئ، تدهور تنفّسي أو نزيف عنيف من فغر القصبة أو الأنف، أو هيماتوما غضروفي لم يُصرف.

3. Diagnostic approach and interpretation

- General ED priorities: ABC, haemodynamic resuscitation, analgesia/antiemetics for vertigo, early senior ENT/anesthesia input for airway/major bleeds.
- Ear:
- Otoscopy: canal debris, TM status (intact vs perforated), granulation tissue. Do not irrigate if TM perforation suspected.
- Ear swab for culture if purulent or malignant OE suspected; check glucose/HbA1c in diabetics; ESR/CRP for osteomyelitis workup.
- Imaging: CT temporal bones for mastoiditis/osseous erosion; MRI skull base for suspected skull-base osteomyelitis.
- Vertigo:
- Dix–Hallpike for positional vertigo (BPPV). If continuous vestibular syndrome: HINTS (Head-Impulse, Nystagmus direction, Test-of-Skew) — use only in ongoing vertigo with nystagmus and by trained clinicians; HINTS suggesting central cause = normal head impulse + direction-changing nystagmus or skew deviation. If HINTS suggests central or examiner uncertain → MRI DWI.
- Throat/neck:
- Peritonsillar abscess: clinical diagnosis; consider needle aspiration; ultrasound or CT neck with contrast if deep space infection suspected.
- Lemierre's: blood cultures, CT neck/chest for IJV thrombosis and septic emboli.
- Nose:
- Anterior rhinoscopy to localise bleeding; posterior bleeds suspected when bleed is heavy and not controlled with anterior measures — consider nasal endoscopy or imaging.
- Tracheostomy bleed:
- Bedside recognition: fresh arterial bleeding from stoma; do not remove a tracheostomy tube casually — follow bleeding algorithm (cuff over-inflation, digital compression).

> شرح عربي موجز:
> - التشخيص يعتمد على فحص الأذن بالمصباح، اختبار Dix–Hallpike للدوار الموضعي، وHINTS للدوار المستمر (يُجرى بواسطة فاحص متمرس). استخدم CT/MRI لتقييم التعقيدات العظمية أو الخِرّاجات العميقة.

4. Emergency management algorithms

- Epistaxis — anterior first aid → escalation:
1. Sit upright, lean forward, pinch soft nasal ala over Little's area for 10–15 min continuous.
2. If persistent: topical vasoconstrictor (oxymetazoline/phenylephrine) ± topical anaesthetic; identify visible source → chemical (silver nitrate) or electrocautery.
3. If still bleeding: anterior packing (absorbable or non-absorbable) — document pack type/time; duration typically 24–48 h.
4. If fails or posterior suspicion: posterior packing (balloon/Foley or commercial device) and urgent ENT/IR for endoscopic ligation or embolisation.
5. Reverse anticoagulation if indicated; consider TXA as adjunct per local protocol.
- Tracheostomy / tracheo-innominate fistula (major haemorrhage):
1. Immediate: over-inflate tracheostomy cuff to tamponade.
2. If bleeding persists: insert finger into stoma, press posteriorly to compress brachiocephalic artery against sternum.
3. Activate major haemorrhage/team, call ENT/vascular surgery/anesthesia, prepare for theatre or OR airway.
4. Do not remove tracheostomy unless required for intubation; if tube displaced, consider oral intubation or tracheostomy revision by expert.
(Algorithm derived from RCEM learning modules) [RCEM].
- Peritonsillar abscess:
1. Airway assessment and analgesia.
2. Needle aspiration or incision & drainage (ENT) ± immediate antibiotics covering streptococci and anaerobes.
3. Single dose dexamethasone reduces symptoms and may ease airway concerns [BMJ j4090].
- Auricular/septal haematoma:
1. Immediate drainage (aspiration or I+D) + compression dressing to prevent cartilage necrosis (cauliflower ear / saddle deformity).
2. Prophylactic antibiotics covering skin flora; urgent ENT/plastic review for packing/suture.
- Otitis externa vs malignant OE:
- Otitis externa: topical therapy, aural toilet.
- Malignant OE: admit, ear swab, IV antipseudomonal antibiotics, ENT/ID input, image to assess skull-base.
- Ear foreign body / live insect:

1. If live insect: instil topical lidocaine 2% to immobilize/kill then remove with suction/forceps.
2. Do not irrigate if suspect TM perforation or hygroscopic objects; ENT review for difficult removals/child not cooperative.
- BPPV:
1. Diagnose with Dix–Hallpike.
2. Treat with Epley canalith repositioning manoeuvre on affected side; repeat/refer to physiotherapy if recurrent.
- SSNHL:
1. Urgent audiometry and ENT referral.
2. Start systemic corticosteroids early (ideally within days; many ENT guidelines within 14 days) or intratympanic steroids per ENT advice.
- Post-tonsillectomy haemorrhage:
1. ABCs, large-bore IV access, suction to visualise, topical vasoconstrictors and direct pressure.
2. Give TXA as adjunctive haemostatic therapy while arranging definitive ENT operative control (paediatric TXA 15 mg/kg IV bolus cited in practice; adults frequently 1 g IV) — verify local protocols.
3. Urgent ENT/anaesthesia for theatre control if ongoing or recurrent bleeding.
- Deep neck infections / Lemierre's:
1. Secure airway early if threatened.
2. Empiric broad IV antibiotics covering anaerobes (metronidazole) and aerobes (e.g., ceftriaxone) pending cultures; urgent imaging (CT neck/chest) and specialist input.
3. Anticoagulation for IJV thrombosis is specialist-dependent.

> شرح عربي موجز:
> - إجراءات عاجلة واضحة: نزيف أنفي → ضغط أمامي أولاً ثم احتراق/حشو/حشْو خلفي أو تدخل إشعاعي؛ نزيف فغر القصبة → نفخ كَفّة الأنبوب ثم ضغط إصبعي داخل الفتحة والإنعاش الجراحي السريع.
> - نزيف ما بعد استئصال اللوزتين = إنعاش + استدعاء ENT فورياً؛ يمكن استخدام التراكicl3اميك أسيد كمعاون وفق البروتوكول المحلي.

5. Drugs, procedures, disposition, and follow-up

- Key drug cautions and choices:
- Avoid topical aminoglycoside drops (e.g., neomycin) in TM perforation —

ototoxicity risk. Use topical fluoroquinolone (ciprofloxacin otic) for infected perforations.
- Malignant otitis externa: systemic antipseudomonal antibiotics (IV) per local formulary; prolonged courses often required.
- SSNHL: systemic corticosteroids (dose per ENT protocol); intratympanic steroids as specialist option.
- TXA: adjunct in post-tonsillectomy bleeding — paediatric bolus commonly 15 mg/kg IV over 10 min; adults often 1 g IV — confirm local dosing.
- Analgesia and antiemetics for vertigo (prochlorperazine/ondansetron; short-term vestibular sedatives only when necessary).
- Procedures commonly required in ED:
- Anterior nasal cautery (silver nitrate) for visible anterior epistaxis.
- Nasal packing (anterior then posterior balloon) — document type/time, provide antibiotics if local policy, and plan removal.
- Auricular/septal haematoma drainage with compression dressing and ENT follow-up.
- Needle aspiration or I&D for peritonsillar abscess; consider steroids to reduce swelling/pain.
- Epley manoeuvre for posterior canal BPPV.
- Tracheostomy stoma management: cuff over-inflation, digital tamponade, urgent specialist transfer.
- Removal of ear foreign body (lidocaine for live insects; suction/curette under visualization).
- Disposition and follow-up:
- Admit: malignant OE, post-tonsillectomy major bleed, uncontrolled epistaxis/posterior bleed, mastoiditis with complications, tracheostomy bleeding, deep neck space infections, SSNHL awaiting ENT review when immediate treatment required.
- Discharge with ENT/GP follow-up: uncomplicated otitis externa (with appropriate drops), resolved BPPV after manoeuvre with outpatient physio, controlled anterior epistaxis after pack removal and ENT review.
- Document instructions: nasal pack removal time, ear-drop restrictions (water precautions with TM perforation), return precautions for airway compromise/fever/worsening hearing.

شرح عربي موجز:
- تجنّب الأمينوغليكوزيدات الموضعية مع ثقب غشاء الطبل؛ استخدم سيبروفلوكساسين موضعياً

للمثقب المصاب.
- إجراءات شائعة: تصريف هيماتومات الغضروف، تفريغ خراج حول اللوزة، مناورة Epley للدوار البوطي.
- اتجاه النقل: الحالات الغازية أو النزيف غير المسيطر عليه أو المهددة للمجرى الهوائي تحتاج قبولًا بالمستشفى وتحويل ENT/جراحة/تخدير عاجل.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية

- FRCEM tricks:
- Peritonsillar abscess: uvula deviates away from affected side; presence of trismus points to quinsy rather than simple tonsillitis.
- Epistaxis first aid: sit up, lean forward, pinch soft nasal ala (not the bony bridge), hold continuously 10–15 min.
- BPPV vs central positional vertigo: BPPV has latency, fatigable and crescendo/decrescendo nystagmus; central is immediate, non-fatigable, may be vertical or direction-changing.
- HINTS (in the right context) can outperform early MRI for posterior circulation stroke detection in continuous vestibular syndrome — but only if examiner is experienced.
- SSNHL definition: ≥30 dB loss over ≥3 contiguous frequencies within 72 h → treat as ENT emergency.
- Ototoxic drops: never use aminoglycoside drops if TM integrity uncertain — examiners love this pitfall.
- Auricular/septal haematomas require drainage; delayed treatment causes permanent deformity — an easy viva point.
- Tracheostomy bleed: over-inflate cuff then digital compression of brachiocephalic artery — a high-yield emergency mnemonic.
- Common pitfalls:
- Misinterpreting head tilt/back-tilt in epistaxis — do not tilt head back.
- Over-reliance on Centor/FeverPAIN scores instead of assessing airway risk in throat infections.
- Using irrigation for ear foreign bodies when TM perforation possible or object is hygroscopic.
- Using HINTS without ongoing spontaneous nystagmus or adequate examiner training — yields false reassurance.

- Delaying ENT for suspected malignant otitis externa in diabetics with severe pain or cranial nerve signs.

شرح عربي موجز:
- نصائح الاختبار: الشرط التعريفي لفقدان السمع المفاجئ، تجنّب الأمينوغليكوزيدات مع ثقب الطبل، تذكّر مناورة Epley للـ BPPV وقيّم HINTS فقط إذا كنت متمرساً وفي حالة دوار مستمر.

7. Arabic rapid-review summary (ملخص عربي سريع)

- الأولويات: تأمين مجرى الهواء، إيقاف النزف الكبير، كشف ومعالجة العدوى الغازية، تفريغ هيماتوما غضروفية فوراً، حفظ السمع.
- نزيف الأنف الأمامي: ضغط للجزء الطري للأذن 10–15 دقيقة، كي موضعي إن ظهر مصدر، حشو أمامي 24–48 ساعة، حشو خلفي أو استدعاء ENT إذا فشل.
- ثقب غشاء الطبل المصاب: لا أمينوغليكوزيدات؛ استخدم قطرات فلوروكينولون موضعية إذا مُصاب.
- دوار: Dix–Hallpike → Epley (BPPV). دوار مستمر → HINTS (بيد خبير)؛ أي إشارة مركزية → تصوير دماغي.
- خراج حول اللوزة: trismus + uvula مبتعدة → سحب بالإبرة أو شق وتصريف + مضاد حيوي ± ستيرويد جرعة واحدة.
- نزيف فغر القصبة الكبير: نفخ كَفّة الأنبوبيّة ثم ضغط إصبعي داخل الفتحة على الشريان، استدعاء جراحي فوري.
- حالات تتطلب دخول مستشفى: OE الخبيث، mastoiditis مع مضاعفات، نزف خلفي أو مستمر، مشكلات ما بعد استئصال اللوزتين، SSNHL إن يحتاج علاج فوري.

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية
- ENT concepts consolidated from supplied notes (no explicit Q-numbers given):
- Epistaxis: anterior vs posterior management, packing, cautery, TXA adjuncts.
- Otitis externa and malignant OE (Pseudomonas) diagnosis and management.
- Ear foreign bodies and live insects removal techniques.
- Tympanic membrane perforation: safe topical choices and water precautions.
- Auricular and nasal septal haematomas: drainage and prevention of cartilage necrosis.
- Peritonsillar abscess (quinsy): recognition, aspiration/I&D, antibiotics, steroid

adjunct.
- Lemierre’s syndrome: recognition, imaging, empiric antibiotics.
- Post-tonsillectomy haemorrhage: resuscitation, use of TXA as adjunct, operative control.
- Tracheostomy bleeding / tracheo-innominate fistula: immediate bedside steps and escalation.
- Vertigo syndromes: BPPV (Dix–Hallpike/Epley), vestibular neuritis, Ménière’s, central vertigo and HINTS.
- Sudden sensorineural hearing loss (SSNHL): diagnostic thresholds, urgent ENT/audiology referral, steroid therapy.
- Mastoiditis and deep neck space infections: imaging, indications for surgical intervention.
- Exam technique pitfalls: HINTS limitations, ear drop ototoxicity, epistaxis first aid errors.

Sources to verify: RCEM learning modules, ENT-UK guidance, BMJ review (dexamethasone in tonsillitis j4090), NUEM discussions on HINTS, local ENT guidelines, NICE and ERC recommendations where applicable.

CHAPTER 30: DERMATOLOGY

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Chapter 30 — Dermatology (final)

1. Clinical overview and pathophysiology

- Organise rash assessment by morphology, distribution, timing and systemic features:
- Morphology: vesiculobullous, maculopapular/morbilliform, pustular, targetoid, ulcerative/necrotic.
- Distribution: localized vs widespread; mucosal involvement always significant.
- Timing: acute (hours–days), delayed (days–weeks) — timing helps distinguish AGEP (rapid), SJS/TEN (1–3 weeks), DRESS (2–6 weeks), drug exanthem (days–weeks).
- Systemic features: fever, lymphadenopathy, organ dysfunction indicate systemic disease/SCAR.
- Pathophysiology highlights:
- SCARs (SJS/TEN/DRESS/AGEP) are drug-triggered immune reactions; epidermal necrosis in SJS/TEN from keratinocyte apoptosis.
- Autoimmune blistering: intraepidermal acantholysis (pemphigus vulgaris) vs subepidermal cleavage (bullous pemphigoid).
- Infectious dermatoses have characteristic patterns (erythema migrans, erythema multiforme, impetigo, eczema herpeticum).
- Pyoderma gangrenosum is neutrophilic dermatosis with pathergy; debridement worsens lesions.
(References: NICE CKS; DermNetNZ)

ملخص بالعربية
- صنف الطفح حسب الشكل، التوزيع والوقت؛ وجود مخاطيات أو علامات جهازية يدل على حالة خطرة.
- حالات فرط التحسس الجلدية الشديدة غالباً دوائية؛ تختلف الآليات بين انفصال البشرة والطفح الحبيبي والآفات النخريّة.

2. Recognition, differential diagnosis, and red flags

EXAM TRICK / تريكة امتحانية

- Red flags (immediate attention):
- Any mucosal erosions/ulceration, progressive epidermal detachment, haemodynamic instability, airway compromise, ocular involvement, high fever, organ failure signs.
- Key recognition cues and DDX:
- Target lesions ± limited mucosa → erythema multiforme (HSV/Mycoplasma trigger).
- Rapid epidermal detachment ± mucosa after drug → SJS/TEN (SJS <10% BSA; overlap 10–30%; TEN >30%).
- Widespread sterile pustules within 48 h of drug → AGEP (neutrophilia).
- Fever + facial oedema + rash 2–6 weeks after drug with eosinophilia/organ involvement → DRESS.
- Tense blisters in elderly → bullous pemphigoid; flaccid blisters with mucosal erosions → pemphigus vulgaris (Nikolsky +).
- Painful undermined ulcer with violaceous border → pyoderma gangrenosum (IBD/paraneoplastic associations).
- Clustered linear pruritic papules after sleep → bed-bug/arthropod bites.
- Honey-coloured crusts on face/children → impetigo (Strep/Staph) [NICE CKS].
- Expanding erythematous annulus with central clearing → tinea corporis (KOH helpful).
- Eczema herpeticum: disseminated vesiculopustules/erosions on atopic skin + systemic signs → treat with antivirals.
- Erythema migrans after tick bite → early Lyme (doxycycline) [NICE CKS].
- Pitfalls that mandate caution:
- Mistaking EM for early SJS; check mucosa and detachment %.
- Surgical debridement of suspected PG → pathergy and worse outcome.
- Assuming morbilliform rash is benign without looking for systemic features (SCAR risk).

الشرح بالعربية

- أي مشاركة مخاطية أو انفصال جلدي تقدّمية أو علامات جهازية هي علامات حمراء؛ تمييز

الأشكال (هدفية، فقاعية، قيحية) يوجّه التشخيص الأولي.
- احذر استئصال قرحات pyoderma gangrenosum وخلط EM مع المراحل المبكرة من SJS.

3. Diagnostic approach and interpretation

- Immediate priorities: ABCs (airway risk from mucosal disease), haemodynamic stabilisation, estimate BSA (palm ≈ 1%).
- Focused history: recent/new drugs (timing), infections, travel, insect exposure, prior dermatoses, immune status.
- Targeted exam: mucosae (oral, ocular, genital), Nikolsky sign, exact morphology, distribution, lymphadenopathy.
- Bedside tests:
- KOH prep for fungal hyphae.
- Bacterial swab/culture if purulent/impetiginised.
- Tzanck smear may show multinucleated giant cells (not specific).
- Consider skin biopsy for histology/immunofluorescence in bullous diseases or unclear SCAR.
- Baseline labs when systemic features: CBC (neutrophilia/eosinophilia), creatinine/urea, electrolytes, LFTs, glucose, blood cultures if febrile.
- Use SCORTEN (day 1) to prognosticate SJS/TEN: age >40, malignancy, HR >120, BSA detached >10%, urea >10 mmol/L, glucose >14 mmol/L, bicarbonate <20 mmol/L.
- Early specialty input: dermatology ± burns/ICU, ophthalmology (ocular mucosal disease), infectious disease, rheumatology as indicated.

الشرح بالعربية
- أقِم ABCs ثم قدّر مساحة الإصابة (كف اليد ≈ 1%).
- فحوصات بسيطة: KOH، مسحات بكتيرية، خزعة جلدية عند الحاجة.
- استخدم SCORTEN لتقدير خطر الوفاة في SJS/TEN واطلب استشارة مبكرة.

4. Emergency management algorithms

A. SJS/TEN (stepwise)
1. Immediate: stop all possible causative drugs.
2. ABCs: monitor airway (early ENT/anaesthesia if mucosal airway risk), oxygen, IV access.
3. Assess BSA (palm ≈ 1%):

- SJS <10% → consider admission; SJS/TEN overlap 10–30% → high-dependency; TEN >30% → burns unit/ICU.
4. Resuscitation: IV fluids, correct electrolytes, monitor urine output; treat as burn patient for fluid/electrolyte/infection risk.
5. Wound care: sterile, non-adhesive dressings; minimal debridement; temperature control and analgesia.
6. Complications: sepsis prevention, nutrition, thromboprophylaxis as appropriate.
7. Ancillary: ophthalmology, dermatologist, burn/ICU input.
8. Disease-modifying therapy: IVIG, systemic corticosteroids, cyclosporine used variably — consult local protocol/specialist (evidence mixed).
(Use SCORTEN for prognosis)

B. Suspected SCARs (AGEP/DRESS/drug exanthem)
- Stop offending drug. Supportive care: fluids, antipyretics, topical emollients/ steroids for pruritus.
- AGEP: usually self-limited; monitor CBC (neutrophilia).
- DRESS: systemic steroids often required; monitor LFTs, renal function, blood counts; admit if organ involvement.

C. Eczema herpeticum
- Start systemic antiviral promptly:
- Mild/moderate: oral acyclovir (per local guidance).
- Severe/disseminated/immunocompromised: IV acyclovir 10 mg/kg IV q8h (adjust for renal function); admit and obtain dermatology/ID/ophthalmology input.

D. Impetigo
- Localised: topical hydrogen peroxide 1% cream first-line (NICE CKS), or topical fusidic acid/mupirocin per local policy.
- Extensive/systemic signs: oral antibiotics (cover Strep/Staph) and consider admission.

E. Lyme disease (erythema migrans)
- Doxycycline 100 mg PO twice daily in adults (duration per local guideline) [NICE CKS]; pregnancy/children: alternatives.

F. Bullous diseases and pyoderma gangrenosum
- Analgesia, wound care, urgent dermatology. Avoid debridement in PG. Systemic immunosuppression under specialist guidance (e.g., steroids, cyclosporine).

خلاصة بالعربية للخطوات الحرجة
- SJS/TEN: أوقف الأدوية فوراً، ABCs، قدّر BSA وأحِل لوحدة الحروق/العناية المركزة حسب العتبات، رعاية جروح ومحافظة على السوائل، اطلب استشارة جلدية/عيون مبكراً.
- Eczema herpeticum: ابدأ acyclovir جهازي مبكراً؛ الحالات الشديدة تستدعي IV acyclovir ودخول المستشفى.
- لا تُجرّب تنظيف/استئصال PG دون إرشاد جلدّي، أوقف الدواء المشتبه في SCARs.

5. Drugs, procedures, disposition, and follow-up

Drugs (well-established thresholds/doses)
- IV acyclovir for severe HSV/eczema herpeticum: commonly 10 mg/kg IV q8h (adjust renal). (Standard practice; verify local protocols.)
- Doxycycline for erythema migrans (adult): 100 mg PO twice daily (duration per guideline) [NICE CKS].
- Topical impetigo first-line: hydrogen peroxide 1% cream [NICE CKS]. Alternatives: topical fusidic acid or mupirocin per formulary.
- Immunomodulatory agents in SJS/TEN (IVIG, systemic steroids, cyclosporine): used variably; specialist decision required.

Procedures
- Estimate BSA (palm ≈ 1%). Use rule of nines for adults where helpful.
- Skin swab/culture for purulent lesions; KOH for suspected tinea.
- Skin biopsy (histology, direct immunofluorescence) for bullous disorders/unclear diagnosis.
- Avoid surgical debridement in pyoderma gangrenosum unless directed.
- Wound care: non-adhesive sterile dressings; preserve slough in SJS/TEN per burn-unit practice.

Disposition and admission criteria
- Admit to ward: extensive rash with moderate systemic features, facial/hand/foot/genital mucosal involvement without airway threat, significant pain or inability to care at home.
- High-dependency / burns unit / ICU: SJS/TEN >10% BSA (high-dependency), >30% (burns/ICU) or airway compromise, sepsis, multi-organ failure.
- Dermatology/ophthalmology follow-up for SJS/TEN even after discharge (ocular sequelae common).
- Admit eczema herpeticum severe/disseminated or immunocompromised for IV

antivirals; outpatient oral antivirals if mild and safe at home.
- Follow-up for dermatomyositis: check CK and malignancy screening; coordinate with rheumatology/oncology.
- DRESS: arrange specialist follow-up and prolonged monitoring of LFTs/renal function (organ involvement can be delayed).

Follow-up testing
- In DRESS: serial LFTs/renal tests, CBC for lymphopenia/eosinophilia.
- In bullous autoimmune disease: autoimmune serology/biopsy, long-term immunosuppression planning.
- In Lyme disease: clinical review at end of therapy and if new systemic symptoms.

الشرح بالعربية
- Aciclovir IV 10 mg/kg q8h شائع للحالات الشديدة (اضبط حسب الكلى).
- دوكسيسيكلين 100 mg مرتين يومياً للـerythema migrans (اتبع مدة المحلية).
- خزعة جلدية ومسوحات البكتيريا مفيدة للتشخيص؛ لا تنظف قرحات PG جراحياً.
- حدد الإحالة لوحدة الحروق/العناية المركزة حسب نسبة BSA ومظاهر فشل الأعضاء.

6. High-yield FRCEM exam tricks and common pitfalls

EXAM TRICK / تريكة امتحانية
Exam tricks
- Palm ≈ 1% BSA — quick check to stage SJS/TEN; rule of nines for larger estimates.
- Timing window for drug reactions: AGEP (within 48 h), SJS/TEN (≈1–3 weeks), DRESS (2–6 weeks) — use to identify culprit drug.
- Look for mucosal involvement and epidermal detachment percent to separate EM from SJS/TEN.
- Nikolsky positive suggests epidermal cleavage (pemphigus, TEN) — correlate clinically.

EXAM TRICK / تريكة امتحانية
Common pitfalls
- Mistaking pyoderma gangrenosum for infective ulcer and debriding it — causes rapid worsening (pathergy).
- Overlooking systemic signs in a “morbilliform” rash — SCARs can present initially as bland exanthem.

- Routine automatic use of IVIG or systemic steroids for TEN — evidence mixed; always seek local/burn unit guidance.
- Assuming absence of fever excludes severe disease — early SJS/TEN or immunosuppressed patients may be afebrile.
- Not adjusting acyclovir dose for renal impairment — risk of toxicity.

النقاط السريعة بالعربية (نصائح للامتحان)
- استخدم كف اليد لتقدير % المساحة؛ تذكّر نوافذ زمنية للتمييز بين AGEP/DRESS/SJS.
- لا تجرّب استئصال PG؛ لا تفترض فاعلية IVIG أو الستيرويدات في TEN دون استشارة.

7. Arabic rapid-review summary

- صنّف الطفح حسب الشكل، التوزيع، الوقت والأعراض الجهازية؛ مشاركة المخاطيات أو انفصال الجلد يتطلب تقييم عاجل.
- أوقف الدواء فوراً، قدّر BSA، وفكر في إحالة SJS <10%، SJS/TEN 10–30%، TEN >30% — لوحدة الحروق/العناية المركزة.
- ابدأ acyclovir جهازي مبكراً في eczema herpeticum؛ استخدم IV acyclovir في الحالات الشديدة (10 mg/kg q8h مع تعديل كلوي).
- لا تنظف أو تزيّل قرحات pyoderma gangrenosum دون استشارة جلدية.
- استخدم SCORTEN لتقدير خطر الوفاة في SJS/TEN واطلب استشارة متخصصة مبكراً.
(مذكرة مراجعة سريعة؛ ليست بديلاً عن إرشادات محلية أو استشارة طبية مخصصة.)

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية
- SCAR classification and acute management: SJS/TEN BSA thresholds, immediate actions, SCORTEN prognostic variables. (Concepts covered)
- Drug timing windows for AGEP / SJS-TEN / DRESS and their distinguishing features. (Concepts covered)
- Eczema herpeticum recognition and antiviral dosing/indications for IV therapy and admission. (Concepts covered)
- Impetigo management: topical hydrogen peroxide 1% (NICE CKS) and indications for systemic antibiotics. (Concepts covered)
- Erythema migrans (Lyme) recognition and doxycycline adult dosing. (Concepts covered)
- Bullous disorders: clinical differences pemphigus vs pemphigoid, biopsy/

serology role. (Concepts covered)
- Pyoderma gangrenosum diagnosis, pathergy, contraindication to debridement, immunosuppression role. (Concepts covered)
- Diagnostic approach in ED: ABCs, BSA estimate (palm ≈ 1%), bedside tests (KOH, swabs, biopsy), baseline labs. (Concepts covered)
- Wound care principles and disposition criteria (ward vs HDU vs burns/ICU). (Concepts covered)
- Exam tricks and common pitfalls (palm rule, timing windows, pathergy, misclassification). (Concepts covered)
(Note: specific source question numbers were not supplied with the chunk notes; the map lists concepts addressed for cross-referencing to exam question banks.)

Sources to verify
- NICE Clinical Knowledge Summaries (CKS) — impetigo, erythema migrans/Lyme.
- DermNetNZ — AGEP and other eruption summaries.
- SCORTEN original descriptions / burn-unit protocols (for prognostication in SJS/TEN).
- Regional/local formularies and specialist (dermatology/burns/ICU/ophthalmology) protocols for IVIG, corticosteroids, cyclosporine use in SJS/TEN and dosing adjustments.
- Royal College of Emergency Medicine (RCEM), European Resuscitation Council (ERC), national dermatology and infectious disease guidelines for final verification and local practice adjustments.

(Study material only — not patient-specific treatment advice.)

CHAPTER 31: CARDIOLOGY

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Chapter 31: Cardiology — final chapter (compact FRCEM study notes)

1. Clinical overview and pathophysiology

- Core principle: haemodynamic state dictates urgency — unstable arrhythmia → immediate electrical therapy (synchronised DC cardioversion/defibrillation); stable → targeted pharmacology or vagal manoeuvres. [Resus Council UK 2021]
- Arrhythmia taxonomy: classify by rate (brady/tachy), QRS width (narrow/wide), regularity (regular/irregular), and stability (stable/unstable).
- Common mechanisms: re-entry (AVNRT/AVRT/VT), enhanced automaticity (sinus tachy/brady), triggered activity (torsades from prolonged QT).
- Ischaemia/infarction causes: acute occlusion → electrical instability (VT/VF, bradyarrhythmias, conduction blocks); STEMI equivalents (De Winter, posterior MI) behave like occlusion.
- Conduction disease: Mobitz II, high-grade AV block, trifascicular block can progress to asystole — pacing likely required.
- Structural/inherited disease: HOCM, ARVC, Brugada → exertional syncope, sudden death risk; ECG clues (HOCM: LVH/abnormal Qs; ARVC: T-wave inversion V1–V3; Brugada: coved V1–V2).
- Devices and mechanical support: ICDs give shocks; LVADs may abolish palpable pulses (use Doppler MAP). Device interrogation is diagnostic/therapeutic.

> Arabic concise:
> المبدأ: شدة حالة المريض تحدد التدخّل—عدم الاستقرار → علاج كهربائى فورى؛ التوصيل:
> تصنيف حسب السرعة، عرض QRS، انتظام، واستقرار. اعتبارات خاصة: أمراض بنيوية/أجهزة
> مزروعة تغير العرض السريرى وطرق القياس.

2. Recognition, differential diagnosis, and red flags

- Red flags (urgent cardioversion/intervention): hypotension, shock, severe chest pain, acute pulmonary oedema, decreased conscious level, syncope with high-risk ECG.
- Tachycardia patterns:

- Regular narrow complex (HR ~150–220): likely AVNRT/AVRT (SVT).
- Irregular wide complex: consider pre-excited AF (WPW) vs polymorphic VT.
- Broad regular tachycardia: VT until proven otherwise.
- Bradycardia causes and red features: symptomatic bradycardia (syncope, hypotension, ischaemia) → consider AV block, drug effect (BB/CCB/digoxin), hyperkalaemia, inferior MI.
- STEMI and equivalents red signs: V4R ST elevation → RV infarct (avoid nitrates); De Winter pattern/Wellens/backwards changes → treat as LAD occlusion.
- Pericardial tamponade: hypotension + raised JVP + muffled heart sounds (Beck's triad), electrical alternans, low voltage ECG — urgent echo and pericardiocentesis if unstable.
- Device/LVAD flags: repeated ICD shocks, absent pulse on LVAD — do not assume no perfusion; call VAD/device team.
- High-risk conduction disease: bifascicular/trifascicular block with syncope → urgent pacing/permanent pacemaker consideration.

> Arabic concise:
> مؤشرات الخطر: هبوط ضغط، وذمة رئوية حادة، إغماء؛ أنماط ECG تحدد التشخيص المبدئي (VT عند QRS واسع منتظم، WPW إذا رجفان أذيني واسع وغير منتظم). التامبوناد والجهاز/الـLVAD يتطلبان استجابة فورية خاصة.

3. Diagnostic approach and interpretation

- Immediate actions: ABCs, continuous monitor, 12-lead ECG within 10 min, capillary glucose, IV access, oxygen if hypoxic.
- ECG classification algorithm: stable vs unstable → narrow/wide → regular/irregular → choose vagal/adenosine, antiarrhythmic, or cardioversion.
- Bradycardia work-up: check drugs, electrolytes (K+, Mg2+, Ca2+), TSH, ischaemia; identify block type on ECG (1°, Mobitz I vs II, complete).
- Pacing capture confirmation: first electrical capture on monitor (pacing spikes + QRS change), then mechanical capture — palpate central pulse (carotid/femoral) or measure MAP. Electrical ≠ mechanical.
- WPW identification: short PR, delta wave, wide QRS; in AF with wide irregular QRS, avoid AV-nodal blockers.
- STEMI localisation: V4R for RV infarct; posterior MI suspected when tall R / ST depression in V1–V3 → record posterior leads V7–V9; high lateral STE in I/aVL ± V2 suggests diagonal/high lateral occlusion; aVR elevation with global depression →

left main/proximal severe disease.
- QTc: use Bazett's formula (exam context): QTc = QT / √RR (s); QTc >500 ms = increased TdP risk.
- Bedside echo: tamponade, large effusion, RV dysfunction, severe LV dysfunction, mechanical complications.
- Labs: troponin, electrolytes, Mg/Ca, blood cultures x3 for suspected infective endocarditis (before antibiotics if feasible).

Arabic concise:
النهج التشخيصى: ECG فوري، تحري الأسباب القابلة للعكس (أدوية، شوارد)، تأكيد الالتقاط الكهربى والميكانيكى عند التحفيز، تسجيل قيادات خلفية (V7–V9) للـMI الخلفى، حساب QTc باستخدام بازيت فى الامتحان.

4. Emergency management algorithms

EXAM TRICK / تريكة امتحانية
- General principle: unstable = electrical therapy first; stable = targeted pharmacology/ manoeuvres.
- Symptomatic bradycardia algorithm:
1. Atropine 0.5 mg IV bolus; repeat q3–5 min to max 3 mg. [Resus Council UK 2021]
2. If inadequate and unstable → immediate transcutaneous pacing (prepare pads, sedation if time), check electrical then mechanical capture.
3. If pacing unavailable or as a bridge → isoprenaline (where local practice) or dopamine/epinephrine infusion while organising transvenous pacing. Correct reversible causes.
- Pitfall: atropine often ineffective in high-degree AV block — do not delay pacing.
- Transcutaneous pacing troubleshooting:
- Start at moderate mA, increase until electrical capture then confirm pulse; if no capture at max, reposition pads (AP), check contacts, treat hyperkalaemia, arrange transvenous pacing.
- Narrow-complex SVT (stable):
1. Modified Valsalva manoeuvre (semi-recumbent + leg lift) first (REVERT trial 2015).
2. If fails → adenosine 6 mg rapid IV bolus; if no effect 12 mg (then 18 mg if

local practice).
3. Unstable → synchronised DC cardioversion (~100 J for SVT; follow device guidance).
- Pre-excited AF (WPW + AF):
- Never give AV-nodal blockers (beta-blockers, non-DHP CCBs, digoxin, adenosine) — risk of rapid accessory pathway conduction/VF.
- Stable → IV procainamide if available.
- Unstable → immediate synchronised DC cardioversion. (WPW teaching; Resus guidance)
- Wide complex tachycardia:
- Unstable → immediate synchronised DC cardioversion.
- Stable monomorphic VT → IV amiodarone (commonly 300 mg over 10–60 min) or procainamide per local practice; torsades → IV magnesium sulfate 2 g.
- VF/pulseless VT: defibrillate per ALS, high-quality CPR, adrenaline and antiarrhythmic per ACLS.
- STEMI reperfusion triage:
- If anticipated primary PCI delay >120 min → give fibrinolysis (if no contraindication) then transfer (pharmacoinvasive strategy). If PCI available within timeframe → primary PCI. Give aspirin 300 mg and follow local antiplatelet/anticoagulant protocols.
- Avoid nitrates if RV infarct suspected (V4R STE) — give cautious IV fluid bolus for preload support.
- Cardiac tamponade: urgent echo; if unstable → pericardiocentesis; use IV fluids as temporary preload support only.
- Cardiogenic pulmonary oedema: oxygen, early CPAP, diuretics, nitrates if not hypotensive.

Arabic concise:
خوارزميات طارئة مختصرة: بطء عرضى → أتروبين 0.5 mg متكرر حتى 3 mg → تحفيز خارجي إذا فشل؛ SVT مستقر → مناورة فالزالفا المعدلة ثم أدينوزين 6→12 mg؛ WPW+AF لا تعطى محصرات العقدة الأذينية البطينية — بروكاييناميد أو صدمة إذا غير مستقر؛ STEMI: قرر PCI مقابل إذابة جلطة بناءً على زمن الوصول (120 دقيقة).

5. Drugs, procedures, disposition, and follow-up

- Key drug doses (well established):
- Atropine: 0.5 mg IV bolus, repeat q3–5 min to max 3 mg. [Resus Council UK 2021]
- Adenosine: 6 mg rapid IV bolus → 12 mg if needed (some protocols use up to 18 mg).
- Amiodarone (acute arrhythmia): 300 mg IV bolus over 10–60 min (ED regimens vary).
- Magnesium sulfate: 2 g IV for torsades.
- Procainamide: used for stable pre-excited AF or some stable VT (local availability determines use).
- Procedures and practical points:
- Transcutaneous pacing: pads AP or as per device, increase current until capture, confirm central pulse; if fail reposition pads then escalate to transvenous pacing.
- Synchronized DC cardioversion: use biphasic device settings per manufacturer; typical exam/ED energies: 100–150 J for SVT/VT with pulse, 200 J common for AF (verify local guidance).
- Pericardiocentesis: emergent for tamponade with instability; perform under echo guidance if possible.
- Device management: urgent ICD interrogation for recurrent shocks; magnet may suspend tachytherapy but device behaviour varies — consult local policy/device team.
- LVAD: measure MAP by Doppler; contact VAD team early.
- Disposition and follow-up:
- Unstable arrhythmias, new high-grade AV block, persistent symptomatic bradycardia requiring pacing → admit/CCU and urgent cardiology ± pacing/EP referral.
- STEMI → PPCI centre or fibrinolysis pathway and cardiology admission.
- WPW with documented pre-excited AF or syncope → cardiology/EP follow-up and consider ablation.
- Pericarditis with high-risk features (fever, large effusion, tamponade, failure of NSAIDs, raised troponin, immunosuppression) → admit and early cardiology.
- Infective endocarditis suspicion → take 3 blood cultures before starting antibiotics when feasible; admit and involve ID/cardiology/CT surgery as needed.

> Arabic concise:
> الأدوية والإجراءات: أتروبين 0.5 mg حتى 3 mg، أدينوزين 6→12 mg؛ أميودارون 300 mg IV

loading شائع؛ ماغنسيوم 2 g لتورسادس. التحفيز الخارجي ثم الوريدى عند فشل، صدمة مزامنة عند عدم الاستقرار. القبول: CCU/PCI/EP حسب الحالة.

6. High-yield FRCEM exam tricks and common pitfalls

- Always check mechanical capture after pacing — electrical capture on ECG is not sufficient (frequently tested).
- Modified Valsalva (semi-recumbent + leg raise) > standard Valsalva — choose modified in exam stems (REVERT 2015).
- Atropine typical exam dose: 0.5 mg bolus, max 3 mg — many pick 2 mg incorrectly.
- Never give AV-nodal blockers (beta-blocker, CCB, digoxin, adenosine) for irregular wide complex tachycardia when WPW suspected — exam trap.
- Treat wide complex regular tachycardia as VT if in doubt — safest answer on exams.
- Posterior MI: tall R / ST depression in V1–V3 → add V7–V9; don't miss (common fault).
- De Winter pattern = LAD occlusion equivalent — treat as STEMI (exam/MCQ trap labelled "not STEMI").
- Nitrates can precipitate severe hypotension in RV infarct — check V4R in inferior STEMI before nitrates.
- ICD magnet use: device behaviour varies; don't assume permanent suspension of therapy — mention device/centre policy in answers.
- Theophylline/aminophylline and other historical therapies for bradycardia are not standard: flag as outdated — verify local guidance.
- Adenosine in asthma: caution in severe/reactive airways disease; not absolute contraindication in mild disease (exam nuance).
- Wellens syndrome = pre-angiography high risk—do not perform exercise stress test.
- BP/pulse in LVAD: absent palpable pulse ≠ no perfusion; use Doppler MAP measurement.

Arabic concise:
خدع الامتحان: تحقق من الالتقاط الميكانيكى بعد التحفيز، اختَر مناورة فالزالفا المعدلة، أتروبين 0.5 mg حتى 3 mg ليس 2 mg، لا تستخدم محصرات العقدة في WPW+AF، تعامل مع QRS واسع كـVT إذا شككت.

7. Arabic rapid-review summary (مختصر سريع بالعربية)

- عدم الاستقرار الإيقاعي → صدمة كهربائية محدثة (مزامنة/عدم مزامنة حسب الحالة).
- بطء عرضي حاد: أتروبين 0.5 mg متكرر حتى 3 mg → إذا فشل: تحفيز خارجي عابر ثم تحضير لتحفيز وريدى أو دائم.
- SVT مستقرة: مناورة فالزالفا المعدلة ثم أديينوزين 6→12 mg؛ SVT غير مستقر → صدمة مزامنة.
- WPW + الرجفان الأذيني: لا تعطِ محصرات عقدة AV؛ مستقِر → بروكايينااميد؛ غير مستقر → صدمة.
- QRS واسع منتظم → افترض VT؛ غير مستقر → صدمة مُزامنة؛ torsades → ماغنسيوم 2 g IV.
- STEMI: إذا انتظار PCI >120 دقيقة وفى مريض ملائم → إذابة جلطة ثم تحويل؛ RV infarct (V4R) → تجنب النتروجليسرين.
- التامبوناد: ثلاثية بيك (ضغط منخفض، JVP مرتفعة، أصوات قلب مكتومة) → صدى عاجل وبزل إن كان غير مستقر.
- أجهزة مثل ICD/LVAD: استدعِ فريق الجهاز، لا تفترض غياب ضغط نابض عند LVAD؛ المغناطيس قد يوقف الصدمات لكن السلوك يختلف حسب الجهاز.
- تحقّق دائمًا من الالتقاط الكهربى والميكانيكى بعد التحفيز، وحافظ على معالجة الأسباب القابلة للانعكاس (شوارد، أدوية، نقص أكسجة).

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية

- Arrhythmia urgency hierarchy; defibrillation/cardioversion vs pharmacologic therapy. [Resus Council UK 2021]
- Symptomatic bradycardia management and atropine dosing; transcutaneous/transvenous pacing, capture confirmation.
- Pacing troubleshooting and pad positioning; causes preventing capture (hyperkalaemia).
- SVT management: modified Valsalva (REVERT 2015), adenosine dosing, cardioversion.
- Wide complex tachycardia management: VT vs SVT with aberrancy; amiodarone/procainamide; torsades and magnesium.
- Pre-excited AF (WPW): contraindicated AV-nodal blockers, use procainamide or cardioversion.
- STEMI triage: PCI timing (120 min), fibrinolysis indication, posterior/V4R leads,

De Winter/Wellens patterns.
- ECG interpretation pearls: QTc calculation, posterior leads V7–V9, De Winter T-waves, Wellens, device ECG signs.
- Pericarditis vs MI vs tamponade: ECG findings, indications for pericardiocentesis.
- Device/LVAD management: ICD shocks, magnet use caveats, MAP by Doppler in LVADs.
- Cardiogenic pulmonary oedema: CPAP early, nitrates caution in RV infarct.
- Infective endocarditis sampling (3 blood cultures) and initial management.
- High-risk conduction disease (Mobitz II, bifascicular/trifascicular) and pacing indications.
- Exam traps and pitfalls: mechanical vs electrical capture, modified Valsalva superiority, atropine max dose, avoid AV-nodal blockers in WPW+AF, treat wide complex as VT if unsure.

Note: no specific question numbers were provided in the supplied material — map lists the concepts/topics represented.

Sources to verify
- Resuscitation Council UK (Adult Tachycardia and Bradycardia Algorithms) 2021; REVERT trial (2015) for modified Valsalva; ESC and AHA/ALS guidelines for cardioversion/arrhythmia therapy; NICE guidance where applicable (e.g., syncope / HOCM referral); RCEM/UK national guidance for ED practice; local hospital/device manufacturer policies for ICD/magnet behaviour; LitFL ECG resources for ECG pattern recognition.

(Study notes — educational only, not patient-specific advice. Verify local protocols and device behaviours before clinical application.)

CHAPTER 32: ALLERGY

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Chapter 32: Allergy

1. Clinical overview and pathophysiology

- Definition: Anaphylaxis = acute, potentially life-threatening systemic allergic reaction, usually IgE-mediated. Allergen cross-links IgE on mast cells → massive mediator release (histamine, tryptase, leukotrienes, prostaglandins) → vasodilation, increased vascular permeability, bronchospasm, mucosal oedema.
- Non-IgE syndromes that mimic allergy:
- Bradykinin-mediated angioedema (ACE-inhibitor) — no urticaria, may be delayed and recur.
- Scombroid poisoning — histamine-like reaction from spoiled fish.
- DRESS (drug reaction with eosinophilia and systemic symptoms) — delayed (2–8 weeks), systemic organ involvement.
- Immune-checkpoint inhibitor toxicity — delayed, organ-specific immune reactions (eg colitis).
- Key principle: diagnosis is clinical; time-critical therapy (adrenaline IM) must not be delayed for tests. [Resus Council UK 2021]

> موجز بالعربية:
> - التأق حالة مناعية حادة تهدِّد الحياة؛ الآلية غالبًا IgE وإفراز وسائط حادة.
> - توجد حالات شبيهة (وذمة ACE-I، تسمم السمك، DRESS، سمية مثبِّطات المناعة) لكن آلياتها مختلفة.
> - القرار سريري والعلاج الفوري لا ينتظر الفحوصات. [Resus Council UK 2021]

2. Recognition, differential diagnosis, and red flags

- Typical presentation: sudden onset after exposure + ≥1:
- Airway: stridor, throat/tongue/facial swelling, hoarseness, progressive dysphagia.
- Breathing: wheeze, severe dyspnoea, hypoxia, poor air entry.
- Circulation: hypotension, syncope, collapse, tachycardia, poor peripheral perfusion.
- Skin: widespread urticaria, angioedema, flushing (may be absent in up to 10–

20%).
- Rapid differentials (distinguish clinically):
- Vasovagal syncope (bradycardia, pallor, precipitant, self-limited).
- Acute severe asthma (history of asthma, no urticaria/angioedema).
- Panic attack (hyperventilation, preserved oxygenation).
- Myocardial ischaemia (age/risk factors, ischaemic ECG changes).
- ACEI angioedema (no urticaria, ongoing ACEI exposure).
- Scombroid (recent fish ingestion).
- DRESS / delayed drug reactions (rash + fever + organ dysfunction days–weeks later).
- Red flags (need immediate escalation):
- Progressive airway compromise (stridor, increasing dysphagia, tongue/facial swelling).
- Refractory hypotension despite initial therapy.
- Rapid deterioration or collapse.
- Hypoxaemia or severe bronchospasm not responding to bronchodilators.
- Pregnant patient with haemodynamic compromise — early senior / obstetric input.

> موجز بالعربية:
> - الأعراض المفاجئة بعد مهيِّج مع علامات جوهرية: مجرى هواء، تنفُّس، دوران أو جلد.
> - التفريق مهم: وذمة ACE-I (بدون شرى)، تسمم السمك، ذبحة صدرية، ذعر.
> - علامات الخطر: ازدياد انسداد المجاري الهوائية، هبوط ضغط مستمر أو نقص أكسجة.

3. Diagnostic approach and interpretation

- Main rule: treat clinically; do not delay adrenaline for investigations.
- Tests that support diagnosis:
- Serum tryptase: peaks ~1–2 hours after onset; take sample as soon as possible and another baseline ≥24 hours later. A normal tryptase does NOT exclude anaphylaxis. [Resus Council UK 2021]
- Bloods/ECG/ABG/peak flow as clinically indicated (to assess organ dysfunction, hypoxia, acidosis).
- Pregnancy test if relevant.
- For delayed reactions: CBC (eosinophilia in DRESS), LFTs/renal panel, and organ-specific tests.
- Documentation: record suspected trigger, timing of exposure, symptom onset,

treatments given and response.
- When to refer for further testing: arrange allergy clinic for skin tests or specific IgE testing after recovery (timing depends on allergen and severity).

> موجز بالعربية:
> - التشخيص سريري؛ فحوصات داعمة: التربتيز (ذروة 1–2 ساعة، أخذ أساسي ≤24 ساعة لاحقًا)؛ دم، ECG حسب الحاجة.
> - العَمليَّة: وثِّق المهيّج، توقيت الأعراض واستجابة العلاج، وأحِل إلى حساسية لاحقًا. [Resus Council UK 2021]

4. Emergency management algorithms

A. Immediate resuscitation (algorithmic steps)
1) Immediate actions: ensure scene safety, remove trigger (stop infusion/stop feed), call for senior/anaesthetic/critical care help early.
2) Positioning: lie flat with legs elevated unless respiratory distress or vomiting (prevents sudden collapse). [Resus Council UK 2021]
3) Oxygen: high-flow oxygen (if hypoxic or respiratory distress).
4) Adrenaline IM: give immediately into mid-anterior lateral thigh (1:1000).
- Adults and children >12 years: 500 µg (0.5 mL)
- Children 6–12 years: 300 µg (0.3 mL)
- Children 6 months–6 years: 150 µg (0.15 mL)
- Infants <6 months: ~100–150 µg (0.1–0.15 mL)
- Repeat every 5 minutes if persistent features.
[Resus Council UK 2021]
5) IV access and fluids: large bore IV; adults 500–1000 mL crystalloid bolus and repeat as required; children 20 mL/kg boluses.
6) If severe bronchospasm: nebulised salbutamol ± nebulised adrenaline as adjuncts (they do not replace IM adrenaline).
7) If persistent hypotension or refractory anaphylaxis despite repeated IM doses → consider IV adrenaline infusion in monitored setting with expert help; IV boluses have high arrhythmia risk. Example infusion regimen (expert use only): 1 mg in 100 mL 0.9% saline (10 µg/mL) and titrate to effect on monitored patient. Use local critical care protocols. [Resus Council UK 2021]
8) Airway management: prepare for early intubation if airway swelling progresses; have equipment for difficult airway and anaesthetic support. Surgical airway if complete obstruction and cannot intubate/ventilate.

9) Adjunct drugs:
- H1 antihistamine (IV/PO) for cutaneous symptoms (not life-saving).
- H2 antagonist optional for additional cutaneous control.
- Steroids (eg IV hydrocortisone) commonly given to potentially reduce protracted/biphasic reactions but do NOT treat immediate airway or circulatory compromise. Evidence for preventing biphasic reactions is limited — practice varies.
10) Observation and reassessment: monitor vitals, urine output, respiratory status; repeat adrenaline and fluids as necessary.

B. Refractory anaphylaxis escalation
- If ongoing shock despite repeated IM adrenaline and fluids:
- Call critical care/anaesthetics.
- Start IV adrenaline infusion (expert) and titrate; use continuous cardiac monitoring.
- Consider vasopressors/inotropes per critical care guidance.
- Consider bronchoscopy for airway oedema assessment if available.
- Consider adjuncts for specific causes (see section 5).

> موجز بالعربية:
> - خوارزمية سريعة: أوقف المهيّج → استدعِ المساعدة → مريض مستلقٍ مع رفع الأرجل → أكسجين → أدرينالين عضلي فورًا (كرر كل 5 دقائق) → سوائل ووسائل داعمة → تحويل لوحدة عناية إذا مقاوم.
> - الأدرينالين الوريدي والجرعات السريعة للحالات المقاومة فقط وبحضور خبرة ومراقبة. [Resus Council UK 2021]

5. Drugs, procedures, disposition, and follow-up

Drugs and doses (exam-important)
- Adrenaline IM (1:1000): adult = 500 µg IM; paeds per age bands above. Repeat every 5 minutes as needed. [Resus Council UK 2021]
- IV adrenaline infusion: expert setting only — example: 1 mg in 100 mL 0.9% saline (10 µg/mL), titrate carefully under monitoring. Avoid unsupervised IV boluses except by experienced clinicians.
- IV fluids: adults 500–1000 mL crystalloids bolus; children 20 mL/kg boluses.
- Nebulised salbutamol for bronchospasm; nebulised adrenaline for upper airway obstruction adjunctively.
- Antihistamines: H1 (eg cetirizine PO/IV) for skin symptoms; H2 optional.

- Steroids: IV hydrocortisone commonly used; not immediate rescue therapy.
- ACEI angioedema specific treatments: stop ACEI. Icatibant or C1 esterase concentrate have been used in severe cases—availability and evidence vary; follow local protocol.

Procedures
- Early airway anticipation and preparation for difficult intubation if swelling or progressive obstruction.
- Monitor on continuous ECG, SpO2, BP; consider arterial line/advanced monitoring in ICU.
- Take blood for tryptase as soon as possible and repeat baseline ≥24 hours later.

Disposition and observation
- Disposition depends on severity and response:
- Discharge criteria (common RCUK approach): complete resolution of symptoms, observed response within 5–10 minutes after single adrenaline dose, reliable follow-up and access to emergency care, patient given adrenaline auto-injector and training — may consider discharge after ~2 hours (local practice varies).
- Observe ≥6 hours if ≥2 adrenaline doses were required.
- Observe ≥12 hours (or admit) if >2 doses required, severe respiratory compromise, protracted absorption risk (eg large intradermal exposure), poor access to emergency care or uncertain cause. [Resus Council UK 2021]
- On discharge: provide adrenaline auto-injector(s) (at least one; two often prescribed), educate on indications and use, give written action plan, counsel on avoidance of trigger, and arrange urgent specialist allergy follow-up (weeks).
- For ACEI angioedema: stop ACE inhibitor, document, advise permanent cessation, arrange follow-up; consider specialist therapies only per local guideline.
- For DRESS or immune-checkpoint toxicities: stop offending drug and arrange urgent specialist (dermatology/haematology/oncology) input; systemic steroids often required.

> **EXAM TRICK / تريكة امتحانية**
> Practical pitfalls and verification flags
> - Antihistamines/steroids do not replace adrenaline (lifesaving drug).
> - IV bolus adrenaline outside monitored/expert setting is risky — wrong answer in many exam vignettes.
> - Icatibant/C1 esterase concentrate for ACEI angioedema: variable evidence and availability—check local policy before use.

- Observation timeframes and discharge criteria vary by jurisdiction—confirm local guidance.

موجز بالعربية:
- أدوية رئيسية: أدرينالين عضلي فورًا (بالجرعات المذكورة)، سوائل، موسعات قصبية نيبولية، مضادات الهيستامين وستيرويد داعم.
- الأدرينالين الوريدي ومداخل الشرايين والإنعاش المتقدِّم للحالات المقاومة فقط.
- المراقبة والمدة تعتمد على شدة الاستجابة: ملاحظة قصيرة إذا استجابة فورية بعد جرعة واحدة؛ ≥6 أو ≥12 ساعة في حالات أكثر خطورة.
- عند الخروج: تعطى حقيبة أدرينالين تلقائي وتعليم ومتابعة حساسية.

6. High-yield FRCEM exam tricks and common pitfalls

- Adrenaline is first-line and time-critical; never choose antihistamine/steroid alone for life-threatening airway/circulatory compromise.
- Always lie the patient flat (unless respiratory distress/vomiting) — tested commonly.
- Know the IM adrenaline adult dose = 500 µg (0.5 mL of 1:1000) and paediatric age bands: 6–12 yrs = 300 µg; 6 months–6 yrs = 150 µg. Many questions test an 11-year-old = 300 µg. [Resus Council UK 2021]
- IV adrenaline is for refractory anaphylaxis under monitoring/expert help — selection of IV bolus in non-monitored scenario is usually wrong.
- Tryptase timing: peak 1–2 hours; baseline ≥24 hours — normal tryptase does not exclude anaphylaxis.
- Absence of urticaria does NOT exclude anaphylaxis (20% may lack skin signs) — don’t be misled by lack of rash.
- ACEI angioedema clues: no urticaria, poor response to antihistamines/steroids, and ongoing ACEI use — consider bradykinin mechanism.
- Beware of distractors: scombroid poisoning (fish history) presents like histamine reaction but management focuses on antihistamines/supportive care; DRESS and immunotherapy toxicities are delayed presentations — timing is key.
- Biphasic reaction risk increases when adrenaline administration is delayed or when initial reaction severe; watch for recurrence during observation period.

نصائح للامتحان بالعربية:
- الأدرينالين أولاً دائمًا؛ لا تستبدله بمضادات الهيستامين/الستيرويد للحالات المهدِّدة.

- احفظ جرعات الأدرينالين العضلي حسب الأعمار؛ تجنّب الأدرينالين الوريدي إلا بحضور خبرة.
- التربتيز داعم لا حاسم؛ غياب الشرَى لا ينفي التأق.

7. Arabic rapid-review summary

- التأق (anaphylaxis) = طارئ مناعي غالبًا IgE → إفراز وسيطات → هبوط ضغط، صفير، وذمة؛ العلاج الفوري أدرينالين عضلي في الفخذ.
- تمييز مهم: وذمة ACE-I (براديكينين) و DRESS وسمك سكامبروید وسمية مثبِّطات المناعة — التاريخ والوقت بعد المهيِّج يوجهان التشخيص.
- خطوات إنعاش سريعة: أوقف المهيِّج، مريض مستلقي، أكسجين، أدرينالين IM فورًا وكرر كل 5 دقائق، سوائل، استعد للمجرى الهوائي، تحويل للعناية المركزة إذا مقاوم.
- اختبارات داعمة: تربتيز (ذروة 1–2 ساعة؛ مكرر أساسى ≤24 ساعة).
- عند الخروج: حقيبة أدرينالين وتعليم ومتابعة حساسية.
- المراجع المحلية (Resuscitation Council UK 2021 وغيرها) للتفاصيل والبروتوكولات المحلية.

ملخّص بالعربية مختصر ومباشر:
- أدرينالين عضلي الآن → سوائل ودعم → مراقبة → تعليم ومتابعة.

8. Question coverage map (concepts covered)

EXAM TRICK / تريكة امتحانية

- No source Q-numbers were supplied. Concepts represented in supplied material (use this map to match likely FRCEM questions):
- Pathophysiology of anaphylaxis (IgE mast cell mediator release).
- Recognition criteria and clinical features (airway, breathing, circulation, skin).
- IM adrenaline dosing (adult and paediatric age bands) and timing (repeat every 5 minutes).
- Positioning of patient (lie flat with legs raised) and scene safety.
- Differential diagnoses: vasovagal, asthma, panic, MI, scombroid, ACEI angioedema, DRESS, checkpoint inhibitor toxicity.
- Diagnostics: clinical diagnosis, tryptase timing/interpretation, supportive blood tests.
- Emergency algorithm and refractory anaphylaxis management (IV adrenaline infusion as expert step).
- Adjunctive therapies: antihistamines, steroids (role and limitations), nebulised

bronchodilators.
- Specific syndromes: ACEI angioedema management considerations, scombroid, DRESS, immune-checkpoint toxicities.
- Observation, disposition and follow-up: auto-injector, allergy referral, observation durations per response.
- Exam-style pitfalls: substituting antihistamines for adrenaline, misinterpretation of tryptase, misuse of IV adrenaline.

Sources to verify
- Resuscitation Council (UK) — Anaphylaxis and Refractory Anaphylaxis algorithms, 2021.
- Local and national guidance: RCEM, ERC, NICE, local paediatric resuscitation and allergy services protocols.

REFERENCES AND GUIDANCE PORTALS

[1] Royal College of Emergency Medicine (RCEM), Clinical Standards and Guidance: https://rcem.ac.uk/clinical-standards-and-guidance/
[2] European Resuscitation Council (ERC), Guidelines 2025 English: https://www.erc.edu/science-research/guidelines/guidelines-2025/guidelines-2025-english/
[3] NICE, Emergency and acute medical care in over 16s (NG94): https://www.nice.org.uk/guidance/NG94
[4] Resuscitation Council UK, Adult and paediatric resuscitation guidance: https://www.resus.org.uk/
[5] Royal College of Paediatrics and Child Health, emergency and acute paediatric resources: https://www.rcpch.ac.uk/
[6] British Thoracic Society, respiratory guidelines: https://www.brit-thoracic.org.uk/
[7] British Society for Antimicrobial Chemotherapy and NICE antimicrobial guidance portals: https://www.nice.org.uk/guidance
[8] Toxbase / National Poisons Information Service access portal: https://www.toxbase.org/
[9] American College of Emergency Physicians, Clinical Policies: https://www.acep.org/patient-care/clinical-policies
[10] International Liaison Committee on Resuscitation (ILCOR): https://www.ilcor.org/

Reference note: The source file itself contains additional topic-specific RCEM, NICE, RCR, ENTUK, and other links. Chapter-level sources should be checked against the newest local and national versions before clinical implementation.